

Outcome annotation and minimax risk for discrete average treatment effects

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Abstract

This paper characterizes the minimax value of outcome annotation in a finite-alphabet causal experiment. A sample of n complete records (X, A, Y) is augmented by m additional records (X, A) from the same population, where X takes d known values, A and Y are binary, and occupied covariate cells satisfy overlap at level ϵ . For the average treatment effect under consistency and conditional exchangeability, the minimax mean-squared risk $R_\epsilon(n, m, d)$ satisfies, uniformly over the unrestricted cellwise model,

$$R_\epsilon(n, m, d) \asymp_\epsilon \min \left\{ 1, \frac{1}{n} + \frac{d^2}{(n+m)^2 \log^2(en)} \right\}.$$

The rate separates outcome information from treatment–covariate marginal information: the n^{-1} term is the labeled-outcome contribution, while the alphabet-dependent term is governed by the combined $n+m$ treatment–covariate records. The upper bound is attained by a fully specified hybrid estimator that uses polynomial correction in light cells and stabilized inverse counts in heavy cells. The lower bound uses common-marginal priors whose induced treatment–covariate tables have the same distribution across hypotheses, so the auxiliary sample is aligned with the marginal-information component. The characterization yields exact consistency, parametric-rate, strict-improvement, binary-alphabet, inverse-count-baseline, and known-marginal-limit consequences for the annotation experiment.

1 Introduction

Chart-reviewed outcomes are often the scarce resource in observational causal studies, while baseline covariates and treatment assignments are available for many more units. This paper studies the resulting allocation problem in a finite-alphabet causal model. There are n outcome-labeled records (X, A, Y) and m additional outcome-unlabeled records (X, A) , drawn independently from the same population. The covariate X takes values in a known alphabet of size d , treatment and outcome are binary, and every occupied covariate cell satisfies overlap at level ϵ . Under the usual potential-outcome consistency and conditional-exchangeability conditions, the target is the finite-alphabet average treatment effect $\tau(P)$ in [Definition 8](#).

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The main result gives the finite-sample minimax frontier for this annotation experiment. For every $0 < \epsilon < 1/2$, [Theorem 4](#) proves that the minimax mean-squared risk in [Definition 11](#) obeys

$$R_\epsilon(n, m, d) \asymp_\epsilon \min \left\{ 1, \frac{1}{n} + \frac{d^2}{(n+m)^2 \log^2(en)} \right\}$$

for all $n \geq 1$, $m \geq 0$, and $d \geq 2$. The rate has two resources. Outcome marks enter through the n^{-1} term. Treatment–covariate frequencies enter through the second term, where the relevant sample size is the pooled marginal-information size $n+m$. Thus the theorem gives a quantitative rule for how additional treatment–covariate records change the attainable accuracy of average-treatment-effect estimation in the unrestricted finite-cell model.

The upper bound is constructive. The estimator in [Algorithm 1](#) splits the labeled sample into outcome, pilot, and factorial pools, splits the auxiliary sample into pilot and factorial pools, and then combines the treatment–covariate records for marginal estimation. Cells with small pilot counts use a Chebyshev polynomial correction and a factorial-moment lift, following the polynomial-approximation ideas used in sharp discrete functional estimation [[Jiao et al., 2015](#), [Wu and Yang, 2016](#)] and the rare-cell perspective developed in sample amplification and unseen-species prediction [[Valiant and Valiant, 2017](#), [Orlitsky et al., 2016](#)]. Cells with larger pilot counts use stabilized inverse-count weights. [Theorem 2](#) shows that this total, finitely calibrated estimator attains the frontier rate uniformly over the model class.

The matching lower bound uses a common-marginal construction. [Theorem 3](#) supplies priors whose induced random treatment–covariate marginal P_{XA} , the full $2d$ -cell table, has the same distribution across the two hypotheses, while the outcome regressions shift the target. This aligns the converse with the two-sample annotation design: auxiliary observations reveal the same random marginal object under both branches, and the remaining testing difficulty produces the label floor and the rare-cell approximation term. A separate parametric label floor in [Lemma 22](#) supplies the n^{-1} component.

The frontier connects several familiar regimes. At $m = 0$, the rate becomes $\min\{1, n^{-1} + d^2/(n^2 \log^2(en))\}$. This refines the bracket of [Zeng et al. \[2026, Theorems 1–2 and the discussion following Theorem 2\]](#): their plug-in upper bound $d^2/n^2 + 1/n$ holds over the unrestricted binary-outcome, discrete-covariate overlap class, while their minimax lower bound $d^2/\{n^2 \log^2 n\} + 1/n$ is stated for $d \lesssim n \log n$; they leave closing the logarithmic gap by a polynomial estimator as future work. The present result supplies that construction and, with auxiliary records, replaces the marginal-information sample size n by $n+m$. With fixed n and d , [Theorem 5](#) shows that $R_\epsilon(n, m, d)$ converges as $m \rightarrow \infty$ to the known-marginal risk in [Definition 15](#), whose order is n^{-1} by [Theorem 1](#). For sequences, [Theorem 4](#) states that consistency is equivalent to $d_n/\{(n+m_n) \log(en)\} \rightarrow 0$, and that parametric risk is equivalent to $d_n = O((n+m_n) \log(en)/\sqrt{n})$. The same theorem gives the exact criterion under which auxiliary treatment–covariate records improve the asymptotic order relative to the labeled-only endpoint.

The paper contributes to causal estimation under potential outcomes [[Rubin, 1974](#), [Rosenbaum and Rubin, 1983](#), [Imbens and Rubin, 2015](#)], semiparametric and doubly robust average-treatment-effect estimation [[Robins et al., 1994](#), [Hahn, 1998](#), [Hirano et al., 2003](#), [Bang and Robins, 2005](#), [Farrell, 2015](#), [Athey et al., 2018](#), [Chernozhukov et al., 2018](#), [Kennedy, 2016](#)], and auxiliary-data methods for causal and distributional estimation, including semi-supervised treatment-effect, surrogate-outcome, missing-outcome, regression, and quantile settings [[Cheng et al., 2021](#), [Chakraborty et al., 2019](#), [Chakraborty and Dai, 2024](#), [Chakraborty et al., 2022](#), [Kallus and Mao, 2025](#), [Anniwaer and Zhang, 2025](#), [Xu et al., 2025](#)]. Those literatures characterize identification, regular asymptotics, efficiency, and gains from structure or auxiliary

measurements. The present analysis gives a finite-alphabet minimax benchmark for the binary-treatment, binary-outcome chart-review design under independent random outcome labeling, occupied-cell overlap, and unrestricted cellwise propensities and outcome regressions.

The appendix records supporting details for the displayed mathematical layer. The paper proceeds as follows. [Section 2](#) positions the result within causal inference, semi-supervised estimation, and non-smooth discrete functional estimation. [Section 3](#) defines the finite-alphabet annotation experiment, the causal target, and minimax risk. [Section 4](#) states the frontier theorem, the known-marginal benchmark, and the common-marginal converse. [Section 5](#) gives the algorithmic construction behind the upper bound. [Section 6](#) discusses the resulting regimes, the inverse-count baseline, the binary-alphabet specialization, and future directions. [Sections A to C](#) collect the upper-bound ingredients, common-marginal lower bounds, known-marginal limit, verification note, and proofs.

2 Related work

This paper studies a two-sample semi-supervised discrete causal experiment: a complete outcome-labeled sample of size n is augmented by an outcome-unlabeled treatment-covariate sample of size m on a finite covariate alphabet of size d . In that experiment, the target is the average treatment effect under the usual potential-outcomes interpretation, while the statistical difficulty is governed by how outcome labels and treatment-covariate cell information are split across the two samples. The contribution is therefore most closely connected to five literatures: identification in potential-outcomes models, semiparametric efficiency for average treatment effects with missing or observed outcomes, high-dimensional causal inference under structure, semi-supervised treatment-effect estimation, and minimax estimation of non-smooth discrete functionals.

The identification framework follows the potential-outcomes tradition of [Rubin \[1974\]](#) and the propensity-score formulation of [Rosenbaum and Rubin \[1983\]](#); see also [Imbens and Rubin \[2015\]](#) for a systematic account. In that tradition, consistency and conditional exchangeability connect observed outcome regressions to the causal estimand. The present analysis specializes this framework to a finite covariate space and asks how accurately the resulting discrete average treatment effect can be estimated when additional observations reveal treatment and covariates but omit the outcome.

A large semiparametric literature characterizes regular estimation of average treatment effects under missingness, inverse weighting, augmentation, and doubly robust structure. Classical efficiency calculations include [Robins et al. \[1994\]](#), [Hahn \[1998\]](#), and [Hirano et al. \[2003\]](#), with augmented and robust estimation developed further by [Bang and Robins \[2005\]](#). Modern high-dimensional versions combine orthogonal scores with flexible first-stage learning, as in [Farrell \[2015\]](#), [Athey et al. \[2018\]](#), and [Chernozhukov et al. \[2018\]](#); [Kennedy \[2016\]](#) gives a broad review of semiparametric causal estimation. Those results identify regular-asymptotic regimes in which nuisance estimation error can be controlled through smoothness, sparsity, sample splitting, or orthogonality. Here the object is the finite-alphabet minimax risk in an unrestricted discrete model, where rare treatment-covariate cells create a non-smooth estimation problem.

Work on high-dimensional and structured semi-supervised inference provides a complementary perspective. [Zhang et al. \[2019\]](#), [Chakraborty and Cai \[2018\]](#), and [Deng et al. \[2024\]](#) develop semi-supervised methods for means and regression targets in high-dimensional settings. Related robustness and prediction-assisted ideas appear in [Angelopoulos et al. \[2023\]](#). These

Work	Auxiliary information		Regime and target		Guarantee	
Zeng et al. [2026]	none		discrete	covariates; ATE	supervised bracket	minimax
Hou et al. [2025]	partially treatment and response	annotated and re-	structured models; ATE	nuisance	root- n inference and efficiency	
Kato [2025]	unlabeled covariates		two-sample model; treatment effects	Riesz	efficiency characterization	
This paper	unlabeled records	(X, A)	unrestricted cells; ATE	finite	exact finite-sample minimax order in n, m, d	

Table 1: Closest comparisons by auxiliary information, structural regime, target, and inferential guarantee.

papers show how auxiliary samples and predictive structure can improve estimation in rich covariate spaces. The present setting uses the finite alphabet itself as the primitive statistical object and characterizes the risk uniformly over the unrestricted cell-level causal model.

A related auxiliary-data literature studies how unlabeled samples, missing-outcome structure, transfer information, regression targets, quantile targets, and surrogate outcomes improve estimation. Cheng et al. [2021], Hou et al. [2025, Sections 2–4], Kato [2025, Sections 3–5], Chakraborty and Dai [2024], Anniwaer and Zhang [2025], and Xu et al. [2025] develop semi-supervised, transfer-style, or regular-asymptotic causal estimators, while Chakraborty et al. [2019], Chakraborty et al. [2022], and Kallus and Mao [2025] analyze missing-outcome, quantile, and surrogate-outcome settings. Hou, Mukherjee, and Cai establish root- n inference and efficiency for partially annotated treatment and response under structured nuisance estimation; Kato studies a two-sample unlabeled-covariate design through generalized Riesz regression and characterizes its efficiency bound. The present paper instead takes an unrestricted finite-cell model and asks for the global finite-sample minimax risk, including rare-cell regimes where regular asymptotics need not apply.

The discrete nature of the model also connects the analysis to minimax estimation of non-smooth functionals of distributions. For entropy, support-size, and related functionals, Jiao et al. [2015] and Wu and Yang [2016] show that polynomial approximation helps determine sharp rates, while Valiant and Valiant [2017] and Orlitsky et al. [2016] develop sample-amplification and unseen-species perspectives on rare cells. The average treatment effect in the annotation experiment has an analogous cell-wise difficulty: estimating outcome means requires labeled observations, while estimating the treatment-covariate weights benefits from the larger treatment-covariate sample. The resulting minimax rate reflects that interaction between labeled outcome information and auxiliary marginal information.

The most direct benchmark is Zeng et al. [2026, Theorems 1–2 and the discussion following Theorem 2], who study the same unrestricted binary-outcome, discrete-covariate overlap class with only complete records. Their plug-in upper bound holds for all d , whereas their minimax lower bound is stated for $d \lesssim n \log n$; the two differ by a logarithmic factor in the alphabet-dependent term. The present paper closes that gap at the supervised endpoint and extends the problem to separate outcome-labeled and outcome-unlabeled samples. Table 1 summarizes the sampling design, structural regime, target, and guarantee for the nearest comparisons.

3 Model and experiment

Throughout the paper, x indexes a finite covariate cell, $a \in \{0, 1\}$ indexes a treatment arm, and sums over cells run over the known alphabet $[d] = \{1, \dots, d\}$. Generic constants may depend on the overlap level ϵ when this dependence is indicated, while sample-size dependence is displayed explicitly. Norms and asymptotic notation are used in their standard finite-dimensional sense, and all probability laws below live on finite measurable spaces unless stated otherwise.

The statistical object is an observed record with a finite covariate, a binary treatment, and a binary outcome. Writing X for the covariate, A for treatment, and Y for the observed outcome, the complete record takes values in the following finite space.

Definition 1. [synth_11] (Complete observed-record space) *For each natural number d , define the complete observed-record space by*

$$\text{Obs}(d) := [d] \times \{0, 1\} \times \{0, 1\}.$$

An element of $\text{Obs}(d)$ is a triple (X, A, Y) , where $X \in [d]$ is the finite-alphabet covariate, $A \in \{0, 1\}$ is the binary treatment indicator, and $Y \in \{0, 1\}$ is the binary observed outcome.

Let P denote a probability law on this observed record space. For a cell x , the covariate mass p_x records how much population probability lies in that cell.

Definition 2. [synth_16] (Covariate cell mass) *For an observed-data law P on $[d] \times \{0, 1\}^2$ and a covariate value $x \in [d]$, define the covariate cell mass*

$$p_x = \sum_{a \in \{0, 1\}} \sum_{y \in \{0, 1\}} P(X = x, A = a, Y = y).$$

We also write s_{ax} for the treatment-covariate mass in arm a . The treatment propensity e_x is the conditional treatment probability in the cell, with the finite-space convention made explicit next.

Definition 3. [synth_10] (Cell treatment propensity) *For an observed-data law P on $[d] \times \{0, 1\}^2$ and a covariate cell $x \in [d]$, define the treatment propensity e_x by*

$$e_x := \frac{\sum_{y \in \{0, 1\}} P(X = x, A = 1, Y = y)}{\sum_{a \in \{0, 1\}} \sum_{y \in \{0, 1\}} P(X = x, A = a, Y = y)}.$$

The ratio is interpreted with the usual totalized real division convention, so it is 0 when the denominator is 0.

Assumption 1. [ass:overlap] (Occupied-cell overlap) *For every covariate cell $x \in [d]$ with cell mass $p_x > 0$, the propensity score satisfies*

$$\epsilon \leq e_x \leq 1 - \epsilon.$$

The occupied-cell overlap condition in [Assumption 1](#) is the standard strong-overlap restriction on cells that occur with positive probability [[Zeng et al., 2026](#)]. It keeps both treatment arms represented wherever the covariate distribution places mass and therefore gives the outcome regressions in the ATE functional a common observed-data interpretation.

The experiment combines complete outcome-labeled records with additional treatment-covariate records. We first record the sampling law for the labeled observations and then define the marginal law governing the auxiliary observations.

Assumption 2. [ass:labeled-iid] (Labeled sampling) *The labeled sample is distributed as*

$$\mathcal{L}_n \sim P^{\otimes n}.$$

The labeled sampling condition in [Assumption 2](#) is the standard independent and identically distributed sampling condition for complete records [[Zeng et al., 2026](#)]. These observations supply the outcome information used to learn the arm-specific regressions.

Definition 4. [synth_14] (Treatment-covariate marginal) *For any d and any observed-data law P on $[d] \times \{0, 1\}^2$, the treatment-covariate marginal P_{XA} is the probability mass function on $[d] \times \{0, 1\}$ obtained by pushing P forward under the projection $(x, a, y) \mapsto (x, a)$:*

$$P_{XA} = P \circ ((x, a, y) \mapsto (x, a))^{-1}.$$

The marginal P_{XA} records the $2d$ treatment-covariate cell probabilities while integrating out the outcome. The auxiliary sample $\mathcal{U}_m$ is drawn from this same population marginal.

Assumption 3. [ass:auxiliary-iid] (Auxiliary sampling) *The auxiliary sample is distributed as*

$$\mathcal{U}_m \sim P_{XA}^{\otimes m}.$$

[Assumption 3](#) is the standard same-population outcome-unlabeled sampling condition [[Chakraborty and Dai, 2024](#)]. It lets the auxiliary records refine information about the treatment-covariate distribution relevant for weighting cell-level outcome contrasts.

Assumption 4. [ass:sample-independence] (Sample independence) *The labeled and auxiliary samples are independent:*

$$\mathcal{L}_n \perp\!\!\!\perp \mathcal{U}_m.$$

The independent two-sample design in [Assumption 4](#) is standard in semisupervised treatment-effect settings with separate labeled and outcome-unlabeled samples [[Chakraborty and Dai, 2024](#)]. Under this design, the product experiment separates outcome information from additional marginal information.

The causal interpretation uses a finite potential-outcome overlay on the same observed-data law. Let $Y(a)$ denote the binary potential outcome under arm a ; the following assumptions connect the observed-data functional to the causal average treatment effect in the usual potential-outcome framework [[Rubin, 1974](#), [Imbens and Rubin, 2015](#)].

Assumption 5. [ass:consistency] (Potential-outcome consistency) *The observed binary outcome equals the potential outcome under the realized treatment:*

$$Y = Y(A)$$

almost surely.

[Assumption 5](#) is the standard consistency condition [[Zeng et al., 2026](#)]. It identifies the observed outcome within each treatment arm with the corresponding potential outcome.

Assumption 6. [ass:conditional-exchangeability] (Conditional exchangeability) *The potential outcomes are conditionally independent of treatment given the finite covariate:*

$$(Y(0), Y(1)) \perp\!\!\!\perp A \mid X.$$

[Assumption 6](#) is the standard conditional exchangeability condition [[Zeng et al., 2026](#)]. Together with [Assumptions 1](#) and [5](#), it gives the observed conditional means their usual causal meaning within each finite covariate cell [[Rosenbaum and Rubin, 1983](#), [Imbens and Rubin, 2015](#)].

More explicitly, let Q be a full-data law for $(X, A, Y(0), Y(1))$, and let P be its observed-data pushforward under

$$(x, a, y_0, y_1) \mapsto (x, a, y_a).$$

Consistency and conditional exchangeability imply, for every occupied cell x and arm a ,

$$\mathbb{E}_Q\{Y(a) \mid X = x\} = \mathbb{E}_Q\{Y(a) \mid X = x, A = a\} = \mathbb{E}_P\{Y \mid X = x, A = a\}.$$

Overlap makes the last conditional mean identified in every occupied cell. Therefore

$$\mathbb{E}_Q\{Y(1) - Y(0)\} = \sum_{x=1}^d p_x [\mathbb{E}_P\{Y \mid X = x, A = 1\} - \mathbb{E}_P\{Y \mid X = x, A = 0\}],$$

which is exactly the observed-data functional defined below. Thus the minimax analysis is over observed laws P , while the causal interpretation is supplied by any compatible full-data law Q satisfying these assumptions.

We now collect the finite-space probability notation, the model class, and the target functional used in the risk analysis.

Definition 5. [[synth_13](#)] (Probability laws on a measurable space) *For a measurable space $\mathcal{Z}$, let $\text{Prob}(\mathcal{Z})$ denote the set of all probability measures on $\mathcal{Z}$. For the finite spaces used below, including $[d] \times \{0, 1\}^2$ and $[d] \times \{0, 1\}$, the measurable structure is the full power set.*

Definition 6. [[def:model-class](#)] (Model class $\mathcal{M}_{d,\epsilon}$) *For an overlap constant ϵ and alphabet size d , define*

$$\mathcal{M}_{d,\epsilon} := \left\{ P \in \text{Prob}([d] \times \{0, 1\}^2) : d \geq 2, 0 < \epsilon < 1/2, P \text{ satisfies } \text{Assumption 1} \right\}.$$

Thus $\mathcal{M}_{d,\epsilon}$ is the class of observed-data laws on the finite alphabet satisfying occupied-cell overlap at level ϵ . All suprema in the main risk statements range over this unrestricted cell-level model.

For each arm and cell, the outcome regression μ_{ax} is the conditional success probability computed from the observed law.

Definition 7. [[synth_17](#)] (Outcome regression) *For a law P on $[d] \times \{0, 1\}^2$, a treatment arm $a \in \{0, 1\}$, and a covariate cell $x \in [d]$, define*

$$\mu_{ax}(P) := \frac{P(X = x, A = a, Y = 1)}{\sum_{y \in \{0, 1\}} P(X = x, A = a, Y = y)}.$$

The quotient is interpreted as total real division, so $\mu_{ax}(P) = 0$ when the treatment-covariate cell mass $\sum_{y \in \{0, 1\}} P(X = x, A = a, Y = y)$ is zero. In particular, $\mu_{1x}(P)$ and $\mu_{0x}(P)$ are obtained by taking $a = 1$ and $a = 0$, respectively.

Definition 8. [def:ate-functional] (Average treatment effect functional) *For a finite alphabet size d and an observed-data law P on $[d] \times \{0, 1\}^2$, define the average treatment effect functional by*

$$\tau(P) := \sum_{x=1}^d p_x(P) (\mu_{1x}(P) - \mu_{0x}(P)).$$

Here $p_x(P)$ is the covariate cell mass at x , and $\mu_{ax}(P)$ is the binary outcome regression in treatment arm a and covariate cell x , totalized by division in the arm cell.

The functional $\tau(P)$ averages the within-cell treated-control contrast using the population covariate distribution. Under the potential-outcome conditions above, it is the finite-alphabet causal average treatment effect.

Finally, the annotation experiment and minimax risk formalize the statistical problem studied in the rest of the paper.

Definition 9. [def:annotation-experiment] (Annotation experiment $\mathcal{E}_{n,m,d,\epsilon}$) *Define the annotation experiment by*

$$\mathcal{E}_{n,m,d,\epsilon} := \left\{ P^{\otimes n} \otimes P_{XA}^{\otimes m} : P \in \mathcal{M}_{d,\epsilon}, n \geq 1, m \geq 0, d \geq 2 \right\}.$$

The product law corresponds to n independent complete outcome-labeled records and m independent additional outcome-unlabeled treatment-covariate records, with independence between the two samples.

Definition 9 defines $\mathcal{E}_{n,m,d,\epsilon}$, the experiment in which outcome labels and treatment-covariate information enter through separate independent samples. This formulation makes the amount of outcome information and the amount of marginal information explicit.

The realized data live on the corresponding product sample space.

Definition 10. [synth_18] (Joint sample space) *For natural numbers n, m, d , the joint sample space is*

$$\text{Sample}(n, m, d) = ([n] \rightarrow [d] \times \{0, 1\} \times \{0, 1\}) \times ([m] \rightarrow [d] \times \{0, 1\}).$$

Equivalently, an element consists of n complete records (X, A, Y) and m outcome-unlabeled records (X, A) .

Definition 11. [def:minimax-risk] (Minimax risk $R_\epsilon(n, m, d)$) *Define the minimax mean-squared risk by*

$$R_\epsilon(n, m, d) := \inf_{T \in \mathcal{T}_{n,m,d}} \sup_{P \in \mathcal{M}_{d,\epsilon}} E_{P^{\otimes n} \otimes P_{XA}^{\otimes m}} \left[(T - \tau(P))^2 \right],$$

where $\mathcal{T}_{n,m,d}$ is the class of estimators measurable with respect to the labeled sample and auxiliary sample.

The risk $R_\epsilon(n, m, d)$ measures the best possible uniform mean-squared accuracy for estimating the ATE functional in Definition 8 under the experiment in Definition 9. The estimator class $\mathcal{T}_{n,m,d}$ contains all measurable procedures based on the two samples, so the minimax risk isolates the statistical difficulty created by finite cells, overlap, and the split between outcome-labeled and outcome-unlabeled information.

4 Main minimax results

The minimax analysis separates two sources of information in the annotation experiment. Outcome marks contribute through the n complete records, while treatment–covariate frequencies contribute through the combined table information carried by the n labeled records and the m auxiliary records. The resulting rate uses the total marginal-information size N and the labeled-sample logarithmic scale.

Definition 12. `[def:frontier-rate-handle]` (Frontier rate $r_\epsilon(n, m, d)$) *Put*

$$N = n + m, \quad \ell_n = \log(en).$$

The unequal-information annotation-frontier rate is

$$r_\epsilon(n, m, d) := \min \left\{ 1, \frac{1}{n} + \frac{d^2}{N^2 \ell_n^2} \right\}.$$

The rate $r_\epsilon(n, m, d)$ has two components. The term $1/n$ is the labeled-outcome contribution. The term $d^2/(N^2 \ell_n^2)$ is the discrete marginal-estimation contribution, where the alphabet dimension is compared with the amount of treatment–covariate information available from both samples.

The upper construction and the converse both use finite calibration devices. For the estimator, the calibration fixes dyadic overlap, polynomial degree, light-cell scale, and the data splits used below.

Definition 13. `[synth_7]` (Finite calibration scales for the mixed estimator) *Set*

$$A_\star = 7056, \quad c^\circ = \frac{1}{256}.$$

For $0 < \epsilon < 1/2$, let j_ϵ be the least integer $j \geq 1$ such that $2^{-j} \leq \epsilon$, and define the dyadic overlap certificate

$$\bar{\epsilon}(\epsilon) = 2^{-j_\epsilon}.$$

Then $0 < \bar{\epsilon}(\epsilon) \leq \epsilon < 2\bar{\epsilon}(\epsilon)$. Let H_ϵ be the least positive integer H satisfying

$$H \geq 4, \quad \frac{H}{4} \geq 15 + \frac{24}{c^\circ}, \quad Hc^\circ \bar{\epsilon}(\epsilon) \geq 672, \quad Hc^\circ \bar{\epsilon}(\epsilon) \geq 24.$$

For $n \geq 1$, put

$$h_n = 1 + \lceil \log_2 n \rceil, \quad L(n) = \max\{2, \lfloor c^\circ h_n \rfloor\}.$$

For $n \geq 24$ and $m \geq 0$, define

$$\begin{aligned} M_0 &= \lfloor n/3 \rfloor, & n_p &= \left\lfloor \frac{n - M_0}{2} \right\rfloor, & n_f &= n - M_0 - n_p, \\ m_p &= \lfloor m/2 \rfloor, & m_f &= m - m_p, & M_p &= n_p + m_p, & M_f &= n_f + m_f, \end{aligned}$$

and

$$t_p = \frac{M_p}{8}, \quad t = \frac{M_f}{8}.$$

The light-cell scale and pilot-count threshold are

$$B(n, m, \epsilon) = \frac{H_\epsilon L(n)}{\min(t_p, t)}, \quad k_0(n, m, \epsilon) = \left\lfloor \frac{t_p B(n, m, \epsilon)}{4} \right\rfloor.$$

For $n < 24$, the calibration predicate using these quantities is declared false.

The lower-bound argument compares two randomized laws through their induced distribution over the treatment–covariate table. The following notation records that marginal distribution for a branch prior.

Definition 14. [synth_12] (Induced law of the treatment–covariate marginal) *Let $b \in \{0, 1\}$ and let Π_b be a probability measure on $\mathcal{M}_{d,\epsilon}$. If $P \sim \Pi_b$, define*

$$\mathcal{L}_{\Pi_b}(P_{XA}) := (P \mapsto P_{XA})_{\#}\Pi_b.$$

Thus, for every measurable set $B \subseteq \text{Prob}([d] \times \{0, 1\})$,

$$\mathcal{L}_{\Pi_b}(P_{XA})(B) = \Pi_b\{P \in \mathcal{M}_{d,\epsilon} : P_{XA} \in B\}.$$

Equivalently, $\mathcal{L}_{\Pi_b}(P_{XA})$ is the distribution of the full $2d$ -cell treatment–covariate marginal table induced when the complete-data law P is drawn from Π_b .

A useful benchmark gives the target risk when the treatment–covariate marginal is observed exactly. In that experiment, procedures may use the labeled sample together with the exact P_{XA} table.

Definition 15. [def:known-marginal-risk] (Known-marginal minimax risk) *For nonnegative integers n and d and a real overlap parameter ϵ , the known-marginal minimax risk is*

$$R_{\epsilon}^{\text{known}}(n, d) := \inf_{T \in \mathcal{T}_{n,d}^{\text{known}}} \sup_{P \in \mathcal{M}_{d,\epsilon}} \int (T(\mathcal{L}_n, P_{XA}) - \tau(P))^2 dP^{\otimes n}.$$

Here $\mathcal{M}_{d,\epsilon}$ is the model class in Definition 6, $\tau(P)$ is the average treatment effect functional in Definition 8, and $\mathcal{T}_{n,d}^{\text{known}}$ is the class of measurable estimators T based on the labeled sample together with the exact treatment-covariate table P_{XA} , with $T(\mathcal{L}_n, P_{XA}) \in [-1, 1]$ for every input.

Here $\mathcal{T}_{n,d}^{\text{known}}$ is the estimator class for the known-marginal problem, and $R_{\epsilon}^{\text{known}}(n, d)$ records the corresponding minimax mean-squared error.

Theorem 1. [thm:known-marginal-boundary] (Known marginal boundary) *Suppose the overlap constant ϵ satisfies $0 < \epsilon < 1/2$. For the known-marginal minimax risk $R_{\epsilon}^{\text{known}}(n, d)$ defined in Definition 15, there exist real constants c_{ϵ} and C_{ϵ} with $0 < c_{\epsilon} \leq C_{\epsilon}$ such that, for every $n, d \in \mathbb{N}$ with $n \geq 1$ and $d \geq 2$,*

$$\frac{c_{\epsilon}}{n} \leq R_{\epsilon}^{\text{known}}(n, d) \leq \frac{C_{\epsilon}}{n}.$$

The benchmark establishes the labeled-sample boundary when the full marginal table is available. Once P_{XA} is known, the remaining uniform risk is of order $1/n$, so additional treatment–covariate observations move the original annotation problem toward this labeled-outcome scale.

The upper bound uses an explicit hybrid estimator. Its calibration first fixes a binary-length scale and a clipping map for the final statistic. The long *Context* item in the next generated definition records formal dependencies and may be skipped on a first reading. Also, the type $\mathcal{T}_{n,m,d}$ in the estimator display denotes precisely the map from the two sample spaces to $\mathbb{R}$ displayed immediately below it.

Definition 16. [synth_8] (Binary length b_n)

- **(Context.)** Work under [Definitions 1 to 15](#) and [17 to 30](#), [Assumptions 1 to 6](#), [Algorithm 1](#), [Theorems 1 to 7](#), and [Lemmas 1 to 29](#).
- **(Integer.)** Let $n \geq 1$ be an integer.

Define

$$b_n := 1 + \lceil \log_2 n \rceil.$$

Equivalently, in the notation of [Definition 13](#), $b_n = h_n$.

Definition 17. `[synth_3]` (Clipping to the unit interval) For $z \in \mathbb{R}$, define

$$\text{clip}_{[-1,1]}(z) = \max\{-1, \min\{1, z\}\}.$$

Equivalently, $\text{clip}_{[-1,1]}(z)$ is the Euclidean projection of z onto the interval $[-1, 1]$.

The estimator itself combines a light-cell polynomial correction with a heavy-cell inverse-count correction. The construction is total: outside its finite calibration region it returns zero, and inside that region it averages a clipped Poisson-prefix statistic. For orientation before the full display, if

$$W_{a,L}(s_0, s_1) = \sum_{r_0, r_1} c_{a, r_0, r_1} s_0^{r_0} s_1^{r_1},$$

then its factorial lift at intensity t is

$$\widehat{W}_{a,L}(K_0, K_1) = \sum_{r_0, r_1} c_{a, r_0, r_1} \frac{(K_0)_{r_0} (K_1)_{r_1}}{t^{r_0+r_1}}.$$

The complete definition and its moment identity appear in [Definition 24](#).

Algorithm 1 (Mixed estimator definition)

For $n, m, d \in \mathbb{N}$ and a real overlap constant ϵ , define

$$\widehat{\tau}_{n,m,d}^{\text{mix}} : \mathcal{T}_{n,m,d}$$

as the following map from

$$([d] \times \{0, 1\} \times \{0, 1\})^n \times ([d] \times \{0, 1\})^m$$

to $\mathbb{R}$. Put

$$\begin{aligned} M_0 &= \lfloor n/3 \rfloor, & n_p &= \left\lfloor \frac{n - M_0}{2} \right\rfloor, & n_f &= n - M_0 - n_p, \\ m_p &= \lfloor m/2 \rfloor, & m_f &= m - m_p, & M_p &= n_p + m_p, & M_f &= n_f + m_f, \end{aligned}$$

and

$$u = \frac{M_0}{8}, \quad t_p = \frac{M_p}{8}, \quad t = \frac{M_f}{8}.$$

Let b_n denote the binary-length calibration quantity used here, and set

$$L = L(n), \quad B = B(n, m, \epsilon), \quad k_0 = k_0(n, m, \epsilon),$$

with $\bar{\epsilon} = \bar{\epsilon}(\epsilon)$, $A_\star = 7056$, and $c^\circ = 1/256$. The calibration predicate $C_\epsilon(n, m)$ holds exactly when all of the following finite inequalities hold:

- (Sample size.) $24 \leq n$.
- (Block proportions.)

$$\frac{1}{32} \leq \frac{u}{n} \leq \frac{1}{24}, \quad \frac{1}{32} \leq \frac{t_p}{n+m} \leq \frac{3}{32}, \quad \frac{1}{32} \leq \frac{t}{n+m} \leq \frac{3}{32}, \quad \frac{1}{3} \leq \frac{t}{t_p} \leq 3.$$

- (Degree and scale.)

$$\frac{c^\circ b_n}{2} \leq L, \quad 8 \leq t_p B, \quad L \leq tB.$$

- (Polynomial size.)

$$L^4 A_\star^{2L} \leq n, \quad L^6 A_\star^{2L} \leq n.$$

- (Tail calibration.)

$$15L + 12b_n \leq \frac{t_p B}{4}, \quad n^{12} \leq \left(1 + \bar{\epsilon} \frac{t}{t_p}\right)^{k_0}, \quad 12b_n \leq \bar{\epsilon} t B.$$

If $C_\epsilon(n, m)$ fails, set

$$\hat{\tau}_{n,m,d}^{\text{mix}}(\mathcal{L}_n, \mathcal{U}_m) = 0.$$

If $C_\epsilon(n, m)$ holds, set

$$\hat{\tau}_{n,m,d}^{\text{mix}}(\mathcal{L}_n, \mathcal{U}_m) = \sum_{r_0=0}^{M_0} \sum_{r_p=0}^{M_p} \sum_{r_f=0}^{M_f} \Pr\{\text{Poi}(u) = r_0\} \Pr\{\text{Poi}(t_p) = r_p\} \Pr\{\text{Poi}(t) = r_f\} \Phi_\epsilon(\mathcal{L}_n, \mathcal{U}_m; r_0, r_p, r_f),$$

where

$$\Phi_\epsilon(\mathcal{L}_n, \mathcal{U}_m; r_0, r_p, r_f) = \text{clip}_{[-1,1]} \left(\sum_{x \in [d]} Z_x \right).$$

For each $x \in [d]$, let J_x be the pilot-prefix count in cell x , let S_{ax} be the outcome-block prefix count with covariate x , treatment a , and outcome 1, and let K_{ax} be the factorial-pool prefix count with covariate x and treatment a . Then

$$Z_x = \begin{cases} \frac{S_{1x}}{u} \widehat{W}_{1,L,x} - \frac{S_{0x}}{u} \widehat{W}_{0,L,x}, & J_x \leq k_0, \\ \frac{S_{1x}}{u} \frac{K_{0x} + K_{1x} + 1}{K_{1x} + 1} - \frac{S_{0x}}{u} \frac{K_{0x} + K_{1x} + 1}{K_{0x} + 1}, & k_0 < J_x. \end{cases}$$

Here $\widehat{W}_{a,L,x}$ is the factorial lift, at degree L , scale B , and intensity t , of the light-cell polynomial weight

$$W_{a,L}(s_0, s_1) = \frac{s_0 + s_1}{B} G_L(s_a/B),$$

where

$$H_L(z) = \frac{1 - T_L(1 - 2z)}{2L^2}, \quad E_L(z) = \frac{H_L(z)}{z}, \quad G_L(z) = \frac{1 - E_L(z)}{z},$$

with polynomial continuation at zero.

The statistic $\hat{\tau}_{n,m,d}^{\text{mix}}$ uses labeled outcomes through the S_{ax} counts and uses the larger treatment–covariate pool through the pilot and factorial counts. Light cells receive the polynomial correction familiar from discrete functional estimation [Jiao et al., 2015, Wu and Yang, 2016], while heavy cells use direct inverse-count weighting.

Theorem 2. [thm:uniform-mixed-upper] (Uniform mixed upper bound) *Fix an overlap constant ϵ with $0 < \epsilon < 1/2$. There exists a constant $C_\epsilon > 0$ such that, for every $n \geq 1$, $m \geq 0$, and $d \geq 2$, the total computable hybrid estimator $\hat{\tau}_{n,m,d}^{\text{mix}}$ of Algorithm 1 satisfies*

$$\sup_{P \in \mathcal{M}_{d,\epsilon}} E_{P^{\otimes n} \otimes P_{XA}^{\otimes m}} \left[\left\{ \hat{\tau}_{n,m,d}^{\text{mix}} - \tau(P) \right\}^2 \right] \leq C_\epsilon r_\epsilon(n, m, d),$$

where $\tau(P)$ is the average treatment effect functional in Definition 8, and $r_\epsilon(n, m, d)$ is the annotation-frontier rate in Definition 12.

The upper bound establishes that the displayed rate is attainable uniformly over the model class. Its form reflects the experiment: increasing m improves the marginal-frequency term through $N = n + m$, while the labeled-outcome term remains governed by n . Here *total computable* means that the displayed finite sums define an everywhere-defined statistic, including at calibration failure; implementation-oriented scope is discussed in Section 6.

The matching converse is organized through priors with a common induced marginal. In the rare-cell regime, the construction below generates a shared random treatment–covariate table and perturbs the treated outcome regression across two branches.

Definition 18. [synth_22] (Common-marginal finite-atom construction) *Fix $0 < \epsilon < 1/2$, an integer $L \geq 2$, scales $B > 0$ and $a > 0$, and write*

$$\kappa = \frac{1 - 2\epsilon}{\epsilon}, \quad I = [a/\kappa, B].$$

Let σ be a finite signed measure on I with total variation one, let $h = d\sigma/d|\sigma|$, and define a probability law ν on $\{0\} \cup I$ by

$$\nu(dp) = \frac{a}{p+a} |\sigma|(dp) \quad (p \in I), \quad \nu(\{0\}) = 1 - \int_I \frac{a}{p+a} |\sigma|(dp).$$

For an integer $k \leq d - 1$, the finite-atom construction has atom-generator law

$$\Gamma = \nu^{\otimes k}$$

on $w = (p_1, \dots, p_k)$. Given w , form k rare cells and one reservoir cell. For each rare cell with $p_i > 0$, set

$$s_i = \epsilon(p_i + a), \quad e_i = \frac{s_i}{p_i},$$

put $\mu_{0i} = 0$, and set the treated outcome regression in branch $b \in \{0, 1\}$ to

$$\mu_{1i}^{(b)} = \frac{1 + (2b - 1)h(p_i)}{2}.$$

At atoms with $p_i = 0$, assign the same legal propensity and outcome values in both branches. The reservoir has raw mass

$$p_*(w) = 1 - kE_\nu[p],$$

propensity $1/2$, and outcome regressions equal to zero. Alphabet cells not used by the k rare cells or the reservoir have mass zero.

The raw total mass is

$$S(w) = p_*(w) + \sum_{i=1}^k p_i.$$

The branch law $P_b(w)$ is obtained by dividing every raw cell mass by $S(w)$ and keeping the propensities and outcome regressions just specified. The corresponding raw target is

$$r_b(w) = \sum_{i=1}^k p_i \mu_{1i}^{(b)},$$

so the normalized target value of the generated law is $r_b(w)/S(w)$. The branch priors are the pushforwards

$$\Pi_b = \Gamma \circ P_b^{-1}, \quad b \in \{0, 1\}.$$

Thus the two branches use the same generated rare-cell masses, reservoir mass, propensities, normalization factor, and treatment-covariate table, while differing only in the branch sign of the treated rare-cell outcome regressions.

The finite-atom construction makes the shared marginal table explicit: the branches draw the same rare-cell masses and propensities, then separate the targets through outcome regressions. The common-marginal handle chooses between this rare-cell recipe and a parametric point-mass regime according to the comparison in the following definition.

Definition 19. `[def:common-marginal-prior-handle]` (Common-marginal prior handle) Fix $n, m, d \in \mathbb{N}$, $\epsilon \in \mathbb{R}$, and a calibration consisting of constants

$$C_\epsilon, \quad C_\epsilon^\circ, \quad b_0, \quad b_\epsilon, \quad \gamma_\epsilon, \quad c_\epsilon^\circ, \quad c_\epsilon, \quad C_d$$

with all eight constants positive and $c_\epsilon \leq 1/4$. A common-marginal prior handle at (n, m, d) consists of integers L, k , real numbers B, a , and a calibrated recipe supplying two induced priors, denoted $\Pi_0(H)$ and $\Pi_1(H)$, on laws on $[d] \times \{0, 1\}^2$, subject to

$$L = \lceil C_\epsilon^\circ \log(en) \rceil, \quad B = \frac{b_0 L}{n + m}, \quad a = \frac{\gamma_\epsilon B}{L^2},$$

and

$$k = \min \{d - 1, \lfloor c_\epsilon^\circ(n + m)L \rfloor\}.$$

The recipe uses the finite-atom construction exactly when

$$\min \left\{ 1, \frac{d^2}{(n + m)^2 \log(en)^2} \right\} > \frac{C_\epsilon}{n}.$$

For a handle H and a branch $b \in \{0, 1\}$, the common-marginal prior is the branch-indexed measure

$$\Pi_b(H) = \begin{cases} \Pi_1(H), & b = 1, \\ \Pi_0(H), & b = 0. \end{cases}$$

The two branches Π_0, Π_1 are calibrated so that the auxiliary sample sees the same random marginal table under both hypotheses. Hence the distinguishing information in the lower-bound experiment must pass through the labeled outcomes.

Theorem 3. [thm:common-marginal-converse] (Common marginal converse) *Let ϵ , the overlap parameter, satisfy $0 < \epsilon < 1/2$. Then there exist a constant $c_\epsilon > 0$ and a collection of ϵ -dependent calibration constants as in Definition 19 such that, for every $n, m, d \in \mathbb{N}$ with $n \geq 1$, $m \geq 0$, and $d \geq 2$, the regime-dependent handle in Definition 19 supplies priors Π_0 and Π_1 satisfying*

$$\mathcal{L}_{\Pi_0}(P_{XA}) = \mathcal{L}_{\Pi_1}(P_{XA})$$

and

$$c_\epsilon r_\epsilon(n, m, d) \leq R_\epsilon(n, m, d),$$

where $r_\epsilon(n, m, d)$ is the frontier rate in Definition 12 and $R_\epsilon(n, m, d)$ is the minimax risk in Definition 11.

The converse gives the complementary half of the sample-size separation. Matching the induced P_{XA} law across the two priors makes the auxiliary records contribute through the marginal-frequency side of the construction; outcome-marked observations remain the resource that controls the $1/n$ component.

Combining the attainable estimator with the common-marginal converse yields the sharp risk statement and its sequence consequences.

Theorem 4. [thm:sharp-annotation-frontier] (Sharp annotation frontier) *For every overlap parameter ϵ satisfying $0 < \epsilon < 1/2$, there exist constants c_ϵ and C_ϵ , depending only on ϵ , with*

$$0 < c_\epsilon \leq C_\epsilon,$$

such that the following assertions hold.

- **(Finite-sample bracket.)** For every $n \geq 1$, $m \geq 0$, and $d \geq 2$,

$$c_\epsilon r_\epsilon(n, m, d) \leq R_\epsilon(n, m, d) \leq C_\epsilon r_\epsilon(n, m, d),$$

where, with $N = n + m$ and $\ell_n = \log(en)$,

$$r_\epsilon(n, m, d) = \min \left\{ 1, \frac{1}{n} + \frac{d^2}{N^2 \ell_n^2} \right\}.$$

- **(Unannotated endpoint.)** For every $n \geq 1$ and every $d \geq 0$,

$$r_\epsilon(n, 0, d) = \min \left\{ 1, \frac{1}{n} + \frac{d^2}{n^2 \log^2(en)} \right\}.$$

- **(Known-marginal limit.)** For every fixed $n \geq 1$ and $d \geq 2$,

$$\lim_{m \rightarrow \infty} R_\epsilon(n, m, d) = R_\epsilon^{\text{known}}(n, d).$$

- **(Sequence regimes.)** For any sequences $(m_n)_{n \geq 1}$ and $(d_n)_{n \geq 1}$ of nonnegative integers satisfying $d_n \geq 2$ for every $n \geq 1$, set $N_n = n + m_n$ and $\ell_n = \log(en)$. Then

$$R_\epsilon(n, m_n, d_n) \rightarrow 0 \iff \frac{d_n}{N_n \ell_n} \rightarrow 0,$$

$$R_\epsilon(n, m_n, d_n) = O(n^{-1}) \iff d_n = O\left(\frac{N_n \ell_n}{\sqrt{n}}\right),$$

and

$$R_\epsilon(n, m_n, d_n) = o(R_\epsilon(n, 0, d_n))$$

if and only if both

$$\frac{d_n}{\sqrt{n} \ell_n} \rightarrow \infty$$

and

$$\frac{d_n/(N_n \ell_n)}{\min\{1, d_n/(n \ell_n)\}} \rightarrow 0.$$

Theorem 4 characterizes the minimax risk up to constants depending only on the overlap parameter ϵ . In its generated statement, the phrase *unannotated endpoint* means $m = 0$, namely the labeled-only endpoint with no auxiliary outcome-unlabeled records. The consistency threshold compares the alphabet size with $N_n \ell_n$; the parametric-rate threshold allows d_n to grow as $N_n \ell_n / \sqrt{n}$; and the strict-improvement criterion identifies when auxiliary treatment-covariate records improve on that labeled-only endpoint. The theorem therefore gives a theoretical sample-size frontier for the two sources of information; its conservative calibrated witness is not an operational allocation rule.

The known-marginal bullet in **Theorem 4** is a summary pointer. The following formally primary statement records that limiting link with the known-treatment-covariate-marginal benchmark experiment and gives the exact large- m endpoint for fixed labeled sample size and alphabet size.

Theorem 5. [thm:known-marginal-limit] (Known marginal limit) *Fix an overlap parameter ϵ , a labeled sample size n , and a finite covariate alphabet size d . Suppose*

- **(Overlap.)** $0 < \epsilon < 1/2$.
- **(Labeled size.)** $n \geq 1$.
- **(Alphabet size.)** $d \geq 2$.

Then the minimax mean-squared risk $R_\epsilon(n, m, d)$ from [Definition 11](#) converges, as the auxiliary sample size $m \rightarrow \infty$, to the known-marginal minimax risk $R_\epsilon^{\text{known}}(n, d)$ from [Definition 15](#):

$$\lim_{m \rightarrow \infty} R_\epsilon(n, m, d) = R_\epsilon^{\text{known}}(n, d).$$

The limit is the formal version of the marginal-information interpretation. As the auxiliary sample grows with n and d fixed, the treatment-covariate table is learned to the known-marginal benchmark, and the risk converges to the n^{-1} boundary in [Theorem 1](#).

5 The hybrid estimator

This section gives an algorithmic reading of the estimator used in [Algorithm 1](#). The construction separates three statistical roles that are intertwined in the target functional: the labeled records supply outcome marks, the combined treatment–covariate records estimate marginal cell frequencies, and the calibration chooses when a cell is treated by polynomial extrapolation rather than by direct inverse-count weighting.

The first step is deterministic sample splitting. The labeled sample is divided into an outcome block of size M_0 , a pilot contribution of size n_p , and a factorial contribution of size n_f . The auxiliary sample is divided into pilot and factorial contributions of sizes m_p and m_f . Pooling the corresponding labeled and auxiliary treatment–covariate records gives pilot and factorial pools of sizes M_p and M_f . This split makes the outcome counts independent of the treatment–covariate counts used to choose and evaluate the weights, matching the independence structure required by the risk analysis.

The finite calibration in [Algorithm 1](#) then fixes the polynomial degree L , the light-cell scale B , and the pilot threshold k_0 . The predicate $C_\epsilon(n, m)$ is a computable list of inequalities involving these quantities, the dyadic overlap certificate $\bar{\epsilon}$, and the prefix intensities u, t_p, t . When the predicate holds, the selected scales keep the polynomial coefficients and Poisson tails within the finite-sample ranges used by the upper-bound argument. When it fails, the estimator is defined by the zero rule, which keeps the procedure total for every admissible n, m, d .

The estimator uses Poisson-prefix derandomization to express an independent-Poisson statistic as a deterministic function of the fixed samples. For each triple of prefix lengths (r_0, r_p, r_f) , the rule keeps the first r_0 observations in the outcome block, the first r_p observations in the pilot pool, and the first r_f observations in the factorial pool, then averages the resulting clipped statistic against the three Poisson probabilities. This finite sum is the conditional expectation of the randomized prefix statistic, so the implemented estimator remains an ordinary measurable statistic of the observed data.

For each covariate cell x , the statistic forms three counts. The pilot count J_x decides the regime. The outcome counts S_{1x} and S_{0x} count labeled successes in the two treatment arms within the outcome prefix. The factorial counts K_{1x} and K_{0x} count treatment–covariate observations in the factorial prefix. Thus outcome information enters only through the success counts, while the larger annotated and unannotated treatment–covariate pool enters through J_x and K_{ax} .

Cells with $J_x \leq k_0$ use the light-cell polynomial weight. The Chebyshev component T_L defines H_L , its continuation E_L , and the correction polynomial G_L , giving the weight polynomial $W_{a,L}$ in [Algorithm 1](#). Its factorial lift $\widehat{W}_{a,L,x}$ replaces monomials in the unknown cell probabilities by falling-factorial count statistics. This applies the polynomial-approximation strategy used in sharp discrete functional estimation [[Jiao et al., 2015](#), [Wu and Yang, 2016](#)] to the inverse propensity-type weights needed for the average treatment effect, with the rare-cell organization motivated by sample-amplification and unseen-species perspectives [[Valiant and Valiant, 2017](#), [Orlitsky et al., 2016](#)].

Cells with $J_x > k_0$ use the heavy-cell inverse-count rule. In those cells, the factorial prefix supplies enough treatment–covariate mass for the ratios $(K_{0x} + K_{1x} + 1)/(K_{ax} + 1)$ to act as stabilized inverse-count weights. The added ones keep the weights finite at every realized count configuration, while the pilot threshold routes cells with limited marginal information to the polynomial branch.

Finally, [Algorithm 1](#) sums the cell contributions and applies $\text{clip}_{[-1,1]}$. The clipping matches

the natural range of the average treatment effect functional and controls exceptional prefix realizations without changing the structure of the light- and heavy-cell rules. The resulting statistic, $\hat{\tau}_{n,m,d}^{\text{mix}}$, is therefore a fully specified finite-sample estimator built from deterministic splitting, calibrated polynomial extrapolation, inverse-count stabilization, and bounded aggregation.

Theorem 2 is the performance guarantee attached to this construction. Under the overlap model and product-sample experiment stated earlier, the calibrated estimator attains the rate in **Definition 12** uniformly over the model class. The two pieces of the rate correspond to the two resources used by the algorithm: labeled outcomes determine the $1/n$ component, while treatment–covariate counts from both samples determine the marginal-frequency component through $n + m$.

Proof roadmap. Poissonization first makes the outcome, pilot, and factorial counts independent. The light-cell argument uses Chebyshev approximation and unbiased factorial moments; the heavy-cell argument uses the exact stabilized inverse-count mean and moment bounds. Pilot tails control routing errors, and summing the cellwise bias and variance bounds yields the Poissonized risk. Prefix conditioning then transfers that bound to the fixed samples. For the converse, moment-matched common-marginal priors keep the auxiliary table indistinguishable while separating the ATE, and a separate binary submodel supplies the $1/n$ label floor. The dependency map at the start of **Section A** links these ideas to the detailed certificates.

6 Interpretation and extensions

The main minimax theorem yields three interpretable sequence consequences. In the notation of **Definition 12**, the risk is consistent precisely along sequences for which the labeled sample grows and the treatment–covariate information is sufficiently rich relative to the alphabet size. The first component, $1/n$, is the irreducible cost of observing binary outcomes; the second component, $d^2/\{(n+m)^2\ell_n^2\}$, is the cost of learning enough about the treatment–covariate distribution to stabilize the inverse-probability structure of **Definition 8**. Thus the auxiliary sample improves risk through the marginal table rather than through direct outcome information.

The parametric regime is the portion of the experiment in which the marginal-frequency component is of order $1/n$ or smaller. There the sharp rate in **Theorem 4** matches the labeled-outcome benchmark up to constants depending only on the overlap constant. In that range, the hybrid estimator of **Algorithm 1** uses the auxiliary treatment–covariate records to make the nuisance marginal problem small enough that the remaining mean-squared error is governed by outcome sampling.

A complementary regime is where annotation yields strict improvement over the supervised endpoint. With $m = 0$, the marginal table is learned only from the labeled sample, and the second term in **Definition 12** is $d^2/(n^2\ell_n^2)$. Adding outcome-unlabeled treatment–covariate records replaces n by $n + m$ in that component. Consequently, when the alphabet is large enough for the marginal-frequency term to matter and the auxiliary sample makes $n + m$ substantially larger than n , **Theorem 4** gives a quantitative risk reduction inside the same overlap model and target functional. This is the sense in which outcome annotation has value here: it converts unlabeled treatment–covariate information into a sharper stabilized estimate of the average treatment effect.

The hybrid rule in **Algorithm 1** is calibrated to attain the logarithmic rate in **Definition 12**. A simpler comparison is obtained by using inverse-count stabilization everywhere, without the

light-cell polynomial correction. The next definition records this baseline in the same sample-splitting and Poisson-prefix format, so that the comparison isolates the role of the polynomial branch. Every quotient in this definition uses totalized real division: when $u = 0$, division by u evaluates to zero. Consequently the clipped prefix statistic remains defined for $n = 1, 2$, where $M_0 = u = 0$.

Definition 20. [synth_5] (Poisson-prefix inverse-count estimator) *For $n \geq 1$, $m \geq 0$, and $d \geq 2$, put*

$$M_0 = \lfloor n/3 \rfloor, \quad n_p = \left\lfloor \frac{n - M_0}{2} \right\rfloor, \quad n_f = n - M_0 - n_p, \quad m_p = \lfloor m/2 \rfloor, \quad m_f = m - m_p,$$

and $u = M_0/8$. Given a labeled sample $\mathcal{L}_n$ and an auxiliary sample $\mathcal{U}_m$, define $\hat{\tau}_{n,m,d}^{\text{IC,prefix}}(r_0, r_f) = 0$ when $r_0 > M_0$ or $r_f > n_f + m_f$. For $0 \leq r_0 \leq M_0$ and $0 \leq r_f \leq n_f + m_f$, let $S_{ax}(r_0)$ be the number of records among the first r_0 labeled observations with $(X, A, Y) = (x, a, 1)$. Let $K_{ax}(r_f)$ be the number of records with $(X, A) = (x, a)$ in the factorial prefix formed from the last n_f labeled records and the last m_f auxiliary records, taking the first r_f records of that pooled factorial block in its fixed order. Define

$$\hat{\tau}_{n,m,d}^{\text{IC,prefix}}(r_0, r_f) = \text{clip}_{[-1,1]} \left[\sum_{x=1}^d \left\{ \frac{S_{1x}(r_0)}{u} \frac{K_{0x}(r_f) + K_{1x}(r_f) + 1}{K_{1x}(r_f) + 1} - \frac{S_{0x}(r_0)}{u} \frac{K_{0x}(r_f) + K_{1x}(r_f) + 1}{K_{0x}(r_f) + 1} \right\} \right].$$

The statistic in Definition 20 uses the outcome prefix to estimate treated and control success masses and the factorial prefix to form stabilized inverse-count weights. The added ones make the expression finite in every cell and for every realized prefix length. Averaging over the Poisson prefix lengths derandomizes the construction while preserving the same count structure.

Theorem 6. [thm:inverse-count-baseline] (Inverse-count baseline bound) *Fix an overlap constant ϵ with $0 < \epsilon < 1/2$. Then there is a constant $C_\epsilon > 0$ such that, for every $n \geq 1$, $m \geq 0$, and $d \geq 2$, the clipped derandomized inverse-count estimator defined for each sample $Z = (\mathcal{L}_n, \mathcal{U}_m)$ by*

$$\hat{\tau}_{n,m,d}^{\text{IC}}(Z) = \sum_{r_0=0}^{M_0} \sum_{r_f=0}^{n_f+m_f} \mathbb{P}\{\text{Poi}(M_0/8) = r_0\} \mathbb{P}\{\text{Poi}((n_f + m_f)/8) = r_f\} \hat{\tau}_{n,m,d}^{\text{IC,prefix}}(Z; r_0, r_f)$$

is measurable. Here $M_0 = \lfloor n/3 \rfloor$, $n_p = \lfloor (n - M_0)/2 \rfloor$, $n_f = n - M_0 - n_p$, $m_p = \lfloor m/2 \rfloor$, and $m_f = m - m_p$ are the block sizes from Algorithm 1, and $\hat{\tau}_{n,m,d}^{\text{IC,prefix}}(Z; r_0, r_f)$ is the sample-and-prefix-count statistic of Definition 20.

Moreover,

$$\sup_{P \in \mathcal{M}_{d,\epsilon}} \mathbb{E}_{P^{\otimes n} \otimes P_{XA}^{\otimes m}} \left[\left\{ \hat{\tau}_{n,m,d}^{\text{IC}} - \tau(P) \right\}^2 \right] \leq C_\epsilon \min \left\{ 1, \frac{1}{n} + \frac{d^2}{(n+m)^2} \right\}.$$

Here $\mathcal{M}_{d,\epsilon}$ is the overlap class in Definition 6, $\tau(P)$ is the average treatment effect functional in Definition 8, and the expectation is under the annotation experiment of Definition 9. Consequently, for the minimax risk in Definition 11,

$$R_\epsilon(n, m, d) \leq C_\epsilon \min \left\{ 1, \frac{1}{n} + \frac{d^2}{(n+m)^2} \right\}.$$

Theorem 6 shows that direct inverse-count stabilization already converts auxiliary treatment–covariate observations into a risk bound with the same two-resource form as the sharp theorem: a labeled-outcome term and a marginal-frequency term. Its marginal term is $d^2/(n+m)^2$, whereas **Theorem 4** attains the logarithmically improved $d^2/\{(n+m)^2\ell_n^2\}$ rate through the calibrated light-cell polynomial component. The comparison identifies the gain supplied by the hybrid construction in sparse cells while keeping the same model, target, and product-sample experiment.

The general rate also has a simple specialization at the smallest nontrivial alphabet size. When the covariate alphabet is binary, the marginal-frequency component is uniformly absorbed into the labeled-outcome scale.

Theorem 7. `[prop:binary-alphabet-parametric-rate]` (Binary parametric rate) *Let $0 < \epsilon < 1/2$. For the minimax risk $R_\epsilon(n, m, d)$ defined in [Definition 11](#), at binary alphabet size $d = 2$ there exist constants $c_\epsilon, C_\epsilon \in \mathbb{R}$ with $0 < c_\epsilon \leq C_\epsilon$ such that, for every integer $n \geq 1$ and every integer $m \geq 0$,*

$$\frac{c_\epsilon}{n} \leq R_\epsilon(n, m, 2) \leq \frac{C_\epsilon}{n}.$$

Theorem 7 gives the exact order of the minimax risk for $d = 2$: the experiment is parametric for every auxiliary sample size. The upper bound follows the same scale as the labeled-outcome term in the main theorem, while the lower bound records that binary outcomes still require order $1/n$ mean-squared error uniformly over the overlap class. Thus, in the binary-alphabet case, auxiliary treatment–covariate records fit naturally into the constant factor rather than changing the rate.

The result complements semisupervised causal and surrogate-outcome analyses in which auxiliary observations improve estimation through nuisance-function, design-distribution, or surrogate-response learning [[Cheng et al., 2021](#), [Chakraborty and Dai, 2024](#), [Kallus and Mao, 2025](#)]. Here the finite-alphabet structure makes that role explicit in minimax form: unlabeled treatment–covariate records sharpen the marginal component of the risk, and the sharp theorem characterizes exactly when that sharpening changes the rate.

Limitations and future work. The analysis treats independent random outcome labeling from a shared population, binary treatment and outcome, a known finite covariate alphabet, occupied-cell overlap, and unrestricted cellwise nuisance functions. These conditions define a transparent discrete benchmark in which the value of outcome annotation can be characterized exactly. The constants in the explicit calibration are deliberately proof-oriented rather than implementation-oriented: for example, already at degree $L = 2$, the stronger displayed polynomial-size condition $L^{67056^{2L}} \leq n$ requires n on the order of 1.6×10^{17} . Thus the theorem establishes a finite-sample minimax rate and an explicit witness, but it does not recommend this conservative calibration as a practical estimator at ordinary sample sizes. Natural extensions include sharper implementable tuning, continuous or growing covariate representations, multi-valued treatments or outcomes, adaptive annotation designs, and structural restrictions on the cellwise outcome regressions or propensity scores.

Appendices

A Upper-bound ingredients

This appendix collects the analytic ingredients used by the upper bounds in [Theorem 2](#) and by the inverse-count comparison in [Theorem 6](#). The statements are organized around the two treatment–covariate masses that enter the estimators, the all-heavy inverse-count calculation, the polynomial certificate for light cells, and the fixed-sample transfer that connects the Poissonized construction to the annotation experiment.

The proof dependency map is as follows. The inverse-count moment lemmas feed the heavy-cell analysis; [Definition 24](#) and [Lemmas 5](#) and [16](#) feed the light-cell analysis; the calibrated mean and second-moment bounds combine in [Lemma 13](#); and [Lemma 14](#) converts that Poissonized bound into [Theorem 2](#). [Section B](#) independently supplies the common-marginal converse, after which the upper and lower bounds meet in [Theorem 4](#). This map is the shortest reading path through the detailed certificates below.

The inverse-count analysis uses one scalar inequality to insert the marginal treatment–covariate intensity into the variance calculation. It is stated first because the later cellwise bounds apply it separately to each covariate cell and treatment arm.

Lemma 1. [`lem:poisson-inverse-marginal-energy-scalar`] (Marginal energy insertion)

Let $u, p, w, \epsilon, \lambda_0, \lambda_1, q_0, q_1, r_{2,0}, r_{2,1}, r_{4,0}, r_{4,1} \in \mathbb{R}$. Suppose that

- (*Basic scales.*) $u > 0, p \geq 0, 0 < \epsilon \leq 1$, and $w \geq 0$.
- (*Outcome masses.*) $0 \leq q_0 \leq p$ and $0 \leq q_1 \leq p$.
- (*Marginal intensities.*) For each $a \in \{0, 1\}$, $\epsilon w \leq \lambda_a \leq w$.
- (*Second-moment rates.*) $r_{2,0}, r_{2,1} \geq 0$, $(1 + \lambda_0)^2 r_{2,0} \leq 8$, and $(1 + \lambda_1)^2 r_{2,1} \leq 8$.
- (*Fourth-moment rates.*) $r_{4,0}, r_{4,1} \geq 0$, $(1 + \lambda_0)^4 r_{4,0} \leq 96$, and $(1 + \lambda_1)^4 r_{4,1} \leq 96$.

Put

$$c_{\text{mar}} = 8\epsilon^{-2} + 96(\epsilon^{-2} + \epsilon^{-3}).$$

Then

$$\lambda_1 \{ (q_1 u^{-1} + q_1^2)(\lambda_0 + \lambda_0^2) r_{4,1} + (q_0 u^{-1} + q_0^2) r_{2,0} \} + \lambda_0 \{ (q_0 u^{-1} + q_0^2)(\lambda_1 + \lambda_1^2) r_{4,0} + (q_1 u^{-1} + q_1^2) r_{2,1} \} \leq (p u^{-1} + p^2) \frac{2c_{\text{mar}}}{1 + w}.$$

Proof of [Lemma 1](#). 1. First derive the scalar bidirectional energy bound. The lower and upper intensity bounds, together with $0 < \epsilon \leq 1$ and $w \geq 0$, imply

$$0 \leq \lambda_0, \quad 0 \leq \lambda_1,$$

and, for the denominator attached to λ_1 ,

$$\epsilon(1 + w) = \epsilon + \epsilon w \leq 1 + \lambda_1, \quad 1 + w \leq \epsilon^{-1}(1 + \lambda_1), \quad w \leq \epsilon^{-1}(1 + \lambda_1).$$

Using also $\lambda_0 \leq w$, we have

$$(1 + w)\lambda_0 \leq (1 + w)w \leq \epsilon^{-2}(1 + \lambda_1)^2,$$

and hence

$$\lambda_0 r_{2,1} \leq 8 \frac{\lambda_0}{(1 + \lambda_1)^2} \leq \frac{8\epsilon^{-2}}{1 + w}.$$

Similarly,

$$(1 + w)\lambda_1\lambda_0 \leq \epsilon^{-2}(1 + \lambda_1)^4, \quad (1 + w)\lambda_1\lambda_0^2 \leq \epsilon^{-3}(1 + \lambda_1)^4,$$

so the fourth-moment rate gives

$$\lambda_1(\lambda_0 + \lambda_0^2)r_{4,1} \leq \frac{96(\epsilon^{-2} + \epsilon^{-3})}{1 + w}.$$

Combining these two estimates yields

$$\lambda_1(\lambda_0 + \lambda_0^2)r_{4,1} + \lambda_0 r_{2,1} \leq \frac{c_{\text{mar}}}{1 + w}.$$

The same calculation with the roles of 0 and 1 interchanged gives

$$\lambda_0(\lambda_1 + \lambda_1^2)r_{4,0} + \lambda_1 r_{2,0} \leq \frac{c_{\text{mar}}}{1 + w}.$$

Therefore

$$\{\lambda_1(\lambda_0 + \lambda_0^2)r_{4,1} + \lambda_0 r_{2,1}\} + \{\lambda_0(\lambda_1 + \lambda_1^2)r_{4,0} + \lambda_1 r_{2,0}\} \leq \frac{2c_{\text{mar}}}{1 + w}.$$

2. Define the three real quantities

$$\eta_0 = q_0 u^{-1} + q_0^2, \quad \eta_1 = q_1 u^{-1} + q_1^2, \quad \eta_p = p u^{-1} + p^2.$$

Since $u > 0$, we have $u^{-1} \geq 0$. For each displayed outcome mass,

$$\eta_p - \eta_0 = (p - q_0)(u^{-1} + p + q_0) \geq 0, \quad \eta_p - \eta_1 = (p - q_1)(u^{-1} + p + q_1) \geq 0,$$

because $0 \leq q_0, q_1 \leq p$. Thus

$$\eta_0 \leq \eta_p, \quad \eta_1 \leq \eta_p, \quad \eta_p \geq 0.$$

3. All coefficients multiplying η_0 and η_1 below are nonnegative, by the nonnegativity of the intensities, the moment rates, and $\lambda_a + \lambda_a^2$. Hence the comparison from the previous step gives

$$\begin{aligned} & \lambda_1 \{ \eta_1(\lambda_0 + \lambda_0^2)r_{4,1} + \eta_0 r_{2,0} \} + \lambda_0 \{ \eta_0(\lambda_1 + \lambda_1^2)r_{4,0} + \eta_1 r_{2,1} \} \\ & \leq \eta_p \left[\{ \lambda_1(\lambda_0 + \lambda_0^2)r_{4,1} + \lambda_0 r_{2,1} \} + \{ \lambda_0(\lambda_1 + \lambda_1^2)r_{4,0} + \lambda_1 r_{2,0} \} \right]. \end{aligned}$$

Applying the scalar bidirectional bound from Step 1 and using $\eta_p \geq 0$, we obtain

$$\lambda_1 \{ (q_1 u^{-1} + q_1^2)(\lambda_0 + \lambda_0^2)r_{4,1} + (q_0 u^{-1} + q_0^2)r_{2,0} \} + \lambda_0 \{ (q_0 u^{-1} + q_0^2)(\lambda_1 + \lambda_1^2)r_{4,0} + (q_1 u^{-1} + q_1^2)r_{2,1} \} \leq (p u^{-1} + p^2) \frac{2}{1}$$

□

Lemma 1 converts arm-level inverse-count moment bounds into a covariate-cell contribution controlled by the total marginal scale w . The overlap inequalities in the statement match the occupied-cell overlap structure of $\mathcal{M}_{d,\epsilon}$, while the final display has the form needed to sum over cells.

We next isolate the cell-level notation used throughout the appendix. The treatment-covariate mass records how often an arm appears in a covariate cell, while the outcome-marked mass records the same cell together with a successful binary outcome.

Definition 21. [synth_2] (Cell arm mass) *For a law P on $[d] \times \{0, 1\}^2$, a covariate cell $x \in [d]$, and a treatment arm $a \in \{0, 1\}$, define the treatment-covariate cell mass*

$$s_{ax} = \sum_{y \in \{0, 1\}} P(X = x, A = a, Y = y).$$

The mass s_{ax} is the denominator scale for inverse treatment-cell weights. Under the overlap model in [Definition 6](#), occupied covariate cells have both arm masses controlled by the same covariate mass, so inverse-count weights can be calibrated uniformly over the model class.

Definition 22. [synth_1] (Outcome-marked cell mass) *For an observed-data law P on the finite record space $[d] \times \{0, 1\} \times \{0, 1\}$, define the outcome-marked cell mass q_{ax} for $x \in [d]$ and $a \in \{0, 1\}$ by*

$$q_{ax} = P(X = x, A = a, Y = 1).$$

Equivalently, q_{ax} is the probability mass assigned by P to the complete atom $(x, a, 1)$.

The outcome-marked mass q_{ax} supplies the numerator scale. Combining q_{ax} with an inverse estimate of s_{ax} gives a cellwise contribution to the average treatment effect functional in [Definition 8](#).

The simplest upper-bound component treats every cell as sufficiently observed and uses inverse counts throughout. This statistic is useful both as a baseline and as the heavy-cell part of the hybrid estimator.

Lemma 2. [lem:poisson-inverse-count-risk] (Poisson inverse-count risk) *Fix an overlap constant ϵ with $0 < \epsilon < 1/2$. There is a constant $C = C_\epsilon > 0$ such that, for every d , every $u, t > 0$, and every observed-data law $P \in \mathcal{M}_{d,\epsilon}$ as in [Definition 6](#), the following holds.*

In the independent Poisson count experiment with independent counts

$$S_{ax} \sim \text{Poi}(uq_{ax}), \quad K_{ax} \sim \text{Poi}(ts_{ax}), \quad a \in \{0, 1\}, \quad x \in [d],$$

define

$$Z^H = \sum_{x=1}^d \left\{ \frac{S_{1x}}{u} \frac{K_{0x} + K_{1x} + 1}{K_{1x} + 1} - \frac{S_{0x}}{u} \frac{K_{0x} + K_{1x} + 1}{K_{0x} + 1} \right\}.$$

Writing $\tau(P)$ for the average treatment effect functional in [Definition 8](#),

$$|E_P Z^H - \tau(P)| \leq C \min\{1, d/t\}, \quad \text{Var}_P(Z^H) \leq C(u^{-1} + t^{-1}).$$

Proof of [Lemma 2](#). Fix $0 < \epsilon < 1/2$. Put

$$c_{\text{out}} := 1 + 10\epsilon^{-1} + 8\epsilon^{-2}, \quad c_{\text{mar}} := 8\epsilon^{-2} + 96(\epsilon^{-2} + \epsilon^{-3}),$$

$$c_{\text{bias}} := 2 \max\{1, (\epsilon e)^{-1}\}, \quad C_\epsilon := c_{\text{bias}} + c_{\text{out}} + 4c_{\text{mar}} + 1.$$

These constants are positive. We prove the two asserted inequalities with $C = C_\epsilon$. Fix $d, u, t > 0$, and $P \in \mathcal{M}_{d,\epsilon}$. For each cell write

$$p_x := \sum_{a \in \{0,1\}} \sum_{y \in \{0,1\}} P(x, a, y), \quad s_{ax} := \sum_{y \in \{0,1\}} P(x, a, y), \quad q_{ax} := P(x, a, 1).$$

Thus $\sum_x p_x = 1$, $q_{ax} \leq s_{ax} \leq p_x$, and $\sum_{x,a} q_{ax} \leq 1$. By [Definition 6](#), the overlap condition gives

$$\epsilon p_x \leq s_{ax} \leq p_x, \quad a \in \{0, 1\}.$$

1. For $a \in \{0, 1\}$, let $\bar{a}$ denote the opposite arm and define the scalar arm mean

$$m_{ax} := E \left[\frac{S_{ax}}{u} \frac{K_{0x} + K_{1x} + 1}{K_{ax} + 1} \right].$$

Independence of the Poisson coordinates and linearity of expectation give

$$E_P Z^H = \sum_x (m_{1x} - m_{0x}).$$

Indeed, after expanding the summand defining Z^H , each term is integrable by the polynomial growth bound

$$\left| \frac{S_{ax}}{u} \frac{K_{0x} + K_{1x} + 1}{K_{ax} + 1} \right| \leq |u^{-1}|(1 + T)^2,$$

where T is the total count over all displayed Poisson coordinates, and all polynomial moments of the finite product Poisson law are finite. The product structure then reduces the expectation to the one-cell arm mean above.

2. For a Poisson variable J with mean $\lambda \geq 0$,

$$E \frac{1}{J+1} = \begin{cases} (1 - e^{-\lambda})/\lambda, & \lambda > 0, \\ 1, & \lambda = 0. \end{cases}$$

Define the arm outcome regression by

$$\mu_{ax} := \begin{cases} q_{ax}/s_{ax}, & s_{ax} > 0, \\ 0, & s_{ax} = 0. \end{cases}$$

Then $q_{ax} = s_{ax}\mu_{ax}$ for all cells and arms, including zero-mass arms, and $0 \leq \mu_{ax} \leq 1$. Using $E(S_{ax}/u) = q_{ax}$ and the preceding identity with $\lambda = ts_{ax}$, one obtains the continuous formula

$$m_{ax} = p_x \mu_{ax} - (p_x - s_{ax}) \mu_{ax} e^{-ts_{ax}}.$$

Since $0 \leq \mu_{ax} \leq 1$, $p_x - s_{ax} \leq p_x$, and $s_{ax} \geq \epsilon p_x$,

$$|m_{ax} - p_x \mu_{ax}| \leq p_x e^{-\epsilon t p_x}.$$

Consequently, using [Definition 8](#),

$$|E_P Z^H - \tau(P)| \leq 2 \sum_x p_x e^{-\epsilon t p_x}.$$

For every $z \geq 0$, $ze^{-\epsilon t z} \leq (\epsilon e t)^{-1}$, while $\sum_x p_x = 1$. Hence

$$|E_P Z^H - \tau(P)| \leq 2 \min \left\{ 1, \frac{d}{\epsilon e t} \right\}.$$

Writing $\xi = d/t$, the elementary inequality

$$2 \min\{1, \xi/(\epsilon e)\} \leq 2 \max\{1, (\epsilon e)^{-1}\} \min\{1, \xi\} = c_{\text{bias}} \min\{1, d/t\}$$

gives

$$|E_P Z^H - \tau(P)| \leq C_\epsilon \min\{1, d/t\}.$$

3. We next bound the variance. For a full count array ω , let $\Delta_{ax}^S F(\omega)$ be the change in F after adding one to S_{ax} , and let $\Delta_{ax}^K F(\omega)$ be the change after adding one to K_{ax} . The finite-product Poisson add-one Poincare inequality gives, for $F = Z^H$,

$$\text{Var}_P(Z^H) \leq \sum_{x,a} u q_{ax} E\{(\Delta_{ax}^S Z^H)^2\} + \sum_{x,a} t s_{ax} E\{(\Delta_{ax}^K Z^H)^2\}.$$

The square-integrability hypotheses required for this inequality follow from the same polynomial growth estimate used above, applied also to the add-one increments.

4. For the outcome-count increments,

$$\Delta_{ax}^S Z^H = \pm u^{-1} \left(1 + \frac{K_{\bar{a}x}}{K_{ax} + 1} \right),$$

where the sign is irrelevant after squaring. If $\lambda = t s_{ax}$ and $\nu = t s_{\bar{a}x}$, then $\epsilon \nu \leq \lambda$. For independent $J \sim \text{Poi}(\lambda)$ and $V \sim \text{Poi}(\nu)$, set

$$r_1 := E \frac{1}{J+1}, \quad r_2 := E \frac{1}{(J+1)^2}.$$

The Poisson reciprocal bounds give

$$\lambda r_1 \leq 1, \quad (1 + \lambda)^2 r_2 \leq 8, \quad r_1 \geq 0, \quad r_2 \geq 0.$$

Moreover $EV = \nu$ and $EV^2 \leq \nu + \nu^2$. Thus product-measure factorization gives

$$E \left(1 + \frac{V}{J+1} \right)^2 \leq 1 + 2\nu r_1 + (\nu + \nu^2) r_2.$$

The overlap comparison $\epsilon \nu \leq \lambda$, together with $\epsilon > 0$, implies $\nu \leq \epsilon^{-1} \lambda$. Hence

$$2\nu r_1 \leq 2\epsilon^{-1} \lambda r_1 \leq 2\epsilon^{-1},$$

and, using $r_2 \leq 8(1 + \lambda)^{-2}$,

$$\nu r_2 \leq 8\epsilon^{-1} \frac{\lambda}{(1 + \lambda)^2} \leq 8\epsilon^{-1}, \quad \nu^2 r_2 \leq 8\epsilon^{-2} \frac{\lambda^2}{(1 + \lambda)^2} \leq 8\epsilon^{-2}.$$

Consequently

$$E \left(1 + \frac{V}{J+1} \right)^2 \leq 1 + 10\epsilon^{-1} + 8\epsilon^{-2} = c_{\text{out}}.$$

Multiplying by the rate uq_{ax} and summing yields

$$\sum_{x,a} uq_{ax} E\{(\Delta_{ax}^S Z^H)^2\} \leq c_{\text{out}} u^{-1} \sum_{x,a} q_{ax} \leq c_{\text{out}} u^{-1}.$$

5. It remains to control the marginal-count increments. In one fixed cell, write

$$\theta_{ax} := ts_{ax}, \quad \Theta_x := tp_x,$$

and, for $\tilde{K}_{ax} \sim \text{Poi}(\theta_{ax})$, write

$$\eta_{2,a} := E \frac{1}{(\tilde{K}_{ax} + 1)^2}, \quad \eta_{4,a} := E \frac{1}{((\tilde{K}_{ax} + 1)(\tilde{K}_{ax} + 2))^2}.$$

The reciprocal estimates are

$$\eta_{2,a} \geq 0, \quad (1 + \theta_{ax})^2 \eta_{2,a} \leq 8, \quad \eta_{4,a} \geq 0, \quad (1 + \theta_{ax})^4 \eta_{4,a} \leq 96.$$

For a full count array ω , define

$$\text{cr}_{ax}(\omega) := \frac{S_{ax}(\omega)}{u} \frac{K_{\bar{a}x}(\omega)}{(K_{ax}(\omega) + 1)(K_{ax}(\omega) + 2)}, \quad \text{rc}_{ax}(\omega) := \frac{S_{ax}(\omega)}{u} \frac{1}{K_{ax}(\omega) + 1}.$$

The add-one formulas for K_{ax} are

$$\Delta_{1x}^K Z^H = -\{\text{cr}_{1x} + \text{rc}_{0x}\}, \quad \Delta_{0x}^K Z^H = \text{cr}_{0x} + \text{rc}_{1x}.$$

The two displayed components are nonnegative. Hence

$$\text{cr}_{ax} \leq |\Delta_{ax}^K Z^H|, \quad \text{rc}_{\bar{a}x} \leq |\Delta_{ax}^K Z^H|,$$

so the component squares are integrable because the add-one increment is square-integrable. Applying $(r + s)^2 \leq 2(r^2 + s^2)$ to the preceding decomposition gives, for each a ,

$$E\{(\Delta_{ax}^K Z^H)^2\} \leq 2(E \text{cr}_{ax}^2 + E \text{rc}_{\bar{a}x}^2).$$

Product-measure factorization gives

$$E \text{cr}_{ax}^2 = E \left(\frac{S_{ax}}{u} \right)^2 \eta_{4,a} E K_{\bar{a}x}^2, \quad E \text{rc}_{ax}^2 = E \left(\frac{S_{ax}}{u} \right)^2 \eta_{2,a}.$$

The Poisson second-moment bounds are

$$E \left(\frac{S_{ax}}{u} \right)^2 \leq q_{ax} u^{-1} + q_{ax}^2, \quad E K_{ax}^2 \leq \theta_{ax} + \theta_{ax}^2.$$

Combining the last three displays yields

$$E\{(\Delta_{ax}^K Z^H)^2\} \leq 2[(q_{ax} u^{-1} + q_{ax}^2)(\theta_{\bar{a}x} + \theta_{\bar{a}x}^2) \eta_{4,a} + (q_{\bar{a}x} u^{-1} + q_{\bar{a}x}^2) \eta_{2,\bar{a}}].$$

Since $q_{ax} \leq p_x$ and $u > 0$,

$$q_{ax}u^{-1} + q_{ax}^2 \leq p_x u^{-1} + p_x^2.$$

The hypotheses of [Lemma 1](#) hold after identifying its generic cell mass and total intensity with

$$p = p_x, \quad w = \Theta_x,$$

with its arm rates equal to θ_{0x}, θ_{1x} , its marked masses equal to q_{0x}, q_{1x} , and its four reciprocal moments equal to $(\eta_{2,0}, \eta_{2,1}, \eta_{4,0}, \eta_{4,1})$. These arm rates are simply ts_{ax} , which are nonnegative because $t > 0$ and $s_{ax} \geq 0$. Also $p_x \geq 0$, $u > 0$, $0 < \epsilon \leq 1$, and $\Theta_x \geq 0$; the outcome masses satisfy $0 \leq q_{ax} \leq p_x$; overlap gives $\epsilon\Theta_x \leq \theta_{ax}$, while $s_{ax} \leq p_x$ gives $\theta_{ax} \leq \Theta_x$; and the displayed reciprocal estimates give the required second- and fourth-moment rate conditions. Therefore [Lemma 1](#) yields

$$\begin{aligned} & \theta_{1x} \{ (q_{1x}u^{-1} + q_{1x}^2)(\theta_{0x} + \theta_{0x}^2)\eta_{4,1} + (q_{0x}u^{-1} + q_{0x}^2)\eta_{2,0} \} \\ & + \theta_{0x} \{ (q_{0x}u^{-1} + q_{0x}^2)(\theta_{1x} + \theta_{1x}^2)\eta_{4,0} + (q_{1x}u^{-1} + q_{1x}^2)\eta_{2,1} \} \\ & \leq (p_x u^{-1} + p_x^2) \frac{2c_{\text{mar}}}{1 + \Theta_x}. \end{aligned}$$

Multiplying the add-one-square bounds by their Poisson rates $\theta_{ax} = ts_{ax}$, summing over a , and using the preceding display gives the cell estimate

$$\sum_a ts_{ax} E\{(\Delta_{ax}^K Z^H)^2\} \leq 4c_{\text{mar}} \frac{p_x u^{-1} + p_x^2}{1 + tp_x}.$$

6. Summing the preceding display over x ,

$$\sum_{x,a} ts_{ax} E\{(\Delta_{ax}^K Z^H)^2\} \leq 4c_{\text{mar}} \sum_x \frac{p_x u^{-1} + p_x^2}{1 + tp_x}.$$

Since $1 + tp_x \geq 1$ and $p_x u^{-1} \geq 0$,

$$\frac{p_x u^{-1} + p_x^2}{1 + tp_x} \leq p_x u^{-1} + \frac{p_x^2}{1 + tp_x}.$$

Thus

$$\sum_{x,a} ts_{ax} E\{(\Delta_{ax}^K Z^H)^2\} \leq 4c_{\text{mar}} \sum_x \left(p_x u^{-1} + \frac{p_x^2}{1 + tp_x} \right).$$

The first sum is u^{-1} . For the second, since $t > 0$ and $p_x \geq 0$,

$$\frac{p_x^2}{1 + tp_x} \leq \frac{p_x}{t},$$

and hence

$$\sum_x \frac{p_x^2}{1 + tp_x} \leq t^{-1} \sum_x p_x = t^{-1}.$$

Thus

$$\sum_{x,a} ts_{ax} E\{(\Delta_{ax}^K Z^H)^2\} \leq 4c_{\text{mar}}(u^{-1} + t^{-1}).$$

7. Combining the Poisson Poincare bound with the outcome and marginal energy estimates,

$$\text{Var}_P(Z^H) \leq c_{\text{out}}u^{-1} + 4c_{\text{mar}}(u^{-1} + t^{-1}).$$

Since $u^{-1} \leq u^{-1} + t^{-1}$ and $C_\epsilon = c_{\text{bias}} + c_{\text{out}} + 4c_{\text{mar}} + 1$,

$$\text{Var}_P(Z^H) \leq C_\epsilon(u^{-1} + t^{-1}).$$

Together with the bias bound from Step 2, this proves the claim. $\square$

Lemma 2 separates the two costs of the all-heavy construction: the bias is governed by the number of cells relative to the treatment–covariate intensity, and the variance is governed by the outcome and count intensities. This is the parametric inverse-count behavior that becomes effective once the pilot stage classifies a cell as heavy.

To use this calculation with sampled prefixes, the Poisson record experiment is identified with the independent count experiment. The next two statements record the count law for two independent prefixes and then name the prefix distribution used by the inverse-count transfer.

Lemma 3. [lem:poisson-prefix-count-law] (Poisson prefix count law) *Let P be a probability law on $[d] \times \{0, 1\}^2$, and let $u, t \geq 0$. Write*

$$q_{ax} = P(x, a, 1), \quad s_{ax} = P(x, a, 0) + P(x, a, 1).$$

Under the prefix law that independently draws an outcome-marked prefix with $\text{Poi}(u)$ records from P and a treatment–covariate prefix with $\text{Poi}(t)$ records from the marginal P_{XA} , define

$$S_{ax} = \#\{\text{outcome-marked prefix records equal to } (x, a, 1)\}, \quad K_{ax} = \#\{\text{treatment–covariate prefix records equal to } (x, a, 0)\}.$$

Then the pushforward of the prefix law under the map sending the two prefixes to the array $(S_{ax}, K_{ax})_{x \in [d], a \in \{0, 1\}}$ is the product measure

$$\bigotimes_{x \in [d]} \bigotimes_{a \in \{0, 1\}} \{\text{Poi}(uq_{ax}) \otimes \text{Poi}(ts_{ax})\}.$$

Equivalently, the coordinates are mutually independent, with

$$S_{ax} \sim \text{Poi}(uq_{ax}), \quad K_{ax} \sim \text{Poi}(ts_{ax})$$

for every $x \in [d]$ and $a \in \{0, 1\}$.

Proof of Lemma 3. Let $\Omega_{\text{obs}} = [d] \times \{0, 1\}^2$ and $\Omega_{\text{aux}} = [d] \times \{0, 1\}$. For a finite prefix $\xi = (z_1, \dots, z_r)$ in a finite space Ω , write

$$H_\xi(z) = \#\{i : z_i = z\}, \quad z \in \Omega.$$

The elementary Poisson splitting identity says that if $R \sim \text{Poi}(\lambda)$ and, conditionally on R , $z_1, \dots, z_R$ are i.i.d. with law ν on Ω , then

$$(H_\xi(z))_{z \in \Omega} \sim \bigotimes_{z \in \Omega} \text{Poi}(\lambda \nu\{z\}).$$

Indeed, for any nonnegative integers $(r_z)_{z \in \Omega}$, with $r_\bullet = \sum_z r_z$,

$$\Pr\{H_\xi(z) = r_z \ \forall z\} = e^{-\lambda} \frac{\lambda^{r_\bullet}}{r_\bullet!} \prod_z \frac{r_z!}{r_z!} \nu\{z\}^{r_z} = \prod_z e^{-\lambda \nu\{z\}} \frac{(\lambda \nu\{z\})^{r_z}}{r_z!}.$$

Applied to the outcome-marked prefix with intensity u and the treatment-covariate prefix with intensity t , and using the independence of the two prefixes, this gives

$$(H_z^{\text{obs}})_{z \in \Omega_{\text{obs}}} \sim \bigotimes_{z \in \Omega_{\text{obs}}} \text{Poi}(uP\{z\}), \quad (H_w^{\text{aux}})_{w \in \Omega_{\text{aux}}} \sim \bigotimes_{w \in \Omega_{\text{aux}}} \text{Poi}(tP_{XA}\{w\}),$$

with the two histogram arrays independent. Mapping the first histogram to

$$E(H^{\text{obs}})(x, a) = H^{\text{obs}}(x, a, 1)$$

selects one coordinate for each (x, a) . Since the selected observed coordinates are distinct, and the auxiliary histogram is independent of the observed histogram, the array

$$(E(H^{\text{obs}})(x, a), H^{\text{aux}}(x, a))_{x \in [d], a \in \{0,1\}}$$

has law

$$\bigotimes_{x \in [d]} \bigotimes_{a \in \{0,1\}} \{\text{Poi}(uP\{(x, a, 1)\}) \otimes \text{Poi}(tP_{XA}\{(x, a)\})\}.$$

This array is exactly $(S_{ax}, K_{ax})_{x,a}$. In the degenerate case $[d] = \emptyset$, the same display is the empty product measure on the one-point array space.

It remains to identify the two singleton masses in the displayed product. By the definition of q_{ax} ,

$$P\{(x, a, 1)\} = P(x, a, 1) = q_{ax}.$$

Hence the first Poisson intensity is uq_{ax} .

For the auxiliary marginal, P_{XA} is the pushforward of P under $(x, a, y) \mapsto (x, a)$. Therefore

$$P_{XA}\{(x, a)\} = \sum_{y \in \{0,1\}} P(x, a, y) = P(x, a, 0) + P(x, a, 1) = s_{ax}.$$

Thus the second Poisson intensity is ts_{ax} . Since $u, t \geq 0$, these products are the corresponding nonnegative Poisson parameters, and substituting the two identities into the preceding product law yields

$$\bigotimes_{x \in [d]} \bigotimes_{a \in \{0,1\}} \{\text{Poi}(uq_{ax}) \otimes \text{Poi}(ts_{ax})\}.$$

The product representation is equivalent to mutual independence of all coordinates and to the stated marginal laws for every x and a . $\square$

Definition 23. [synth_19] (Poisson-prefix law) *For $P \in \mathcal{M}_{d,\epsilon}$ and intensities $u, t \geq 0$, let $\mathbb{P}_{P,u,t}^{\text{pref}}$ denote the following probability law. Draw two prefix lengths independently,*

$$R_0 \sim \text{Poi}(u), \quad R_f \sim \text{Poi}(t).$$

Conditional on $(R_0, R_f) = (r_0, r_f)$, draw an outcome prefix

$$((X_i, A_i, Y_i))_{i=1}^{r_0} \sim P^{\otimes r_0}$$

and, independently, a factorial treatment-covariate prefix

$$((\tilde{X}_j, \tilde{A}_j))_{j=1}^{r_f} \sim P_{XA}^{\otimes r_f}.$$

Thus the two prefixes are independent, the outcome prefix has complete-record sampling law P , and the factorial prefix has treatment-covariate sampling law P_{XA} . Under $\mathbb{P}_{P,u,t}^{\text{pref}}$, the associated counts

$$S_{ax} = \sum_{i=1}^{R_0} \mathbf{1}\{X_i = x, A_i = a, Y_i = 1\}, \quad K_{ax} = \sum_{j=1}^{R_f} \mathbf{1}\{\tilde{X}_j = x, \tilde{A}_j = a\}$$

are independent across the two prefixes, with

$$S_{ax} \sim \text{Poi}(uq_{ax}), \quad K_{ax} \sim \text{Poi}(ts_{ax}).$$

[Lemma 3](#) and [Definition 23](#) justify applying the count-risk calculation directly to Poisson prefixes of the original records. The prefix representation is convenient for derandomization because every admissible prefix length is a finite statistic of the fixed sample pools.

Lemma 4. [[lem:inverse-count-poisson-prefix-transfer](#)] (Poisson prefix transfer) *Let n, m, d be nonnegative integers, let $\epsilon \in \mathbb{R}$, and let $P \in \mathcal{M}_{d,\epsilon}$ as in [Definition 6](#). Define*

$$M_0 = \lfloor n/3 \rfloor, \quad n_p = \lfloor (n - M_0)/2 \rfloor, \quad n_f = n - M_0 - n_p, \quad m_f = m - \lfloor m/2 \rfloor,$$

and set $u = M_0/8$ and $t = (n_f + m_f)/8$. Let $\hat{\tau}_{n,m,d}^{\text{IC}}$ be the clipped derandomized inverse-count estimator obtained by averaging, over $0 \leq r_0 \leq M_0$ and $0 \leq r_f \leq n_f + m_f$, the finite-prefix inverse-count output with weights

$$\Pr\{\text{Poi}(u) = r_0\} \Pr\{\text{Poi}(t) = r_f\}.$$

Let $\hat{\tau}_{u,t}^{\text{IC,pref}}$ be the clipped inverse-count statistic computed from two independent Poisson prefixes: an outcome-marked prefix with law induced by P and length $\text{Poi}(u)$, and a treatment-covariate prefix with law P_{XA} and length $\text{Poi}(t)$. Then, for the annotation experiment in [Definition 9](#) and the ATE functional $\tau(P)$ in [Definition 8](#),

$$E_{P^{\otimes n} \otimes P_{XA}^{\otimes m}} \left[(\hat{\tau}_{n,m,d}^{\text{IC}} - \tau(P))^2 \right] \leq E_{\mathbb{P}_{P,u,t}^{\text{pref}}} \left[(\hat{\tau}_{u,t}^{\text{IC,pref}} - \tau(P))^2 \right] + 4 \left[\Pr\{\text{Poi}(u) > M_0\} + \Pr\{\text{Poi}(t) > n_f + m_f\} \right].$$

Proof of [Lemma 4](#). Put

$$M_f := n_f + m_f, \quad m_p := \lfloor m/2 \rfloor.$$

All lists below use the empty-list convention when one of the displayed lengths is zero.

1. For finite prefixes ω_0 of outcome-marked observations and ω_f of treatment-covariate observations, write

$$\Phi_u(\omega_0, \omega_f) := \text{the clipped finite-prefix inverse-count output at scale } u$$

so that, by clipping,

$$-1 \leq \Phi_u(\omega_0, \omega_f) \leq 1 \quad \text{for every } (\omega_0, \omega_f).$$

Also,

$$|\tau(P)| = \left| \sum_{x=1}^d p_x (\mu_{1x} - \mu_{0x}) \right| \leq \sum_{x=1}^d p_x |\mu_{1x} - \mu_{0x}| \leq \sum_{x=1}^d p_x = 1,$$

using [Definition 8](#), the fact that P is a probability law in [Definition 6](#), and $\mu_{ax} \in [0, 1]$, with the standard zero-mass-cell convention.

For fixed pools $\mathbf{o} = (o_1, \dots, o_{M_0})$ and $\mathbf{v} = (v_1, \dots, v_{M_f})$, define the capped prefix value

$$\mathcal{C}(\mathbf{o}, \mathbf{v}, r_0, r_f) := \begin{cases} \Phi_u((o_1, \dots, o_{r_0}), (v_1, \dots, v_{r_f})), & r_0 \leq M_0, r_f \leq M_f, \\ 0, & \text{otherwise.} \end{cases}$$

Let $R_0 \sim \text{Poi}(u)$ and $R_f \sim \text{Poi}(t)$ be independent, and set

$$\mathcal{R}_{u,t}(\mathbf{o}, \mathbf{v}) := \mathbb{E}[\mathcal{C}(\mathbf{o}, \mathbf{v}, R_0, R_f)].$$

If $(\mathbf{O}, \mathbf{V}) \sim P^{\otimes M_0} \otimes P_{XA}^{\otimes M_f}$, Jensen's inequality for the conditional expectation over (R_0, R_f) gives

$$\mathbb{E}[(\mathcal{R}_{u,t}(\mathbf{O}, \mathbf{V}) - \tau(P))^2] \leq \mathbb{E}[(\mathcal{C}(\mathbf{O}, \mathbf{V}, R_0, R_f) - \tau(P))^2].$$

On the event $\Omega_{\text{fit}} = \{R_0 \leq M_0, R_f \leq M_f\}$, the selected prefixes have the same law as the independent Poisson prefixes restricted to their lengths fitting in the two pools. Hence

$$\mathbb{E}[(\mathcal{C} - \tau(P))^2 \mathbf{1}_{\Omega_{\text{fit}}}] \leq \mathbb{E}_{\mathbb{P}_{P,u,t}^{\text{pref}}}[(\Phi_u - \tau(P))^2].$$

On Ω_{fit}^c , the capped value is 0, and the interval bound above gives

$$(0 - \tau(P))^2 \leq (1 - (-1))^2 = 4.$$

The union bound gives

$$\Pr(\Omega_{\text{fit}}^c) \leq \Pr\{\text{Poi}(u) > M_0\} + \Pr\{\text{Poi}(t) > M_f\}.$$

Combining these displays,

$$\mathbb{E}[(\mathcal{R}_{u,t}(\mathbf{O}, \mathbf{V}) - \tau(P))^2] \leq \mathbb{E}_{\mathbb{P}_{P,u,t}^{\text{pref}}}[(\Phi_u - \tau(P))^2] + 4[\Pr\{\text{Poi}(u) > M_0\} + \Pr\{\text{Poi}(t) > M_f\}].$$

2. For a full sample $z = ((O_1, \dots, O_n), (V_1, \dots, V_m))$, with $O_i = (X_i, A_i, Y_i)$ and $V_j \in [d] \times \{0, 1\}$, define the fixed-pool projection

$$\Psi(z) := ((O_1, \dots, O_{M_0}), ((X_{M_0+n_p+1}, A_{M_0+n_p+1}), \dots, (X_n, A_n), V_{m_p+1}, \dots, V_m)).$$

For every z ,

$$\hat{\tau}_{n,m,d}^{\text{IC}}(z) = \mathcal{R}_{u,t}(\Psi(z)).$$

Indeed, expanding $\mathcal{R}_{u,t}$ gives

$$\sum_{r_0, r_f \geq 0} \Pr\{\text{Poi}(u) = r_0\} \Pr\{\text{Poi}(t) = r_f\} \mathcal{C}(\Psi(z), r_0, r_f),$$

and the terms with $r_0 > M_0$ or $r_f > M_f$ are zero, leaving exactly the finite average over $0 \leq r_0 \leq M_0$ and $0 \leq r_f \leq M_f$ in the derandomized inverse-count estimator.

3. Under the annotation experiment in [Definition 9](#),

$$\Psi_{\#}(P^{\otimes n} \otimes P_{XA}^{\otimes m}) = P^{\otimes M_0} \otimes P_{XA}^{\otimes M_f}.$$

The first component selects the first M_0 labeled records. The second component concatenates the treatment–covariate projections of the final n_f labeled records with the final m_f auxiliary records; independence of the original product samples and the definition of P_{XA} give the displayed product law. Therefore the change of variables formula yields

$$\begin{aligned} \mathbb{E}_{P^{\otimes n} \otimes P_{XA}^{\otimes m}} \left[\left(\hat{\tau}_{n,m,d}^{\text{IC}} - \tau(P) \right)^2 \right] \\ = \mathbb{E}_{P^{\otimes M_0} \otimes P_{XA}^{\otimes M_f}} \left[\left(\mathcal{R}_{u,t} - \tau(P) \right)^2 \right]. \end{aligned}$$

Measurability is immediate because all sample spaces involved are finite products and the statistic is clipped.

4. Applying the transfer bound from the first step to the fixed-pool risk in the last display gives

$$\begin{aligned} \mathbb{E}_{P^{\otimes n} \otimes P_{XA}^{\otimes m}} \left[\left(\hat{\tau}_{n,m,d}^{\text{IC}} - \tau(P) \right)^2 \right] \\ \leq \mathbb{E}_{\mathbb{P}_{P,u,t}^{\text{pref}}} \left[\left(\Phi_u - \tau(P) \right)^2 \right] + 4 \left[\Pr\{\text{Poi}(u) > M_0\} + \Pr\{\text{Poi}(t) > M_f\} \right]. \end{aligned}$$

Under $\mathbb{P}_{P,u,t}^{\text{pref}}$ the random value Φ_u is precisely $\hat{\tau}_{u,t}^{\text{IC,pref}}$, and $M_f = n_f + m_f$. This is the claimed inequality. □

[Lemma 4](#) converts the Poisson-prefix risk into a fixed-sample inverse-count estimator, with the excess risk controlled by the two prefix-overflow probabilities. This is the transfer step behind the inverse-count comparison in [Theorem 6](#).

For light cells, the hybrid estimator replaces reciprocal weights by polynomial weights. The next definition records the factorial-moment lift used to turn the polynomial into an unbiased Poisson statistic, following the same polynomial-estimation principle used in discrete functional estimation [[Jiao et al., 2015](#), [Wu and Yang, 2016](#), [Valiant and Valiant, 2017](#), [Orlitsky et al., 2016](#)].

Definition 24. [[synth_4](#)] (Factorial lift of the light-cell polynomial) *Fix $a \in \{0, 1\}$, an integer $L \geq 2$, and constants $B, t > 0$. Let*

$$W_{a,L}(s_0, s_1) = \frac{s_0 + s_1}{B} G_L(s_a/B) = \sum_{r_0, r_1 \geq 0} c_{a,r_0,r_1} s_0^{r_0} s_1^{r_1},$$

where G_L is the Chebyshev correction polynomial and only finitely many coefficients c_{a,r_0,r_1} are nonzero. For integer counts $K_0, K_1 \geq 0$, define the factorial lift

$$\widehat{W}_{a,L}(K_0, K_1) = \sum_{r_0, r_1 \geq 0} c_{a,r_0,r_1} \frac{(K_0)_{r_0} (K_1)_{r_1}}{t^{r_0+r_1}},$$

with $(k)_r = k(k-1)\cdots(k-r+1)$ and $(k)_0 = 1$. Thus each monomial of $W_{a,L}$ is replaced by the corresponding unbiased Poisson factorial-moment statistic.

The lift in [Definition 24](#) is tailored to independent Poisson counts: each falling factorial has expectation equal to the corresponding power of the intensity. The remaining task is to control the polynomial and the second moment of its lift uniformly over the light-cell range.

Lemma 5. [\[lem:chebyshev-factorial-certificate\]](#) (Chebyshev factorial certificate) *There is a universal constant $A > 1$ such that, for every integer $L \geq 2$, the polynomial continuations*

$$E_L(z) = H_L(z)/z, \quad G_L(z) = \{1 - E_L(z)\}/z$$

satisfy the following properties:

- **(Degree.)** *The polynomial G_L has degree at most $L - 2$.*
- **(Chebyshev certificate.)** *For every $z \in [0, 1]$,*

$$0 \leq E_L(z) \leq 1, \quad zG_L(z) + E_L(z) = 1,$$

and, whenever $z > 0$,

$$E_L(z) \leq (L^2 z)^{-1}.$$

- **(Factorial-lift moment.)** *For every $a \in \{0, 1\}$ and every $t, B, s_0, s_1 \in \mathbb{R}$ with*

$$t > 0, \quad B > 0, \quad s_0 \geq 0, \quad s_1 \geq 0, \quad L \leq tB,$$

if $K_0 \sim \text{Poi}(ts_0)$ and $K_1 \sim \text{Poi}(ts_1)$ are independent, then the factorial lift $\widehat{W}_{a,L}$ satisfies

$$\mathbb{E}[\widehat{W}_{a,L}(K_0, K_1)^2] \leq A^L \left(1 + \frac{s_0 + s_1}{B}\right)^{2L}.$$

Moreover, if $s_0 + s_1 \leq B$, then

$$\mathbb{E}[\widehat{W}_{a,L}(K_0, K_1)^2] \leq A^L.$$

Proof of [Lemma 5](#). Choose $A = 7056$, so $A > 1$.

1. Fix an integer $L \geq 2$. Write

$$Q_L(z) = T_L(1-2z), \quad H_L(z) = \frac{1 - Q_L(z)}{2L^2}, \quad E_L(z) = H_L(z)/z, \quad G_L(z) = \{1 - E_L(z)\}/z.$$

Since T_L has degree L and $1 - 2z$ has degree at most 1,

$$\deg Q_L \leq L.$$

Multiplication by the scalar $(2L^2)^{-1}$ does not increase degree, and subtraction from 1 gives

$$\deg H_L \leq \deg(1 - Q_L) \leq \max\{0, \deg Q_L\} \leq L.$$

Because E_L is the quotient of H_L by z , its degree is at most $L - 1$. Hence

$$\deg(1 - E_L) \leq \max\{0, \deg E_L\} \leq L - 1,$$

using $L \geq 2$. Dividing once more by z gives

$$\deg G_L \leq L - 2.$$

2. By [Lemma 16](#), for every $z \in [0, 1]$,

$$0 \leq E_L(z) \leq 1, \quad zG_L(z) + E_L(z) = 1,$$

and, for $z > 0$,

$$E_L(z) \leq \frac{1}{L^2 z}.$$

These are exactly the Chebyshev-certificate assertions.

3. It remains to check the moment assertions with the same constant. Let $a \in \{0, 1\}$, $t > 0$, $B > 0$, $s_0, s_1 \geq 0$, and $L \leq tB$, and let $K_0 \sim \text{Poi}(ts_0)$ and $K_1 \sim \text{Poi}(ts_1)$ be independent. Applying [Lemma 16](#) with $p = s_0 + s_1$ gives

$$\mathbb{E}[\widehat{W}_{a,L}(K_0, K_1)^2] \leq 7056^L \left(1 + \frac{s_0 + s_1}{B}\right)^{2L}.$$

If in addition $s_0 + s_1 \leq B$, the same calibration yields

$$\mathbb{E}[\widehat{W}_{a,L}(K_0, K_1)^2] \leq 7056^L.$$

Since $A = 7056$, these are the two required factorial-lift moment bounds. $\square$

[Lemma 5](#) provides the approximation identity and moment growth bound needed for the light-cell part of the hybrid statistic. The identity $zG_L(z) + E_L(z) = 1$ expresses the reciprocal-weight correction, and the displayed decay of E_L quantifies the approximation error at positive mass. The factorial-lift moment bound then keeps the variance compatible with the polynomial degree used in [Algorithm 1](#).

The hybrid mean analysis separates the deterministic light-cell approximation error from the pilot-classification error on heavy cells. The next statements give the corresponding cell-summed bounds and the envelope calculation that puts them on the target rate scale.

Lemma 6. [`lem:calibrated-envelope-absorption`] (Calibrated envelope absorption) *Let $\epsilon > 0$ and $C_0 > 0$. There is a constant $C > 0$, depending on ϵ and C_0 , such that the following holds.*

- **(Sample sizes.)** n, m, d are integers with $n \geq 1$, $m \geq 0$, and $d \geq 2$.
- **(Calibration.)** The finite calibration predicate $C_\epsilon(n, m)$ from [Algorithm 1](#) holds.
- **(Estimator scales.)** With the block sizes and scales from [Algorithm 1](#),

$$M_0 = \lfloor n/3 \rfloor, \quad n_p = \lfloor (n - M_0)/2 \rfloor, \quad n_f = n - M_0 - n_p, \quad m_f = m - \lfloor m/2 \rfloor,$$

and

$$u = M_0/8, \quad t = (n_f + m_f)/8, \quad L = L(n), \quad B = B(n, m, \epsilon), \quad A_\star = 7056.$$

Let $N = n + m$ and $\ell_n = \log(en)$, as in [Definition 12](#).

Then

$$4A_\star^L \left\{ \frac{\min\{1, dB\}}{u} + dB^2 \right\} + C_0 (u^{-1} + t^{-1}) + \frac{8dB^2}{\epsilon^2 L^4} + \frac{4}{\epsilon t} + 414n^{-12} + \left(\frac{2dB}{\epsilon L^2} + 6n^{-12} \right)^2 \leq C \left(\frac{1}{n} + \frac{d^2}{N^2 \ell_n^2} \right).$$

Proof of Lemma 6. 1. Let H_ϵ be the calibration constant appearing in Algorithm 1, viewed as a real number, and set

$$K_\epsilon(C_0) = 4096H_\epsilon 512^2 + 4096H_\epsilon^2 512 + 64C_0 + \frac{8192H_\epsilon^2 512^2}{\epsilon^2} + \frac{128}{\epsilon} + 414 + 2 \left(\frac{64H_\epsilon 512}{\epsilon} \right)^2 + 72.$$

We take

$$C = 1 + K_\epsilon(C_0).$$

Since $\epsilon > 0$, $C_0 > 0$, and $H_\epsilon \geq 0$, every displayed summand in $K_\epsilon(C_0)$ is nonnegative except for none, and hence $C > 0$.

2. Fix $n, m, d \in \mathbb{N}$ with $n \geq 1$, $d \geq 2$, and $C_\epsilon(n, m)$. Write

$$N = n + m, \quad \ell_n = \log(en), \quad A = A_\star^L, \quad \eta = n^{-12},$$

where $L = L(n)$, and define the comparison scale

$$\Xi = \frac{1}{n} + \frac{d^2}{N^2 \ell_n^2}.$$

All quantities in the displayed inequalities below are understood as real numbers. The sample-size assumptions give

$$1 \leq n, \quad n \leq N, \quad 1 \leq d.$$

Moreover,

$$\ell_n = \log(en) = 1 + \log n \geq 1,$$

because $n \geq 1$. From $L \geq 2$ in Algorithm 1, $L > 0$, and $A = A_\star^L \geq 0$. The calibration inequalities also give $u > 0$, $t_p > 0$, $t > 0$, and $B > 0$: indeed $u/n \geq 1/32$, $t_p/N \geq 1/32$, $t/N \geq 1/32$, and $8 \leq t_p B$.

3. The block-proportion part of $C_\epsilon(n, m)$ supplies

$$\frac{n}{32} \leq u, \quad \frac{N}{32} \leq t.$$

The scale definition $B = H_\epsilon L / \min\{t_p, t\}$, together with

$$\frac{N}{32} \leq t_p, \quad \frac{N}{32} \leq t,$$

gives

$$B = \frac{H_\epsilon L}{\min\{t_p, t\}} \leq \frac{32H_\epsilon L}{N}.$$

The same calculation with the upper block bounds $t_p \leq 3N/32$ and $t \leq 3N/32$ yields the full auxiliary estimate

$$\frac{32H_\epsilon L}{3N} \leq B \leq \frac{32H_\epsilon L}{N}.$$

Only the upper bound is used below.

4. Let b_n be the binary-length calibration quantity from [Algorithm 1](#). The logarithmic estimate gives

$$\ell_n \leq b_n.$$

The degree calibration inequality is

$$\frac{c^\circ b_n}{2} \leq L, \quad c^\circ = \frac{1}{256},$$

so $b_n \leq 512L$. Therefore

$$\ell_n \leq 512L.$$

The polynomial-size condition $L^6 A_\star^{2L} \leq n$ gives

$$(L^3 A_\star^L)^2 = L^6 A_\star^{2L} \leq n.$$

Finally, since $n \geq 1$,

$$0 \leq \eta = n^{-12} \leq \frac{1}{n}.$$

5. Put

$$\Theta = L^3 A = L^3 A_\star^L.$$

From $\Theta^2 \leq n$, the elementary inequality $(x - y)^2 \geq 0$, applied with

$$x = \frac{\Theta}{n}, \quad y = \frac{d}{N\ell_n},$$

gives

$$\frac{\Theta d}{Nn\ell_n} \leq \frac{1}{2} \left\{ \left(\frac{\Theta}{n} \right)^2 + \left(\frac{d}{N\ell_n} \right)^2 \right\} \leq \frac{1}{2} \left(\frac{1}{n} + \frac{d^2}{N^2\ell_n^2} \right) = \frac{\Xi}{2}.$$

Also, from $\ell_n \leq 512L$ and positivity of ℓ_n, L ,

$$\frac{1}{L} \leq \frac{512}{\ell_n}, \quad \frac{1}{L^2} \leq \frac{512^2}{\ell_n^2}.$$

6. The reciprocal block bounds are

$$u^{-1} \leq \frac{32}{n}, \quad t^{-1} \leq \frac{32}{n},$$

because $u \geq n/32$ and $t \geq N/32 \geq n/32$. Hence

$$C_0(u^{-1} + t^{-1}) \leq 64C_0 \frac{1}{n},$$

$$\frac{4}{\epsilon t} = \frac{4}{\epsilon} t^{-1} \leq \frac{128}{\epsilon} \frac{1}{n},$$

and

$$414\eta \leq 414 \frac{1}{n}.$$

Since $1/n \leq \Xi$, these estimates combine to

$$C_0(u^{-1} + t^{-1}) + \frac{4}{\epsilon t} + 414\eta \leq \left(64C_0 + \frac{128}{\epsilon} + 414 \right) \Xi.$$

7. The two core normalized products obey

$$\frac{\Theta d}{NnL^2} \leq \frac{\Theta d}{Nn} \frac{512^2}{\ell_n^2} \leq 512^2 \frac{\Theta d}{Nn\ell_n} \leq 512^2 \frac{\Xi}{2},$$

where the middle step uses $\ell_n \geq 1$, and

$$\frac{\Theta d}{N^2 L} \leq \frac{\Theta d}{N^2} \frac{512}{\ell_n} \leq 512 \frac{\Theta d}{Nn\ell_n} \leq 512 \frac{\Xi}{2},$$

where $N \geq n$ is used in the second inequality.

8. The first light-envelope term satisfies

$$4A \frac{\min\{1, dB\}}{u} \leq 4A \frac{dB}{u} \leq 4A \frac{d(32H_\epsilon L/N)}{n/32} = 4096H_\epsilon \frac{\Theta d}{NnL^2} \leq (4096H_\epsilon 512^2) \frac{\Xi}{2}.$$

The second light-envelope term satisfies

$$4AdB^2 \leq 4Ad \left(\frac{32H_\epsilon L}{N} \right)^2 = 4096H_\epsilon^2 \frac{\Theta d}{N^2 L} \leq (4096H_\epsilon^2 512) \frac{\Xi}{2}.$$

Adding the two bounds and using $\Xi \geq 0$ gives

$$4A \left\{ \frac{\min\{1, dB\}}{u} + dB^2 \right\} \leq (4096H_\epsilon 512^2 + 4096H_\epsilon^2 512) \Xi.$$

9. Since $d \geq 1$,

$$d \leq d^2.$$

Consequently,

$$\frac{8dB^2}{\epsilon^2 L^4} \leq \frac{8d}{\epsilon^2 L^4} \left(\frac{32H_\epsilon L}{N} \right)^2 = \frac{8192H_\epsilon^2}{\epsilon^2} \frac{d}{N^2 L^2}.$$

Using $1/L^2 \leq 512^2/\ell_n^2$ and $d \leq d^2$,

$$\frac{d}{N^2 L^2} \leq \frac{512^2 d^2}{N^2 \ell_n^2}.$$

Because $d^2/(N^2 \ell_n^2) \leq \Xi$, we obtain

$$\frac{8dB^2}{\epsilon^2 L^4} \leq \frac{8192H_\epsilon^2 512^2}{\epsilon^2} \Xi.$$

10. For the squared bias term,

$$\frac{2dB}{\epsilon L^2} \leq \frac{2d}{\epsilon L^2} \frac{32H_\epsilon L}{N} = \frac{64H_\epsilon}{\epsilon} \frac{d}{NL} \leq \frac{64H_\epsilon 512}{\epsilon} \frac{d}{N\ell_n}.$$

Both sides are nonnegative, so squaring gives

$$\left(\frac{2dB}{\epsilon L^2} \right)^2 \leq \left(\frac{64H_\epsilon 512}{\epsilon} \right)^2 \frac{d^2}{N^2 \ell_n^2} \leq \left(\frac{64H_\epsilon 512}{\epsilon} \right)^2 \Xi.$$

Also $0 \leq \eta \leq 1/n \leq 1$, hence

$$(6\eta)^2 = 36\eta^2 \leq 36\eta \leq \frac{36}{n}.$$

The inequality $(x + y)^2 \leq 2x^2 + 2y^2$, applied to

$$x = \frac{2dB}{\epsilon L^2}, \quad y = 6\eta,$$

therefore yields

$$\left(\frac{2dB}{\epsilon L^2} + 6\eta\right)^2 \leq \left\{2\left(\frac{64H_\epsilon 512}{\epsilon}\right)^2 + 72\right\} \Xi.$$

11. Combining the light-envelope, reciprocal, variance-polynomial, and squared-bias estimates gives

$$4A \left\{ \frac{\min\{1, dB\}}{u} + dB^2 \right\} + C_0(u^{-1} + t^{-1}) + \frac{8dB^2}{\epsilon^2 L^4} + \frac{4}{\epsilon t} + 414\eta + \left(\frac{2dB}{\epsilon L^2} + 6\eta\right)^2 \leq K_\epsilon(C_0) \Xi.$$

Since $0 \leq \Xi$ and $K_\epsilon(C_0) \leq C$, the right-hand side is at most $C\Xi$. Substituting back

$$A = A_\star^L, \quad \eta = n^{-12}, \quad \Xi = \frac{1}{n} + \frac{d^2}{N^2 \ell_n^2},$$

with $N = n + m$ and $\ell_n = \log(en)$ from [Definition 12](#), is precisely the asserted inequality. $\square$

[Lemma 7](#) compares the light polynomial contribution with the inverse-count contribution on cells whose covariate mass is at most B . The bound has a Chebyshev approximation term and a factorial-pool intensity term, matching the light-branch mean terms that later enter the calibrated envelope.

Lemma 7. [[lem:hybrid-light-branch-mean-bound](#)] (Light branch mean bound) *Let $P \in \mathcal{M}_{d,\epsilon}$ be a law in the finite model class of [Definition 6](#). Let $u, t, B > 0$ and let $L \geq 2$ be an integer. For each $x \in [d]$, write*

$$p_x = P(X = x), \quad q_{ax} = P(X = x, A = a, Y = 1), \quad s_{ax} = P(X = x, A = a).$$

For each $x \in [d]$, let $S_{0x}, S_{1x}, K_{0x}, K_{1x}$ be independent Poisson counts with

$$S_{ax} \sim \text{Poi}(uq_{ax}), \quad K_{ax} \sim \text{Poi}(ts_{ax}), \quad a \in \{0, 1\}.$$

Using the light factorial-lift weights and heavy inverse-count weights from [Algorithm 1](#), set

$$P_x = \frac{S_{1x}}{u} \widehat{W}_{1,L,x} - \frac{S_{0x}}{u} \widehat{W}_{0,L,x}, \quad H_x = \frac{S_{1x}}{u} \widehat{W}_{1,x}^H - \frac{S_{0x}}{u} \widehat{W}_{0,x}^H,$$

where the light weights use degree L , scale B , and intensity t . Then

$$\sum_{x: p_x \leq B} (\mathbb{E}P_x - \mathbb{E}H_x)^2 \leq \frac{8dB^2}{\epsilon^2 L^4} + \frac{4}{\epsilon t}.$$

Proof of Lemma 7. Let

$$\theta_x := p_x(\mu_{1x} - \mu_{0x}), \quad \ell_x := \mathbb{E}P_x, \quad h_x := \mathbb{E}H_x,$$

where $\mu_{ax} = P(Y = 1 \mid X = x, A = a)$, with the usual harmless value when $s_{ax} = 0$. Let $S_B = \{x \in [d] : p_x \leq B\}$. The cell masses are nonnegative and $\sum_{x \in [d]} p_x = 1$, since P is a probability law. We prove the bound by first controlling each summand and then summing over S_B .

Step 1. Fix $x \in S_B$. The light-branch expectation equals

$$\ell_x = q_{1x} \frac{p_x}{B} G_L(s_{1x}/B) - q_{0x} \frac{p_x}{B} G_L(s_{0x}/B).$$

To verify the first equality, write the light polynomial as

$$W_{a,L}(z_0, z_1) = \sum_{r_0, r_1} c_{a,r_0,r_1} z_0^{r_0} z_1^{r_1}.$$

The factorial lift in Definition 24 is

$$\widehat{W}_{a,L,x} = \sum_{r_0, r_1} c_{a,r_0,r_1} \frac{(K_{0x})_{r_0} (K_{1x})_{r_1}}{t^{r_0+r_1}}.$$

Independence and the Poisson factorial-moment identity $\mathbb{E}[(K_{ax})_r] = (ts_{ax})^r$ therefore give $\mathbb{E}\widehat{W}_{a,L,x} = W_{a,L}(s_{0x}, s_{1x})$. Independently, $\mathbb{E}(S_{ax}/u) = q_{ax}$, which proves the displayed formula for ℓ_x . The light Chebyshev estimate then gives

$$|\ell_x - \theta_x| \leq \frac{2B}{\epsilon L^2}.$$

Indeed $p_x \leq B$, $s_{0x} + s_{1x} = p_x$, and Assumption 1 gives $s_{ax} \geq \epsilon p_x$ for each arm a whenever $p_x > 0$. If $p_x = 0$, then $s_{0x} = s_{1x} = q_{0x} = q_{1x} = 0$, so the displayed bound is immediate. If $p_x > 0$, the shifted Chebyshev certificate in Lemma 5, together with the definitions of E_L and G_L in Algorithm 1, gives, for $z_a = s_{ax}/B \in [0, 1]$,

$$0 \leq E_L(z_a) \leq \frac{1}{L^2 z_a}, \quad 1 - z_a G_L(z_a) = E_L(z_a).$$

Writing $q_{ax} = s_{ax}\mu_{ax}$, the light-branch error becomes a linear combination of the two terms $p_x\mu_{ax}E_L(s_{ax}/B)$. Since $0 \leq \mu_{ax} \leq 1$ and $s_{ax} \geq \epsilon p_x$, each arm contributes at most $B/(\epsilon L^2)$, giving the stated two-arm bound.

Step 2. The heavy inverse-count branch satisfies

$$|h_x - \theta_x| \leq 2p_x \exp(-\epsilon t p_x).$$

Write $u_{\text{out}} := u$ for the outcome-block intensity. For each arm $a \in \{0, 1\}$, set

$$\eta_{ax} := \mathbb{E} \left[\frac{S_{ax}}{u_{\text{out}}} \frac{K_{0x} + K_{1x} + 1}{K_{ax} + 1} \right].$$

Since the outcome pool is independent of the factorial pool and $\mathbb{E}S_{ax} = u_{\text{out}}q_{ax}$,

$$\eta_{ax} = q_{ax} \mathbb{E} \left[\frac{K_{0x} + K_{1x} + 1}{K_{ax} + 1} \right].$$

The armwise inverse-count expectation formula in [Lemma 8](#), applied to the independent factorial counts, gives

$$\eta_{ax} = p_x \mu_{ax} - (p_x - s_{ax}) \mu_{ax} \exp(-ts_{ax}).$$

Consequently

$$|\eta_{ax} - p_x \mu_{ax}| = (p_x - s_{ax}) \mu_{ax} \exp(-ts_{ax}) \leq p_x \exp(-\epsilon t p_x),$$

because $0 \leq p_x - s_{ax} \leq p_x$, $0 \leq \mu_{ax} \leq 1$, and occupied-cell overlap gives $s_{ax} \geq \epsilon p_x$; when $p_x = 0$, both sides vanish. Since

$$h_x - \theta_x = (\eta_{1x} - p_x \mu_{1x}) - (\eta_{0x} - p_x \mu_{0x}),$$

the triangle inequality gives the displayed heavy-branch bound.

Step 3. Combining the two preceding bounds,

$$(\ell_x - h_x)^2 \leq \frac{8B^2}{\epsilon^2 L^4} + 8p_x^2 \exp(-2\epsilon t p_x).$$

To see this, write

$$\ell_x - h_x = (\ell_x - \theta_x) - (h_x - \theta_x)$$

and use $(r - s)^2 \leq 2r^2 + 2s^2$ with $|r| \leq 2B/(\epsilon L^2)$ and $|s| \leq 2p_x \exp(-\epsilon t p_x)$.

Step 4. For every x ,

$$p_x^2 \exp(-2\epsilon t p_x) \leq \frac{p_x}{2\epsilon e t}.$$

Indeed,

$$p_x^2 \exp(-2\epsilon t p_x) = p_x \{p_x \exp(-(2\epsilon) t p_x)\}.$$

The elementary envelope $z \exp(-\alpha t z) \leq (\alpha e t)^{-1}$, applied with $z = p_x$ and $\alpha = 2\epsilon$, gives

$$p_x \exp(-(2\epsilon) t p_x) \leq \frac{1}{2\epsilon e t}.$$

Multiplying by the nonnegative factor p_x gives the displayed bound. The envelope itself follows from $e(\alpha t z) \leq \exp(\alpha t z)$, multiplied by $\exp(-\alpha t z)$, and then divided by the positive number $\alpha e t$.

Step 5. Summing the pointwise bound over S_B gives

$$\sum_{x \in S_B} (\ell_x - h_x)^2 \leq \sum_{x \in S_B} \left\{ \frac{8B^2}{\epsilon^2 L^4} + 8 \frac{p_x}{2\epsilon e t} \right\}.$$

Since $|S_B| \leq d$ and $\sum_{x \in S_B} p_x \leq \sum_{x \in [d]} p_x = 1$,

$$\sum_{x \in S_B} (\ell_x - h_x)^2 \leq \frac{8dB^2}{\epsilon^2 L^4} + \frac{8}{2\epsilon e t}.$$

Finally $1 \leq e$, so $8/(2\epsilon e t) \leq 4/(\epsilon t)$. Substituting $\ell_x = \mathbb{E}P_x$ and $h_x = \mathbb{E}H_x$ yields

$$\sum_{x: p_x \leq B} (\mathbb{E}P_x - \mathbb{E}H_x)^2 \leq \frac{8dB^2}{\epsilon^2 L^4} + \frac{4}{\epsilon t},$$

as claimed. □

The comparison in [Lemma 7](#) is anchored by an exact arm-level identity for the heavy inverse-count mean. The identity expresses the inverse-count arm contribution as the desired cell term plus an exponentially damped residual.

Lemma 8. `[lem:poisson-inverse-arm-mean-formula]` (Inverse-count arm mean) *Let P be an observed-data law on $[d] \times \{0, 1\}^2$, let $x \in [d]$, and let $a \in \{0, 1\}$. Write p_x for the covariate-cell mass, s_{ax} for the treatment-covariate cell mass, q_{ax} for the outcome-marked mass, and $\mu_{ax} := q_{ax}/s_{ax}$ for the totalized outcome regression. If $u > 0$ and $t > 0$, and if $K_{ax} \sim \text{Poi}(ts_{ax})$ and $K_{1-a,x} \sim \text{Poi}(ts_{1-a,x})$ are independent, then*

$$\frac{q_{ax}}{u/u} \mathbb{E} \left[1 + \frac{K_{1-a,x}}{K_{ax} + 1} \right] = p_x \mu_{ax} - (p_x - s_{ax}) \mu_{ax} \exp(-ts_{ax}).$$

Proof of Lemma 8. 1. Put $\bar{a} := 1 - a$ and write

$$\Lambda_a := uq_{ax}, \quad \Theta_a := ts_{ax}, \quad \Theta_{\bar{a}} := ts_{\bar{a}x}.$$

The cell probabilities are nonnegative, so $s_{ax} \geq 0$, $s_{\bar{a}x} \geq 0$, and $q_{ax} \geq 0$. Since $u, t > 0$, the nonnegative rates above are the real values of the corresponding Poisson intensities. Also,

$$s_{ax} + s_{\bar{a}x} = p_x,$$

because the two treatment arms partition the covariate cell x .

2. Define

$$\Omega_a := \mathbb{E} \left[1 + \frac{K_{\bar{a}x}}{K_{ax} + 1} \right], \quad K_{ax} \sim \text{Poi}(\Theta_a), \quad K_{\bar{a}x} \sim \text{Poi}(\Theta_{\bar{a}}),$$

with the two counts independent. The product-Poisson identity used for this expectation is

$$\Theta_a \Omega_a = \Theta_a + \Theta_{\bar{a}} (1 - e^{-\Theta_a}).$$

Indeed, for $r(k) := (k + 1)^{-1}$,

$$\Omega_a = 1 + \int v r(k) d(\text{Poi}(\Theta_a) \otimes \text{Poi}(\Theta_{\bar{a}}))(k, v).$$

Fubini factorizes the cross term as

$$\int v r(k) d(\text{Poi}(\Theta_a) \otimes \text{Poi}(\Theta_{\bar{a}})) = \left(\int r(k) d\text{Poi}(\Theta_a)(k) \right) \left(\int v d\text{Poi}(\Theta_{\bar{a}})(v) \right).$$

The Poisson first moment gives

$$\int v d\text{Poi}(\Theta_{\bar{a}})(v) = \Theta_{\bar{a}},$$

and the shifted reciprocal identity gives

$$\Theta_a \int \frac{1}{k + 1} d\text{Poi}(\Theta_a)(k) = 1 - e^{-\Theta_a}.$$

Multiplying the displayed formula for Ω_a by Θ_a yields the claimed identity.

3. Substituting $\Theta_a = ts_{ax}$, $\Theta_{\bar{a}} = ts_{\bar{a}x}$, and $s_{\bar{a}x} = p_x - s_{ax}$, the preceding display becomes

$$(ts_{ax})\Omega_a = ts_{ax} + t(p_x - s_{ax})(1 - e^{-ts_{ax}}).$$

Because $t > 0$, cancellation of t gives

$$s_{ax}\Omega_a = s_{ax} + (p_x - s_{ax})(1 - e^{-ts_{ax}}).$$

Multiplying by μ_{ax} and rearranging,

$$(s_{ax}\mu_{ax})\Omega_a = p_x\mu_{ax} - (p_x - s_{ax})\mu_{ax}e^{-ts_{ax}}.$$

4. Finally,

$$q_{ax} = s_{ax}\mu_{ax}.$$

This is immediate from $\mu_{ax} = q_{ax}/s_{ax}$ when $s_{ax} \neq 0$. When $s_{ax} = 0$, the bound $0 \leq q_{ax} \leq s_{ax}$, obtained by summing the nonnegative outcome masses in the arm cell, gives $q_{ax} = 0$, and the same identity holds under the totalized division convention. Hence, since the normalized outcome count has mean q_{ax} ,

$$q_{ax} \mathbb{E} \left[1 + \frac{K_{\bar{a}x}}{K_{ax} + 1} \right] = (s_{ax}\mu_{ax})\Omega_a = p_x\mu_{ax} - (p_x - s_{ax})\mu_{ax}e^{-ts_{ax}},$$

as claimed. $\square$

Lemma 8 supplies the cellwise bias formula used when the heavy branch is compared with the target ATE contribution. The exponential factor is controlled by the treatment–covariate intensity, so the identity connects the heavy branch to the same marginal scale that appears in [Lemma 2](#).

A second auxiliary bound controls the event that a truly heavy cell is nevertheless routed to the light polynomial branch. It combines the pilot probability of such a routing event with the polynomial second moment under the factorial-pool law.

Lemma 9. [lem:heavy-light-second-moment-sum] (Heavy light second moment) *Let d be a finite alphabet size, let P be an observed-data law on $[d] \times \{0, 1\}^2$, and let n, m be integers. Suppose that:*

- **(Positive overlap parameter.)** *The overlap parameter satisfies $\epsilon > 0$.*
- **(Sample size.)** *The labeled sample size satisfies $n \geq 1$.*
- **(Calibration.)** *The finite calibration predicate $C_\epsilon(n, m)$ from [Algorithm 1](#) holds.*

Set

$$u = \frac{M_0}{8}, \quad t_p = \frac{M_p}{8}, \quad t = \frac{M_f}{8}, \quad L = L(n), \quad B = B(n, m, \epsilon), \quad k_0 = k_0(n, m, \epsilon),$$

with these quantities defined as in [Algorithm 1](#). For each $x \in [d]$, let

$$\pi_x = \mathbb{P}\{J_x \leq k_0\},$$

where, conditionally on P , the pilot counts have independent components $J_{ax} \sim \text{Poi}(t_p s_{ax})$, $a \in \{0, 1\}$. Let μ_x be the product law under which the outcome and factorial counts are independent with

$$S_{ax} \sim \text{Poi}(u q_{ax}), \quad K_{ax} \sim \text{Poi}(t s_{ax}), \quad a \in \{0, 1\},$$

and define the light polynomial cell contribution

$$P_x = \frac{S_{1x}}{u} \widehat{W}_{1,L,x} - \frac{S_{0x}}{u} \widehat{W}_{0,L,x},$$

where the factorial lift is taken at degree L , scale B , and intensity t . Then

$$\sum_{x: p_x > B} \pi_x \mathbb{E}_{\mu_x} [P_x^2] \leq 132 n^{-12}.$$

Proof of Lemma 9. Let

$$\text{Heavy}_B = \{x \in [d] : p_x > B\}, \quad p_x = \mathbb{P}_P\{X = x\}.$$

The quantities u, t_p, t, L, B, k_0 are those fixed in Algorithm 1. For each x , put

$$\Xi_x = \int \left[\frac{S_{1x}}{u} \widehat{W}_{1,L,x} - \frac{S_{0x}}{u} \widehat{W}_{0,L,x} \right]^2 d\mu_x.$$

Thus the left-hand side is $\sum_{x \in \text{Heavy}_B} \pi_x \Xi_x$.

1. The cell masses satisfy

$$0 \leq p_x \leq 1 \quad \text{for every } x \in [d], \quad \sum_{x \in [d]} p_x = 1.$$

Hence, for the finite set Heavy_B ,

$$\sum_{x \in \text{Heavy}_B} p_x \leq 1, \quad \sum_{x \in \text{Heavy}_B} p_x^2 \leq \sum_{x \in \text{Heavy}_B} p_x \leq 1.$$

The calibration predicate gives $u, t_p, t, B > 0$, $L \geq 2$, $L \leq tB$, and $1/32 \leq u/n$. Since $n \geq 1$,

$$u \geq \frac{n}{32} \geq \frac{1}{32}, \quad u^{-1} \leq 32.$$

These inequalities also cover the case $\text{Heavy}_B = \emptyset$, where the sums are zero.

2. Fix $x \in \text{Heavy}_B$, and define

$$y_x = \frac{p_x}{B}.$$

Because $B > 0$ and $p_x > B$, $y_x \geq 1$. Also $p_x = B y_x$. The pilot count is

$$J_x = J_{0x} + J_{1x}, \quad J_{ax} \sim \text{Poi}(t_p s_{ax}),$$

with independent arms and $s_{0x} + s_{1x} = p_x$. Since $k_0 = \lfloor t_p B/4 \rfloor$ and $t_p B \geq 0$,

$$k_0 \leq \frac{t_p B}{4} < \frac{t_p p_x}{4}.$$

The standard Poisson lower-tail bound therefore gives

$$\pi_x = \mathbb{P}\{J_x \leq k_0\} \leq \exp\left(-\frac{t_p p_x}{4}\right).$$

For the light-polynomial second moment, [Lemma 16](#) supplies the factorial-lift bound

$$\mathbb{E}\left[\widehat{W}_{a,L,x}^2\right] \leq A_\star^L \left(1 + \frac{p_x}{B}\right)^{2L}, \quad a \in \{0, 1\}.$$

Writing $q_{ax} = \mathbb{P}_P\{X = x, A = a, Y = 1\}$, independence of S_{ax} and the factorial counts gives

$$\mathbb{E}\left[\left(\frac{S_{ax}}{u} \widehat{W}_{a,L,x}\right)^2\right] = \left(\frac{q_{ax}}{u} + q_{ax}^2\right) \mathbb{E}\left[\widehat{W}_{a,L,x}^2\right].$$

Since $0 \leq q_{ax} \leq p_x$, and using $(r - s)^2 \leq 2r^2 + 2s^2$,

$$\Xi_x \leq 4 \left(\frac{p_x}{u} + p_x^2\right) A_\star^L \left(1 + \frac{p_x}{B}\right)^{2L}.$$

Consequently,

$$\pi_x \Xi_x \leq 4 \left(\frac{p_x}{u} + p_x^2\right) A_\star^L (1 + y_x)^{2L} \exp\left(-\frac{t_p p_x}{4}\right).$$

3. The definition $B = H_\epsilon L / \min(t_p, t)$, together with positivity of t_p, t , gives

$$H_\epsilon L \leq t_p B.$$

Therefore

$$\exp\left(-\frac{t_p p_x}{4}\right) \leq \exp\left(-\frac{H_\epsilon L y_x}{4}\right).$$

The numerical choice of H_ϵ in [Algorithm 1](#) includes

$$15 + \frac{24}{c^\circ} \leq \frac{H_\epsilon}{4}.$$

Using $\log A_\star \leq 13$, $\log(1 + y_x) \leq y_x$, and $y_x \geq 1$,

$$\begin{aligned} \log\left[A_\star^L (1 + y_x)^{2L} \exp\left(-\frac{H_\epsilon L y_x}{4}\right)\right] &= L \log A_\star + 2L \log(1 + y_x) - \frac{H_\epsilon L y_x}{4} \\ &\leq 15L y_x - \frac{H_\epsilon L y_x}{4} \\ &\leq -\frac{24L y_x}{c^\circ}. \end{aligned}$$

Thus

$$A_\star^L (1 + y_x)^{2L} \exp\left(-\frac{t_p p_x}{4}\right) \leq \exp\left(-\frac{24L y_x}{c^\circ}\right).$$

The calibration predicate also gives $c^\circ b_n / 2 \leq L$, where $b_n = 1 + \lceil \log_2 n \rceil$. Since $y_x \geq 1$,

$$12b_n \leq \frac{24L y_x}{c^\circ}.$$

Finally b_n dominates $\log n$ for $n \geq 1$, so

$$\exp(-12b_n) \leq n^{-12}.$$

Combining the last three displays yields

$$A_\star^L (1 + y_x)^{2L} \exp\left(-\frac{t_p p_x}{4}\right) \leq n^{-12}.$$

Substituting this into the bound from the previous step gives, for every $x \in \text{Heavy}_B$,

$$\pi_x \Xi_x \leq 4 \left(\frac{p_x}{u} + p_x^2 \right) n^{-12}.$$

4. Summing the preceding cellwise inequality over $x \in \text{Heavy}_B$,

$$\sum_{x \in \text{Heavy}_B} \pi_x \Xi_x \leq 4 \left(\frac{\sum_{x \in \text{Heavy}_B} p_x}{u} + \sum_{x \in \text{Heavy}_B} p_x^2 \right) n^{-12}.$$

By the bounds from the first step,

$$\frac{\sum_{x \in \text{Heavy}_B} p_x}{u} \leq u^{-1} \leq 32, \quad \sum_{x \in \text{Heavy}_B} p_x^2 \leq 1.$$

Therefore

$$\sum_{x \in \text{Heavy}_B} \pi_x \Xi_x \leq 4(32 + 1)n^{-12} = 132 n^{-12}.$$

This is the claimed bound. □

Lemma 9 makes the heavy-cell misclassification contribution negligible on the calibrated scale. Its summand contains both the pilot routing probability and the light-branch second moment, so it is the variance-side counterpart of the heavy-branch mean control.

Lemma 10 controls the contribution from heavy cells near the pilot decision boundary. The product factor $\pi_x \eta_x$ records that only disagreement between the light and heavy branches matters, and the calibrated split in [Algorithm 1](#) makes the summed contribution polynomially small in the labeled sample size.

Lemma 10. [lem:hybrid-heavy-branch-mean-bound] (Heavy branch mean bound) *Let d be a finite alphabet size, let $\epsilon > 0$, and let $P \in \mathcal{M}_{d,\epsilon}$ as in [Definition 6](#). Let $n \geq 1$ and $m \geq 0$ be integers. Assume:*

- **(Calibration.)** *The hybrid-estimator calibration predicate $C_\epsilon(n, m)$ in [Algorithm 1](#) holds.*
- **(Block intensities.)** *With M_0, n_p, n_f, m_p, m_f as in [Algorithm 1](#), set*

$$u = \frac{M_0}{8}, \quad t_p = \frac{n_p + m_p}{8}, \quad t = \frac{n_f + m_f}{8}.$$

- **(Cell laws.)** For each $x \in [d]$, let μ_x be the product law under which the outcome-block success counts S_{ax} are independent $\text{Poi}(uq_{ax})$ variables and the factorial-pool counts K_{ax} are independent $\text{Poi}(ts_{ax})$ variables, for $a \in \{0, 1\}$.

Let $L = L(n)$, $B = B(n, m, \epsilon)$, and $k_0 = k_0(n, m, \epsilon)$ be the degree, light-cell scale, and pilot threshold from [Algorithm 1](#). For each $x \in [d]$, let

$$\pi_x = \mathbb{P}\{J_x \leq k_0\}, \quad \eta_x = \mathbb{P}\{J_x > k_0\},$$

where $J_x = J_{0x} + J_{1x}$ has independent components $J_{ax} \sim \text{Poi}(t_p s_{ax})$. Let P_x be the light polynomial cell contribution and H_x the heavy inverse-count cell contribution from [Algorithm 1](#), evaluated under μ_x . Then

$$\sum_{x \in [d]: p_x > B} \pi_x \eta_x (\mathbb{E}_{\mu_x} P_x - \mathbb{E}_{\mu_x} H_x)^2 \leq 282 n^{-12}.$$

Proof of [Lemma 10](#). Let

$$\mathcal{H} = \{x \in [d] : p_x > B\}, \quad E_x^L = \mathbb{E}_{\mu_x} P_x, \quad E_x^H = \mathbb{E}_{\mu_x} H_x.$$

The calibration inequalities give $u > 0$, $t_p > 0$, $t > 0$, and $B > 0$. Also $0 \leq p_x \leq 1$ for every x , and $\sum_x p_x = 1$, hence

$$\sum_{x \in \mathcal{H}} p_x \leq 1.$$

Fix $x \in \mathcal{H}$. We first establish the calibrated pilot-tail estimate

$$\pi_x \leq n^{-12}.$$

For $a \in \{0, 1\}$, write s_{ax} for the treatment-covariate mass in arm a , so $p_x = s_{0x} + s_{1x}$. Under the pilot law, $J_x = J_{0x} + J_{1x}$ has Poisson mean

$$\lambda_x = t_p(s_{0x} + s_{1x}) = t_p p_x,$$

because the two pilot-arm counts are independent Poisson variables. The definition of k_0 gives $k_0 \leq t_p B/4$, and $x \in \mathcal{H}$ gives $B < p_x$. Since $t_p > 0$,

$$k_0 < \frac{t_p p_x}{4} = \frac{\lambda_x}{4}.$$

For a Poisson variable with mean λ_x , the following lower-tail estimate applies whenever the cutoff is below one quarter of the mean:

$$\mathbb{P}\{J_x \leq k_0\} \leq \exp\{-\lambda_x/4\}.$$

Indeed, for every $\alpha > 0$, Markov's inequality applied to $e^{-\alpha J_x}$ gives

$$\mathbb{P}\{J_x \leq k_0\} \leq \exp\{\lambda_x(e^{-\alpha} - 1) + \alpha k_0\}.$$

If $k_0 = 0$, this is $e^{-\lambda_x} \leq e^{-\lambda_x/4}$. If $k_0 \geq 1$, take $\alpha = \log(\lambda_x/k_0)$. With $\zeta_x = k_0/\lambda_x \leq 1/4$, the exponent becomes

$$\lambda_x\{-1 + \zeta_x - \zeta_x \log \zeta_x\} \leq -\lambda_x/4,$$

since $z - z \log z \leq 1/4 + (\log 4)/4 < 3/4$ on $0 \leq z \leq 1/4$. Therefore

$$\pi_x = \mathbb{P}\{J_x \leq k_0\} \leq \exp\{-t_p p_x/4\}.$$

Let $h_n = 1 + \lceil \log_2 n \rceil$ be the binary-length calibration quantity in [Algorithm 1](#). The tail calibration inequality

$$15L + 12h_n \leq \frac{t_p B}{4},$$

together with $p_x > B$, yields

$$\frac{t_p p_x}{4} \geq 12h_n.$$

Thus

$$\pi_x \leq \exp\{-12h_n\} \leq n^{-12},$$

where the last inequality follows from $n \geq 1$ and $h_n \geq \log n$.

The heavy inverse-count mean satisfies

$$|E_x^H| \leq 3p_x.$$

To prove this, first write

$$\theta_x = p_x(\mu_{1x} - \mu_{0x}),$$

where, for $a \in \{0, 1\}$, q_{ax} is the outcome-marked mass, s_{ax} is the treatment-covariate mass, and the arm outcome regression is characterized by

$$q_{ax} = s_{ax}\mu_{ax}, \quad 0 \leq \mu_{ax} \leq 1.$$

For the arm-specific inverse-count means set

$$\overline{H}_{a,x} = \mathbb{E}_{\mu_x} \left[\frac{S_{ax}}{u} \frac{K_{0x} + K_{1x} + 1}{K_{ax} + 1} \right], \quad a \in \{0, 1\}.$$

By the definition of the heavy cell contribution,

$$E_x^H = \overline{H}_{1,x} - \overline{H}_{0,x}.$$

We next compute each arm mean. Since S_{ax} is independent of (K_{0x}, K_{1x}) and $\mathbb{E}S_{ax} = uq_{ax}$,

$$\overline{H}_{a,x} = q_{ax} \mathbb{E} \left[\frac{K_{0x} + K_{1x} + 1}{K_{ax} + 1} \right].$$

For the chosen arm a , the fraction inside the expectation is

$$\frac{K_{0x} + K_{1x} + 1}{K_{ax} + 1} = 1 + \frac{K_{1-a,x}}{K_{ax} + 1},$$

where $1 - a$ denotes the opposite arm. Therefore [Lemma 8](#), applied with this arm and its opposite arm, gives the exact identity

$$\overline{H}_{a,x} = p_x \mu_{ax} - (p_x - s_{ax}) \mu_{ax} \exp\{-ts_{ax}\}.$$

Since $x \in \mathcal{H}$, the cell is occupied. Hence $P \in \mathcal{M}_{d,\epsilon}$ and [Definition 6](#) and [Assumption 1](#) give, for both arms,

$$\epsilon p_x \leq s_{ax} \leq p_x.$$

Using this, $0 \leq \mu_{ax} \leq 1$, and $t > 0$, the exact identity yields the armwise envelope

$$\begin{aligned} |\overline{H}_{a,x} - p_x \mu_{ax}| &= |-(p_x - s_{ax}) \mu_{ax} \exp\{-ts_{ax}\}| \\ &= (p_x - s_{ax}) \mu_{ax} \exp\{-ts_{ax}\} \\ &\leq p_x \exp\{-\epsilon t p_x\}. \end{aligned}$$

The first equality uses the displayed mean formula, the second uses nonnegativity of every factor, and the last step uses $p_x - s_{ax} \leq p_x$ and $\exp\{-ts_{ax}\} \leq \exp\{-\epsilon t p_x\}$. Hence

$$\begin{aligned} |E_x^H - \theta_x| &= |(\overline{H}_{1,x} - p_x \mu_{1x}) - (\overline{H}_{0,x} - p_x \mu_{0x})| \\ &\leq |\overline{H}_{1,x} - p_x \mu_{1x}| + |\overline{H}_{0,x} - p_x \mu_{0x}| \\ &\leq 2p_x \exp\{-\epsilon t p_x\}. \end{aligned}$$

Since $0 \leq p_x \leq 1$ and $\mu_{1x}, \mu_{0x} \in [0, 1]$,

$$|\theta_x| \leq p_x.$$

Also, because $\epsilon > 0$, $t > 0$, and $p_x \geq 0$,

$$\exp\{-\epsilon t p_x\} \leq 1.$$

Thus

$$|E_x^H| \leq |E_x^H - \theta_x| + |\theta_x| \leq 2p_x + p_x = 3p_x.$$

Now $\eta_x \leq 1$, $\pi_x \geq 0$, and $\eta_x \geq 0$. Using

$$(E_x^L - E_x^H)^2 \leq 2(E_x^L)^2 + 2(E_x^H)^2$$

and the preceding heavy-mean bound,

$$(E_x^H)^2 \leq 9p_x^2 \leq 9p_x,$$

where the last inequality uses $0 \leq p_x \leq 1$. Hence

$$\begin{aligned} \pi_x \eta_x (E_x^L - E_x^H)^2 &\leq \pi_x (E_x^L - E_x^H)^2 \\ &\leq 2\pi_x (E_x^L)^2 + 2\pi_x (E_x^H)^2 \\ &\leq 2\pi_x (E_x^L)^2 + 18p_x n^{-12}. \end{aligned}$$

Summing this pointwise estimate over $x \in \mathcal{H}$ gives

$$\sum_{x \in \mathcal{H}} \pi_x \eta_x (E_x^L - E_x^H)^2 \leq 2 \sum_{x \in \mathcal{H}} \pi_x (E_x^L)^2 + 18n^{-12} \sum_{x \in \mathcal{H}} p_x.$$

It remains to bound the first sum. For each x , nonnegativity of variance under μ_x gives

$$(E_x^L)^2 \leq \mathbb{E}_{\mu_x} [P_x^2].$$

Consequently,

$$\sum_{x \in \mathcal{H}} \pi_x (E_x^L)^2 \leq \sum_{x \in \mathcal{H}} \pi_x \mathbb{E}_{\mu_x} [P_x^2].$$

The heavy-light second-moment bound in [Lemma 9](#), applied with the same values of u, t_p, t, L, B, k_0 , gives

$$\sum_{x \in \mathcal{H}} \pi_x \mathbb{E}_{\mu_x} [P_x^2] \leq 132n^{-12}.$$

Combining the last two displays with $\sum_{x \in \mathcal{H}} p_x \leq 1$,

$$\sum_{x \in \mathcal{H}} \pi_x \eta_x (E_x^L - E_x^H)^2 \leq 2 \cdot 132 n^{-12} + 18 n^{-12} = 282 n^{-12}.$$

This is the desired bound. $\square$

The preceding local estimates are assembled into separate variance and mean-bias bounds for the full Poisson hybrid statistic. These two summaries expose exactly which terms enter the final rate calculation.

Lemma 11. [\[lem:calibrated-hybrid-variance-bound\]](#) (Hybrid variance bound) *Let ϵ , the overlap parameter, satisfy $0 < \epsilon < 1/2$. There is a constant $C_{\text{var}} > 0$, depending only on ϵ , such that the following holds. Let n, m, d be integers and let P be an observed-data law on $[d] \times \{0, 1\}^2$. Assume:*

- *(Sample size.)* $n \geq 1$.
- *(Model class.)* $P \in \mathcal{M}_{d, \epsilon}$, as in [Definition 6](#).
- *(Calibration.)* The finite calibration predicate $C_\epsilon(n, m)$ in [Algorithm 1](#) holds.

Let

$$M_0 = \lfloor n/3 \rfloor, \quad n_p = \lfloor (n - M_0)/2 \rfloor, \quad n_f = n - M_0 - n_p, \quad m_p = \lfloor m/2 \rfloor, \quad m_f = m - m_p,$$

and set

$$u = M_0/8, \quad t_p = (n_p + m_p)/8, \quad t = (n_f + m_f)/8.$$

Let $L = L(n)$, $B = B(n, m, \epsilon)$, $k_0 = k_0(n, m, \epsilon)$, and $A_\star = 7056$ be the degree, scale, pilot threshold, and factorial-moment constant from [Algorithm 1](#). Under the independent Poisson count law in which, independently over $x \in [d]$ and $a \in \{0, 1\}$,

$$S_{ax} \sim \text{Poi}(uq_{ax}), \quad J_{ax} \sim \text{Poi}(t_p s_{ax}), \quad K_{ax} \sim \text{Poi}(t s_{ax}),$$

let

$$\tilde{Z} = \sum_{x \in [d]} (\mathbf{1}\{J_{0x} + J_{1x} \leq k_0\} P_x + \mathbf{1}\{J_{0x} + J_{1x} > k_0\} H_x),$$

where P_x and H_x are the light-cell polynomial and heavy inverse-count cell contributions specified in [Algorithm 1](#). Then

$$\text{Var}(\tilde{Z}) \leq 4A_\star^L \left\{ \frac{\min\{1, dB\}}{u} + dB^2 \right\} + C_{\text{var}} (u^{-1} + t^{-1}) + \frac{8dB^2}{\epsilon^2 L^4} + \frac{4}{\epsilon t} + 414 n^{-12}.$$

Proof of Lemma 11. Choose C_{var} to be the positive constant supplied, at the present overlap parameter ϵ , by Lemma 2. Fix n, m, d and P satisfying the hypotheses. Write

$$p_x = P(X = x), \quad q_{ax} = P(X = x, A = a, Y = 1), \quad s_{ax} = P(X = x, A = a),$$

and let $J_x = J_{0x} + J_{1x}$. The calibration predicate in Algorithm 1 gives

$$u > 0, \quad t_p > 0, \quad t > 0, \quad B > 0, \quad L \geq 2, \quad L \leq tB.$$

For each $x \in [d]$, let $\mathbb{Q}_x$ be the product law of the outcome and factorial counts

$$S_{ax} \sim \text{Poi}(uq_{ax}), \quad K_{ax} \sim \text{Poi}(ts_{ax}), \quad a \in \{0, 1\},$$

with all four variables independent. Define

$$\lambda_x = \mathbb{P}\{J_x \leq k_0\}, \quad \theta_x = \mathbb{P}\{J_x > k_0\} = 1 - \lambda_x,$$

and, under $\mathbb{Q}_x$,

$$\bar{P}_x = \mathbb{E}_{\mathbb{Q}_x} P_x, \quad \bar{H}_x = \mathbb{E}_{\mathbb{Q}_x} H_x, \quad V_x^P = \text{Var}_{\mathbb{Q}_x}(P_x), \quad V_x^H = \text{Var}_{\mathbb{Q}_x}(H_x).$$

The probabilities satisfy $0 \leq \lambda_x, \theta_x \leq 1$.

1. Since the pilot counts are independent of the outcome and factorial counts, the selected cell is a two-branch mixture. For each cell,

$$\text{Var}(\mathbf{1}\{J_x \leq k_0\}P_x + \mathbf{1}\{J_x > k_0\}H_x) = \lambda_x V_x^P + \theta_x V_x^H + \lambda_x \theta_x (\bar{P}_x - \bar{H}_x)^2.$$

The cells are independent under the product Poisson law, so the variance of the sum is the sum of the cell variances:

$$\text{Var}(\tilde{Z}) = \sum_{x \in [d]} \{ \lambda_x V_x^P + \theta_x V_x^H + \lambda_x \theta_x (\bar{P}_x - \bar{H}_x)^2 \}.$$

2. Bound the light-branch variance part by splitting the cells at the scale B . Put

$$\text{Light}_B = \{x \in [d] : p_x \leq B\}, \quad \text{Heavy}_B = \{x \in [d] : p_x > B\}.$$

For $x \in \text{Light}_B$, $\lambda_x \leq 1$ and

$$V_x^P \leq \mathbb{E}_{\mathbb{Q}_x} P_x^2.$$

The factorial-lift second-moment estimate in Lemma 16, together with $(r-s)^2 \leq 2r^2 + 2s^2$, gives

$$\mathbb{E}_{\mathbb{Q}_x} P_x^2 \leq 4A_\star^L \left(\frac{p_x}{u} + p_x^2 \right), \quad x \in \text{Light}_B.$$

Indeed, for each arm a , the light factorial weight has second moment at most $A_\star^L$, while

$$\mathbb{E} \left[\left(\frac{S_{ax}}{u} \right)^2 \right] = \frac{q_{ax}}{u} + q_{ax}^2 \leq \frac{p_x}{u} + p_x^2.$$

Summing over Light_B and using

$$\sum_{x \in \text{Light}_B} p_x \leq \min\{1, dB\}, \quad \sum_{x \in \text{Light}_B} p_x^2 \leq dB^2,$$

yields

$$\sum_{x \in \text{Light}_B} \lambda_x V_x^P \leq 4A_\star^L \left\{ \frac{\min\{1, dB\}}{u} + dB^2 \right\}.$$

On the complementary cells, [Lemma 9](#) gives

$$\sum_{x \in \text{Heavy}_B} \lambda_x \mathbb{E}_{\mathbb{Q}_x} P_x^2 \leq 132 n^{-12}.$$

Consequently,

$$\sum_{x \in [d]} \lambda_x V_x^P \leq 4A_\star^L \left\{ \frac{\min\{1, dB\}}{u} + dB^2 \right\} + 132 n^{-12}.$$

3. For the heavy-branch variance part, [Lemma 2](#) applied to the all-heavy statistic gives

$$\text{Var} \left(\sum_{x \in [d]} H_x \right) \leq C_{\text{var}} (u^{-1} + t^{-1}).$$

Independence across cells gives

$$\text{Var} \left(\sum_{x \in [d]} H_x \right) = \sum_{x \in [d]} V_x^H,$$

and since $0 \leq \theta_x \leq 1$,

$$\sum_{x \in [d]} \theta_x V_x^H \leq C_{\text{var}} (u^{-1} + t^{-1}).$$

4. It remains to bound the branch-selection mean term. On Light_B , the inequality $\lambda_x \theta_x \leq 1$ and [Lemma 7](#) give

$$\sum_{x \in \text{Light}_B} \lambda_x \theta_x (\bar{P}_x - \bar{H}_x)^2 \leq \sum_{x \in \text{Light}_B} (\bar{P}_x - \bar{H}_x)^2 \leq \frac{8dB^2}{\epsilon^2 L^4} + \frac{4}{\epsilon t}.$$

On Heavy_B , [Lemma 10](#) gives

$$\sum_{x \in \text{Heavy}_B} \lambda_x \theta_x (\bar{P}_x - \bar{H}_x)^2 \leq 282 n^{-12}.$$

Thus

$$\sum_{x \in [d]} \lambda_x \theta_x (\bar{P}_x - \bar{H}_x)^2 \leq \frac{8dB^2}{\epsilon^2 L^4} + \frac{4}{\epsilon t} + 282 n^{-12}.$$

5. Combining the three displayed bounds in the variance decomposition gives

$$\begin{aligned} \text{Var}(\tilde{Z}) &\leq 4A_\star^L \left\{ \frac{\min\{1, dB\}}{u} + dB^2 \right\} + C_{\text{var}}(u^{-1} + t^{-1}) \\ &\quad + \frac{8dB^2}{\epsilon^2 L^4} + \frac{4}{\epsilon t} + (132 + 282)n^{-12}. \end{aligned}$$

Since $132 + 282 = 414$, this is the asserted bound. □

Lemma 11 combines the light-cell factorial moment bound, the inverse-count variance contribution, and the pilot-routing error into a single variance inequality. Its display is arranged to match the envelope calculation in [Lemma 6](#) term by term.

Lemma 12. [lem:calibrated-hybrid-mean-bias] (Calibrated hybrid mean bias) *Fix an overlap constant $0 < \epsilon < 1/2$. For every $n, m, d \in \mathbb{N}$ and every observed-data law P on $[d] \times \{0, 1\}^2$, suppose that:*

- *(Sample size.)* $1 \leq n$.
- *(Model class.)* $P \in \mathcal{M}_{d, \epsilon}$, as in [Definition 6](#).
- *(Calibration.)* The finite calibration predicate $C_\epsilon(n, m)$ in [Algorithm 1](#) holds.

Let M_0, n_p, n_f, m_p, m_f be the block sizes from [Algorithm 1](#), and set

$$u = \frac{M_0}{8}, \quad t_p = \frac{n_p + m_p}{8}, \quad t = \frac{n_f + m_f}{8}.$$

Let $L = L(n)$, $B = B(n, m, \epsilon)$, and $k_0 = k_0(n, m, \epsilon)$ be the degree, light-cell scale, and pilot threshold from [Algorithm 1](#). For each $x \in [d]$, let the hybrid selected cell mean be the expectation of the cell contribution selected at threshold k_0 , degree L , scale B , and intensities (u, t_p, t) under the independent Poisson count law in [Algorithm 1](#). Then

$$\left| \sum_{x \in [d]} \mathbb{E}_P[Z_x] - \tau(P) \right| \leq \frac{2dB}{\epsilon L^2} + 6n^{-12},$$

where $\tau(P)$ is the average treatment effect functional in [Definition 8](#).

Proof of [Lemma 12](#). For $x \in [d]$ and $a \in \{0, 1\}$, write

$$p_x = P\{X = x\}, \quad s_{ax} = P\{X = x, A = a\}, \quad q_{ax} = P\{X = x, A = a, Y = 1\},$$

and define the totalized outcome regression by the paper's total real-division convention

$$\mu_{ax} := \frac{q_{ax}}{s_{ax}},$$

so that $\mu_{ax} = 0$ when $s_{ax} = 0$, because real division is totalized with $r/0 = 0$. Put

$$\vartheta_x := p_x(\mu_{1x} - \mu_{0x}).$$

Then $\tau(P) = \sum_x \vartheta_x$, $\sum_x p_x = 1$, and

$$|\vartheta_x| \leq p_x,$$

because $p_x \geq 0$ and the two binary outcome regressions lie in $[0, 1]$. Write b_n for the binary-length calibration quantity appearing in $C_\epsilon(n, m)$. Throughout this proof, $\log$ denotes the natural logarithm; since $b_n = 1 + \lceil \log_2 n \rceil$, $n \geq 1$ implies $\log n = (\log 2) \log_2 n \leq b_n$, and hence $\exp(-12b_n) \leq n^{-12}$. We record the positivity consequences of the finite inequalities defining $C_\epsilon(n, m)$ in [Algorithm 1](#). Since $24 \leq n$, we have $n > 0$ and $n + m > 0$, so the denominators in the block-proportion bounds are positive. Those bounds give

$$\frac{1}{32} \leq \frac{u}{n}, \quad \frac{1}{32} \leq \frac{t_p}{n+m}, \quad \frac{1}{32} \leq \frac{t}{n+m},$$

and hence $u > 0$, $t_p > 0$, and $t > 0$. The scale inequality $8 \leq t_p B$, together with $t_p > 0$, then gives $B > 0$. The calibration-scale definition in [Definition 13](#) makes $L = L(n)$ a maximum whose first entry is 2; in particular $L \geq 2$. The model condition in [Definition 6](#) places P in the occupied-cell overlap class, and [Assumption 1](#) gives, whenever $p_x > 0$,

$$\epsilon p_x \leq s_{0x}, \quad \epsilon p_x \leq s_{1x}.$$

When $p_x = 0$, the identity $s_{0x} + s_{1x} = p_x$ and nonnegativity of the two cell masses imply $s_{0x} = s_{1x} = 0$, so the same inequalities remain valid.

For each cell, let J_{0x}^P and J_{1x}^P be the two independent pilot arm counts, with

$$J_{0x}^P \sim \text{Poi}(t_p s_{0x}), \quad J_{1x}^P \sim \text{Poi}(t_p s_{1x}), \quad J_x := J_{0x}^P + J_{1x}^P.$$

Define

$$\pi_x := \Pr\{J_x \leq k_0\}, \quad \eta_x := \Pr\{J_x > k_0\}.$$

Thus $0 \leq \pi_x, \eta_x \leq 1$ and $\eta_x = 1 - \pi_x$. Let $\mathbb{Q}_x$ be the independent product law of four armwise Poisson counts

$$S_{ax} \sim \text{Poi}(u q_{ax}), \quad K_{ax} \sim \text{Poi}(t s_{ax}), \quad a \in \{0, 1\},$$

with all four variables mutually independent. These are ordinary nonnegative Poisson intensities, because $u, t > 0$ and the cell masses q_{ax} and s_{ax} are nonnegative. Define the light and heavy branch means

$$\Lambda_x := \mathbb{E}_{\mathbb{Q}_x} \left[\frac{S_{1x}}{u} \widehat{W}_{1,L,x} - \frac{S_{0x}}{u} \widehat{W}_{0,L,x} \right],$$

and

$$\Gamma_x := \mathbb{E}_{\mathbb{Q}_x} \left[\frac{S_{1x}}{u} \frac{K_{0x} + K_{1x} + 1}{K_{1x} + 1} - \frac{S_{0x}}{u} \frac{K_{0x} + K_{1x} + 1}{K_{0x} + 1} \right].$$

Let M_x denote the selected cell mean under the independent Poisson count law described in the lemma: the pilot pool selects the light branch on $\{J_x \leq k_0\}$ and the heavy branch on $\{J_x > k_0\}$, while the outcome and factorial pools are integrated under $\mathbb{Q}_x$. Thus

$$M_x = \mathbb{E} \left[\mathbf{1}_{\{J_x \leq k_0\}} \left(\frac{S_{1x}}{u} \widehat{W}_{1,L,x} - \frac{S_{0x}}{u} \widehat{W}_{0,L,x} \right) + \mathbf{1}_{\{J_x > k_0\}} \left(\frac{S_{1x}}{u} \frac{K_{0x} + K_{1x} + 1}{K_{1x} + 1} - \frac{S_{0x}}{u} \frac{K_{0x} + K_{1x} + 1}{K_{0x} + 1} \right) \right].$$

Independence of the pilot pool from the two evaluation pools gives the exact identity

$$M_x = \pi_x \Lambda_x + \eta_x \Gamma_x.$$

We now derive the first-moment identity from the armwise Poisson factorial moments, using the factorial-lift definition in [Definition 24](#). For $a \in \{0, 1\}$, set

$$\Lambda_{a,x} := \mathbb{E}_{\mathbb{Q}_x} \left[\frac{S_{ax}}{u} \widehat{W}_{a,L,x} \right], \quad \Lambda_x = \Lambda_{1,x} - \Lambda_{0,x}.$$

Under $\mathbb{Q}_x$, the outcome count S_{ax} is independent of the factorial-pool counts, and

$$\mathbb{E}_{\mathbb{Q}_x} S_{ax} = u q_{ax}.$$

Write the light polynomial as in [Definition 24](#):

$$W_{a,L}(r_0, r_1) = \sum_{\rho_0, \rho_1} c_{\rho_0, \rho_1}^{(a)} r_0^{\rho_0} r_1^{\rho_1},$$

with only finitely many nonzero coefficients. Its factorial lift is

$$\widehat{W}_{a,L,x}(K_{0x}, K_{1x}) = \sum_{\rho_0, \rho_1} c_{\rho_0, \rho_1}^{(a)} \frac{(K_{0x})_{\rho_0} (K_{1x})_{\rho_1}}{t^{\rho_0 + \rho_1}},$$

where $(k)_\rho$ is the falling factorial. If $K \sim \text{Poi}(ts)$ with $s \geq 0$, then, for every $\rho \geq 0$,

$$\mathbb{E}(K)_\rho = \sum_{\ell=\rho}^{\infty} \frac{\ell!}{(\ell-\rho)!} e^{-ts} \frac{(ts)^\ell}{\ell!} = (ts)^\rho \sum_{j=0}^{\infty} e^{-ts} \frac{(ts)^j}{j!} = (ts)^\rho.$$

Since $t > 0$ and the two factorial-pool counts are independent Poisson variables with means ts_{0x} and ts_{1x} , this gives

$$\mathbb{E}_{\mathbb{Q}_x} \frac{(K_{0x})_{\rho_0} (K_{1x})_{\rho_1}}{t^{\rho_0 + \rho_1}} = s_{0x}^{\rho_0} s_{1x}^{\rho_1}.$$

Termwise summation over the finite coefficient expansion yields the first-moment identity

$$\mathbb{E}_{\mathbb{Q}_x} \widehat{W}_{a,L,x} = W_{a,L}(s_{0x}, s_{1x}),$$

and therefore

$$\Lambda_{a,x} = q_{ax} W_{a,L}(s_{0x}, s_{1x}).$$

Using $s_{0x} + s_{1x} = p_x$ and

$$W_{a,L}(s_{0x}, s_{1x}) = \frac{p_x}{B} G_L\left(\frac{s_{ax}}{B}\right),$$

we obtain

$$\Lambda_x = q_{1x} \frac{p_x}{B} G_L\left(\frac{s_{1x}}{B}\right) - q_{0x} \frac{p_x}{B} G_L\left(\frac{s_{0x}}{B}\right).$$

Consider first a light cell, $p_x \leq B$. If $p_x = 0$, then $s_{0x} = s_{1x} = q_{0x} = q_{1x} = 0$, so $\Lambda_x = \vartheta_x = 0$ and the light-branch bias bound is immediate. It remains in this paragraph to handle the case $p_x > 0$. For $a \in \{0, 1\}$, set

$$z_{ax} := \frac{s_{ax}}{B}, \quad E_{ax} := E_L(z_{ax}).$$

The overlap inequalities give $s_{ax} \geq \epsilon p_x > 0$ for $a = 0, 1$, so $z_{ax} > 0$; the light-cell condition $p_x \leq B$, together with $s_{ax} \leq p_x$ and $B > 0$, gives $z_{ax} \leq 1$. Hence $z_{ax} \in (0, 1]$. The factorial-moment calculation above has already identified the light-pool first moment. Since $L \geq 2$ and $z_{ax} \in (0, 1]$, the interval part of [Lemma 16](#) applies and gives

$$z_{ax} G_L(z_{ax}) + E_{ax} = 1, \quad 0 \leq E_{ax}, \quad E_{ax} \leq \frac{1}{L^2 z_{ax}}.$$

Since $q_{ax} = s_{ax}\mu_{ax}$, the displayed formula for Λ_x becomes, in the case $p_x > 0$,

$$\Lambda_x = p_x\mu_{1x}(1 - E_{1x}) - p_x\mu_{0x}(1 - E_{0x}),$$

and therefore

$$|\Lambda_x - \vartheta_x| \leq p_x\mu_{1x}E_{1x} + p_x\mu_{0x}E_{0x} \leq \frac{p_x B}{L^2 s_{1x}} + \frac{p_x B}{L^2 s_{0x}} \leq \frac{2B}{\epsilon L^2}.$$

Together with the already settled case $p_x = 0$, this proves the light-branch bound for every light cell.

For the heavy branch, decompose the branch mean armwise by setting

$$\Gamma_{a,x} := \mathbb{E}_{\mathbb{Q}_x} \left[\frac{S_{ax}}{u} \left(1 + \frac{K_{1-a,x}}{K_{ax} + 1} \right) \right], \quad a \in \{0, 1\},$$

so that $\Gamma_x = \Gamma_{1,x} - \Gamma_{0,x}$. By independence of S_{ax} from the factorial-pool counts and $\mathbb{E}_{\mathbb{Q}_x} S_{ax} = uq_{ax}$,

$$\Gamma_{a,x} = q_{ax} \mathbb{E} \left[1 + \frac{K_{1-a,x}}{K_{ax} + 1} \right].$$

Substituting the cell quantities $p_x = s_{0x} + s_{1x}$, s_{ax} , q_{ax} , and $\mu_{ax} = q_{ax}/s_{ax}$ in [Lemma 8](#), and using the total real-division simplification $u/u = 1$ that follows from $u > 0$, gives

$$q_{ax} \mathbb{E} \left[1 + \frac{K_{1-a,x}}{K_{ax} + 1} \right] = p_x \mu_{ax} - (p_x - s_{ax}) \mu_{ax} \exp(-ts_{ax}),$$

whose left side is $\Gamma_{a,x}$. The totalized zero-arm cases are covered as follows: if $s_{ax} = 0$, then $0 \leq q_{ax} \leq s_{ax}$ gives $q_{ax} = 0$, so $\Gamma_{a,x} = 0$, while $\mu_{ax} = q_{ax}/s_{ax} = 0$ and the right side is also zero. If $s_{ax} > 0$, the quotient is the ordinary outcome regression and the same identity applies directly. Thus, for each arm,

$$|\Gamma_{a,x} - p_x \mu_{ax}| = (p_x - s_{ax}) \mu_{ax} \exp(-ts_{ax}).$$

If $p_x = 0$, this expression is zero. If $p_x > 0$, occupied-cell overlap gives $s_{ax} \geq \epsilon p_x$; together with $0 \leq p_x - s_{ax} \leq p_x$ and $0 \leq \mu_{ax} \leq 1$, this yields

$$|\Gamma_{a,x} - p_x \mu_{ax}| \leq p_x \exp(-\epsilon t p_x).$$

Summing the two armwise errors gives

$$|\Gamma_x - \vartheta_x| \leq 2p_x \exp(-\epsilon t p_x).$$

We next derive the pilot-selection estimate used on light cells. Since the two pilot arm counts are independent Poisson variables with means $t_p s_{0x}$ and $t_p s_{1x}$, their sum satisfies

$$J_x = J_{0x}^p + J_{1x}^p \sim \text{Poi}(t_p(s_{0x} + s_{1x})) = \text{Poi}(t_p p_x).$$

Let

$$\lambda_x^p := t_p p_x.$$

For any $\alpha_{\text{tail}} > 0$, the exponential Markov bound applied to $(1 + \alpha_{\text{tail}})^{J_x}$ gives

$$\begin{aligned} \Pr\{J_x > k_0\} \exp(-\alpha_{\text{tail}} \lambda_x^{\text{p}}) &\leq \frac{\mathbb{E}(1 + \alpha_{\text{tail}})^{J_x}}{(1 + \alpha_{\text{tail}})^{k_0}} \exp(-\alpha_{\text{tail}} \lambda_x^{\text{p}}) \\ &= (1 + \alpha_{\text{tail}})^{-k_0}, \end{aligned}$$

because the Poisson probability generating function gives $\mathbb{E}(1 + \alpha_{\text{tail}})^{J_x} = \exp(\alpha_{\text{tail}} \lambda_x^{\text{p}})$. Taking $\alpha_{\text{tail}} = \epsilon t / t_p$, which is positive since the calibration predicate gives $t_p > 0$ and $t > 0$, yields

$$\eta_x \exp(-\epsilon t p_x) \leq \left(1 + \epsilon \frac{t}{t_p}\right)^{-k_0}.$$

By $\bar{\epsilon} \leq \epsilon$ from [Definition 13](#) and $t/t_p \geq 0$,

$$\left(1 + \bar{\epsilon} \frac{t}{t_p}\right)^{k_0} \leq \left(1 + \epsilon \frac{t}{t_p}\right)^{k_0}.$$

The calibration inequality in [Algorithm 1](#) gives

$$n^{12} \leq \left(1 + \bar{\epsilon} \frac{t}{t_p}\right)^{k_0},$$

and hence, with positive denominators,

$$\left(1 + \epsilon \frac{t}{t_p}\right)^{-k_0} \leq n^{-12}.$$

Combining these estimates with the mixture identity yields, for $p_x \leq B$,

$$|M_x - \vartheta_x| \leq \frac{2B}{\epsilon L^2} + 2p_x n^{-12}.$$

Now consider a heavy cell, $B < p_x$. Put

$$\lambda_x^{\text{p}} := t_p p_x.$$

We derive the light-pilot lower-tail estimate directly. By the definition above, $J_x = J_{0x}^{\text{p}} + J_{1x}^{\text{p}}$, and independence of the two pilot arm counts gives

$$J_x \sim \text{Poi}(t_p(s_{0x} + s_{1x})) = \text{Poi}(\lambda_x^{\text{p}}).$$

Moreover, [Algorithm 1](#) defines the pilot threshold by

$$k_0 = \left\lfloor \frac{t_p B}{4} \right\rfloor.$$

Because $t_p B / 4 \geq 0$, the defining floor inequality, after casting the natural-valued threshold k_0 to the reals, gives $k_0 \leq t_p B / 4$. The heavy-cell inequality $B < p_x$, multiplied by the positive number $t_p / 4$, gives $t_p B / 4 < t_p p_x / 4 = \lambda_x^{\text{p}} / 4$. Thus

$$k_0 \leq \frac{t_p B}{4} < \frac{\lambda_x^{\text{p}}}{4}.$$

With $\theta = \log 2$, apply exponential Markov directly to $e^{-\theta J_x}$. Since the event $J_x \leq k_0$ is the same as $e^{-\theta J_x} \geq e^{-\theta k_0}$, with the natural number k_0 cast to a real in the exponential, the floor inequality above gives

$$\begin{aligned}\pi_x &= \Pr\{J_x \leq k_0\} \\ &\leq \exp\left(\theta k_0 + \lambda_x^p(e^{-\theta} - 1)\right) \\ &\leq \exp\left(\frac{\theta \lambda_x^p}{4} + \lambda_x^p(e^{-\theta} - 1)\right) \\ &= \exp\left(\left(\frac{\theta}{4} - \frac{1}{2}\right) \lambda_x^p\right) \leq \exp\left(-\frac{\lambda_x^p}{4}\right),\end{aligned}$$

since $e^{-\theta} = 1/2$ and $\theta \leq 1$. The tail-calibration inequalities in [Algorithm 1](#) include

$$15L + 12b_n \leq \frac{t_p B}{4},$$

so

$$12b_n \leq 15L + 12b_n \leq \frac{t_p B}{4} < \frac{\lambda_x^p}{4}.$$

Consequently

$$\pi_x \leq \exp(-12b_n) \leq n^{-12},$$

where the last comparison is the binary-length consequence recorded at the start of the proof.

We next prove the auxiliary estimate for the wrong-light contribution on the same heavy cell:

$$\pi_x |\Lambda_x| \leq p_x n^{-12}.$$

Put

$$y_x := \frac{p_x}{B},$$

so $y_x \geq 1$. First we bound the absolute light-branch mean. The first-moment identity gives

$$\Lambda_x = q_{1x} \frac{p_x}{B} G_L\left(\frac{s_{1x}}{B}\right) - q_{0x} \frac{p_x}{B} G_L\left(\frac{s_{0x}}{B}\right).$$

For each arm, $0 \leq s_{ax} \leq p_x$ and $0 \leq q_{ax} \leq p_x$. If

$$G_L(z) = \sum_{\rho=0}^{L-2} g_\rho z^\rho,$$

then the degree bound in [Lemma 5](#) justifies the displayed finite expansion with no terms beyond degree $L - 2$, while [Lemma 16](#) gives $2\|G_L\|_1 \leq 14^L$. Since $0 \leq s_{ax}/B \leq y_x$ and $y_x \geq 1$, each displayed exponent satisfies

$$\left(\frac{s_{ax}}{B}\right)^\rho \leq y_x^\rho \leq (1 + y_x)^\rho \leq (1 + y_x)^L.$$

Multiplying this monomial bound by $|g_\rho|$ and summing over the displayed finite support gives

$$\left|G_L\left(\frac{s_{ax}}{B}\right)\right| \leq \sum_{\rho=0}^{L-2} |g_\rho| \left(\frac{s_{ax}}{B}\right)^\rho \leq \|G_L\|_1 (1 + y_x)^L.$$

Also $y_x \leq (1 + y_x)^L$, because $L \geq 2$ and $y_x \geq 0$. Therefore

$$|\Lambda_x| \leq 2p_x y_x \|G_L\|_1 (1 + y_x)^L \leq p_x 14^L (1 + y_x)^{2L}.$$

This coefficient-envelope argument uses the exact first-moment identity above, together with the degree and coefficient bounds for G_L , and yields the displayed bound without invoking any second-moment estimate. The lower-tail calculation above, kept in exponential form, gives

$$\pi_x \leq \exp\left(-\frac{t_p p_x}{4}\right).$$

The explicit scale definition in [Definition 13](#) gives

$$B = \frac{H_\epsilon L}{\min(t_p, t)}.$$

Since $t_p > 0$ and $t > 0$, their minimum is positive and the scale identity gives $H_\epsilon L = B \min(t_p, t)$. As $\min(t_p, t) \leq t_p$ and $B > 0$, multiplication preserves the inequality, so $H_\epsilon L \leq t_p B$. Thus, with $p_x = B y_x$,

$$\exp\left(-\frac{t_p p_x}{4}\right) \leq \exp\left(-\frac{H_\epsilon L y_x}{4}\right).$$

The environment [Definition 13](#) gives $A_\star = 7056$, $c^\circ = 1/256$, and

$$\frac{H_\epsilon}{4} \geq 15 + \frac{24}{c^\circ}.$$

Elementary arithmetic applied to the displayed value of $A_\star$ gives $14 \leq A_\star < e^{13}$. Using these consequences, $y_x \geq 1$, and $\log(1 + y_x) \leq y_x$, we obtain

$$L \log A_\star + 2L \log(1 + y_x) - \frac{H_\epsilon L y_x}{4} \leq 13L y_x + 2L y_x - \left(15 + \frac{24}{c^\circ}\right) L y_x = -\frac{24L y_x}{c^\circ}.$$

Exponentiating gives the calibrated absorption bound

$$A_\star^L (1 + y_x)^{2L} \exp\left(-\frac{H_\epsilon L y_x}{4}\right) \leq \exp\left(-\frac{24L y_x}{c^\circ}\right).$$

Combining the envelope for Λ_x , the pilot-tail display, and $14^L \leq A_\star^L$,

$$\pi_x |\Lambda_x| \leq p_x \exp\left(-\frac{24L y_x}{c^\circ}\right).$$

Finally, the degree lower bound $c^\circ b_n / 2 \leq L$ in $\mathbf{C}_\epsilon(n, m)$, listed in [Algorithm 1](#), together with $y_x \geq 1$, implies

$$12b_n \leq \frac{24L y_x}{c^\circ}.$$

The binary-length consequence recorded at the start of the proof gives

$$\exp\left(-\frac{24L y_x}{c^\circ}\right) \leq \exp(-12b_n) \leq n^{-12}.$$

Combining the last two displays proves

$$\pi_x |\Lambda_x| \leq p_x n^{-12}.$$

The intrinsic heavy-branch exponential is also absorbed on a heavy cell:

$$\exp(-\epsilon t p_x) \leq n^{-12}.$$

Indeed, $B < p_x$, $\bar{\epsilon} \leq \epsilon$, and the tail-calibration inequality in [Algorithm 1](#),

$$12b_n \leq \bar{\epsilon} t B,$$

give $12b_n \leq \epsilon t p_x$, and hence, by the binary-length consequence recorded at the start of the proof,

$$\exp(-\epsilon t p_x) \leq \exp(-12b_n) \leq n^{-12}.$$

Therefore

$$\eta_x |\Gamma_x - \vartheta_x| \leq 2p_x n^{-12}.$$

Together with $\pi_x |\vartheta_x| \leq p_x n^{-12}$, the decomposition

$$M_x - \vartheta_x = \pi_x \Lambda_x - \pi_x \vartheta_x + \eta_x (\Gamma_x - \vartheta_x)$$

gives, for $B < p_x$,

$$|M_x - \vartheta_x| \leq 4p_x n^{-12}.$$

The alternatives $p_x \leq B$ and $B < p_x$ are disjoint and exhaustive, so they partition $[d]$. Summing the two cellwise bounds over these two regimes,

$$\begin{aligned} \left| \sum_{x \in [d]} M_x - \tau(P) \right| &= \left| \sum_{x \in [d]} (M_x - \vartheta_x) \right| \\ &\leq \sum_{x \in [d]} |M_x - \vartheta_x| \\ &\leq \sum_{x: p_x \leq B} \left(\frac{2B}{\epsilon L^2} + 2p_x n^{-12} \right) + \sum_{x: B < p_x} 4p_x n^{-12} \\ &\leq \frac{2dB}{\epsilon L^2} + 4n^{-12} \sum_{x \in [d]} p_x \\ &= \frac{2dB}{\epsilon L^2} + 4n^{-12} \leq \frac{2dB}{\epsilon L^2} + 6n^{-12}. \end{aligned}$$

This is the claimed bound. $\square$

[Lemma 12](#) gives the matching aggregate bias control for the selected cell means. The bound contains the light-cell Chebyshev approximation term and a polynomially small pilot-boundary residual, which together form the squared bias term in [Lemma 6](#).

[Lemma 6](#) is the bookkeeping step that turns the local hybrid analysis into the global rate scale. Its left-hand side collects the light-cell factorial moment contribution, the heavy-cell inverse-count contribution, the Chebyshev approximation residual, the pilot-boundary term, and the finite prefix terms, while the right-hand side matches the two summands in [Definition 12](#).

The calibrated hybrid statistic combines the light-cell polynomial with the all-heavy inverse-count weight. The construction below is the Poissonized counterpart of the finite-sample mixed estimator introduced in [Algorithm 1](#).

Definition 25. [synth_9] (Calibrated Poisson hybrid statistic) Use the split sizes, intensities, constants, dyadic overlap certificate, and calibration predicate $C_\epsilon(n, m)$ from [Algorithm 1](#). In the independent Poisson experiment with outcome-prefix length $D_0 \sim \text{Poi}(u)$, pilot-prefix length $D_p \sim \text{Poi}(t_p)$, and factorial-prefix length $D_f \sim \text{Poi}(t)$, define Z^{mix} as follows. If $C_\epsilon(n, m)$ is false, set

$$Z^{\text{mix}} := 0.$$

If $C_\epsilon(n, m)$ is true, return zero on the event that any Poisson prefix exceeds its corresponding pool. On the complementary valid-prefix event, let J_x be the pilot count in cell x , let K_{ax} be the factorial-pool count in arm a and cell x , and let S_{ax} be the outcome-block success count in arm a and cell x . With

$$H_L(z) = \frac{1 - T_L(1 - 2z)}{2L^2}, \quad E_L(z) = \frac{H_L(z)}{z}, \quad G_L(z) = \frac{1 - E_L(z)}{z},$$

under polynomial continuation at zero, expand

$$W_{a,L}(s_0, s_1) = \frac{s_0 + s_1}{B} G_L(s_a/B) = \sum_{r_0, r_1} c_{a,r_0,r_1} s_0^{r_0} s_1^{r_1}.$$

For falling factorials $(k)_r$, put

$$\widehat{W}_{a,L,x} = \sum_{r_0, r_1} c_{a,r_0,r_1} \frac{(K_{0x})_{r_0} (K_{1x})_{r_1}}{t^{r_0+r_1}}, \quad \widehat{W}_{a,x}^H = \frac{K_{0x} + K_{1x} + 1}{K_{ax} + 1},$$

and define the cell contribution

$$Z_x = \mathbf{1}\{J_x \leq k_0\} \left(\frac{S_{1x}}{u} \widehat{W}_{1,L,x} - \frac{S_{0x}}{u} \widehat{W}_{0,L,x} \right) + \mathbf{1}\{J_x > k_0\} \left(\frac{S_{1x}}{u} \widehat{W}_{1,x}^H - \frac{S_{0x}}{u} \widehat{W}_{0,x}^H \right).$$

The calibrated Poisson hybrid statistic is the clipped aggregate

$$Z^{\text{mix}} := \text{clip}_{[-1,1]} \left(\sum_{x=1}^d Z_x \right)$$

on the valid-prefix event, together with the preceding zero values on invalid prefixes or failed calibration.

The pilot threshold in [Definition 25](#) chooses between two weight regimes at the cell level. Cells with small pilot counts use the bounded-degree Chebyshev lift, while cells with larger pilot counts use the inverse-count weight from [Lemma 2](#); clipping keeps the aggregate on the natural scale of the treatment effect.

Lemma 13. [lem:poisson-hybrid-risk] (Hybrid Poisson risk) Fix an overlap constant ϵ with $0 < \epsilon < 1/2$. There is a constant $C_\epsilon \in (0, \infty)$, depending only on ϵ , such that the following holds for every n, m, d and every observed-data law P :

- (Sample size.) $n \geq 1$.
- (Alphabet size.) $d \geq 2$.
- (Model class.) $P \in \mathcal{M}_{d,\epsilon}$, as in [Definition 6](#).

Let $M_0 = \lfloor n/3 \rfloor$, $n_p = \lfloor (n - M_0)/2 \rfloor$, $n_f = n - M_0 - n_p$, $m_p = \lfloor m/2 \rfloor$, and $m_f = m - m_p$, and set

$$u = \frac{M_0}{8}, \quad t_p = \frac{n_p + m_p}{8}, \quad t = \frac{n_f + m_f}{8}.$$

In the independent Poisson count experiment with these intensities, let Z^{mix} be the calibrated hybrid statistic when the finite calibration predicate $C_\epsilon(n, m)$ holds and the constant 0 otherwise. Then its mean-squared error satisfies

$$\mathbb{E}_P \left[(Z^{\text{mix}} - \tau(P))^2 \right] \leq C_\epsilon r_\epsilon(n, m, d),$$

where $r_\epsilon(n, m, d)$ is the frontier rate in [Definition 12](#), equivalently

$$r_\epsilon(n, m, d) = \min \left\{ 1, \frac{1}{n} + \frac{d^2}{(n + m)^2 \log^2(en)} \right\}.$$

Proof of [Lemma 13](#). Fix $\epsilon \in (0, 1/2)$. Throughout, let $M_0, n_p, n_f, m_p, m_f, u, t_p, t, L, B, k_0$ be the block sizes, Poisson intensities, degree, scale, and pilot threshold from [Algorithm 1](#). Write

$$\text{rad}_{n,m,d} = \frac{1}{n} + \frac{d^2}{(n + m)^2 \log^2(en)},$$

so that $r_\epsilon(n, m, d) = \min\{1, \text{rad}_{n,m,d}\}$ by [Definition 12](#).

1. First record the uniform bounded-loss estimate. For $P \in \mathcal{M}_{d,\epsilon}$, the functional $\tau(P)$ lies in $[-1, 1]$, since it is a cell-mass average of differences of binary outcome regressions. On the calibrated branch, the hybrid statistic is clipped to $[-1, 1]$, so its squared distance from $\tau(P)$ is at most 4 pointwise. On the uncalibrated branch, the risk term is the deterministic quantity $\tau(P)^2$, which is at most 1 and hence at most 4. Therefore, for all n, m, d ,

$$\mathbb{E}_P \left[(Z^{\text{mix}} - \tau(P))^2 \right] \leq 4.$$

2. We next record the calibrated variance envelope used when $C_\epsilon(n, m)$ holds. For each $x \in [d]$, let S_{ax} , J_{ax} , and K_{ax} be the independent outcome-block, pilot-pool, and factorial-pool Poisson counts from [Algorithm 1](#), and put $J_x = J_{0x} + J_{1x}$. Let P_x and H_x be the light-cell polynomial and heavy inverse-count cell contributions from [Algorithm 1](#), and define the unclipped selected hybrid sum

$$\tilde{Z} = \sum_{x=1}^d (\mathbf{1}\{J_x \leq k_0\} P_x + \mathbf{1}\{J_x > k_0\} H_x).$$

By [Lemma 11](#), there is a constant $C_{\text{var}} > 0$, depending only on ϵ , such that, whenever $n \geq 1$, $P \in \mathcal{M}_{d,\epsilon}$, and $C_\epsilon(n, m)$ holds,

$$\text{Var}_P(\tilde{Z}) \leq 4A_\star^L \left\{ \frac{\min\{1, dB\}}{u} + dB^2 \right\} + C_{\text{var}} (u^{-1} + t^{-1}) + \frac{8dB^2}{\epsilon^2 L^4} + \frac{4}{\epsilon t} + 414n^{-12}.$$

3. Still under $C_\epsilon(n, m)$, projection onto $[-1, 1]$ cannot increase squared distance from $\tau(P) \in [-1, 1]$. Hence

$$\mathbb{E}_P \left[(Z^{\text{mix}} - \tau(P))^2 \right] \leq \mathbb{E}_P \left[(\tilde{Z} - \tau(P))^2 \right] = \text{Var}_P(\tilde{Z}) + (\mathbb{E}_P \tilde{Z} - \tau(P))^2.$$

For the same hybrid branches, [Lemma 12](#) gives

$$\left| \mathbb{E}_P \tilde{Z} - \tau(P) \right| \leq \frac{2dB}{\epsilon L^2} + 6n^{-12}.$$

Consequently,

$$\mathbb{E}_P [(Z^{\text{mix}} - \tau(P))^2] \leq \text{Var}_P(\tilde{Z}) + \left(\frac{2dB}{\epsilon L^2} + 6n^{-12} \right)^2.$$

4. Apply [Lemma 6](#) with $C_0 = C_{\text{var}}$. That result gives a constant depending on ϵ and C_{var} ; since Step 2 chose C_{var} depending only on ϵ , we may write this constant as $C_{\text{abs}} > 0$, depending only on ϵ . Under $C_\epsilon(n, m)$,

$$\begin{aligned} 4A_\star^L \left\{ \frac{\min\{1, dB\}}{u} + dB^2 \right\} + C_{\text{var}} (u^{-1} + t^{-1}) + \frac{8dB^2}{\epsilon^2 L^4} + \frac{4}{\epsilon t} + 414n^{-12} + \left(\frac{2dB}{\epsilon L^2} + 6n^{-12} \right)^2 \\ \leq C_{\text{abs}} \text{rad}_{n,m,d}. \end{aligned}$$

Set

$$C_{\text{cal}} = \max\{4, C_{\text{abs}}\}.$$

If $1 \leq \text{rad}_{n,m,d}$, then $r_\epsilon(n, m, d) = 1$, and Step 1 gives

$$\mathbb{E}_P [(Z^{\text{mix}} - \tau(P))^2] \leq 4 \leq C_{\text{cal}} r_\epsilon(n, m, d).$$

If $\text{rad}_{n,m,d} \leq 1$, then $r_\epsilon(n, m, d) = \text{rad}_{n,m,d}$. Steps 2 and 3, followed by the absorption display, give

$$\mathbb{E}_P [(Z^{\text{mix}} - \tau(P))^2] \leq C_{\text{abs}} \text{rad}_{n,m,d} \leq C_{\text{cal}} r_\epsilon(n, m, d).$$

Thus the desired bound holds on the calibrated branch with constant C_{cal} .

5. It remains to cover the finite set of uncalibrated sample sizes. The uniform calibration-failure bound gives an integer $n_\epsilon^\dagger \geq 1$, depending only on ϵ , such that

$$\neg C_\epsilon(n, m) \implies n < n_\epsilon^\dagger \quad \text{for all } n, m.$$

Now set

$$C_\epsilon = \max\{C_{\text{cal}}, 4n_\epsilon^\dagger\}.$$

If $C_\epsilon(n, m)$ holds, Step 4 gives the result because $C_{\text{cal}} \leq C_\epsilon$. If $C_\epsilon(n, m)$ fails, then $n < n_\epsilon^\dagger$, so $n \leq n_\epsilon^\dagger$. Since $n \geq 1$, the frontier rate is positive and dominates n^{-1} : the second entry in the minimum is

$$\frac{1}{n} + \frac{d^2}{(n+m)^2 \log^2(en)} \geq \frac{1}{n},$$

and $1/n \leq 1$, so

$$\frac{1}{n} \leq r_\epsilon(n, m, d).$$

Multiplying by the positive number n gives $1 \leq n r_\epsilon(n, m, d)$. Since $n \leq n_\epsilon^\dagger$ and $r_\epsilon(n, m, d) > 0$,

$$1 \leq n r_\epsilon(n, m, d) \leq n_\epsilon^\dagger r_\epsilon(n, m, d).$$

Combining this with Step 1 gives

$$\mathbb{E}_P [(Z^{\text{mix}} - \tau(P))^2] \leq 4 \leq 4n_\epsilon^\dagger r_\epsilon(n, m, d) \leq C_\epsilon r_\epsilon(n, m, d).$$

This proves the asserted bound for every n, m, d and every $P \in \mathcal{M}_{d,\epsilon}$. $\square$

Lemma 13 establishes the Poisson risk bound at exactly the split sizes and intensities used by the finite estimator. The bound has the same two components as the rate in [Definition 12](#): labeled outcomes contribute the $1/n$ term, and treatment–covariate information contributes the alphabet-dependent term through the pilot and factorial pools.

Uniform finite-calibration input. The uncalibrated branch uses the formally checked helper `calibrationPredicate_failure_bounded`: for fixed $0 < \epsilon < 1/2$, there is $n_\epsilon^\dagger \geq 1$ such that

$$\neg C_\epsilon(n, m) \implies n < n_\epsilon^\dagger \quad \text{for every } m.$$

This is the contrapositive of eventual validity of all finitely many calibration inequalities, uniformly over the auxiliary sample size. It reduces calibration failure to finitely many labeled sizes, where clipping gives risk at most 4 and $r_\epsilon(n, m, d) \geq 1/n$, so $4 \leq 4n_\epsilon^\dagger r_\epsilon(n, m, d)$.

It remains to connect the independent Poisson count analysis to the fixed sample sizes in the annotation experiment. The next definition packages the three independent Poisson pools used in that transfer.

Definition 26. `[synth_6]` (Three-pool Poisson count law) *Fix an observed-data law P , intensities $u, t_p, t \geq 0$, and write*

$$q_{ax} = P(X = x, A = a, Y = 1), \quad s_{ax} = P(X = x, A = a).$$

The law $\mathbb{P}_{P, u, t_p, t}^{\text{Pois}}$ is the joint distribution of independent count triples

$$(S_{ax}, J_{ax}, K_{ax})_{a \in \{0,1\}, x \in [d]},$$

where

$$S_{ax} \sim \text{Poi}(uq_{ax}), \quad J_{ax} \sim \text{Poi}(t_p s_{ax}), \quad K_{ax} \sim \text{Poi}(t s_{ax}).$$

Here S_{ax} is the outcome-success count from the outcome pool, J_{ax} is the pilot count, and K_{ax} is the factorial-pool treatment-covariate count. The product law makes all these counts independent across cells, arms, and pools.

The three-pool law in [Definition 26](#) mirrors the deterministic sample split in [Algorithm 1](#). Each pool has its own intensity, so the Poisson calculation can track outcome information, pilot classification, and polynomial weight estimation separately.

Lemma 14. `[lem:fixed-sample-hybrid-transfer]` (Fixed-sample hybrid transfer) *There is a universal constant $C > 0$ such that the following holds. Let*

- **(Overlap level.)** *The overlap level ϵ satisfies $0 < \epsilon < 1/2$.*
- **(Sample sizes and alphabet.)** *$n \geq 1$, $m \geq 0$, and $d \geq 2$.*
- **(Model class.)** *$P \in \mathcal{M}_{d, \epsilon}$, with $\mathcal{M}_{d, \epsilon}$ as in [Definition 6](#).*

For the mixed estimator $\hat{\tau}_{n, m, d}^{\text{mix}}$ in [Algorithm 1](#), the estimator is measurable as a function of the annotation sample and

$$\mathbb{E}_{P^{\otimes n} \otimes P_{XA}^{\otimes m}} \left[\left\{ \hat{\tau}_{n, m, d}^{\text{mix}} - \tau(P) \right\}^2 \right] \leq 2 \mathcal{R}_{\text{hyb}}^{\text{Pois}}(P; n, m, \epsilon) + \frac{C}{n}.$$

Here $\tau(P)$ is the average treatment effect functional in [Definition 8](#). Writing $M_0 = \lfloor n/3 \rfloor$, $n_p = \lfloor (n - M_0)/2 \rfloor$, $n_f = n - M_0 - n_p$, $m_p = \lfloor m/2 \rfloor$, $m_f = m - m_p$, $M_p = n_p + m_p$, $M_f = n_f + m_f$, $u = M_0/8$, $t_p = M_p/8$, and $t = M_f/8$, the Poisson hybrid risk is

$$\mathcal{R}_{\text{hyb}}^{\text{Pois}}(P; n, m, \epsilon) = \begin{cases} \int \{Z^{\text{mix}}(K) - \tau(P)\}^2 d\mathbb{P}_{P,u,t_p,t}^{\text{Pois}}(K), & \text{C}_\epsilon(n, m), \\ \tau(P)^2, & \text{otherwise,} \end{cases}$$

where Z^{mix} is the calibrated hybrid statistic of [Definition 25](#).

Proof of [Lemma 14](#). Take $C = 384$. Fix $0 < \epsilon < 1/2$, $n \geq 1$, $m \geq 0$, $d \geq 2$, and $P \in \mathcal{M}_{d,\epsilon}$. The estimator is measurable because it is either the constant zero map or, on the calibrated branch of [Algorithm 1](#), a finite linear combination of measurable prefix-count statistics. It remains to prove the displayed risk bound.

1. Suppose first that $\text{C}_\epsilon(n, m)$ holds. Put

$$M_0 = \lfloor n/3 \rfloor, \quad M_p = n_p + m_p, \quad M_f = n_f + m_f,$$

and

$$u = \frac{M_0}{8}, \quad t_p = \frac{M_p}{8}, \quad t = \frac{M_f}{8}.$$

Let $\mathbb{P}_3$ be the three-pool product law whose complete-record coordinate has law P on $[d] \times \{0, 1\} \times \{0, 1\}$, while the two shared coordinates have law P_{XA} on $[d] \times \{0, 1\}$. Let $\mathbf{s} = (u, t_p, t)$ and let the capacity vector be $\mathbf{M} = (M_0, M_p, M_f)$. For a three-prefix array

$$\mathbf{s} = (\mathbf{s}_0, \mathbf{s}_p, \mathbf{s}_f),$$

define

$$\Psi_{n,m,d,\epsilon}(\mathbf{s}) = Z_{n,m,\epsilon,u,t}^{\text{mix}}(\text{hist}(\mathbf{s}_0), \text{hist}(\mathbf{s}_p), \text{hist}(\mathbf{s}_f)),$$

where the first histogram records complete outcome-labeled observations and the last two record treatment-covariate observations. This prefix statistic is measurable and satisfies

$$-1 \leq \Psi_{n,m,d,\epsilon}(\mathbf{s}) \leq 1 \quad \text{for every } \mathbf{s}, \quad -1 \leq \tau(P) \leq 1,$$

the first bound coming from clipping in [Algorithm 1](#) and the second from [Definition 8](#) together with occupied-cell overlap in [Definition 6](#).

2. For fixed pools

$$\mathbf{p} = (\mathbf{p}_0, \mathbf{p}_p, \mathbf{p}_f) \in ([d] \times \{0, 1\}^2)^{M_0} \times ([d] \times \{0, 1\})^{M_p} \times ([d] \times \{0, 1\})^{M_f},$$

define the Rao–Blackwellized prefix statistic by

$$\text{RB}(\mathbf{p}) = \sum_{r_0, r_p, r_f \geq 0} \Pr\{\text{Poi}(u) = r_0\} \Pr\{\text{Poi}(t_p) = r_p\} \Pr\{\text{Poi}(t) = r_f\} \Psi^\circ(\mathbf{p}; r_0, r_p, r_f),$$

where

$$\Psi^\circ(\mathbf{p}; r_0, r_p, r_f) = \begin{cases} \Psi_{n,m,d,\epsilon}(\mathbf{p}_0[1:r_0], \mathbf{p}_p[1:r_p], \mathbf{p}_f[1:r_f]), & r_0 \leq M_0, r_p \leq M_p, r_f \leq M_f, \\ 0, & \text{otherwise.} \end{cases}$$

Under $C_\epsilon(n, m)$, expanding the finite sums in [Algorithm 1](#) gives the pointwise identity

$$\widehat{\tau}_{n,m,d}^{\text{mix}}(\mathcal{L}_n, \mathcal{U}_m) = \text{RB}(\mathfrak{P}(\mathcal{L}_n, \mathcal{U}_m)),$$

where $\mathfrak{P}(\mathcal{L}_n, \mathcal{U}_m)$ is the deterministic three-pool split formed from the first M_0 complete labeled records, the next n_p labeled records projected to (X, A) together with the first m_p auxiliary records, and the remaining n_f projected labeled records together with the remaining m_f auxiliary records. By [Definition 9](#), this split has law equal to the fixed-pool law with coordinate law $\mathbb{P}_3$ and capacities $\mathbf{M}$. Hence

$$\mathbb{E}\left[\left\{\widehat{\tau}_{n,m,d}^{\text{mix}} - \tau(P)\right\}^2\right] = \int \left\{\text{RB}(\mathfrak{p}) - \tau(P)\right\}^2 d\mathbb{P}_{\text{fix}}(\mathfrak{p}).$$

3. The finite-pool-to-Poisson comparison used here is the following. For any prefix statistic with values in an interval $[a, b]$, any target and overflow value in $[a, b]$, and independent Poisson requested lengths with means $\boldsymbol{\lambda}$,

$$\int \left\{\text{RB}(\mathfrak{p}) - \theta\right\}^2 d\mathbb{P}_{\text{fix}}(\mathfrak{p}) \leq \int \left\{\Psi(\mathfrak{s}) - \theta\right\}^2 d\mathbb{P}_{\text{Pois}}(\mathfrak{s}) + (b-a)^2 \sum_{\iota \in \{0,p,f\}} \Pr\{\text{Poi}(\lambda_\iota) > M_\iota\}.$$

Indeed, RB is the conditional expectation of the capped randomized-prefix statistic given the fixed pools, so Jensen's inequality bounds its squared risk by the capped randomized-prefix risk. Off the overflow event the capped and uncapped prefixes agree; on overflow both the capped value and the target lie in $[a, b]$, so the extra squared loss is bounded by $(b-a)^2$ times the overflow probability. Applying this with $a = -1$, $b = 1$, $\theta = \tau(P)$, overflow value 0, and $\Psi = \Psi_{n,m,d,\epsilon}$ gives

$$\int \left\{\text{RB}(\mathfrak{p}) - \tau(P)\right\}^2 d\mathbb{P}_{\text{fix}}(\mathfrak{p}) \leq \int \left\{\Psi_{n,m,d,\epsilon}(\mathfrak{s}) - \tau(P)\right\}^2 d\mathbb{P}_{\text{Pois}}(\mathfrak{s}) + 4[\pi_0 + \pi_p + \pi_f],$$

where

$$\pi_0 = \Pr\{\text{Poi}(u) > M_0\}, \quad \pi_p = \Pr\{\text{Poi}(t_p) > M_p\}, \quad \pi_f = \Pr\{\text{Poi}(t) > M_f\}.$$

The image of the independent Poisson-prefix law under the three histograms is exactly $\mathbb{P}_{P,u,t_p,t}^{\text{Pois}}$, and therefore

$$\int \left\{\Psi_{n,m,d,\epsilon}(\mathfrak{s}) - \tau(P)\right\}^2 d\mathbb{P}_{\text{Pois}}(\mathfrak{s}) = \mathcal{R}_{\text{hyb}}^{\text{Pois}}(P; n, m, \epsilon)$$

on the calibrated branch.

4. It remains to bound the three overflow probabilities. The calibration predicate gives $24 \leq n$. From the definitions of the integer blocks,

$$n \leq 4M_0, \quad n \leq 4M_p, \quad n \leq 4M_f,$$

and all three capacities are positive. For any positive integer $M_\bullet$ satisfying $n \leq 4M_\bullet$,

$$\Pr\{\text{Poi}(M_\bullet/8) > M_\bullet\} \leq \frac{8}{M_\bullet} \leq \frac{32}{n}.$$

To prove the first inequality, let $X_\bullet \sim \text{Poi}(M_\bullet/8)$. The overflow event is contained in the self-normalized Bernstein event at parameter $M_\bullet/4$, because

$$\sqrt{2 \cdot \frac{M_\bullet}{8} \cdot \frac{M_\bullet}{4}} = \frac{M_\bullet}{4}$$

and $X_\bullet > M_\bullet$ implies

$$X_\bullet - \frac{M_\bullet}{8} > \frac{7M_\bullet}{8} > \frac{M_\bullet}{4} + 2 \cdot \frac{M_\bullet}{4}.$$

The Poisson Bernstein bound for this larger event gives

$$\Pr\{\text{Poi}(M_\bullet/8) > M_\bullet\} \leq \exp(-M_\bullet/4) = \frac{1}{\exp(M_\bullet/4)} \leq \frac{4}{M_\bullet} \leq \frac{8}{M_\bullet},$$

where $1 + M_\bullet/4 \leq \exp(M_\bullet/4)$ gives the penultimate inequality. Consequently

$$\pi_0 + \pi_p + \pi_f \leq \frac{96}{n},$$

and the preceding display yields

$$\mathbb{E}\left[\{\hat{\tau}_{n,m,d}^{\text{mix}} - \tau(P)\}^2\right] \leq \mathcal{R}_{\text{hyb}}^{\text{Pois}}(P; n, m, \epsilon) + \frac{384}{n}.$$

Since $\mathcal{R}_{\text{hyb}}^{\text{Pois}}(P; n, m, \epsilon) \geq 0$, this is at most

$$2\mathcal{R}_{\text{hyb}}^{\text{Pois}}(P; n, m, \epsilon) + \frac{384}{n}.$$

5. If $\mathcal{C}_\epsilon(n, m)$ fails, then [Algorithm 1](#) sets $\hat{\tau}_{n,m,d}^{\text{mix}} = 0$, while the definition of the hybrid Poisson risk gives

$$\mathcal{R}_{\text{hyb}}^{\text{Pois}}(P; n, m, \epsilon) = \tau(P)^2.$$

Because the annotation law is a probability measure,

$$\mathbb{E}\left[\{\hat{\tau}_{n,m,d}^{\text{mix}} - \tau(P)\}^2\right] = \tau(P)^2 \leq 2\tau(P)^2 + \frac{384}{n} = 2\mathcal{R}_{\text{hyb}}^{\text{Pois}}(P; n, m, \epsilon) + \frac{384}{n},$$

using $n \geq 1$. The two cases prove the asserted bound with the universal constant $C = 384$. $\square$

[Lemma 14](#) converts the Poisson risk calculation into the fixed- n, m annotation experiment of [Definition 9](#). The additional C/n term is on the labeled-outcome scale already present in the rate, so combining [Lemmas 13](#) and [14](#) yields the finite-sample upper-bound ingredient used in [Theorem 2](#).

The final calibration statements make the Chebyshev constants explicit. The first controls the coefficient norm of the shifted quotient, and the second records the numerical bounds used in the finite calibration predicate of [Algorithm 1](#).

Lemma 15. [`lem:shifted-chebyshev-quotient-coefficient-bound`] (Shifted quotient coefficient bound) *Let L be a positive integer, and let G_L be the polynomial continuation at zero of $(1 - E_L(z))/z$ from [Algorithm 1](#). For a real polynomial $p(z) = \sum_i a_i z^i$, write*

$$\|p\|_1 = \sum_i |a_i|$$

for its coefficient ℓ^1 -norm. Then

$$\|G_L\|_1 \leq 6^L.$$

Proof of Lemma 15. Let $L \geq 1$. We use the notation for H_L, E_L, G_L from Algorithm 1. The phrase “polynomial continuation at zero” is used through the quotient identities

$$zE_L(z) = H_L(z), \quad zG_L(z) = 1 - E_L(z),$$

with equality in $\mathbb{R}[z]$; the cancellations below use only that multiplication by the nonzero indeterminate z is injective on $\mathbb{R}[z]$.

1. Define, for every $\alpha \in \mathbb{N}$,

$$\theta_{L,\alpha} = (-1)^\alpha \frac{2^{2\alpha+3}}{L(L+\alpha+2)} \binom{L+\alpha+2}{2\alpha+4},$$

and put

$$\tilde{G}_L(z) = \sum_{0 \leq \alpha < L-1} \theta_{L,\alpha} z^\alpha.$$

We first derive the shifted-Chebyshev coefficient expansion used below. Write $Q_r(z) = T_r(1-2z)$. The initial polynomials and the defining Chebyshev recurrence give

$$Q_0(z) = 1, \quad Q_1(z) = 1 - 2z, \quad Q_{r+2}(z) = 2(1-2z)Q_{r+1}(z) - Q_r(z).$$

For $r \in \mathbb{N}$ and $\nu \in \mathbb{Z}$, define

$$\beta_{r,\nu} := \begin{cases} 1, & \nu = 0, \\ (-1)^\nu \frac{r}{r+\nu} \binom{r+\nu}{2\nu} 4^\nu, & 1 \leq \nu \leq r, \\ 0, & \nu < 0 \text{ or } r < \nu. \end{cases}$$

Comparing coefficients in the displayed recurrence reduces the induction to

$$\beta_{r+2,\nu} = 2\beta_{r+1,\nu} - 4\beta_{r+1,\nu-1} - \beta_{r,\nu}.$$

For $1 \leq \nu \leq r$, write

$$A_{r,\nu} := \frac{r}{r+\nu} \binom{r+\nu}{2\nu}.$$

Set $A_{r,0} := 1$. For $\nu \geq 1$, the factorial form is

$$A_{r,\nu} = \frac{r(r+\nu-1)!}{(2\nu)!(r-\nu)!},$$

After cancelling the common factor $(-1)^\nu 4^\nu$, the coefficient recurrence is

$$A_{r+2,\nu} = 2A_{r+1,\nu} + A_{r+1,\nu-1} - A_{r,\nu}.$$

For $1 \leq \nu \leq r$, multiply this equality by $(2\nu)!(r+2-\nu)!/(r+\nu-1)!$. It becomes

$$\begin{aligned} & (r+2)(r+\nu+1)(r+\nu) \\ &= 2(r+1)(r+\nu)(r+2-\nu) + (r+1)(2\nu)(2\nu-1) \\ & \quad - r(r+2-\nu)(r+1-\nu), \end{aligned}$$

and expanding both sides proves it. At $\nu = 0$, the term $\beta_{r+1,-1}$ is zero by the outside-range convention, and the identity becomes $1 = 2 \cdot 1 - 0 - 1$. At $\nu = r + 1$ and $\nu = r + 2$, the same zero convention leaves, respectively,

$$\begin{aligned} (-1)^{r+1}(r+2)4^{r+1} &= 2 \left((-1)^{r+1} \frac{4^{r+1}}{2} \right) - 4((-1)^r(r+1)4^r), \\ (-1)^{r+2} \frac{4^{r+2}}{2} &= -4 \left((-1)^{r+1} \frac{4^{r+1}}{2} \right), \end{aligned}$$

so the recurrence holds there as well. Since the formula agrees with Q_0, Q_1 , induction proves, for every real z ,

$$T_L(1-2z) = 1 + \sum_{1 \leq \nu \leq L} (-1)^\nu \frac{L}{L+\nu} \binom{L+\nu}{2\nu} 4^\nu z^\nu.$$

The $\nu = 1$ term is $-2L^2z$. For the remaining terms write $\nu = \alpha + 2$. Then

$$(-1)^{\alpha+2} \frac{L}{L+\alpha+2} \binom{L+\alpha+2}{2\alpha+4} 4^{\alpha+2} z^{\alpha+2} = 2L^2 \theta_{L,\alpha} z^{\alpha+2},$$

so, with $Q_L(z) = T_L(1-2z)$,

$$Q_L(z) = 1 - 2L^2z + 2L^2z^2 \tilde{G}_L(z).$$

Using these quotient identities, since $Q_L(0) = 1$, the polynomial

$$H_L(z) = \frac{1 - Q_L(z)}{2L^2}$$

has zero constant term. Moreover $Q'_L(0) = -2T'_L(1) = -2L^2$, so $H'_L(0) = 1$, and hence the quotient identity for E_L gives $E_L(0) = 1$. Thus $1 - E_L$ has zero constant term, as required for the quotient identity defining G_L . Using the displayed formula for Q_L ,

$$H_L(z) = \frac{1 - Q_L(z)}{2L^2} = z - z^2 \tilde{G}_L(z) = z(1 - z \tilde{G}_L(z)).$$

Thus

$$zE_L(z) = z(1 - z \tilde{G}_L(z)).$$

Because multiplication by the nonzero indeterminate z is cancellative in $\mathbb{R}[z]$,

$$E_L(z) = 1 - z \tilde{G}_L(z).$$

Consequently

$$zG_L(z) = 1 - E_L(z) = z \tilde{G}_L(z),$$

and the same cancellation gives

$$G_L(z) = \tilde{G}_L(z)$$

as polynomials.

2. Since

$$\tilde{G}_L(z) = \sum_{0 \leq \alpha < L-1} \theta_{L,\alpha} z^\alpha,$$

every coefficient of $\tilde{G}_L$ outside the range $0 \leq \alpha < L-1$ is zero. Hence

$$\|G_L\|_1 = \|\tilde{G}_L\|_1 = \sum_{\alpha \in \text{supp}(\tilde{G}_L)} |[z^\alpha] \tilde{G}_L| \leq \sum_{0 \leq \alpha < L-1} |\theta_{L,\alpha}|.$$

3. The last sum is the positive coefficient sum at 1:

$$\sum_{0 \leq \alpha < L-1} |\theta_{L,\alpha}| = \sum_{0 \leq \alpha < L-1} |\theta_{L,\alpha}| 1^\alpha.$$

The coefficient formula also gives

$$|\theta_{L,\alpha}| = \frac{2^{2\alpha+3}}{L(L+\alpha+2)} \binom{L+\alpha+2}{2\alpha+4},$$

because $L > 0$ and all factors except $(-1)^\alpha$ are nonnegative.

4. It remains to bound this positive coefficient sum. Write

$$\Theta_L^+ = \sum_{0 \leq \alpha < L-1} |\theta_{L,\alpha}|.$$

This is the positive coefficient sum at the argument 1. The sign identity $(-1)^\alpha \theta_{L,\alpha} = |\theta_{L,\alpha}|$ gives

$$\Theta_L^+ = \sum_{0 \leq \alpha < L-1} \theta_{L,\alpha} (-1)^\alpha = \tilde{G}_L(-1).$$

Substituting $z = -1$ in the identity

$$Q_L(z) = 1 - 2L^2 z + 2L^2 z^2 \tilde{G}_L(z)$$

and using $Q_L(-1) = T_L(3)$ gives

$$T_L(3) = 1 + 2L^2 + 2L^2 \Theta_L^+.$$

Since $\Theta_L^+ \geq 0$ and $L \geq 1$,

$$\Theta_L^+ \leq 2L^2 \Theta_L^+ \leq 1 + 2L^2 + 2L^2 \Theta_L^+ = T_L(3).$$

It remains to bound $T_L(3)$. Put $a_\nu = T_\nu(3)$. The Chebyshev recurrence gives

$$a_0 = 1, \quad a_1 = 3, \quad a_{\nu+2} = 6a_{\nu+1} - a_\nu.$$

Let $\lambda_+ = 3 + \sqrt{8}$ and $\lambda_- = 3 - \sqrt{8}$. Since $\lambda_+ + \lambda_- = 6$ and $\lambda_+ \lambda_- = 1$, both roots satisfy $x^2 = 6x - 1$. Consequently

$$b_\nu := \frac{\lambda_+^\nu + \lambda_-^\nu}{2}$$

satisfies $b_{\nu+2} = 6b_{\nu+1} - b_\nu$, while

$$b_0 = 1 = a_0, \quad b_1 = \frac{\lambda_+ + \lambda_-}{2} = 3 = a_1.$$

Induction in the common recurrence gives $a_\nu = b_\nu$, so

$$a_\nu = \frac{(3 + \sqrt{8})^\nu + (3 - \sqrt{8})^\nu}{2},$$

and in particular $a_\nu \geq 0$, because $2 < \sqrt{8} < 3$ makes both λ_+ and λ_- positive. A two-step induction now gives $a_\nu \leq 6^\nu$: the cases $\nu = 0, 1$ are immediate, and if the bound holds for ν and $\nu + 1$, then

$$a_{\nu+2} = 6a_{\nu+1} - a_\nu \leq 6a_{\nu+1} \leq 6 \cdot 6^{\nu+1} = 6^{\nu+2}.$$

Thus $T_L(3) = a_L \leq 6^L$, and therefore

$$\sum_{0 \leq \alpha < L-1} |\theta_{L,\alpha}| = \Theta_L^+ \leq 6^L.$$

5. Combining the preceding inequalities yields

$$\|G_L\|_1 \leq \sum_{0 \leq \alpha < L-1} |\theta_{L,\alpha}| \leq 6^L,$$

which proves the claim. □

Lemma 15 gives a direct coefficient bound for the quotient polynomial G_L . This estimate is the coefficient-growth input behind the explicit factorial second-moment calibration.

Lemma 16. [lem:explicit-chebyshev-calibration] (Explicit Chebyshev calibration) *Let T_j be the Chebyshev polynomial of degree j , set $Q_j(z) = T_j(1-2z)$, and let $\|\cdot\|_1$ denote coefficient ℓ^1 -norm. For every integer $L \geq 2$, the polynomial continuations E_L and G_L satisfy the following calibration.*

- **(Coefficient bounds.)**

$$\|Q_L\|_1 \leq 7^L, \quad 2\|G_L\|_1 \leq 14^L.$$

- **(Factorial-lift second moment.)** *For every $a \in \{0, 1\}$, every $t > 0$, $B > 0$, and every $s_0, s_1 \geq 0$ with $L \leq tB$, set $p = s_0 + s_1$. If $K_b \sim \text{Poi}(ts_b)$ independently and $\widehat{W}_{a,L}$ denotes the factorial lift of the light-cell polynomial based on G_L , then*

$$E\widehat{W}_{a,L}^2 \leq 7056^L \left(1 + \frac{p}{B}\right)^{2L},$$

and, whenever $p \leq B$,

$$E\widehat{W}_{a,L}^2 \leq 7056^L.$$

- (**Interval identities.**) For every $z \in [0, 1]$,

$$0 \leq E_L(z), \quad E_L(z) \leq 1, \quad zG_L(z) + E_L(z) = 1.$$

Moreover, for every $z \in (0, 1]$,

$$E_L(z) \leq \frac{1}{L^2 z}.$$

Proof of Lemma 16. Fix $L \geq 2$, so $L > 0$.

1. Write

$$Q_L(z) = 1 + \sum_{j=1}^L q_{L,j} z^j, \quad q_{L,j} = [z^j]Q_L.$$

The shifted Chebyshev expansion used for $Q_L(z) = T_L(1 - 2z)$ gives

$$\|Q_L\|_1 = 1 + \sum_{j=1}^L |q_{L,j}| = T_L(3).$$

The standard growth bound for Chebyshev polynomials at 3 gives $T_L(3) \leq 6^L$, and hence

$$\|Q_L\|_1 \leq 6^L \leq 7^L.$$

For the quotient polynomial, Lemma 15 gives

$$\|G_L\|_1 \leq 6^L.$$

Since $L \geq 2$,

$$2\|G_L\|_1 \leq 2 \cdot 6^L \leq 2^L 7^L = 14^L.$$

This proves the two coefficient bounds.

2. Fix $a \in \{0, 1\}$, $t > 0$, $B > 0$, and $s_0, s_1 \geq 0$, and assume $L \leq tB$. Put

$$\mathcal{D} = tB, \quad p = s_0 + s_1, \quad \eta = 1 + \frac{p}{B}.$$

Then $\mathcal{D} > 0$, $\mathcal{D}(s_b/B) = ts_b$ for $b = 0, 1$, and $\eta \geq 1$. Let

$$\mathcal{J}_L = \{j \in \mathbb{N} : [z^j]G_L \neq 0\}.$$

To identify this index set, write

$$Q_L(z) = 1 + q_{L,1}z + \sum_{r=2}^L q_{L,r}z^r, \quad q_{L,1} = -2L^2.$$

The definitions of H_L and E_L give the scaled identity

$$zE_L(z) = \frac{1 - Q_L(z)}{2L^2}.$$

Using the displayed value of $q_{L,1}$, this identity gives

$$E_L(z) = 1 - \sum_{r=2}^L \frac{q_{L,r}}{2L^2} z^{r-1},$$

and the quotient definition of G_L gives

$$G_L(z) = \frac{1 - E_L(z)}{z} = \sum_{r=2}^L \frac{q_{L,r}}{2L^2} z^{r-2}.$$

Thus

$$\mathcal{J}_L \subseteq \{0, \dots, L-2\},$$

with the assertion empty if the displayed sum has no nonzero terms, and every $j \in \mathcal{J}_L$ satisfies $j+1 \leq L$. For $K = (K_0, K_1)$, with $K_a = K_1$ when $a = 1$ and $K_a = K_0$ when $a = 0$, define

$$X_j(K) = \frac{(K_a)_{j+1}}{\mathcal{D}^{j+1}} + \frac{K_{1-a}(K_a)_j}{\mathcal{D}^{j+1}}, \quad j \in \mathcal{J}_L.$$

If $\mathcal{J}_L$ is empty, the following sum is zero. With $K_b \sim \text{Poi}(ts_b)$ independently, the identities $\mathcal{D}(s_b/B) = ts_b$ give the factorial lift as

$$\widehat{W}_{a,L} = \sum_{j \in \mathcal{J}_L} [z^j] G_L X_j(K).$$

For the moment bound, first use the one-dimensional Poisson falling-factorial estimate: if $\mathfrak{N}_z \sim \text{Poi}(\mathcal{D}z)$, $z \geq 0$, and $r, v \leq L$, then

$$\mathbb{E} \frac{(\mathfrak{N}_z)_r (\mathfrak{N}_z)_v}{\mathcal{D}^{r+v}} = \sum_{h=0}^{\min(r,v)} \binom{r}{h} \binom{v}{h} h! \frac{(\mathcal{D}z)^{r+v-h}}{\mathcal{D}^{r+v}}.$$

For $0 \leq h \leq \min(r, v)$, the corresponding summand satisfies

$$\binom{r}{h} \binom{v}{h} h! \frac{(\mathcal{D}z)^{r+v-h}}{\mathcal{D}^{r+v}} \leq \frac{(rv)^h}{h! \mathcal{D}^h} (1+z)^{r+v} \leq \frac{L^h}{h!} (1+z)^{r+v},$$

because $rv/\mathcal{D} \leq L$, which follows from $r, v \leq L$ and $L \leq \mathcal{D}$, and because $z^{r+v-h} \leq (1+z)^{r+v}$. Therefore

$$\mathbb{E} \frac{(\mathfrak{N}_z)_r (\mathfrak{N}_z)_v}{\mathcal{D}^{r+v}} \leq \sum_{h=0}^L \frac{L^h}{h!} (1+z)^{r+v} \leq 3^L (1+z)^{r+v},$$

where the last inequality is the elementary finite-series bound $\sum_{h=0}^L L^h/h! \leq 3^L$.

This estimate gives the same-arm square bound. For every $0 \leq r \leq L$, integrating out the unused arm in the product law and applying the preceding estimate with $z = s_a/B$ and orders (r, r) gives

$$\mathbb{E} \left(\frac{(K_a)_r}{\mathcal{D}^r} \right)^2 \leq 3^L \left(1 + \frac{s_a}{B} \right)^{2r}.$$

Since $1 + s_a/B \leq \eta$, $\eta \geq 1$, and $2r \leq 2L$,

$$\mathbb{E} \left(\frac{(K_a)_r}{\mathcal{D}^r} \right)^2 \leq 3^L \eta^{2L}.$$

In particular, for $j \in \mathcal{J}_L$, using $j+1 \leq L$,

$$\mathbb{E} \left(\frac{(K_a)_{j+1}}{\mathcal{D}^{j+1}} \right)^2 \leq 3^L \eta^{2L}.$$

The cross-arm square bound follows by the product structure. For $j \in \mathcal{J}_L$, independence and $(K_{1-a})_1 = K_{1-a}$ give

$$\mathbb{E} \left(\frac{K_{1-a}(K_a)_j}{\mathcal{D}^{j+1}} \right)^2 = \mathbb{E} \left(\frac{(K_a)_j}{\mathcal{D}^j} \right)^2 \mathbb{E} \left(\frac{(K_{1-a})_1}{\mathcal{D}} \right)^2.$$

Applying the one-dimensional estimate to arm a with orders (j, j) and to arm $1-a$ with orders $(1, 1)$ yields

$$\mathbb{E} \left(\frac{K_{1-a}(K_a)_j}{\mathcal{D}^{j+1}} \right)^2 \leq 3^L \left(1 + \frac{s_a}{B} \right)^{2j} \cdot 3^L \left(1 + \frac{s_{1-a}}{B} \right)^2.$$

Now $1 + s_a/B \leq \eta$, $1 + s_{1-a}/B \leq \eta$, $\eta \geq 1$, and $2j+2 \leq 2L$. Hence

$$\left(1 + \frac{s_a}{B} \right)^{2j} \left(1 + \frac{s_{1-a}}{B} \right)^2 \leq \eta^{2j+2} \leq \eta^{2L},$$

and therefore

$$\mathbb{E} \left(\frac{K_{1-a}(K_a)_j}{\mathcal{D}^{j+1}} \right)^2 \leq 9^L \eta^{2L}.$$

Since $(\xi + v)^2 \leq 2\xi^2 + 2v^2$,

$$\mathbb{E} X_j(K)^2 \leq 2 \cdot 3^L \eta^{2L} + 2 \cdot 9^L \eta^{2L} \leq 36^L \eta^{2L} = (6^L \eta^L)^2.$$

The finite-sum L^2 inequality

$$\mathbb{E} \left(\sum_{j \in \mathcal{J}_L} c_j X_j \right)^2 \leq \left(\sum_{j \in \mathcal{J}_L} |c_j| \right)^2 \max_{j \in \mathcal{J}_L} \mathbb{E} X_j^2$$

therefore yields

$$\widehat{W}_{a,L}^2 \leq \|G_L\|_1^2 (6^L \eta^L)^2 \leq (6^L)^2 (6^L \eta^L)^2 = 1296^L \eta^{2L}.$$

Because $1296 \leq 7056$,

$$\widehat{W}_{a,L}^2 \leq 7056^L \left(1 + \frac{p}{B} \right)^{2L}.$$

If $p \leq B$, then $\eta \leq 2$, and hence

$$\widehat{W}_{a,L}^2 \leq 1296^L 2^{2L} = 5184^L \leq 7056^L.$$

This proves both second-moment assertions.

3. Let $z \in [0, 1]$. The scaled identity for the continuation is

$$zE_L(z) = (2L^2)^{-1}\{1 - Q_L(z)\} = \frac{1 - Q_L(z)}{2L^2}.$$

At $z = 0$, the derivative identity $T'_L(1) = L^2$ gives $E_L(0) = 1$. For $z > 0$, put

$$\theta = \arccos(1 - 2z).$$

Then $Q_L(z) = \cos(L\theta)$, so $|Q_L(z)| \leq 1$. Hence $1 - Q_L(z) \geq 0$, and the scaled identity gives $E_L(z) \geq 0$. Also

$$1 - Q_L(z) = 1 - \cos(L\theta) \leq L^2(1 - \cos \theta) = 2L^2z,$$

so $zE_L(z) \leq z$, whence $E_L(z) \leq 1$. Finally, $|Q_L(z)| \leq 1$ gives $1 - Q_L(z) \leq 2$, and the normalization by $2L^2$ in the scaled identity gives

$$zE_L(z) = \frac{1 - Q_L(z)}{2L^2} \leq \frac{1}{L^2},$$

and division by $z > 0$ gives

$$E_L(z) \leq \frac{1}{L^2z}.$$

The quotient definition gives the polynomial identity

$$zG_L(z) + E_L(z) = 1$$

for every real z , and therefore on $[0, 1]$.

□

[Lemma 16](#) supplies the numerical coefficient bounds used to instantiate [Lemma 5](#) in the calibrated estimator. In particular, the constant 7056 gives the explicit factorial-lift second-moment scale that appears in the finite calibration of [Algorithm 1](#).

B Common-marginal lower bounds

The converse in [Theorem 3](#) rests on priors whose treatment–covariate marginal is shared across branches while their average treatment effect centers are separated. This appendix collects the finite approximation and transfer statements that make that construction compatible with the annotation experiment in [Definition 9](#). The first object is a two-cell calculation that isolates the statistical issue in its smallest form.

Definition 27. `[def:finite-small-instance]` (Finite small instance $\mathfrak{C}_{\text{small}}$) *Let*

$$\epsilon = \frac{1}{4}, \quad d = 2, \quad p_1 = p_2 = \frac{1}{2}, \quad e_1 = e_2 = \frac{1}{2}, \quad \mu_{01} = \mu_{02} = 0.$$

For $n = m = 1$, independence of the samples gives

$$\mathbb{E}[2\mathbf{1}\{\mathcal{L}_1 = ((1, 1, 1))\}(\mathbf{1}\{\mathcal{U}_1 = ((1, 0))\} + \mathbf{1}\{\mathcal{U}_1 = ((1, 1))\})] = 2q_{11}p_1,$$

where q_{11} is the outcome-marked mass for treatment-covariate cell $(1, 1)$. At $\mu_{11} = 1$, this equals $1/4$, while the treated-cell target is

$$p_1 \mu_{11} = \frac{1}{2}.$$

Define Π_0^{small} to put unit mass on the law with

$$\mu_{11} = \mu_{12} = 0,$$

and define Π_1^{small} to put unit mass on the law with

$$\mu_{11} = 1, \quad \mu_{12} = 0.$$

Under both priors,

$$P_{XA}(x, a) = \frac{1}{4} \quad \text{for all } (x, a),$$

whereas

$$\tau(P) = 0$$

under Π_0^{small} , and

$$\tau(P) = \frac{1}{2}$$

under Π_1^{small} . Consequently, the auxiliary laws agree for every m , and the total variation distance between the two labeled laws is

$$1 - \left(\frac{3}{4}\right)^n.$$

These identities constitute $\mathfrak{C}_{\text{small}}$.

The instance in [Definition 27](#) makes concrete the role of outcome labels. The two priors Π_0^{small} and Π_1^{small} share the same treatment-covariate marginal P_{XA} , so auxiliary observations identify the common table while the target separation remains attached to the outcome-marked mass q_{ax} . The calculation $\mathfrak{C}_{\text{small}}$ is finite, but it has the same common-marginal structure used in the high-dimensional lower bound.

The next statement supplies the approximation-duality ingredient. It constructs a signed finite measure on a short interval whose first L moments vanish while the inverse-weight functional retains a fixed gap. This is the same polynomial approximation mechanism that appears in nonsmooth functional lower bounds for discrete distributions [[Jiao et al., 2015](#), [Wu and Yang, 2016](#), [Valiant and Valiant, 2017](#), [Orlitsky et al., 2016](#)].

Lemma 17. [[lem:finite-alternation-duality](#)] (Finite alternation duality) *Fix an overlap constant ϵ with $0 < \epsilon < 1/2$, and set*

$$\kappa = \frac{1 - 2\epsilon}{\epsilon}.$$

There exist constants $\gamma_\epsilon > 0$ and $c_\epsilon > 0$ such that, for every integer $L \geq 2$ and every $B > 0$, with

$$a = \frac{\gamma_\epsilon B}{L^2},$$

one has $a/\kappa \leq B$, and there are distinct points $p_0, \dots, p_{L+1} \in [a/\kappa, B]$ and real weights $w_0, \dots, w_{L+1}$ satisfying

$$\sum_{i=0}^{L+1} |w_i| = 1, \quad \sum_{i=0}^{L+1} w_i p_i^j = 0 \quad \text{for every } 0 \leq j \leq L,$$

and

$$c_\epsilon \leq \sum_{i=0}^{L+1} w_i \frac{p_i}{p_i + a}.$$

Proof of Lemma 17. 1. Set

$$D_\epsilon = \frac{\kappa}{(1+\kappa)(2+\kappa)}, \quad \gamma_\epsilon = \min \left\{ \frac{\kappa}{4}, \frac{D_\epsilon \kappa}{32} \right\}, \quad c_\epsilon = \frac{D_\epsilon}{8}.$$

Since $0 < \epsilon < 1/2$, we have $\kappa > 0$. Hence $D_\epsilon > 0$, $\gamma_\epsilon > 0$, and $c_\epsilon > 0$. Also $D_\epsilon < 1$, because

$$\kappa < (1+\kappa)(2+\kappa)$$

for $\kappa > 0$.

2. Fix $L \geq 2$ and $B > 0$, and put

$$a = \frac{\gamma_\epsilon B}{L^2}.$$

Then $a > 0$. Since $\gamma_\epsilon \leq \kappa/4$ and $L^2 \geq 4$,

$$\frac{a}{\kappa} = \frac{\gamma_\epsilon B}{\kappa L^2} \leq B.$$

Moreover,

$$\frac{a}{\kappa} \leq \frac{B}{16},$$

because $\gamma_\epsilon \leq \kappa/4$ and $L^2 \geq 4$. In particular $a/\kappa < B$.

3. Let

$$I = \left[\frac{a}{\kappa}, B \right], \quad f(p) = \frac{p}{p+a}.$$

The pole $-a$ lies to the left of I , since $a > 0$ and $a/\kappa > 0$. Thus f is continuous on I . Let E be the best uniform error for approximating f on I by polynomials of degree at most L . The finite alternation theorem supplies an alternation dual certificate: distinct nodes in I , normalized weights annihilating all monomials of degree at most L , and an associated degree- L best approximating polynomial.

4. We claim

$$c_\epsilon \leq E.$$

Suppose instead that $E < c_\epsilon$. Define

$$p_- = \frac{a}{\kappa}, \quad p_+ = \frac{2a}{\kappa}.$$

Then $p_- < p_+$, and both points belong to I . Indeed $p_- = a/\kappa$, while

$$p_+ = \frac{2\gamma_\epsilon B}{\kappa L^2} \leq B$$

because $\gamma_\epsilon \leq \kappa$ and $L^2 \geq 4$. At these two points,

$$f(p_-) = \frac{1}{1+\kappa}, \quad f(p_+) = \frac{2}{2+\kappa},$$

so

$$f(p_+) - f(p_-) = \frac{\kappa}{(1 + \kappa)(2 + \kappa)} = D_\epsilon.$$

Let Q be the best approximating polynomial from the alternation certificate. For every $p \in I$,

$$|f(p) - Q(p)| \leq E.$$

Consequently,

$$D_\epsilon - 2E \leq Q(p_+) - Q(p_-).$$

By the mean-value theorem there is a point $\xi \in [p_-, p_+] \subset I$ such that

$$Q'(\xi) = \frac{Q(p_+) - Q(p_-)}{p_+ - p_-},$$

and therefore

$$\frac{D_\epsilon - 2E}{p_+ - p_-} \leq |Q'(\xi)|.$$

Since $E < c_\epsilon = D_\epsilon/8$, the numerator is positive.

5. On I , the target satisfies $0 \leq f(p) \leq 1$. Combining this with $|f(p) - Q(p)| \leq E$ gives

$$|Q(p)| \leq 1 + E \quad (p \in I),$$

or equivalently

$$\|Q\|_{\infty, I} \leq 1 + E.$$

Markov's inequality on the interval $I = [a/\kappa, B]$, applied to the degree- L polynomial Q , gives

$$\|Q'\|_{\infty, I} \leq \frac{2L^2}{B - a/\kappa} \|Q\|_{\infty, I} \leq \frac{2L^2}{B - a/\kappa} (1 + E).$$

Because $a/\kappa \leq B/16$, we have $B/2 \leq B - a/\kappa$. Also $E < c_\epsilon < 1$. Hence

$$|Q'(\xi)| \leq \frac{2L^2}{B - a/\kappa} (1 + E) \leq \frac{2L^2}{B/2} \cdot 2 = \frac{8L^2}{B}.$$

Combining this with the previous lower bound and using

$$p_+ - p_- = \frac{a}{\kappa} = \frac{\gamma_\epsilon B}{\kappa L^2}$$

yields

$$\frac{(D_\epsilon - 2E)\kappa L^2}{\gamma_\epsilon B} \leq \frac{8L^2}{B}.$$

After cancellation,

$$(D_\epsilon - 2E)\kappa \leq 8\gamma_\epsilon.$$

But $E < D_\epsilon/8$, so $D_\epsilon - 2E > 3D_\epsilon/4$, while

$$8\gamma_\epsilon \leq \frac{D_\epsilon \kappa}{4}.$$

This is impossible. Therefore $c_\epsilon \leq E$.

6. Let the alternation certificate have nodes $p_0, \dots, p_{L+1} \in I$ and weights $\alpha_0, \dots, \alpha_{L+1}$. They satisfy

$$\sum_{i=0}^{L+1} |\alpha_i| = 1, \quad \sum_{i=0}^{L+1} \alpha_i p_i^j = 0 \quad (0 \leq j \leq L),$$

and

$$\left| \sum_{i=0}^{L+1} \alpha_i f(p_i) \right| = E.$$

Choose

$$s = \begin{cases} 1, & \text{if } \sum_i \alpha_i f(p_i) \geq 0, \\ -1, & \text{if } \sum_i \alpha_i f(p_i) < 0, \end{cases} \quad w_i = s \alpha_i.$$

Then the nodes remain distinct and lie in I , and

$$\sum_{i=0}^{L+1} |w_i| = \sum_{i=0}^{L+1} |\alpha_i| = 1.$$

For every $0 \leq j \leq L$,

$$\sum_{i=0}^{L+1} w_i p_i^j = s \sum_{i=0}^{L+1} \alpha_i p_i^j = 0.$$

Finally,

$$\sum_{i=0}^{L+1} w_i \frac{p_i}{p_i + a} = s \sum_{i=0}^{L+1} \alpha_i f(p_i) = \left| \sum_{i=0}^{L+1} \alpha_i f(p_i) \right| = E \geq c_\epsilon.$$

These nodes and weights give the required finite moment dual certificate on $[a/\kappa, B]$. $\square$

Lemma 17 turns polynomial indistinguishability into target separation. The moment cancellations control likelihood differences after Poissonization, while the final display gives a fixed gap for the inverse-weight expression that enters the treatment-effect contrast.

The rare-cell prior packages this dual certificate into a common-marginal experiment. For the transfer arguments, it is useful to state the construction as a recipe separate from any particular choice of sample intensities. The next definition records the raw rare-cell masses, the two branch laws, and the raw target centers that the lower-bound reductions manipulate; the later uniform-intensity lemma supplies the exact Poisson witness actually used by the converse proof.

Definition 28. [synth_20] (Common-marginal rare-cell recipe) *A common-marginal rare-cell recipe with k rare cells at scale (a, B, L) is a finite random construction $\mathcal{R}$ consisting of a generator law $\text{generator}_{\mathcal{R}}$ for a latent draw w , together with two branch laws indexed by $b \in \{0, 1\}$. For each generated w , the recipe specifies rare-cell raw masses $p_i(w) \in [0, B]$, $i = 1, \dots, k$, branch-specific outcome regressions, and any reservoir mass needed to normalize the construction. The two branches have the same treatment-covariate raw masses and treatment propensities, so after normalization they induce the same P_{XA} table, while their outcome regressions may differ.*

For each branch $b \in \{0, 1\}$, let $P_b(w)$ be the normalized law on $[d] \times \{0, 1\}^2$ induced by the recipe, and let

$$\text{rawTarget}_{\mathcal{R}}(b, w)$$

be the unnormalized target contribution assigned by the rare cells in branch b . The branch center is

$$\theta_b = \int \text{rawTarget}_{\mathcal{R}}(b, w) d\text{generator}_{\mathcal{R}}(w),$$

and the raw-center separation is $\Delta = |\theta_1 - \theta_0|$. The recipe induces, for each branch b , the mixture experiment obtained by first drawing $w \sim \text{generator}_{\mathcal{R}}$, then sampling from $P_b(w)$ for labeled observations and from the common marginal $P_{b, X_A}(w)$ for auxiliary observations. Its raw Poisson version uses labeled intensity u and auxiliary intensity v with the same generated law $P_b(w)$, so the treatment-covariate component is common across the two branches and the likelihood discrepancy is carried by the labeled outcome-bearing component.

Definition 28 fixes the bookkeeping used by the lower-bound priors. The random raw masses determine a common treatment-covariate table, and the branch-specific outcome regressions create a raw-center separation before normalization. This separation is the quantity later converted into minimax risk.

The raw Poisson comparison enters several concentration and transfer statements, so the notation for the branch predictive laws is named explicitly before those estimates.

Lemma 18. [lem:common-marginal-recipe-target-concentration] (Prior target concentration) *Fix $\epsilon \in \mathbb{R}$, a common-marginal calibration at overlap level ϵ with dual-gap constant c_ϵ , and parameters $n, m, d, L, k \in \mathbb{N}$ and $B, a, u, v, C, \rho \in \mathbb{R}$. Let $\mathcal{R}$ be a common-marginal rare-cell recipe with atom-generator measure Γ , branch laws $P_b(w)$, raw targets $r_b(w)$, raw total mass $S(w)$, and branch priors*

$$\Pi_b = \Gamma \circ P_b^{-1}, \quad \theta_b = \int r_b(w) d\Gamma(w), \quad b \in \{0, 1\}.$$

Assume the recipe bounds hold at intensities (u, v) with constants (C, ρ) and dual gap c_ϵ :

- **(Finite-atom construction.)** *The recipe uses the finite-atom construction.*
- **(Model support.)** *Each prior is supported on $\mathcal{M}_{d, \epsilon}$ of Definition 6.*
- **(Common table.)** *The two priors induce the same law of the random treatment-covariate table.*
- **(Separation.)** *The raw centers satisfy*

$$c_\epsilon k a \leq |\theta_1 - \theta_0|.$$

- **(Mass and variance bounds.)** *Under the usual measurability and integrability conditions,*

$$\int S(w) d\Gamma(w) = 1, \quad \text{Var}_{\Gamma}(S) \leq CkBa, \quad \max_{b \in \{0, 1\}} \text{Var}_{\Gamma}(r_b) \leq CkBa.$$

- **(Mixture proximity.)** *The two raw mixture laws at labeled intensity u and auxiliary intensity v have total variation distance at most*

$$Cukap^L.$$

Then for every branch $b \in \{0, 1\}$ and every Δ with $0 < \Delta \leq 2$, the branch prior concentrates its average treatment effect $\tau(P)$, defined in [Definition 8](#), around its raw center:

$$\Pi_b \left\{ P : \frac{\Delta}{4} < |\tau(P) - \theta_b| \right\} \leq \frac{512 CkBa}{\Delta^2}.$$

Proof of [Lemma 18](#). Fix $b \in \{0, 1\}$. Define the two exceptional events in the atom space by

$$\mathbf{A}_{\text{tar}} := \left\{ w : \frac{\Delta}{16} < |r_b(w) - \theta_b| \right\}, \quad \mathbf{A}_{\text{mass}} := \left\{ w : \frac{\Delta}{16} < |S(w) - 1| \right\}.$$

We first record the normalization implication used below. Outside a Γ -null set the recipe gives

$$r_b(w) = S(w)\tau(P_b(w)), \quad S(w) > 0.$$

Suppose

$$\frac{\Delta}{4} < |\tau(P_b(w)) - \theta_b|, \quad w \notin \mathbf{A}_{\text{tar}} \cup \mathbf{A}_{\text{mass}}.$$

The second condition means

$$|r_b(w) - \theta_b| \leq \frac{\Delta}{16}, \quad |S(w) - 1| \leq \frac{\Delta}{16}.$$

The raw-center bound supplied by the recipe normalization is

$$|\theta_b| \leq 1.$$

Hence

$$|S(w)\theta_b - \theta_b| = |(S(w) - 1)\theta_b| \leq \frac{\Delta}{16}.$$

Using $r_b(w) = S(w)\tau(P_b(w))$, the triangle inequality gives

$$S(w)|\tau(P_b(w)) - \theta_b| = |S(w)\tau(P_b(w)) - S(w)\theta_b| \leq \frac{\Delta}{8}.$$

Also $|S(w) - 1| \leq \Delta/16$ and $\Delta \leq 2$ imply

$$S(w) \geq 1 - \frac{\Delta}{16} \geq \frac{7}{8}.$$

Therefore

$$\frac{7}{8} |\tau(P_b(w)) - \theta_b| \leq S(w)|\tau(P_b(w)) - \theta_b| \leq \frac{\Delta}{8},$$

which contradicts $\Delta/4 < |\tau(P_b(w)) - \theta_b|$. Thus, for Γ -almost every w ,

$$\frac{\Delta}{4} < |\tau(P_b(w)) - \theta_b| \implies w \in \mathbf{A}_{\text{tar}} \cup \mathbf{A}_{\text{mass}}.$$

By the definition of the branch prior as the pushforward of Γ under P_b , with the two Boolean branches handled separately,

$$\Pi_b \left\{ P : \frac{\Delta}{4} < |\tau(P) - \theta_b| \right\} = \Gamma \left\{ w : \frac{\Delta}{4} < |\tau(P_b(w)) - \theta_b| \right\}.$$

The implication above yields

$$\Pi_b \left\{ P : \frac{\Delta}{4} < |\tau(P) - \theta_b| \right\} \leq \Gamma(\mathbf{A}_{\text{tar}} \cup \mathbf{A}_{\text{mass}}).$$

We next bound the two exceptional probabilities. Chebyshev's inequality, applied under Γ with mean

$$\theta_b = \int r_b(w) d\Gamma(w),$$

and the assumed variance bound

$$\text{Var}_\Gamma(r_b) \leq CkBa,$$

gives

$$\Gamma(\mathbf{A}_{\text{tar}}) \leq \frac{CkBa}{(\Delta/16)^2}.$$

Similarly, since

$$\int S(w) d\Gamma(w) = 1, \quad \text{Var}_\Gamma(S) \leq CkBa,$$

Chebyshev's inequality gives

$$\Gamma(\mathbf{A}_{\text{mass}}) \leq \frac{CkBa}{(\Delta/16)^2}.$$

Subadditivity of measure gives

$$\Gamma(\mathbf{A}_{\text{tar}} \cup \mathbf{A}_{\text{mass}}) \leq \Gamma(\mathbf{A}_{\text{tar}}) + \Gamma(\mathbf{A}_{\text{mass}}).$$

Combining the preceding displays,

$$\begin{aligned} \Pi_b \left\{ P : \frac{\Delta}{4} < |\tau(P) - \theta_b| \right\} &\leq \frac{CkBa}{(\Delta/16)^2} + \frac{CkBa}{(\Delta/16)^2} \\ &= \frac{512 CkBa}{\Delta^2}, \end{aligned}$$

where the final equality uses $\Delta > 0$. This is the desired bound for the fixed branch b , and hence for both branches. $\square$

Definition 29 separates the prior randomness in the raw masses from the Poisson sampling randomness. With this notation, total variation between the two branch mixtures measures the experiment-level testing difficulty after the common treatment–covariate table has been integrated over.

The next two lemmas isolate the normalization estimates used by the transfer. The first converts raw target control into concentration of the average treatment effect, and the second records the corresponding lower-tail control for the raw scale.

Definition 29. [synth_21] (Raw Poisson prior-predictive mixture laws) *Fix a common-marginal rare-cell recipe $\mathcal{R}$ with atom generator Γ , branch laws $P_b(w)$ for $b \in \{0, 1\}$, and branch priors $\Pi_b = \Gamma \circ P_b^{-1}$. At labeled and auxiliary Poisson intensities u and v , define $\mathbf{Q}_b$ to be the branch- b raw prior-predictive law obtained by first drawing $w \sim \Gamma$, forming the raw branch law $P_b(w)$, and then sampling the independent raw Poisson experiment with labeled intensity u and auxiliary intensity v from that raw law and its treatment–covariate marginal. Equivalently,*

$$\mathbf{Q}_b = \int \left\{ \text{PoiExp}_{u,v}^{\text{raw}}(P_b(w)) \right\} d\Gamma(w) = \int \text{PoiExp}_{u,v}^{\text{raw}}(P) d\Pi_b(P), \quad b \in \{0, 1\},$$

where $\text{PoiExp}_{u,v}^{\text{raw}}(P)$ denotes the product Poisson law for the outcome-labeled raw cell counts at intensity u and the auxiliary treatment-covariate raw cell counts at intensity v .

Lemma 19. [lem:common-marginal-recipe-scale-tail] (Small-scale prior tail) Fix $\epsilon \in \mathbb{R}$, a common-marginal calibration with positive constants including dual-gap constant $c_\epsilon \leq 1/4$, integers $n, m, d, L, k \geq 0$, and reals B, a, u, v, C, ρ . Let $\mathcal{R}$ be a common-marginal rare-cell recipe with atom-generator Γ , branch laws $P_b(w)$, branch priors $\Pi_b = \Gamma \circ P_b^{-1}$, raw total mass $S(w)$, raw targets $r_b(w)$, and measurable raw scale $s_{\mathcal{R}}$ satisfying $s_{\mathcal{R}}(P_b(w)) = S(w)$ for Γ -almost every w .

Assume that the recipe bounds hold for $\mathcal{R}$ at $(u, v, c_\epsilon, C, \rho)$:

- **(Finite-atom construction.)** The recipe uses the finite-atom construction.
- **(Model support and common table.)** The two priors are supported on $\mathcal{M}_{d,\epsilon}$ from [Definition 6](#), and they induce the same distribution of the random treatment-covariate table P_{XA} .
- **(Center separation.)** If θ_b denotes the raw prior center for branch b , then

$$c_\epsilon k a \leq |\theta_1 - \theta_0|.$$

- **(Mass and variance bounds.)** Under the usual measurability and integrability conditions,

$$\int S d\Gamma = 1, \quad \text{Var}_\Gamma(S) \leq CkBa, \quad \max\{\text{Var}_\Gamma(r_0), \text{Var}_\Gamma(r_1)\} \leq CkBa.$$

- **(Mixture distance.)** The two raw Poisson mixture laws generated by the recipe satisfy

$$\text{TV}(\mathbf{Q}_0, \mathbf{Q}_1) \leq Cukap^L.$$

Then, for each branch $b \in \{0, 1\}$,

$$\Pi_b \left\{ P : \max\{s_{\mathcal{R}}(P), 0\} < \frac{1}{2} \right\} \leq 4CkBa.$$

Proof of Lemma 19. 1. Define the mass-tail event on the atom space by

$$\text{MassTail} := \left\{ w : \frac{1}{2} < |S(w) - 1| \right\}.$$

The mass assumptions give

$$\int S d\Gamma = 1, \quad \text{Var}_\Gamma(S) \leq CkBa.$$

Since Γ is a probability law, Chebyshev's inequality at radius $1/2$ yields

$$\Gamma(\text{MassTail}) = \Gamma \left\{ w : \frac{1}{2} < |S(w) - 1| \right\} \leq \frac{\text{Var}_\Gamma(S)}{(1/2)^2} \leq 4CkBa.$$

2. For a fixed branch $b \in \{0, 1\}$, set

$$\text{SmallScale}_b := \left\{ P : \max\{s_{\mathcal{R}}(P), 0\} < \frac{1}{2} \right\}.$$

Because $\Pi_b = \Gamma \circ P_b^{-1}$,

$$\Pi_b(\text{SmallScale}_b) = \Gamma\{w : P_b(w) \in \text{SmallScale}_b\}.$$

For Γ -almost every w , the recipe supplies both identities

$$s_{\mathcal{R}}(P_b(w)) = S(w) \quad \text{and} \quad 0 < S(w).$$

On this full-measure set, $P_b(w) \in \text{SmallScale}_b$ implies

$$S(w) = \max\{S(w), 0\} < \frac{1}{2}.$$

Hence $S(w) - 1 < -1/2$, and therefore

$$|S(w) - 1| = 1 - S(w) > \frac{1}{2}.$$

Thus

$$\{w : P_b(w) \in \text{SmallScale}_b\} \subseteq \text{MassTail} \quad \text{up to a } \Gamma\text{-null set,}$$

and consequently

$$\Pi_b(\text{SmallScale}_b) \leq \Gamma(\text{MassTail}).$$

3. The preceding containment is applied to the two branch cases separately: for $b = 0$ it uses

$$\Pi_0 = \Gamma \circ P_0^{-1},$$

and for $b = 1$ it uses

$$\Pi_1 = \Gamma \circ P_1^{-1}.$$

Combining the appropriate branch identity with the bound from the first step gives, for each $b \in \{0, 1\}$,

$$\Pi_b \left\{ P : \max\{s_{\mathcal{R}}(P), 0\} < \frac{1}{2} \right\} \leq \Gamma(\text{MassTail}) \leq 4CkBa.$$

□

[Lemmas 18](#) and [19](#) make the raw construction usable after normalization. The target concentration bound relates the branch centers θ_b to the actual functional values under the induced priors, while the scale-tail bound keeps the normalizing mass away from the region where raw-to-normalized conversion would amplify the rare-cell perturbation.

The next transfer lemma applies the recipe at fixed Poisson intensities large enough to dominate the fixed sample sizes. Its assumptions combine a total-variation bound with a scale condition that controls the random normalization.

Lemma 20. [lem:common-marginal-recipe-transfer] (Recipe transfer bound) *Let $0 < \epsilon < 1/2$, where ϵ is the overlap level. Fix a common-marginal calibration at level ϵ , consisting of positive constants including γ_ϵ and dual gap $\delta_\epsilon \leq 1/4$, a constant $C_\epsilon > 0$, and a real number ρ_ϵ . Let n, m, d, L, k be nonnegative integers, let $B, a \in \mathbb{R}$, and let $\mathcal{R}$ be a common-marginal rare-cell recipe with k rare cells at scale (a, B, L) . For each branch $b \in \{0, 1\}$, define the raw prior center*

$$\theta_b = \int \text{rawTarget}_{\mathcal{R}}(b, w) dG_{\mathcal{R}}(w),$$

where $G_{\mathcal{R}}$ is the atom-generator measure of the recipe, and put

$$\Delta = |\theta_1 - \theta_0|.$$

Assume:

- **(Sample and dimension.)** $n \geq 1$ and $d \geq 2$.
- **(Poisson-prior data.)** With

$$u = 65536 n \quad \text{and} \quad v = 65536 (n + m),$$

the recipe has bandwidth $B > 0$, shift

$$a = \frac{\gamma_\epsilon B}{L^2},$$

rare-shift size

$$ka \leq \delta_\epsilon,$$

and the recipe intensities agree with u and v .

- **(Bandwidth and likelihood bounds.)** The realized scale satisfies

$$(u + v)B \leq b_\epsilon L,$$

where b_ϵ is the calibration's uniform-bandwidth constant, and the recipe satisfies the common-marginal Poisson-prior bounds at gap δ_ϵ , intensity pair (u, v) , variance constant C_ϵ , and decay constant ρ_ϵ .

- **(Mixture affinity.)** The two prior-predictive raw Poisson-count laws generated by the two recipe priors and the recipe's experiment kernel have total variation distance at most

$$\frac{1}{8}.$$

- **(Variance scale.)** The recipe scale obeys

$$kBa \leq \frac{\Delta^2}{65536 C_\epsilon}.$$

Then the minimax risk of [Definition 11](#) satisfies

$$R_\epsilon(n, m, d) \geq \frac{\Delta^2}{65536}.$$

Proof of Lemma 20. Let

$$\Pi_b := \text{the branch-}b \text{ prior induced by } \mathcal{R}, \quad \theta_b := \int r_b(w) d\Gamma(w), \quad b \in \{0, 1\},$$

where Γ is the atom-generator measure and r_b is the raw target of the recipe. Thus

$$\Delta = |\theta_1 - \theta_0|.$$

Also put

$$u := 65536 n, \quad v := 65536 (n + m),$$

and, for every law P , define the totalized nonnegative recipe scale

$$s_+(P) := \max\{s_{\mathcal{R}}(P), 0\}.$$

1. The witness bounds imply

$$|\theta_0| \leq 1, \quad |\theta_1| \leq 1.$$

Indeed, each θ_b is the Γ -mean of the raw target, the raw target is $S(w)\tau(P_b(w))$, $|\tau(P_b(w))| \leq 1$ on $\mathcal{M}_{d,\epsilon}$, and the recipe bounds give $\int S d\Gamma = 1$. Hence

$$0 \leq \Delta \leq 2.$$

The intensities satisfy $u \geq 0$ and $v \geq 0$. The branch priors are probability measures, and the support part of the witness gives the required nonempty model class.

2. Let $\mathcal{K}_{\text{raw}}$ be the ordered raw Poisson-count experiment which, at law P , draws independent ordered Poisson samples from the labeled mark law and the auxiliary mark law with intensities

$$u s_+(P) \quad \text{and} \quad v s_+(P).$$

Let $\mathbf{Q}_b$ be the prior-predictive law obtained from Π_b through $\mathcal{K}_{\text{raw}}$. We now identify this law as a Markov image of the raw count law in the statement. Let $\mathcal{K}_{\text{cnt}}$ denote the recipe's raw Poisson-count kernel before ordering. Let reindex be the deterministic map that rewrites a raw count table as labeled-mark and auxiliary-mark count functions, and let order be the paired-histogram kernel that, conditional on those two count functions, orders the corresponding multiset of labeled marks and the corresponding multiset of auxiliary marks uniformly and independently. Put

$$\text{Rec} := \text{order} \circ \text{reindex}.$$

For every law P , the raw count kernel factors through this reconstruction as

$$\text{Rec} \circ \mathcal{K}_{\text{cnt}}(P) = \mathcal{K}_{\text{raw}}(P).$$

Indeed, after the deterministic reindexing, the labeled and auxiliary count functions have independent Poisson coordinates with means $u s_+(P)$ times the labeled mark law and $v s_+(P)$ times the auxiliary mark law; ordering each histogram gives exactly the ordered independent Poisson samples with those two intensities. Therefore, for each branch,

$$\mathcal{R}_{\text{raw},b} := \int \mathcal{K}_{\text{cnt}}(P) d\Pi_b(P) \quad \text{satisfies} \quad \mathbf{Q}_b = \int \mathcal{K}_{\text{raw}}(P) d\Pi_b(P) = (\text{Rec})_{\#} \mathcal{R}_{\text{raw},b}.$$

Total variation contracts under a Markov kernel, and the assumed raw mixture affinity gives

$$\text{TV}(\mathbf{Q}_0, \mathbf{Q}_1) \leq \text{TV}(\mathcal{R}_{\text{raw},0}, \mathcal{R}_{\text{raw},1}) \leq \frac{1}{8}.$$

3. Set

$$\alpha_{\text{floor}} := \frac{1}{100}.$$

The fixed experiment has the parametric two-point floor

$$\frac{\alpha_{\text{floor}}}{n} \leq R_{\epsilon}(n, m, d).$$

To see this, take

$$\delta_n := \frac{2}{5\sqrt{n}}.$$

Choose a cell $x_{\star} \in [d]$. Under both laws below, let $X = x_{\star}$ almost surely and let the treatment propensity be $1/2$. Let the control outcome mean be $1/2$ under both laws; let the treated outcome mean be $1/2$ for $P_{\text{par},0}$ and $1/2 + \delta_n$ for $P_{\text{par},1}$. Since $n \geq 1$,

$$0 \leq \delta_n \leq \frac{2}{5},$$

so these are valid binary outcome laws, and since $0 < \epsilon < 1/2$, both belong to $\mathcal{M}_{d,\epsilon}$. Their treatment-covariate marginals agree, their average treatment effects are

$$\tau(P_{\text{par},0}) = 0, \quad \tau(P_{\text{par},1}) = \delta_n,$$

and hence the auxiliary samples have identical laws. For the labeled samples, the standard Bernoulli product testing bound applies because

$$n \cdot \frac{(1/2)\delta_n^2}{(1/2)(1-1/2)} = \frac{8}{25} \leq \log 2,$$

and therefore

$$\text{TV}(P_{\text{par},0}^{\otimes n}, P_{\text{par},1}^{\otimes n}) \leq \frac{1}{2}.$$

Le Cam's two-point bound with separation δ_n yields

$$R_{\epsilon}(n, m, d) \geq \frac{(\delta_n/2)^2}{4} = \frac{1}{100n}.$$

For the fixed-sample decision problem with the class-gated clipped squared loss, let

$$\mathfrak{R}_{\text{dec}} := \inf_{\psi} \sup_P \int \ell_{\epsilon}(P, \psi) d\mathcal{K}_{\text{fix}}(P),$$

where the infimum is over measurable fixed-sample rules, $\mathcal{K}_{\text{fix}}$ is the fixed labeled-and-auxiliary experiment, and

$$\ell_{\epsilon}(P, z) := \begin{cases} (\text{clip}_{[-1,1]}(z) - \tau(P))^2, & P \in \mathcal{M}_{d,\epsilon}, \\ 0, & P \notin \mathcal{M}_{d,\epsilon}. \end{cases}$$

We use the following elementary comparison between [Definition 11](#) and the class-gated decision problem:

$$R_{\epsilon}(n, m, d) \leq \mathfrak{R}_{\text{dec}}.$$

Indeed, every measurable fixed-sample rule ψ induces the measurable estimator

$$\tilde{T}_\psi := \text{clip}_{[-1,1]} \circ \psi.$$

For every $P \in \mathcal{M}_{d,\epsilon}$, the class-gated loss evaluates the same clipped rule, so

$$\int \ell_\epsilon(P, \psi(y)) d\mathcal{K}_{\text{fix}}(P)(y) = E_{P^{\otimes n} \otimes P_{XA}^{\otimes m}} \left[(\tilde{T}_\psi - \tau(P))^2 \right].$$

Thus the worst-case on-class decision risk of ψ is exactly the worst-case mean-squared risk of $\tilde{T}_\psi$. Since $\tilde{T}_\psi$ is one admissible estimator in $\mathcal{T}_{n,m,d}$, [Definition 11](#) is no larger than that worst-case risk; taking the infimum over ψ gives the displayed comparison. Combining it with the parametric floor gives

$$\frac{\alpha_{\text{floor}}}{n} \leq \mathfrak{R}_{\text{dec}}.$$

4. If $\Delta = 0$, then the desired lower bound is $0 \leq R_\epsilon(n, m, d)$, which follows from the preceding floor since $n \geq 1$. Hence assume for the rest of the proof that

$$0 < \Delta.$$

The variance-scale hypothesis and $C_\epsilon > 0$ give

$$C_\epsilon k B a \leq \frac{\Delta^2}{65536}.$$

By [Lemma 18](#), for each $b \in \{0, 1\}$,

$$\Pi_b \left\{ P : \frac{\Delta}{4} < |\tau(P) - \theta_b| \right\} \leq \frac{512 C_\epsilon k B a}{\Delta^2} \leq \frac{1}{128} \leq \frac{1}{16}.$$

The assumed common-marginal Poisson-prior bounds also include the raw-center separation

$$\delta_\epsilon k a \leq |\theta_1 - \theta_0| = \Delta,$$

which is the center-separation hypothesis required below. By [Lemma 19](#), again for each $b \in \{0, 1\}$,

$$\Pi_b \{ P : s_+(P) < 1/2 \} \leq 4 C_\epsilon k B a \leq \frac{\Delta^2}{16384}.$$

5. The two fuzzy hypotheses with centers θ_0, θ_1 , radius $\Delta/4$, prior tail bounds $1/16, 1/16$, and predictive total variation bound $1/8$ give the raw lower bound as follows. For a measurable raw rule φ , define

$$\mathcal{B}_b(\varphi) := \int^+ \int^+ \ell_\epsilon(P, \varphi(y)) d\mathcal{K}_{\text{raw}}(P)(y) d\Pi_b(P) \in \overline{\mathbb{R}}_+, \quad b \in \{0, 1\}.$$

These are the branch Bayes decision risks for the class-gated clipped loss. Since the two branch priors are supported on $\mathcal{M}_{d,\epsilon}$, the integrand on the support is

$$\ell_\epsilon(P, \varphi(y)) = (\text{clip}_{[-1,1]}(\varphi(y)) - \tau(P))^2.$$

Let $\mathbf{Q}_b$ be the branch predictive laws from step 2, and first suppose $\theta_0 \leq \theta_1$. Put

$$\theta_{\text{mid}} := \frac{\theta_0 + \theta_1}{2}, \quad \mathbf{A}_\varphi := \{y : \text{clip}_{[-1,1]}(\varphi(y)) \geq \theta_{\text{mid}}\},$$

and define the good prior events

$$\mathbf{G}_0 := \{P : |\tau(P) - \theta_0| \leq \Delta/4\}, \quad \mathbf{G}_1 := \{P : |\tau(P) - \theta_1| \leq \Delta/4\}.$$

Because $\theta_1 - \theta_0 = |\theta_1 - \theta_0| \geq \Delta$, on $\mathbf{G}_0 \cap \mathbf{A}_\varphi$ the clipped decision is at least $\Delta/4$ above $\tau(P)$, while on $\mathbf{G}_1 \cap \mathbf{A}_\varphi^c$ it is at least $\Delta/4$ below $\tau(P)$. Hence

$$\mathcal{B}_0(\varphi) \geq \frac{\Delta^2}{16} \Pr_{\Pi_0 \otimes \mathcal{K}_{\text{raw}}}(\mathbf{G}_0 \cap \mathbf{A}_\varphi), \quad \mathcal{B}_1(\varphi) \geq \frac{\Delta^2}{16} \Pr_{\Pi_1 \otimes \mathcal{K}_{\text{raw}}}(\mathbf{G}_1 \cap \mathbf{A}_\varphi^c).$$

The concentration bounds from step 4 give

$$\Pr_{\Pi_0 \otimes \mathcal{K}_{\text{raw}}}(\mathbf{G}_0^c) \leq \frac{1}{16}, \quad \Pr_{\Pi_1 \otimes \mathcal{K}_{\text{raw}}}(\mathbf{G}_1^c) \leq \frac{1}{16},$$

and therefore

$$\mathcal{B}_0(\varphi) \geq \frac{\Delta^2}{16} \left(\mathbf{Q}_0(\mathbf{A}_\varphi) - \frac{1}{16} \right), \quad \mathcal{B}_1(\varphi) \geq \frac{\Delta^2}{16} \left(1 - \mathbf{Q}_1(\mathbf{A}_\varphi) - \frac{1}{16} \right).$$

Using $\max\{x_0, x_1\} \geq (x_0 + x_1)/2$ and the total-variation bound,

$$\begin{aligned} \max\{\mathcal{B}_0(\varphi), \mathcal{B}_1(\varphi)\} &\geq \frac{\Delta^2}{32} \left(1 - (\mathbf{Q}_1(\mathbf{A}_\varphi) - \mathbf{Q}_0(\mathbf{A}_\varphi)) - \frac{1}{16} - \frac{1}{16} \right) \\ &\geq \frac{\Delta^2}{32} \left(1 - \frac{1}{8} - \frac{1}{16} - \frac{1}{16} \right) = \frac{3\Delta^2}{128}. \end{aligned}$$

If $\theta_1 \leq \theta_0$, the same calculation is applied with the two branches interchanged. Thus, for every measurable raw rule φ ,

$$\frac{3\Delta^2}{128} \leq \max\{\mathcal{B}_0(\varphi), \mathcal{B}_1(\varphi)\}$$

in $\overline{\mathbb{R}}_+$.

6. The loss is bounded by

$$0 \leq \ell_\epsilon(P, z) \leq 4$$

because both $\text{clip}_{[-1,1]}(z)$ and $\tau(P)$ lie in $[-1, 1]$ on the model class, and the loss is 0 off the model class. The Poisson retention tails used to transfer raw rules to fixed-sample rules are as follows. For any integer $\mathbf{n} \geq 1$ and any Poisson mean $\lambda \geq 32768 \mathbf{n}$,

$$\Pr\{\text{Poi}(\lambda) < \mathbf{n}\} \leq \exp(-\lambda/8) \leq \frac{1}{4096 \mathbf{n}}.$$

Indeed, with $z_\lambda = \lambda/8$,

$$\sqrt{2\lambda z_\lambda} = \lambda/2,$$

and $\{\text{Poi}(\lambda) < \mathbf{n}\} \subseteq \{\lambda - \text{Poi}(\lambda) > \lambda/2\}$; the Poisson Bernstein lower-tail bound gives the exponential term, and $\exp(-z_\lambda) \leq 1/z_\lambda \leq 1/(4096\mathbf{n})$.

On the event $s_+(P) \geq 1/2$, the labeled Poisson mean satisfies

$$u s_+(P) \geq 32768 n,$$

so the labeled retention failure probability is at most $1/(4096n)$. Similarly,

$$v s_+(P) \geq 32768 (n + m),$$

and the event of retaining fewer than m auxiliary observations is contained in the event of retaining fewer than $n + m$ observations; hence the auxiliary retention failure probability is at most

$$\frac{1}{4096(n + m)} \leq \frac{1}{4096n}.$$

7. We next transfer the raw fuzzy lower bound to the fixed experiment by the prefix coupling. Fix any measurable fixed-sample decision rule ψ . Define its transferred raw rule φ_ψ as follows. Given an ordered raw output y_{raw} , set

$$\varphi_\psi(y_{\text{raw}}) = \begin{cases} \psi(\text{the first } n \text{ labeled marks and the first } m \text{ auxiliary marks in } y_{\text{raw}}), & \text{if those marks are present} \\ \zeta_\star, & \text{otherwise.} \end{cases}$$

The rule uses only the first m auxiliary marks, while the raw auxiliary Poisson process has intensity $v s_+(P) = 65536(n + m)s_+(P)$; the auxiliary tail estimate below therefore bounds the event of having fewer than m auxiliary marks by the larger-threshold event of having fewer than $n + m$. Here $\zeta_\star \in \mathbb{R}$ is any fixed real-valued action. For a law P , let

$$\mathbf{F}_P := \{\text{the ordered raw draw contains fewer than } n \text{ labeled marks or fewer than } m \text{ auxiliary marks}\}.$$

On $\mathbf{F}_P^c$, the first n labeled marks and first m auxiliary marks have exactly the fixed-sample law $\mathcal{K}_{\text{fix}}(P)$, because the ordered raw experiment is generated by independent Poisson samples from the same labeled and auxiliary mark kernels and the fixed experiment keeps the first prescribed counts.

For branch b , the exceptional probability under the branch prior predictive is bounded by integrating the conditional failure probability. Splitting according to the scale event gives

$$\begin{aligned} \Pr_{\Pi_b \otimes \mathcal{K}_{\text{raw}}}(\mathbf{F}_P) &\leq \Pi_b\{P : s_+(P) < 1/2\} \\ &\quad + \int_{\{P : s_+(P) \geq 1/2\}} [\Pr\{\text{Poi}(u s_+(P)) < n\} + \Pr\{\text{Poi}(v s_+(P)) < m\}] d\Pi_b(P). \end{aligned}$$

The scale-tail bound from step 4 and the Poisson retention bounds from step 6 therefore yield, for each $b \in \{0, 1\}$,

$$\Pr_{\Pi_b \otimes \mathcal{K}_{\text{raw}}}(\mathbf{F}_P) \leq \frac{\Delta^2}{16384} + \frac{1}{4096n} + \frac{1}{4096n}.$$

Since $0 \leq \ell_\epsilon \leq 4$, the Bayes risk of the transferred raw rule satisfies

$$\begin{aligned} \mathcal{B}_b(\varphi_\psi) &\leq \int \int \ell_\epsilon(P, \psi(y)) d\mathcal{K}_{\text{fix}}(P)(y) d\Pi_b(P) \\ &\quad + 4 \left(\frac{\Delta^2}{16384} + \frac{1}{4096n} + \frac{1}{4096n} \right) \\ &\leq \sup_P \int \ell_\epsilon(P, \psi(y)) d\mathcal{K}_{\text{fix}}(P)(y) + 4 \left(\frac{\Delta^2}{16384} + \frac{1}{4096n} + \frac{1}{4096n} \right). \end{aligned}$$

Combining this with the raw fuzzy bound above gives, for every measurable ψ ,

$$\frac{3\Delta^2}{128} \leq \sup_P \int \ell_\epsilon(P, \psi(y)) d\mathcal{K}_{\text{fix}}(P)(y) + 4 \left(\frac{\Delta^2}{16384} + \frac{1}{4096n} + \frac{1}{4096n} \right).$$

Equivalently,

$$\frac{3\Delta^2}{128} - 4 \left(\frac{\Delta^2}{16384} + \frac{1}{4096n} + \frac{1}{4096n} \right) \leq \sup_P \int \ell_\epsilon(P, \psi(y)) d\mathcal{K}_{\text{fix}}(P)(y).$$

Taking the infimum over ψ gives

$$\frac{3\Delta^2}{128} - 4 \left(\frac{\Delta^2}{16384} + \frac{1}{4096n} + \frac{1}{4096n} \right) \leq \mathfrak{R}_{\text{dec}}.$$

The floor inequality from step 3 also gives $\alpha_{\text{floor}}/n \leq \mathfrak{R}_{\text{dec}}$, so

$$\max \left\{ \frac{3\Delta^2}{128} - 4 \left(\frac{\Delta^2}{16384} + \frac{1}{4096n} + \frac{1}{4096n} \right), \frac{\alpha_{\text{floor}}}{n} \right\} \leq \mathfrak{R}_{\text{dec}}.$$

For the reverse comparison to step 3, start with any measurable estimator T in [Definition 11](#) and use the same function as the fixed-sample decision rule ψ_T . On $\mathcal{M}_{d,\epsilon}$, clipping cannot increase squared error because $\tau(P) \in [-1, 1]$, while off the model class the class-gated loss is zero. Hence the worst-case decision risk of ψ_T is no larger than the worst-case mean-squared risk of T . Taking the infimum over T gives

$$\mathfrak{R}_{\text{dec}} \leq R_\epsilon(n, m, d),$$

and consequently

$$\max \left\{ \frac{3\Delta^2}{128} - 4 \left(\frac{\Delta^2}{16384} + \frac{1}{4096n} + \frac{1}{4096n} \right), \frac{\alpha_{\text{floor}}}{n} \right\} \leq R_\epsilon(n, m, d).$$

8. It remains only to read off the deterministic arithmetic. From the floor bound,

$$R_\epsilon(n, m, d) \geq \frac{\alpha_{\text{floor}}}{n} > 0.$$

First suppose

$$\frac{1}{n} \leq \Delta^2.$$

Then

$$4 \left(\frac{\Delta^2}{16384} + \frac{1}{4096n} + \frac{1}{4096n} \right) \leq 4 \left(\frac{\Delta^2}{16384} + \frac{2\Delta^2}{4096} \right) = \frac{9\Delta^2}{4096}.$$

Therefore

$$\frac{3\Delta^2}{128} - 4 \left(\frac{\Delta^2}{16384} + \frac{1}{4096n} + \frac{1}{4096n} \right) \geq \frac{87\Delta^2}{4096} \geq \frac{\Delta^2}{65536}.$$

The first branch of the maximum in the preceding display gives

$$R_\epsilon(n, m, d) \geq \frac{\Delta^2}{65536}.$$

Now suppose

$$\Delta^2 < \frac{1}{n}.$$

Since $\alpha_{\text{floor}} = 1/100$,

$$\frac{\Delta^2}{65536} < \frac{1}{65536n} \leq \frac{1}{100n} = \frac{\alpha_{\text{floor}}}{n}.$$

The second branch of the maximum gives the same conclusion. Combining the two cases,

$$R_\epsilon(n, m, d) \geq \frac{\Delta^2}{65536}.$$

□

Lemma 20 converts the raw recipe separation into a lower bound for $R_\epsilon(n, m, d)$ under the fixed annotation sample sizes. The total-variation requirement supplies the testing affinity, and the variance-scale requirement keeps the normalization fluctuations small relative to the squared raw gap.

The Poisson construction is then converted to a general fixed-size annotation experiment. The transfer statement tracks the raw normalization explicitly, so the resulting lower bound applies to the minimax risk in [Definition 11](#).

Lemma 21. `[lem:fixed-sample-common-marginal-transfer]` (Fixed-sample transfer) *There exist numerical constants $C_{\text{fixed}} > 0$ and $c_{\text{risk}} > 0$ such that the following holds. For every $0 < \epsilon < 1/2$ and every $C_\epsilon > 0$, there is a constant $c_{\epsilon, C_\epsilon} > 0$ such that, for every common-marginal calibration at overlap level ϵ with positive calibration constants and dual gap $\delta_\epsilon \leq 1/4$, every $\rho \in \mathbb{R}$, every $n, m, d, L, k \in \mathbb{N}$, every $B, a \in \mathbb{R}$, and every common-marginal recipe R at these indices, the implication below holds.*

Write G_R for the recipe's rare-cell generator measure and $\theta_{R,b}(w)$ for its raw target on branch b . Let

$$\Delta = \int \theta_{R,1}(w) dG_R(w) - \int \theta_{R,0}(w) dG_R(w).$$

Assume:

- **(Sample and dimension.)** $n \geq 1$ and $d \geq 2$.
- **(Witness data.)** The recipe carries a common-marginal Poisson-prior witness at labeled and auxiliary intensities

$$u = C_{\text{fixed}}n, \quad v = C_{\text{fixed}}(n + m).$$

Equivalently, the witness supplies

$$(u + v)B \leq b_\epsilon L, \quad B > 0, \quad a = \gamma_\epsilon B/L^2, \quad ka \leq \delta_\epsilon,$$

identifies the recipe's labeled and auxiliary intensities with u and v , and satisfies the corresponding common-marginal bounds at dual gap δ_ϵ , variance-bound constant C_ϵ , and likelihood-decay parameter ρ .

- **(Testing distance.)** The raw prior-predictive mixture laws M_0 and M_1 generated by the two branches of the recipe satisfy

$$\text{TV}(M_0, M_1) \leq \frac{1}{8}.$$

- **(Concentration scale.)** The realized scale obeys

$$kBa \leq c_{\epsilon, C_\epsilon} \Delta^2.$$

Then the minimax risk $R_\epsilon(n, m, d)$ from [Definition 11](#) satisfies

$$c_{\text{risk}} \Delta^2 \leq R_\epsilon(n, m, d).$$

Proof of [Lemma 21](#). Take

$$C_{\text{fixed}} = 65536, \quad c_{\text{risk}} = \frac{1}{65536}.$$

Fix $0 < \epsilon < 1/2$ and a positive variance-bound constant C_ϵ , and set

$$c_{\epsilon, C_\epsilon} := \frac{1}{65536 C_\epsilon} > 0.$$

Let a calibration, a recipe, and all parameters satisfy the hypotheses of the statement. Put

$$\bar{\theta}_b := \int \theta_b(w) dG(w), \quad b \in \{0, 1\},$$

so that the raw-center separation in the statement is

$$\Delta = \bar{\theta}_1 - \bar{\theta}_0.$$

The recipe's raw prior centers are exactly these two numbers:

$$\bar{\theta}_1 - \bar{\theta}_0 = \int \theta_1(w) dG(w) - \int \theta_0(w) dG(w).$$

The Poisson-prior witness in the hypotheses identifies the recipe's labeled and auxiliary intensities as

$$u = 65536 n, \quad v = 65536 (n + m).$$

Separately, the concentration-scale hypothesis, the definition of c_{ϵ, C_ϵ} , and $\Delta = \bar{\theta}_1 - \bar{\theta}_0$ give

$$kBa \leq c_{\epsilon, C_\epsilon} \Delta^2 = \frac{\Delta^2}{65536 C_\epsilon} = \frac{1}{65536 C_\epsilon} |\bar{\theta}_1 - \bar{\theta}_0|^2.$$

Together with $n \geq 1$, $d \geq 2$, and $\text{TV}(M_0, M_1) \leq 1/8$, these are precisely the hypotheses of [Lemma 20](#), applied with the present calibration, recipe, variance-bound constant C_ϵ , likelihood-decay parameter ρ , and intensities $u = 65536n$, $v = 65536(n + m)$. That result yields, for the minimax risk $R_\epsilon(n, m, d)$ of [Definition 11](#),

$$R_\epsilon(n, m, d) \geq \frac{1}{65536} |\bar{\theta}_1 - \bar{\theta}_0|^2.$$

Since $|\bar{\theta}_1 - \bar{\theta}_0|^2 = \Delta^2$ and $c_{\text{risk}} = 1/65536$, this is exactly

$$c_{\text{risk}}\Delta^2 \leq R_\epsilon(n, m, d).$$

□

Lemma 21 is the bridge from a Poisson fuzzy-hypothesis calculation to the fixed- n, m risk. The testing-distance condition supplies indistinguishability, the concentration-scale condition controls the random raw normalization, and the conclusion converts the raw-center separation Δ into a lower bound for $R_\epsilon(n, m, d)$.

A separate one-dimensional ingredient records the unavoidable parametric contribution of the labeled sample. It is used together with the rare-cell construction to cover regimes in which the lower-bound rate is governed by outcome labels.

Lemma 22. [lem:parametric-label-floor] (Parametric label floor) *Let $R_\epsilon(n, m, d)$ be the minimax risk in Definition 11. There exists a universal constant $c > 0$ such that, whenever*

- *(Overlap level.)* $0 < \epsilon < 1/2$,
- *(Labeled sample size.)* $n \geq 1$,
- *(Auxiliary sample size.)* m is a nonnegative integer,
- *(Alphabet size.)* $d \geq 2$,

the minimax risk satisfies

$$R_\epsilon(n, m, d) \geq \frac{c}{n}.$$

Proof of Lemma 22. Set $c = 1/100$. Fix $0 < \epsilon < 1/2$, $n \geq 1$, $m \geq 0$, and $d \geq 2$. Define

$$\eta_n := \frac{2}{5\sqrt{n}}.$$

Then $\eta_n \geq 0$, and

$$\eta_n^2 = \frac{4}{25n}.$$

Consider two laws P_{null} and P_{pert} in $\mathcal{M}_{d,\epsilon}$ with uniform covariate distribution on $[d]$, propensity $e_x = 1/2$ in every cell, control regression $1/2$ in every cell, and treated regression respectively $1/2$ and $1/2 + \eta_n$. Since $n \geq 1$, $\eta_n \leq 2/5$, hence $1/2 + \eta_n \leq 1$. The overlap condition follows from $0 < \epsilon < 1/2$, so both laws belong to $\mathcal{M}_{d,\epsilon}$.

For the labeled samples, the parametric total-variation bound applies once

$$n \cdot \frac{(1/2)\eta_n^2}{(1/2)(1 - 1/2)} \leq \log 2.$$

Using the displayed value of η_n^2 , the left side is $8/25$, and $8/25 \leq \log 2$. Therefore

$$\text{TV}(P_{\text{null}}^{\otimes n}, P_{\text{pert}}^{\otimes n}) \leq \frac{1}{2}.$$

The auxiliary treatment-covariate marginals of the two laws are equal:

$$(P_{\text{null}})_{XA} = (P_{\text{pert}})_{XA}.$$

Indeed, both laws have the same uniform covariate distribution and the same propensity $e_x = 1/2$ in every cell; summing over the binary outcome coordinate gives the same mass for each (x, a) . Hence their m -fold auxiliary product laws are identical. The product total-variation inequality gives

$$\text{TV}(P_{\text{null}}^{\otimes n} \otimes (P_{\text{null}})_{XA}^{\otimes m}, P_{\text{pert}}^{\otimes n} \otimes (P_{\text{pert}})_{XA}^{\otimes m}) \leq \text{TV}(P_{\text{null}}^{\otimes n}, P_{\text{pert}}^{\otimes n}) + 0 \leq \frac{1}{2}.$$

The two average treatment effects are

$$\tau(P_{\text{null}}) = 0, \quad \tau(P_{\text{pert}}) = \eta_n.$$

The first identity follows because the two outcome regressions coincide in every cell, and the second because every cell has treated-minus-control contrast η_n and the cell masses sum to one. Thus, with $s_n := \eta_n/2$,

$$2s_n = \eta_n \leq |\tau(P_{\text{null}}) - \tau(P_{\text{pert}})|.$$

We now record the two-point risk consequence used for [Definition 11](#). Let $T \in \mathcal{T}_{n,m,d}$ be any estimator, and let

$$\mathbf{A}_0 := \{z : s_n \leq |T(z) - \tau(P_{\text{null}})|\}, \quad \mathbf{A}_1 := \{z : s_n \leq |T(z) - \tau(P_{\text{pert}})|\}.$$

The testing bound at separation $2s_n$, together with the preceding total-variation bound, yields

$$\max \left\{ \Pr_{P_{\text{null}}^{\otimes n} \otimes (P_{\text{null}})_{XA}^{\otimes m}}(\mathbf{A}_0), \Pr_{P_{\text{pert}}^{\otimes n} \otimes (P_{\text{pert}})_{XA}^{\otimes m}}(\mathbf{A}_1) \right\} \geq \frac{1}{4}.$$

Moreover, by the elementary inequality

$$s_n^2 \Pr\{s_n \leq |T - \theta|\} \leq \mathbb{E}[(T - \theta)^2],$$

applied under each of the two laws, the larger of the two mean-squared risks is at least $s_n^2/4$. Taking the supremum over $P \in \mathcal{M}_{d,\epsilon}$ and then the infimum over $T \in \mathcal{T}_{n,m,d}$ gives

$$\frac{s_n^2}{4} \leq R_\epsilon(n, m, d).$$

Finally,

$$\frac{s_n^2}{4} = \frac{(\eta_n/2)^2}{4} = \frac{1}{100n}.$$

Consequently $R_\epsilon(n, m, d) \geq 1/(100n)$, which proves the claim with the universal constant $c = 1/100$. $\square$

[Lemma 22](#) fixes the labeled-outcome scale uniformly over auxiliary sample sizes and alphabet sizes. This lower bound matches the $1/n$ component in [Definition 12](#), so the remaining common-marginal construction can focus on the alphabet-dependent contribution.

One final finite-product identity records how independent coordinates can be embedded into a larger product experiment while adding a distinguished coordinate and deterministic padding. This bookkeeping is useful when the rare-cell construction is placed inside a larger finite alphabet with reservoir coordinates.

Lemma 23. [`lem:finite-product-padding-identity`] (Finite product padding identity) *Let I and J be finite index sets, and let $\mathbf{Y}$ be a measurable space whose singletons are measurable. Suppose the following conditions hold:*

- **(Embedding.)** $f : I \rightarrow J$ is injective.
- **(Distinguished coordinate.)** $r \in J$ satisfies $r \neq f(i)$ for every $i \in I$.
- **(Coordinate laws.)** $\mu_i, i \in I$, and ν are probability measures on Y , and $y_0 \in \mathsf{Y}$.

Define $A : \mathsf{Y}^I \times \mathsf{Y} \rightarrow \mathsf{Y}^J$ by

$$A(z, y)(j) = \begin{cases} y, & j = r, \\ z_i, & j = f(i) \text{ for the unique } i \in I, \\ y_0, & j \in J \setminus (\{r\} \cup f(I)). \end{cases}$$

Define the coordinate laws $H_j, j \in J$, by

$$H_j = \begin{cases} \nu, & j = r, \\ \mu_i, & j = f(i) \text{ for the unique } i \in I, \\ \delta_{y_0}, & j \in J \setminus (\{r\} \cup f(I)). \end{cases}$$

Then

$$A_{\#} \left(\left(\bigotimes_{i \in I} \mu_i \right) \otimes \nu \right) = \bigotimes_{j \in J} H_j.$$

Proof of Lemma 23. 1. Write

$$\Lambda_{\text{in}} := \left(\bigotimes_{i \in I} \mu_i \right) \otimes \nu \quad \text{on } \mathsf{Y}^I \times \mathsf{Y}, \quad \Lambda_{\text{out}} := \bigotimes_{j \in J} H_j \quad \text{on } \mathsf{Y}^J.$$

The map A is measurable. Indeed, for each $j \in J$, the j -th coordinate of $A(z, y)$ is the second projection if $j = r$, is the i -th coordinate projection from z if $j = f(i)$, and is the constant y_0 on the remaining coordinates. The condition $r \neq f(i)$ separates the distinguished coordinate from the embedded ones, and injectivity of f makes the embedded coordinate i unique. Since I and J are finite, coordinatewise measurability gives measurability of A . It is therefore enough, by uniqueness of finite product measures on measurable rectangles, to prove that for every family of measurable sets

$$(\mathsf{G}_j)_{j \in J}, \quad \mathsf{G}_j \in \mathcal{V},$$

one has

$$\Lambda_{\text{in}}(A^{-1}\{w \in \mathsf{Y}^J : \forall j \in J, w_j \in \mathsf{G}_j\}) = \prod_{j \in J} H_j(\mathsf{G}_j).$$

2. First suppose that there is an unused coordinate $j_{\circ} \in J$ such that

$$j_{\circ} \neq r, \quad j_{\circ} \neq f(i) \text{ for every } i \in I, \quad y_0 \notin \mathsf{G}_{j_{\circ}}.$$

For every $(z, y) \in \mathsf{Y}^I \times \mathsf{Y}$, the definition of A gives

$$A(z, y)_{j_{\circ}} = y_0.$$

Thus

$$\mathbf{A}^{-1}\{w : \forall j, w_j \in \mathbf{G}_j\} = \emptyset,$$

so the left-hand side is 0. On the right-hand side, the same coordinate has

$$\mathbf{H}_{j_0} = \delta_{y_0}, \quad \mathbf{H}_{j_0}(\mathbf{G}_{j_0}) = \delta_{y_0}(\mathbf{G}_{j_0}) = 0,$$

because $y_0 \notin \mathbf{G}_{j_0}$. Hence

$$\prod_{j \in J} \mathbf{H}_j(\mathbf{G}_j) = 0,$$

and the required rectangle identity holds in this case.

3. It remains to consider the complementary case:

$$\forall j \in J, \quad j \neq r \text{ and } (\forall i \in I, j \neq f(i)) \implies y_0 \in \mathbf{G}_j.$$

Then the preimage of the rectangle is exactly

$$\mathbf{A}^{-1}\{w : \forall j, w_j \in \mathbf{G}_j\} = \{z \in \mathbf{Y}^I : \forall i \in I, z_i \in \mathbf{G}_{f(i)}\} \times \mathbf{G}_r.$$

For the forward inclusion, membership in the rectangle gives the r -coordinate condition $y \in \mathbf{G}_r$, and for each $i \in I$ it gives

$$z_i = \mathbf{A}(z, y)_{f(i)} \in \mathbf{G}_{f(i)},$$

using $r \neq f(i)$ and injectivity of f . Conversely, if $z_i \in \mathbf{G}_{f(i)}$ for all i and $y \in \mathbf{G}_r$, then each coordinate $j \in J$ is checked by the three alternatives in the definition of $\mathbf{A}$: the coordinate r gives y , a coordinate $f(i)$ gives z_i , and an unused coordinate gives $y_0 \in \mathbf{G}_j$ by the displayed complementary condition.

4. Using the preceding preimage identity and the finite product formula,

$$\begin{aligned} \Lambda_{\text{in}}(\mathbf{A}^{-1}\{w : \forall j, w_j \in \mathbf{G}_j\}) &= \left(\bigotimes_{i \in I} \mu_i \right) \{z : \forall i, z_i \in \mathbf{G}_{f(i)}\} \nu(\mathbf{G}_r) \\ &= \left(\prod_{i \in I} \mu_i(\mathbf{G}_{f(i)}) \right) \nu(\mathbf{G}_r). \end{aligned}$$

The product over J has the same value. The distinguished coordinate contributes

$$\mathbf{H}_r(\mathbf{G}_r) = \nu(\mathbf{G}_r),$$

each embedded coordinate contributes, for $i \in I$,

$$\mathbf{H}_{f(i)}(\mathbf{G}_{f(i)}) = \mu_i(\mathbf{G}_{f(i)}),$$

again by $r \neq f(i)$ and injectivity of f , and every unused coordinate j contributes

$$\mathbf{H}_j(\mathbf{G}_j) = \delta_{y_0}(\mathbf{G}_j) = 1$$

by the complementary condition. Since f is injective, the embedded coordinates are indexed without repetition, and therefore

$$\prod_{j \in J} H_j(\mathbf{G}_j) = \nu(\mathbf{G}_r) \prod_{i \in I} \mu_i(\mathbf{G}_{f(i)}).$$

This agrees with the preceding expression for the pushforward on every measurable rectangle, so

$$\mathbf{A}_{\#} \Lambda_{\text{in}} = \Lambda_{\text{out}},$$

which is the desired identity. □

Lemma 23 formalizes the product-measure preservation behind this padding step. Injectivity keeps the rare coordinates distinct, the distinguished coordinate carries the added law, and the remaining coordinates are fixed at point masses, so the assembled law is the intended product over the larger index set.

The final calibration statements choose constants so that the rare-cell prior works uniformly across the intensity range needed in [Theorem 3](#). The first gives a ready-to-use recipe with explicit counting, variance, and total-variation controls; the second packages these ingredients into the common-marginal certificate family used by the main converse.

Lemma 24. [lem:common-marginal-uniform-intensity] (Uniform intensity common marginal) *For every overlap level $0 < \epsilon < 1/2$, there are positive constants*

$$r_{\epsilon}, \quad D_{\epsilon}, \quad b_{0,\epsilon}, \quad b_{\epsilon}, \quad \gamma_{\epsilon}, \quad q_{\epsilon}, \quad c_{\epsilon}, \quad \delta_{\epsilon}, \quad C_{\epsilon}$$

and a number $0 < \rho_{\epsilon} < 1$, with $c_{\epsilon} \leq 1/4$, such that the following calibration facts hold:

- **(Bandwidth slack.)** $131072 b_{0,\epsilon} \leq b_{\epsilon}$.
- **(Degree floor.)** $2 \leq D_{\epsilon}$.
- **(Rare-cell scale.)**

$$q_{\epsilon} \gamma_{\epsilon} b_{0,\epsilon} \leq c_{\epsilon}, \quad q_{\epsilon} = \frac{c_{\epsilon}}{2\gamma_{\epsilon} b_{0,\epsilon}}, \quad 4 \leq q_{\epsilon}.$$

- **(Dimension scale.)**

$$\delta_{\epsilon} r_{\epsilon} \leq \frac{1}{65536}, \quad \delta_{\epsilon} \leq \frac{c_{\epsilon}^4}{16}, \quad \delta_{\epsilon} \leq \left(\frac{c_{\epsilon} \gamma_{\epsilon} b_{0,\epsilon}}{2(D_{\epsilon} + 1)} \right)^2.$$

- **(Variance and counting calibration.)**

$$8(D_{\epsilon} + 1)b_{0,\epsilon} \leq \frac{c_{\epsilon}^3}{65536 C_{\epsilon}},$$

$$\frac{4(D_{\epsilon} + 1)}{(1/(65536 C_{\epsilon}))c_{\epsilon}^2 \gamma_{\epsilon}} \leq q_{\epsilon},$$

and

$$\left\{ \frac{4(D_{\epsilon} + 1)^2}{(1/(65536 C_{\epsilon}))c_{\epsilon}^2 \gamma_{\epsilon}} \right\}^2 \leq r_{\epsilon}.$$

- (*Mixture decay.*) For every integer $n \geq 1$,

$$C_\epsilon(65536n)c_\epsilon\rho_\epsilon^{[D_\epsilon \log(en)]} \leq \frac{1}{8}.$$

For every n, m, L, d, k and every u, v, B, a , if

- $n \geq 1$, $L \geq 2$, and $d \geq 2$;
- $u \geq 0$, $v \geq 0$, and $u + v > 0$;
- $B > 0$, $(u + v)B \leq b_\epsilon L$, and

$$a = \frac{\gamma_\epsilon B}{L^2};$$

- $ka \leq c_\epsilon$ and $k + 1 \leq d$,

then there is a common-marginal rare-cell recipe with k rare cells at scale (a, B, L) whose labeled and auxiliary Poisson intensities are u and v . Its null and alternative priors are the two branch priors of the recipe, and they satisfy:

- the recipe is in the finite-atom construction regime, and both priors are supported on $\mathcal{M}_{d,\epsilon}$;
- the induced random treatment-covariate tables P_{XA} have exactly the same law under the two priors;
- the raw total mass has mean one;
- the raw target centers obey

$$c_\epsilon ka \leq |\theta_1 - \theta_0|,$$

where θ_1 and θ_0 are the alternative and null raw prior centers;

- the raw total-mass variance is at most

$$C_\epsilon kBa,$$

and the larger of the two raw target variances is at most

$$C_\epsilon kBa;$$

- in the raw Poisson experiment with labeled intensity u and auxiliary intensity v , the two prior-predictive mixture laws have total variation distance at most

$$C_\epsilon uka \rho_\epsilon^L.$$

Proof of Lemma 24. Fix $0 < \epsilon < 1/2$.

1. By Lemma 17, with

$$\kappa_\epsilon := \frac{1 - 2\epsilon}{\epsilon} > 0,$$

there are $\gamma_\epsilon > 0$ and $\bar{c}_\epsilon > 0$ such that every $L \geq 2$ and $B > 0$, with $a = \gamma_\epsilon B/L^2$, admit a finite signed certificate on $[a/\kappa_\epsilon, B]$, annihilating monomials through degree L ,

whose signed integral of $p/(p+a)$ is at least $\bar{c}_\epsilon$. The one-coordinate marked-Poisson moment-matching estimate supplies constants $\bar{b}_\epsilon > 0$, $\bar{C}_\epsilon > 0$, and $\rho_\epsilon \in (0, 1)$ with the following uniform property for the complete normalized certificate object that generates the coordinate mixtures. For every finite index set $\mathcal{I}$ and every normalized finite signed moment certificate

$$\mathfrak{C} = \{(p_\iota, \omega_\iota) : \iota \in \mathcal{I}\}$$

of order L , meaning that its signed measure $\sum_{\iota \in \mathcal{I}} \omega_\iota \delta_{p_\iota}$ has normalized variation $\sum_{\iota \in \mathcal{I}} |\omega_\iota| = 1$ and the certificate data include the moment equations through degree L , if its variation measure is supported on $[a/\kappa_\epsilon, B]$ and the support scale satisfies $(u+v)B \leq \bar{b}_\epsilon L$, then the two branch marked-Poisson predictive laws $\mathbf{Q}_0, \mathbf{Q}_1$ associated with that same certificate obey

$$\text{TV}(\mathbf{Q}_0, \mathbf{Q}_1) \leq \bar{C}_\epsilon u a \rho_\epsilon^L.$$

Set

$$C_\epsilon := \max\{\bar{C}_\epsilon, 1\}, \quad c_\epsilon := \min\{\bar{c}_\epsilon, 1/4\}, \quad b_\epsilon := \bar{b}_\epsilon.$$

Let

$$D_\epsilon := \max\left\{2, \frac{\max\{1, \log(8C_\epsilon 65536 c_\epsilon) + 2\}}{-\log \rho_\epsilon}\right\}, \quad s_\epsilon := \frac{1}{65536 C_\epsilon}.$$

Define

$$b_{0,\epsilon} := \min\left\{\frac{b_\epsilon}{131072}, \frac{c_\epsilon}{8\gamma_\epsilon}, \frac{s_\epsilon c_\epsilon^3}{8(D_\epsilon + 1)}\right\},$$

$$q_\epsilon := \frac{c_\epsilon}{2\gamma_\epsilon b_{0,\epsilon}}, \quad r_\epsilon := \left(\max\left\{1, \frac{4(D_\epsilon + 1)^2}{s_\epsilon c_\epsilon^2 \gamma_\epsilon}\right\}\right)^2,$$

and

$$\delta_\epsilon := \min\left\{\frac{c_\epsilon^4}{16}, \left(\frac{c_\epsilon \gamma_\epsilon b_{0,\epsilon}}{2(D_\epsilon + 1)}\right)^2, \frac{1}{65536 r_\epsilon}\right\}.$$

All these constants are positive, $0 < \rho_\epsilon < 1$, and $c_\epsilon \leq 1/4$. The first five displayed calibrations follow immediately from the three minima and the definitions: $131072 b_{0,\epsilon} \leq b_\epsilon$, $D_\epsilon \geq 2$,

$$q_\epsilon \gamma_\epsilon b_{0,\epsilon} = \frac{c_\epsilon}{2} \leq c_\epsilon, \quad q_\epsilon = \frac{c_\epsilon}{2\gamma_\epsilon b_{0,\epsilon}}, \quad q_\epsilon \geq 4,$$

and

$$\delta_\epsilon r_\epsilon \leq \frac{1}{65536}, \quad \delta_\epsilon \leq \frac{c_\epsilon^4}{16}, \quad \delta_\epsilon \leq \left(\frac{c_\epsilon \gamma_\epsilon b_{0,\epsilon}}{2(D_\epsilon + 1)}\right)^2.$$

Moreover,

$$8(D_\epsilon + 1)b_{0,\epsilon} \leq \frac{c_\epsilon^3}{65536 C_\epsilon},$$

and the same bound on $b_{0,\epsilon}$ gives

$$\frac{4(D_\epsilon + 1)}{(1/(65536 C_\epsilon))c_\epsilon^2 \gamma_\epsilon} \leq q_\epsilon.$$

The definition of r_ϵ gives

$$\left\{\frac{4(D_\epsilon + 1)^2}{(1/(65536 C_\epsilon))c_\epsilon^2 \gamma_\epsilon}\right\}^2 \leq r_\epsilon.$$

Finally, put $\lambda_\epsilon := -\log \rho_\epsilon > 0$ and $M_\epsilon := 8C_\epsilon 65536 c_\epsilon$. Since

$$\lambda_\epsilon D_\epsilon \geq \max\{1, \log M_\epsilon + 2\},$$

for every $n \geq 1$,

$$\rho_\epsilon^{\lceil D_\epsilon \log(en) \rceil} \leq \frac{1}{M_\epsilon n}.$$

Thus

$$C_\epsilon(65536n)c_\epsilon\rho_\epsilon^{\lceil D_\epsilon \log(en) \rceil} \leq \frac{C_\epsilon 65536 n c_\epsilon}{8C_\epsilon 65536 c_\epsilon n} = \frac{1}{8}.$$

2. Now fix $n, m, L, d, k, u, v, B, a$ satisfying the hypotheses in the statement. By the bandwidth slack,

$$(u + v)B \leq b_\epsilon L = \bar{b}_\epsilon L,$$

and $a = \gamma_\epsilon B/L^2 > 0$. Applying [Lemma 17](#) at this L, B gives distinct points $p_j \in [a/\kappa_\epsilon, B]$ and weights ω_j with

$$\sum_j |\omega_j| = 1, \quad \sum_j \omega_j p_j^\ell = 0 \quad (0 \leq \ell \leq L), \quad c_\epsilon \leq \sum_j \omega_j \frac{p_j}{p_j + a}.$$

Let

$$\sigma := \sum_j \omega_j \delta_{p_j}, \quad |\sigma| := \sum_j |\omega_j| \delta_{p_j},$$

and define the polar sign on all $p \in \mathbb{R}$ by

$$h(p) := \sum_j \mathbf{1}_{\{p=p_j\}} \operatorname{sgn}(\omega_j).$$

Then $|\sigma|$ is a probability measure, $|h(p)| \leq 1$, and

$$\sigma = h |\sigma|, \quad c_\epsilon \leq \int \frac{p}{p + a} d\sigma(p).$$

Also, every certificate node satisfies $p_j \geq a/\kappa_\epsilon > 0$, so 0 is not one of the nodes. Hence the displayed definition gives $h(0) = 0$. Define the zero-inflated scalar law

$$\nu(E) := \int_E \frac{a}{p + a} d|\sigma|(p) + \left(1 - \int \frac{a}{p + a} d|\sigma|(p)\right) \mathbf{1}_E(0).$$

Since $p \geq a/\kappa_\epsilon > 0$ on the support of $|\sigma|$,

$$0 \leq \frac{a}{p + a} \leq 1 \quad |\sigma|\text{-a.e.}$$

and $|\sigma|$ has total mass one. Hence

$$0 \leq \int \frac{a}{p + a} d|\sigma|(p) \leq 1,$$

so the coefficient of δ_0 is nonnegative and ν is a probability measure supported on $\{0\} \cup [a/\kappa_\epsilon, B]$. Its mean $\bar{p} := \int p d\nu(p)$ is computed directly from the definition of ν : the atom at zero contributes nothing, and therefore

$$\bar{p} = \int p \frac{a}{p + a} d|\sigma|(p) = a \int \frac{p}{p + a} d|\sigma|(p).$$

Because $0 \leq p/(p+a) \leq 1$ on the support, this gives

$$0 \leq \bar{p} \leq a.$$

Let $\mathbf{G} := \nu^{\otimes k}$. For every vector $\boldsymbol{\xi} \in \mathbb{R}^k$, set

$$\widehat{\xi}_i := \begin{cases} \xi_i, & \xi_i \in \{p_j\}_j, \\ 0, & \xi_i \notin \{p_j\}_j. \end{cases}$$

Under $\mathbf{G}$, $\widehat{\boldsymbol{\xi}} = \boldsymbol{\xi}$ almost surely, and the range of $\widehat{\boldsymbol{\xi}}$ is finite. Choose one reservoir cell $x_\star \in [d]$ and distinct rare cells $x_1, \dots, x_k \in [d] \setminus \{x_\star\}$, possible because $k+1 \leq d$. Put

$$p_\star := 1 - k\bar{p}.$$

Since $ka \leq c_\epsilon \leq 1/4$, one has $p_\star \geq 3/4 > 0$. Define the raw cell masses and raw total mass by

$$\Omega_{\boldsymbol{\xi}}(x) := p_\star \mathbf{1}_{\{x=x_\star\}} + \sum_{i=1}^k \widehat{\xi}_i \mathbf{1}_{\{x=x_i\}}, \quad S(\boldsymbol{\xi}) := p_\star + \sum_{i=1}^k \widehat{\xi}_i.$$

Then $S(\boldsymbol{\xi}) > 0$ and

$$S(\boldsymbol{\xi}) = \sum_{x=1}^d \Omega_{\boldsymbol{\xi}}(x).$$

3. Define the propensity field on all cells by

$$e_{\boldsymbol{\xi}}(x) := \begin{cases} 1/2, & x = x_\star, \\ 1/2, & x \neq x_\star \text{ and } \sum_i \widehat{\xi}_i \mathbf{1}_{\{x=x_i\}} = 0, \\ \epsilon \frac{\sum_i \widehat{\xi}_i \mathbf{1}_{\{x=x_i\}} + a}{\sum_i \widehat{\xi}_i \mathbf{1}_{\{x=x_i\}}}, & \text{otherwise.} \end{cases}$$

If the last line applies, the denominator is some nonzero $\widehat{\xi}_i \in [a/\kappa_\epsilon, B]$, so

$$\epsilon \leq \epsilon \left(1 + \frac{a}{\widehat{\xi}_i} \right) \leq \epsilon(1 + \kappa_\epsilon) = 1 - \epsilon.$$

The other cases give $e_{\boldsymbol{\xi}}(x) = 1/2 \in [\epsilon, 1 - \epsilon]$. For branches $b \in \{0, 1\}$, define control outcome means to be zero and treated outcome means by

$$\mu_{1,\boldsymbol{\xi}}^{(0)}(x_i) := \frac{1 - h(\widehat{\xi}_i)}{2}, \quad \mu_{1,\boldsymbol{\xi}}^{(1)}(x_i) := \frac{1 + h(\widehat{\xi}_i)}{2},$$

with $\mu_{1,\boldsymbol{\xi}}^{(b)}(x_\star) = 0$ and $\mu_{1,\boldsymbol{\xi}}^{(b)}(x) = 0$ on unused cells. Since $|h| \leq 1$, all outcome means lie in $[0, 1]$. Let $P_b(\boldsymbol{\xi})$ be the law on $[d] \times \{0, 1\}^2$ with cell mass

$$p_{\boldsymbol{\xi}}(x) := \frac{\Omega_{\boldsymbol{\xi}}(x)}{S(\boldsymbol{\xi})},$$

propensity $e_{\xi}(x)$, zero control outcome mean, and treated outcome mean $\mu_{1,\xi}^{(b)}(x)$. This is a probability law because each raw mass is nonnegative and

$$\sum_{x=1}^d p_{\xi}(x) = \frac{\sum_x \Omega_{\xi}(x)}{S(\xi)} = 1.$$

For the reservoir cell, the raw mass is $p_{\star} > 0$ and the propensity is $1/2$, which lies in $[\epsilon, 1 - \epsilon]$. For an unused cell, the raw mass is zero, so the occupied-cell overlap condition has no constraint; the totalized propensity there is $1/2$. For a rare cell x_i , if $\hat{\xi}_i = 0$ then its raw mass is zero. If $\hat{\xi}_i \neq 0$, then $\hat{\xi}_i \in [a/\kappa_{\epsilon}, B]$, and the calculation above gives

$$\epsilon \leq e_{\xi}(x_i) = \epsilon \left(1 + \frac{a}{\hat{\xi}_i} \right) \leq 1 - \epsilon.$$

The control outcome means are zero, and the treated outcome means are either zero or of the form $(1 \pm h(\hat{\xi}_i))/2$, hence lie in $[0, 1]$. Thus every occupied cell satisfies the overlap bounds and all outcome means are legal binary means, so $P_b(\xi) \in \mathcal{M}_{d,\epsilon}$ by [Definition 6](#). Define

$$\Pi_b := (P_b)_{\#} \mathbf{G}, \quad b \in \{0, 1\}.$$

These are the two branch priors of the recipe, and the finite range of $\hat{\xi}$ places the recipe in the finite-atom construction regime. The labeled and auxiliary intensities stored in the recipe are, by construction, u and v . Since $p_{\xi}(x)$ and $e_{\xi}(x)$ are independent of b ,

$$(P \mapsto P_{XA})_{\#} \Pi_0 = (P \mapsto P_{XA})_{\#} \Pi_1.$$

4. The raw total mass has mean one:

$$\int S(\xi) d\mathbf{G}(\xi) = p_{\star} + \sum_{i=1}^k \int \xi_i d\mathbf{G}(\xi) = 1 - k\bar{p} + k\bar{p} = 1.$$

For branch b , the raw target is

$$T_b(\xi) := \sum_{i=1}^k \hat{\xi}_i \frac{1 + (-1)^{1-b} h(\hat{\xi}_i)}{2}.$$

It satisfies the raw version of [Definition 8](#). Indeed, the control outcome means are zero, so

$$\tau(P_b(\xi)) = \sum_{x=1}^d \frac{\Omega_{\xi}(x)}{S(\xi)} \mu_{1,\xi}^{(b)}(x).$$

Multiplying by $S(\xi)$ cancels the normalization. The reservoir and unused cells contribute zero, and the rare-cell terms give

$$S(\xi) \tau(P_b(\xi)) = \sum_{i=1}^k \hat{\xi}_i \frac{1 + (-1)^{1-b} h(\hat{\xi}_i)}{2} = T_b(\xi).$$

Moreover,

$$\int \left[p \frac{1 + h(p)}{2} - p \frac{1 - h(p)}{2} \right] d\nu(p) = \int p h(p) d\nu(p).$$

The atom of ν at zero contributes zero. On the nonzero support, $d\nu = (a/(p+a)) d|\sigma|$ and $d\sigma = h d|\sigma|$, so

$$\int ph(p) d\nu(p) = a \int \frac{p}{p+a} d\sigma(p).$$

Therefore the raw prior centers $\theta_b := \int T_b dG$ obey

$$\theta_1 - \theta_0 = ka \int \frac{p}{p+a} d\sigma(p).$$

The alternation gap gives

$$c_\epsilon \leq \int \frac{p}{p+a} d\sigma(p),$$

so the displayed difference is nonnegative and at least $c_\epsilon ka$. Hence

$$c_\epsilon ka \leq |\theta_1 - \theta_0|.$$

5. The support and mean bounds for ν give

$$\text{Var}_G(S) = \text{Var}_G\left(\sum_{i=1}^k \xi_i\right) \leq \sum_{i=1}^k \mathbb{E}_\nu[p^2] \leq kB \mathbb{E}_\nu[p] \leq kB a \leq C_\epsilon kBa.$$

For each branch define

$$g_b(p) := p \frac{1 + (-1)^{1-b} h(p)}{2}.$$

Then $0 \leq g_b(p) \leq p \leq B$ on the support of ν , and

$$T_b(\boldsymbol{\xi}) = \sum_{i=1}^k g_b(\xi_i) \quad \text{almost surely under } G.$$

Consequently,

$$\text{Var}_G(T_b) \leq \sum_{i=1}^k \mathbb{E}_\nu[g_b(p)^2] \leq kB \mathbb{E}_\nu[g_b(p)] \leq kB \mathbb{E}_\nu[p] \leq kB a \leq C_\epsilon kBa.$$

Taking the maximum over $b = 0, 1$ gives the asserted raw target-variance bound. The same bounded-support estimates also give the required square integrability of S and of both T_b .

6. It remains to bound the prior-predictive distance in the raw Poisson experiment. For a single rare cell, let $\mathcal{Z}_{\text{rare}}$ be the finite product space consisting of the two labeled treated-outcome counts, the auxiliary treated count, and the labeled and auxiliary aggregate control counts, and let $z_\emptyset$ be its all-zero element. For $b \in \{0, 1\}$ and $p \in \mathbb{R}$, define the raw rare-cell kernel by

$$\mathbf{Q}_{b,p}^{\text{raw}} := \begin{cases} \delta_{z_\emptyset}, & p = 0, \\ \mathbf{S}_{\alpha_{u,v}} \mathbf{M}_{b,p}, & p \neq 0, \end{cases} \quad \alpha_{u,v} := \frac{u}{u+v} \in [0, 1],$$

where $\mathbf{M}_{b,p}$ is the branch- b marked-Poisson law associated with the certificate at latent mass p , and $\mathbf{S}_{\alpha_{u,v}}$ is the fixed splitting kernel that separates the marked treated counts into

labeled and auxiliary pieces and carries along the corresponding control-count components. Let

$$Q_b := \int Q_{b,p}^{\text{raw}} d\nu(p), \quad \tilde{Q}_b := \int S_{\alpha_{u,v}} M_{b,p} d\nu(p).$$

For every measurable event $E \subseteq \mathcal{Z}_{\text{rare}}$ and every $p \in \mathbb{R}$,

$$Q_{1,p}^{\text{raw}}(E) - Q_{0,p}^{\text{raw}}(E) = (S_{\alpha_{u,v}} M_{1,p})(E) - (S_{\alpha_{u,v}} M_{0,p})(E).$$

Indeed, this is the definition when $p \neq 0$. When $p = 0$, the left side is zero because both raw kernels are $\delta_{z_\emptyset}$. The right side is also zero: as shown above, 0 is not a certificate node, so $h(0) = 0$, and the two branch marked-Poisson laws agree at the zero fiber before the fixed splitting is applied. Integrating the displayed identity with respect to ν gives, for every measurable E ,

$$(Q_1 - Q_0)(E) = (\tilde{Q}_1 - \tilde{Q}_0)(E).$$

Thus the signed finite measures $Q_1 - Q_0$ and $\tilde{Q}_1 - \tilde{Q}_0$ are equal, and their total variations are equal. Hence

$$\text{TV}(Q_0, Q_1) = \text{TV}(\tilde{Q}_0, \tilde{Q}_1).$$

Data processing under the splitting kernel and the one-coordinate marked-Poisson estimate give

$$\text{TV}(Q_0, Q_1) \leq \bar{C}_\epsilon u a \rho_\epsilon^L.$$

The certificate used here is precisely the finite-node certificate constructed above: its node set is finite, its signed weights have total variation one, all nodes lie in $[a/\kappa_\epsilon, B]$, and its moments through degree L vanish. Thus the normalized finite-certificate and support hypotheses of the one-coordinate estimate are all discharged by the displayed construction, rather than by an additional regularity convention. Independence of the k rare coordinates yields

$$\text{TV}(Q_0^{\otimes k}, Q_1^{\otimes k}) \leq k \text{TV}(Q_0, Q_1) \leq \bar{C}_\epsilon u k a \rho_\epsilon^L.$$

It remains to identify this product experiment with the raw Poisson experiment generated by the constructed laws. Let $R_b(\xi)$ be the conditional raw Poisson law generated by $P_b(\xi)$ at raw scale $S(\xi)$. Let

$$\mathcal{Y}_{\text{cell}} := \{y : \{0, 1\} \rightarrow (\mathbb{N} \times \mathbb{N}) \times \mathbb{N}\}$$

be the single-cell raw-count space; $y(\alpha) = ((y_\alpha^s, y_\alpha^f), y_\alpha^{\text{aux}})$ records labeled successes, labeled failures, and auxiliary counts in arm α . Let $z_{\text{cell},0} \in \mathcal{Y}_{\text{cell}}$ be the all-zero cell observation, and let

$$\mathbf{r} : \mathcal{Z}_{\text{rare}} \rightarrow \mathcal{Y}_{\text{cell}}$$

send the five rare-cell counts to the two arm-indexed cell observations: in the treated arm it records the treated success count, treated failure count, and auxiliary treated count, while in the control arm it records zero successes, the labeled control count, and the auxiliary control count. The common reservoir law is

$$Q_\star := \bigotimes_{\alpha \in \{0,1\}} \left\{ \text{Poi}(0) \otimes \text{Poi}\left(\frac{up_\star}{2}\right) \otimes \text{Poi}\left(\frac{vp_\star}{2}\right) \right\}.$$

For fixed ξ , define rare-coordinate cell laws by

$$H_{b,\xi,i} := (Q_{b,\hat{\xi}_i}^{\text{raw}})_{\#} \mathbf{r}, \quad i = 1, \dots, k,$$

and define a zero law $H_0 := \delta_{z_{\text{cell},0}}$. For $\mathbf{y} = (y_1, \dots, y_k) \in \mathcal{Y}_{\text{cell}}^k$ and $y_\star \in \mathcal{Y}_{\text{cell}}$, define the deterministic assembly map by

$$[A(\mathbf{y}, y_\star)](x, \alpha) := \begin{cases} y_\star(\alpha), & x = x_\star, \\ y_i(\alpha), & x = x_i \text{ for the unique } i, \\ z_{\text{cell},0}(\alpha), & x \notin \{x_\star, x_1, \dots, x_k\}. \end{cases}$$

The middle line is unambiguous because the rare cells are distinct and none is the reservoir. By Lemma 23, applied with the embedding $i \mapsto x_i$, distinguished coordinate $x_\star$, coordinate laws $H_{b,\xi,i}$, reservoir law $Q_\star$, and filler $z_{\text{cell},0}$,

$$A_\# \left(\left(\bigotimes_{i=1}^k H_{b,\xi,i} \right) \otimes Q_\star \right) = \bigotimes_{x=1}^d J_{b,\xi,x},$$

where

$$J_{b,\xi,x} := \begin{cases} Q_\star, & x = x_\star, \\ H_{b,\xi,i}, & x = x_i \text{ for the unique } i, \\ H_0, & x \notin \{x_\star, x_1, \dots, x_k\}. \end{cases}$$

Equivalently, if A_{rare} denotes the map

$$A_{\text{rare}}(z, z_\star) := A((\mathbf{r}(z_1), \dots, \mathbf{r}(z_k)), z_\star),$$

then

$$(A_{\text{rare}})_\# \left(\left(\bigotimes_{i=1}^k Q_{b,\hat{\xi}_i}^{\text{raw}} \right) \otimes Q_\star \right) = \bigotimes_{x=1}^d J_{b,\xi,x},$$

because pushforward by $\mathbf{r}$ in each rare coordinate turns $\bigotimes_i Q_{b,\hat{\xi}_i}^{\text{raw}}$ into $\bigotimes_i H_{b,\xi,i}$. More explicitly, coordinatewise pushforward gives

$$\left(\bigotimes_{i=1}^k Q_{b,\hat{\xi}_i}^{\text{raw}} \right)_\# [(z_i)_i \mapsto (\mathbf{r}(z_i))_i] = \bigotimes_{i=1}^k H_{b,\xi,i}.$$

Taking the product with $Q_\star$, then composing this coordinate map with A , gives A_{rare} . Functoriality of pushforward therefore turns the left side of the padding identity into the asserted assembled rare-coordinate law.

We now compare these coordinate laws with the conditional raw Poisson law cell by cell. At the reservoir, raw scale multiplication cancels the normalization:

$$S(\xi) p_\xi(x_\star) = p_\star, \quad e_\xi(x_\star) = \frac{1}{2}, \quad \mu_{1,\xi}^{(b)}(x_\star) = \mu_{0,\xi}^{(b)}(x_\star) = 0.$$

Thus each arm has raw treatment-covariate mass $p_\star/2$, zero labeled-success rate, labeled-failure rate $up_\star/2$, and auxiliary rate $vp_\star/2$. Therefore the reservoir coordinate law of $R_b(\xi)$ is

$$J_{b,\xi,x_\star} = Q_\star.$$

At a rare cell x_i , put $p = \hat{\xi}_i$. If $p = 0$, then

$$S(\xi) p_\xi(x_i) = 0,$$

so all labeled and auxiliary Poisson rates at this cell are zero. The conditional raw Poisson coordinate law is $\delta_{z_{\text{cell},0}}$. The raw rare-cell kernel is also $\delta_{z_{\emptyset}}$, and $\mathbf{r}(z_{\emptyset}) = z_{\text{cell},0}$, hence

$$(\mathbf{Q}_{b,0}^{\text{raw}})_{\#}\mathbf{r} = \delta_{z_{\text{cell},0}} = \mathbf{J}_{b,\boldsymbol{\xi},x_i}.$$

If $p > 0$, define

$$s_{\text{tr}}(p) := \epsilon(p + a), \quad s_{\text{ct}}(p) := p - s_{\text{tr}}(p), \quad \eta_1 := 1, \quad \eta_0 := -1.$$

The construction gives

$$S(\boldsymbol{\xi}) p_{\boldsymbol{\xi}}(x_i) = p, \quad e_{\boldsymbol{\xi}}(x_i) = \frac{s_{\text{tr}}(p)}{p}, \quad \mu_{1,\boldsymbol{\xi}}^{(b)}(x_i) = \frac{1 + \eta_b h(p)}{2}, \quad \mu_{0,\boldsymbol{\xi}}^{(b)}(x_i) = 0.$$

Consequently the raw treatment-covariate masses in the treated and control arms are

$$p e_{\boldsymbol{\xi}}(x_i) = s_{\text{tr}}(p), \quad p\{1 - e_{\boldsymbol{\xi}}(x_i)\} = s_{\text{ct}}(p).$$

The conditional raw Poisson coordinate law at x_i is therefore the product law on $\mathcal{Y}_{\text{cell}}$ with treated-arm rates

$$\frac{us_{\text{tr}}(p)(1 + \eta_b h(p))}{2}, \quad \frac{us_{\text{tr}}(p)(1 - \eta_b h(p))}{2}, \quad vs_{\text{tr}}(p)$$

for labeled successes, labeled failures, and auxiliary counts, and with control-arm rates

$$0, \quad us_{\text{ct}}(p), \quad vs_{\text{ct}}(p)$$

for labeled successes, labeled failures, and auxiliary counts. The split marked rare-cell law $\mathbf{Q}_{b,p}^{\text{raw}}$ is the fivefold product law whose coordinates have exactly the nonzero rates just displayed: treated success, treated failure, auxiliary treated, labeled control, and auxiliary control. Applying $\mathbf{r}$ places the first three counts in the treated arm and the last two in the control arm, with zero control successes. Product Poisson laws are determined by their coordinate rates, so

$$(\mathbf{Q}_{b,p}^{\text{raw}})_{\#}\mathbf{r} = \mathbf{J}_{b,\boldsymbol{\xi},x_i} \quad (p = \hat{\xi}_i > 0).$$

Combining the two rare-cell cases gives

$$(\mathbf{Q}_{b,\hat{\xi}_i}^{\text{raw}})_{\#}\mathbf{r} = \mathbf{J}_{b,\boldsymbol{\xi},x_i} \quad (1 \leq i \leq k).$$

At an unused cell the raw mass is zero, so all rates are zero and its coordinate law is

$$\mathbf{J}_{b,\boldsymbol{\xi},x} = \mathbf{H}_0.$$

Thus the coordinate laws of the conditional raw Poisson experiment are precisely the padded product coordinate laws: reservoir law at $x_{\star}$, the pushed-forward rare laws at the rare cells, and the zero law elsewhere. Since conditional raw Poisson counts are independent across cells,

$$\mathbf{R}_b(\boldsymbol{\xi}) = \bigotimes_{x=1}^d \mathbf{J}_{b,\boldsymbol{\xi},x}.$$

The preceding coordinate identification, together with the padding identity

$$A_{\#} \left(\left(\bigotimes_{i=1}^k H_{b,\xi,i} \right) \otimes Q_{\star} \right) = \bigotimes_{x=1}^d J_{b,\xi,x},$$

which follows from [Lemma 23](#), yields the pointwise fiber identity

$$R_b(\xi) = (A_{\text{rare}})_{\#} \left(\left(\bigotimes_{i=1}^k Q_{b,\hat{\xi}_i}^{\text{raw}} \right) \otimes Q_{\star} \right).$$

This equality uses more than the abstract padding identity: the reservoir calculation fixes the distinguished coordinate, the two rare-cell cases identify every pushed-forward rare law, and the unused-cell calculation supplies the Dirac fillers. Independence of the raw Poisson counts then identifies the conditional experiment with the resulting finite product. Hence every coordinate law on the two sides is the same before finite-product uniqueness is invoked. The identity holds for every ξ . Integrating it against G and using $\hat{\xi} = \xi$ almost surely gives

$$\int R_b(\xi) dG(\xi) = (A_{\text{rare}})_{\#} \left(\left[\int \bigotimes_{i=1}^k Q_{b,\xi_i}^{\text{raw}} d\nu^{\otimes k}(\xi) \right] \otimes Q_{\star} \right).$$

For cylinder events $E_1 \times \cdots \times E_k \subseteq \mathcal{Z}_{\text{rare}}^k$, independence under $\nu^{\otimes k}$ gives

$$\int \prod_{i=1}^k Q_{b,\xi_i}^{\text{raw}}(E_i) d\nu^{\otimes k}(\xi) = \prod_{i=1}^k \int Q_{b,p}^{\text{raw}}(E_i) d\nu(p) = \prod_{i=1}^k Q_b(E_i),$$

because each map $p \mapsto Q_{b,p}^{\text{raw}}(E_i)$ is measurable and takes values in $[0, 1]$. Finite Fubini therefore factors the integral of their product under $\nu^{\otimes k}$ into the product of the coordinate integrals. Both sides define probability measures on the finite product space, and measurable rectangles form a generating π -system; the π - λ uniqueness theorem extends the displayed equality from rectangles to the whole product σ -algebra. Thus

$$\int \bigotimes_{i=1}^k Q_{b,\xi_i}^{\text{raw}} d\nu^{\otimes k}(\xi) = Q_b^{\otimes k}.$$

Consequently the branch- b raw prior-predictive law is

$$R_b := \int R_b(\xi) dG(\xi) = (A_{\text{rare}})_{\#} \left(Q_b^{\otimes k} \otimes Q_{\star} \right).$$

Appending the common reservoir law and applying the deterministic assembly map cannot increase total variation, so

$$\text{TV}(R_0, R_1) \leq \text{TV}(Q_0^{\otimes k}, Q_1^{\otimes k}) \leq \bar{C}_{\epsilon} u k a \rho_{\epsilon}^L \leq C_{\epsilon} u k a \rho_{\epsilon}^L.$$

The constructed recipe with priors Π_0, Π_1 therefore satisfies all displayed calibration, support, common-table, mean, separation, variance, and mixture-distance conclusions. □

Lemma 25. [lem:common-marginal-canonical-certificate-family] (Canonical certificate family) *There exist real constants C_{fixed} and c_{risk} with*

$$0 < C_{\text{fixed}}, \quad 0 < c_{\text{risk}},$$

such that, for every overlap level ϵ with $0 < \epsilon < 1/2$, there are a common-marginal calibration in the sense of [Definition 19](#) and real constants

$$C_\epsilon, \rho_\epsilon, c_{\text{small}}, c_{\text{floor}}$$

with the following properties. Write b_0 for the calibration bandwidth constant, b_ϵ for the uniform bandwidth constant, γ_ϵ for the shift constant, c_ϵ for the dual-gap constant, and C_d for the dimension constant supplied by the calibration. These calibration constants are positive, and

$$c_\epsilon \leq \frac{1}{4}.$$

The auxiliary constants satisfy

$$0 < C_\epsilon, \quad \rho_\epsilon \in (0, 1), \quad 0 < c_{\text{small}}, \quad 0 < c_{\text{floor}}, \quad c_{\text{floor}} \leq c_{\text{risk}}.$$

Moreover:

- **(Fixed-sample transfer.)** *For every n, m, d with $n \geq 1$ and $d \geq 2$, and every common-marginal prior handle H at (n, m, d) in the sense of [Definition 19](#), let L, B, a, k denote the degree, bandwidth, shift, and rare-cell count components of H , and let $\theta_0(H)$ and $\theta_1(H)$ denote the two raw prior centers of its recipe. If the recipe of H carries the exact common-marginal Poisson-prior construction at bandwidth constant b_0 , information size $n + m$, labeled intensity*

$$u = C_{\text{fixed}}n,$$

auxiliary intensity

$$v = C_{\text{fixed}}(n + m),$$

and constants $(C_\epsilon, \rho_\epsilon)$, including the corresponding finite-support prior data, exact common treatment-covariate marginal table, moment-matching identities through degree L , raw-center separation data, and Poisson likelihood-ratio bound at rate $C_\epsilon u k a \rho_\epsilon^L$, if the two raw prior-predictive Poisson experiment laws supplied by that recipe have total variation distance at most $1/8$, and if

$$kBa \leq c_{\text{small}} |\theta_1(H) - \theta_0(H)|^2,$$

then

$$c_{\text{risk}} |\theta_1(H) - \theta_0(H)|^2 \leq R_\epsilon(n, m, d),$$

where $R_\epsilon(n, m, d)$ is the minimax risk from [Definition 11](#).

- **(Canonical handles.)** *For every n, m, d with $n \geq 1$ and $d \geq 2$, there exists a common-marginal prior handle H at (n, m, d) . Let L, B, a, k denote its degree, bandwidth, shift, and rare-cell count components, set*

$$\Delta_H = c_\epsilon k a, \quad D(n, m, d) = \min \left\{ 1, \frac{d^2}{(n + m)^2 \log(en)^2} \right\},$$

and let $\theta_0(H)$ and $\theta_1(H)$ denote the two raw prior centers of its recipe. The same constants satisfy

$$\begin{aligned} L &\geq 2, & 2C_{\text{fixed}}b_0 &\leq b_\epsilon, \\ (C_{\text{fixed}}n + C_{\text{fixed}}(n+m))B &\leq b_\epsilon L, \\ \mathcal{L}(P_{XA} \mid \Pi_0(H)) &= \mathcal{L}(P_{XA} \mid \Pi_1(H)), & \frac{c_{\text{floor}}}{n} &\leq R_\epsilon(n, m, d). \end{aligned}$$

On the finite-atom branch, the handle also satisfies

$$ka \leq c_\epsilon,$$

and its recipe carries the exact common-marginal Poisson-prior construction at bandwidth constant b_0 , information size $n+m$, labeled intensity $u = C_{\text{fixed}}n$, auxiliary intensity $v = C_{\text{fixed}}(n+m)$, and constants $(C_\epsilon, \rho_\epsilon)$, including the finite-support prior data, exact common treatment-covariate marginal table, moment-matching identities through degree L , raw-center separation data, and Poisson likelihood-ratio bound at rate $C_\epsilon u k a \rho_\epsilon^L$. The recipe bounds hold at these intensities: both priors are supported on $\mathcal{M}_{d,\epsilon}$ from [Definition 6](#), they induce the same random P_{XA} law, their raw-center separation is at least $c_\epsilon ka$, the raw total mass of the generator has mean one, the raw total mass and raw targets are square-integrable with generator variances at most $C_\epsilon k B a$, and the raw mixture total variation is at most $C_\epsilon u k a \rho_\epsilon^L$. Moreover, the two raw prior-predictive Poisson experiment laws supplied by the recipe have total variation distance at most $1/8$, and

$$\begin{aligned} \Delta_H &\leq |\theta_1(H) - \theta_0(H)|, & k B a &\leq c_{\text{small}} \Delta_H^2, \\ c_{\text{floor}} \Delta_H^2 &\leq R_\epsilon(n, m, d), & C_d D(n, m, d) &\leq \Delta_H^2. \end{aligned}$$

On the parametric branch,

$$C_d D(n, m, d) \leq \frac{c_{\text{floor}}}{n}.$$

Proof of [Lemma 25](#). Set

$$C_{\text{fixed}} = 65536, \quad c_{\text{risk}} = \frac{1}{65536}.$$

These constants are positive. Fix $\epsilon \in (0, 1/2)$. By [Lemma 24](#), choose a common-marginal calibration and constants $C_\epsilon, \rho_\epsilon$ with

$$C_\epsilon > 0, \quad \rho_\epsilon \in (0, 1),$$

together with the uniform construction and closure bounds supplied there. Define

$$c_{\text{small}} = \frac{1}{65536 C_\epsilon}, \quad c_{\text{floor}} = \frac{1}{65536}.$$

Then

$$c_{\text{small}} > 0, \quad c_{\text{floor}} > 0, \quad c_{\text{floor}} \leq c_{\text{risk}},$$

and [Lemma 24](#) supplies positive $b_0, b_\epsilon, \gamma_\epsilon, q_\epsilon, c_\epsilon, C_d$, with $c_\epsilon \leq 1/4$. We retain that statement's notation: q_ϵ is the rare-count coefficient, c_ϵ is the dual-gap constant, and C_d is the dimension constant.

First consider the fixed-sample transfer assertion. Let n, m, d, H satisfy the hypotheses in that assertion, and write

$$\Delta = |\theta_1(H) - \theta_0(H)|.$$

By the hypotheses of the fixed-sample assertion, the recipe carried by H comes with an exact Poisson-prior witness at

$$u = C_{\text{fixed}}n, \quad v = C_{\text{fixed}}(n + m),$$

with constants $C_\epsilon, \rho_\epsilon$. The same hypotheses give total variation at most $1/8$ for its prior-predictive laws, and the assumed variance absorption is

$$kBa \leq c_{\text{small}}\Delta^2 = \frac{1}{65536 C_\epsilon}\Delta^2.$$

Applying the fixed-sample recipe transfer bound [Lemma 20](#) at these intensities gives

$$\frac{1}{65536}\Delta^2 \leq R_\epsilon(n, m, d).$$

Since $c_{\text{risk}} = 1/65536$, this is exactly the fixed-sample transfer conclusion.

It remains to construct the canonical handle for arbitrary n, m, d with $n \geq 1$ and $d \geq 2$. Write

$$N = n + m, \quad \ell_n = \log(en), \quad D(n, m, d) = \min \left\{ 1, \frac{d^2}{N^2 \ell_n^2} \right\}.$$

Let r_ϵ denote the calibration regime constant, D_ϵ the calibration degree constant, and q_ϵ the calibration rare-count constant. The proof splits according to whether the parametric regime holds:

$$D(n, m, d) \leq \frac{r_\epsilon}{n}.$$

In both regimes set

$$L = \lceil D_\epsilon \ell_n \rceil, \quad B = \frac{b_0 L}{N}, \quad a = \frac{\gamma_\epsilon B}{L^2},$$

and

$$k = \min \{d - 1, \lfloor q_\epsilon N L \rfloor\}.$$

The calibration closure in [Lemma 24](#) gives

$$L \geq 2$$

and

$$2C_{\text{fixed}}b_0 = 131072 b_0 \leq b_\epsilon.$$

Moreover, since $n \leq N$,

$$(C_{\text{fixed}}n + C_{\text{fixed}}N)B \leq (131072 N) \frac{b_0 L}{N} = 131072 b_0 L \leq b_\epsilon L.$$

Thus the degree and bandwidth conclusions hold for either branch.

Suppose first that

$$D(n, m, d) \leq \frac{r_\epsilon}{n}.$$

The local testing gap used by the parametric handle is admissible because

$$\frac{c_\epsilon}{\sqrt{n}} \leq c_\epsilon \leq \frac{1}{4},$$

where $\sqrt{n} \geq 1$. Let H be the parametric common-marginal prior handle with the above values of L, B, a, k . Its branch condition makes the finite-atom predicate false in this case. Consequently all conclusions whose hypotheses include the finite-atom predicate are vacuous. The common treatment-covariate marginal identity is the handle identity

$$\mathcal{L}(P_{XA} \mid \Pi_0(H)) = \mathcal{L}(P_{XA} \mid \Pi_1(H)).$$

The risk floor follows from the parametric two-point floor:

$$\frac{c_{\text{floor}}}{n} = \frac{1/65536}{n} \leq \frac{1/100}{n} \leq R_\epsilon(n, m, d).$$

Finally, the calibration closure gives

$$C_d r_\epsilon \leq \frac{1}{65536} = c_{\text{floor}},$$

and hence, using the parametric-regime inequality,

$$C_d D(n, m, d) \leq C_d \frac{r_\epsilon}{n} \leq \frac{c_{\text{floor}}}{n}.$$

This proves all required canonical-handle properties on the parametric branch.

Now suppose that the parametric regime fails:

$$D(n, m, d) > \frac{r_\epsilon}{n}.$$

With L, B, a, k as above, the quantities B and a are positive. From

$$k = \min \{d - 1, \lfloor q_\epsilon N L \rfloor\}$$

and the elementary inequality $\lfloor q_\epsilon N L \rfloor \leq q_\epsilon N L$, valid since $q_\epsilon N L \geq 0$, we have

$$k \leq q_\epsilon N L.$$

Therefore

$$ka \leq q_\epsilon N L \cdot \frac{\gamma_\epsilon b_0}{N L} = q_\epsilon \gamma_\epsilon b_0 \leq c_\epsilon,$$

where the last inequality is the rare-scale calibration inequality. Hence the rare shift is small.

Apply the finite-atom part of [Lemma 24](#) with

$$u = 65536 n, \quad v = 65536 N.$$

The preceding bandwidth bound, positivity of B , the identity $a = \gamma_\epsilon B / L^2$, the rare-shift bound $ka \leq c_\epsilon$, and $k+1 \leq d$ produce a recipe R at these intensities. The bounds supplied for this recipe include the finite-atom construction flag, support in $\mathcal{M}_{d,\epsilon}$, equality of the induced treatment-covariate marginal table, the raw total-mass mean-one identity, the raw-center separation, the raw total-mass and target variance bounds, and the raw mixture total-variation bound

$$\text{TV}(\mathbf{P}_{0,R}^{\text{Pois}}, \mathbf{P}_{1,R}^{\text{Pois}}) \leq C_\epsilon u k a \rho_\epsilon^L.$$

Together with $n \geq 1$, $L \geq 2$, $d \geq 2$, the nonnegativity of u, v , the positivity of $u + v$, the bandwidth and shift identities, and the rare-count fit $k + 1 \leq d$, these same bounds form the exact Poisson-prior witness used below. Let H be the corresponding handle. Its branch condition identifies this recipe with the finite-atom branch. The common marginal identity again gives

$$\mathcal{L}(P_{XA} \mid \Pi_0(H)) = \mathcal{L}(P_{XA} \mid \Pi_1(H)).$$

The recipe bounds also give

$$|\theta_1(H) - \theta_0(H)| \geq c_\epsilon k a = \Delta_H,$$

the raw total mass has mean one, the raw total mass and raw targets are square-integrable with variances at most $C_\epsilon k B a$, and the raw mixture total variation is bounded by

$$C_\epsilon u k a \rho_\epsilon^L.$$

The calibration decay inequality and the bound $k a \leq c_\epsilon$ imply

$$\text{TV}(\mathbf{P}_{0,H}^{\text{Pois}}, \mathbf{P}_{1,H}^{\text{Pois}}) \leq C_\epsilon (65536 n) k a \rho_\epsilon^L \leq C_\epsilon (65536 n) c_\epsilon \rho_\epsilon^L \leq \frac{1}{8}.$$

Thus the prior-predictive total variation condition required by the transfer rule holds.

It remains to record the two absorptions used by the handle. The counting estimate used in the finite-atom branch is

$$\frac{L^2}{c_{\text{small}} c_\epsilon^2 \gamma_\epsilon} \leq k.$$

Indeed, write $N = n + m$ and $\ell_n = \log(en)$. Since the branch condition is

$$D(n, m, d) > \frac{r_\epsilon}{n},$$

the minimum in $D(n, m, d)$ forces

$$\frac{r_\epsilon}{n} < \frac{d^2}{N^2 \ell_n^2}.$$

Put

$$K = c_{\text{small}} c_\epsilon^2 \gamma_\epsilon > 0.$$

The calibration inequality

$$\left(\frac{4(D_\epsilon + 1)^2}{K} \right)^2 \leq r_\epsilon$$

implies

$$\frac{4(D_\epsilon + 1)^2}{K} \leq \sqrt{r_\epsilon}.$$

Meanwhile, the failed-regime inequality above gives

$$\frac{\sqrt{r_\epsilon} N \ell_n}{\sqrt{n}} < d.$$

Using the ceiling bound $L \leq (D_\epsilon + 1) \ell_n$ and

$$\ell_n \sqrt{n} \leq 2n \leq 2N,$$

we obtain the complete cross-multiplication

$$\begin{aligned}\frac{2L^2}{K} &\leq \frac{2(D_\epsilon + 1)^2 \ell_n^2}{K} \\ &= \left(\frac{4(D_\epsilon + 1)^2}{K} \right) \frac{N \ell_n}{\sqrt{n}} \frac{\ell_n \sqrt{n}}{2N} \\ &\leq \frac{\sqrt{r_\epsilon} N \ell_n}{\sqrt{n}} < d.\end{aligned}$$

Equivalently, the proof has compared the nonnegative quantities through the strict square inequality

$$\left(\frac{4(D_\epsilon + 1)^2}{K} N \ell_n \right)^2 < (d\sqrt{n})^2,$$

and then divided by the positive factor $\sqrt{n}$. Weakening the strict bound gives

$$\frac{2L^2}{K} \leq d.$$

Now split according to the actual comparison defining the minimum in k . If

$$d - 1 \leq \lfloor q_\epsilon N L \rfloor,$$

then

$$k = d - 1.$$

Since $d \geq 2$,

$$\frac{d}{2} \leq d - 1,$$

and hence

$$\frac{L^2}{K} \leq \frac{d}{2} \leq d - 1 = k.$$

If instead

$$\lfloor q_\epsilon N L \rfloor < d - 1,$$

then

$$k = \lfloor q_\epsilon N L \rfloor.$$

The calibration gives

$$\frac{4(D_\epsilon + 1)}{K} \leq q_\epsilon,$$

and, since $q_\epsilon N L \geq 1$, the elementary floor bound $\lfloor q_\epsilon N L \rfloor \geq q_\epsilon N L / 2$ gives

$$k = \lfloor q_\epsilon N L \rfloor \geq \frac{q_\epsilon N L}{2} \geq \frac{2(D_\epsilon + 1) N L}{K}.$$

Using $L \leq (D_\epsilon + 1) \ell_n$ and $\ell_n \leq N$, this is at least

$$\frac{L^2}{K}.$$

Thus the displayed counting estimate holds in both subcases. Recall here that c_ϵ denotes the calibration dual-gap constant; the rare-count coefficient is q_ϵ . Using $a = \gamma_\epsilon B / L^2$, the counting estimate yields

$$kBa = \left(\frac{ka^2}{\gamma_\epsilon} \right) L^2 \leq \frac{ka^2}{\gamma_\epsilon} (kc_{\text{small}}c_\epsilon^2\gamma_\epsilon) = c_{\text{small}}(c_\epsilon ka)^2 = c_{\text{small}}\Delta_H^2.$$

The recipe separation bound gives

$$\Delta_H = c_\epsilon ka \leq |\theta_1(H) - \theta_0(H)|.$$

Consequently

$$kBa \leq c_{\text{small}} |\theta_1(H) - \theta_0(H)|^2.$$

The fixed-sample transfer already proved above therefore gives

$$c_{\text{floor}} |\theta_1(H) - \theta_0(H)|^2 \leq R_\epsilon(n, m, d),$$

and hence

$$c_{\text{floor}}\Delta_H^2 \leq R_\epsilon(n, m, d).$$

As before, the parametric floor gives

$$\frac{c_{\text{floor}}}{n} \leq R_\epsilon(n, m, d).$$

Finally we verify the dimension absorption for the above L, k, a . Since

$$a = \frac{\gamma_\epsilon B}{L^2} = \frac{\gamma_\epsilon b_0}{NL},$$

it is enough to compare $C_d D(n, m, d)$ with

$$\left(c_\epsilon k \frac{\gamma_\epsilon b_0}{NL} \right)^2.$$

Split according to which term realizes

$$k = \min \{d - 1, \lfloor q_\epsilon NL \rfloor\}.$$

On the cap-active subcase, $k = d - 1$. Because $d \geq 2$,

$$\frac{d}{2} \leq d - 1 = k.$$

The definition of $D(n, m, d)$ and the dimension calibration give

$$\begin{aligned} C_d D(n, m, d) &\leq C_d \frac{d^2}{N^2 \ell_n^2} \\ &\leq \left(\frac{c_\epsilon \gamma_\epsilon b_0}{2(D_\epsilon + 1)} \right)^2 \frac{d^2}{N^2 \ell_n^2}. \end{aligned}$$

All factors here are nonnegative. The two bounds $d \leq 2k$ and $L \leq (D_\epsilon + 1)\ell_n$ therefore give, after cross-multiplication by the positive denominators,

$$0 \leq \frac{c_\epsilon \gamma_\epsilon b_0 d}{2(D_\epsilon + 1)N\ell_n} \leq c_\epsilon k \frac{\gamma_\epsilon b_0}{NL} = c_\epsilon ka.$$

Squaring this nonnegative comparison proves

$$C_d D(n, m, d) \leq (c_\epsilon ka)^2$$

on the cap-active subcase. On the floor-active subcase, $k = \lfloor q_\epsilon NL \rfloor$. The calibration includes the rare-count identity

$$q_\epsilon = \frac{c_\epsilon}{2\gamma_\epsilon b_0}.$$

The calibration has $q_\epsilon \geq 4$, while $N \geq 1$ and $L \geq 2$, so $q_\epsilon NL \geq 1$. The elementary floor bound therefore gives

$$k \geq \frac{q_\epsilon NL}{2}.$$

Thus, with every factor nonnegative,

$$ka \geq \frac{q_\epsilon NL}{2} \cdot \frac{\gamma_\epsilon b_0}{NL} = \frac{c_\epsilon}{4}.$$

Since $D(n, m, d) \leq 1$, the calibration gives $C_d \leq c_\epsilon^4/16$, and squaring preserves the preceding comparison,

$$C_d D(n, m, d) \leq C_d \leq \frac{c_\epsilon^4}{16} = \left(c_\epsilon \frac{c_\epsilon}{4}\right)^2 \leq (c_\epsilon ka)^2.$$

Thus in both subcases

$$C_d D(n, m, d) \leq (c_\epsilon ka)^2 = \Delta_H^2.$$

This proves every required canonical-handle property on the finite-atom branch, and completes the construction for all n, m, d . $\square$

The constants in [Lemma 24](#) coordinate the degree, bandwidth, rare-cell count, and mixture decay. The shared-law conclusion for the random treatment–covariate tables is the structural feature that makes the prior compatible with auxiliary annotation data, and the final total-variation display supplies the testing bound used by [Lemmas 20](#) and [21](#). [Lemma 25](#) then assembles the transfer constants, the canonical handle, the parametric label floor, and the finite-atom branch into the certificate invoked by [Theorem 3](#).

One-coordinate marked-Poisson input. The load-bearing analytic estimate is a one-coordinate marked-Poisson total-variation bound for normalized signed certificates. For a normalized signed certificate supported on $[a/\kappa_\epsilon, B]$, with moments through degree L equal to zero and $(u + v)B \leq b_\epsilon L$, its two marked-Poisson predictive laws satisfy

$$\text{TV}(\mathbf{Q}_0, \mathbf{Q}_1) \leq C_\epsilon u a \rho_\epsilon^L, \quad 0 < \rho_\epsilon < 1.$$

Expanding each marked-Poisson likelihood in the latent mass gives the intuition behind the bound: moment matching cancels the polynomial terms through degree L , and the bandwidth condition leaves a geometrically controlled exponential-series tail. Tensorization over rare cells contributes the factor k , while the common reservoir and deterministic splitting and assembly maps preserve or contract total variation.

C Known-marginal experiment and verification note

The benchmark in [Definition 15](#) and [Theorems 1](#) and [5](#) uses the same labeled experiment as the annotation problem, but supplies the treatment–covariate marginal as exact side information. This appendix records the corresponding experiment explicitly so that the comparison with the auxiliary-sample experiment in [Definition 9](#) has a fixed statistical object.

Definition 30. `[def:known-marginal-experiment]` (Known-marginal experiment $\mathcal{E}_{n,d,\epsilon}^{\text{known}}$) *For $n \geq 1$ and $d \geq 2$, the known-marginal experiment is*

$$\mathcal{E}_{n,d,\epsilon}^{\text{known}} := \{(P^{\otimes n}, P_{XA}) : P \in \mathcal{M}_{d,\epsilon}\}.$$

The first coordinate generates $\mathcal{L}_n$, and the second coordinate is the exact 2d-cell table observed as deterministic side information.

The deterministic table P_{XA} contains the full joint distribution of treatment and covariates under the same observed-data law P used for the labeled sample. Thus the benchmark isolates the contribution of outcome labels after the treatment–covariate frequencies are known exactly. This is the population analogue of having unlimited annotation data for (X, A) , and it is the object compared with finite- m annotation in [Theorem 5](#).

The risk $R_\epsilon^{\text{known}}(n, d)$ in [Definition 15](#) takes the infimum over $\mathcal{T}_{n,d}^{\text{known}}$, the estimators measurable with respect to the labeled sample and this exact table. The boundary result in [Theorem 1](#) therefore describes the benchmark rate for the known-marginal problem, while [Theorem 5](#) identifies the annotation regimes in which the finite auxiliary sample attains the same limiting benchmark. The construction is aligned with the finite-alphabet causal model in [Definition 6](#) and the average treatment effect functional in [Definition 8](#); the overlap and sampling conditions remain those stated in [Assumptions 1](#) and [2](#).

The next statements record the finite side-information comparison underlying that benchmark. The finite-cover statement is the compactness input that turns exact side information into finitely many stable decision rules; the causal specialization then yields convergence of empirical side samples to the exact-side benchmark; and the measurable comparison formalizes exact side information as at least as informative as an empirical side sample.

Lemma 26. `[lem:finite-side-cover-approximate-decision]` (Finite side cover) *Let Θ be a nonempty compact topological space, let Z be a finite set, let C be a finite nonempty set, and let $l \leq u$. Suppose the following conditions hold.*

- **(Laws.)** *For each $\theta \in \Theta$, Λ_θ is a probability vector on Z and Ξ_θ is a probability vector on C .*
- **(Target.)** *For each $\theta \in \Theta$, $v_\theta \in [l, u]$.*
- **(Continuity.)** *For every $z \in Z$ and $c \in C$, the maps $\theta \mapsto \Lambda_\theta(z)$, $\theta \mapsto \Xi_\theta(c)$, and $\theta \mapsto v_\theta$ are continuous.*

For probability vectors ν, ν' on C , set

$$D_\infty(\nu, \nu') = \max_{c \in C} |\nu(c) - \nu'(c)|.$$

Define

$$\mathfrak{R}_\star = \inf_{\psi} \sup_{\theta \in \Theta} \sum_{z \in Z} \Lambda_\theta(z) (\psi(z, \Xi_\theta) - v_\theta)^2,$$

where ψ ranges over all maps

$$\psi : Z \times \text{Prob}(\mathbf{C}) \rightarrow [l, u].$$

Then for every $\delta > 0$, there exist an integer $k \geq 1$, centers $\theta_1^\circ, \dots, \theta_k^\circ \in \Theta$, decisions $\beta_1, \dots, \beta_k : Z \rightarrow [l, u]$, radii $\text{rad}_1, \dots, \text{rad}_k$, and a margin $\eta > 0$ such that

$$\eta < \text{rad}_j$$

for every j , every $\theta \in \Theta$ admits some $j \in \{1, \dots, k\}$ with

$$D_\infty(\Xi_\theta, \Xi_{\theta_j^\circ}) < \text{rad}_j - \eta,$$

and, for every $j \in \{1, \dots, k\}$ and every $\theta \in \Theta$,

$$D_\infty(\Xi_\theta, \Xi_{\theta_j^\circ}) < \text{rad}_j \implies \sum_{z \in Z} \Lambda_\theta(z) (\beta_j(z) - v_\theta)^2 \leq \mathfrak{R}_\star + \delta.$$

Proof of Lemma 26. 1. For a decision $b : Z \rightarrow [l, u]$, write

$$R_\theta(b) = \sum_{z \in Z} \Lambda_\theta(z) (b(z) - v_\theta)^2.$$

Since $b(z), v_\theta \in [l, u]$, each squared term is at most $(u - l)^2$, and since Λ_θ is a probability vector,

$$0 \leq R_\theta(b) \leq (u - l)^2 \quad \text{for every } \theta \in \Theta.$$

In particular the constant decision $b_0(z) = l$ is admissible.

2. Fix $\theta_0 \in \Theta$. Let

$$\mathcal{F}(\theta_0) = \{\theta \in \Theta : \Xi_\theta = \Xi_{\theta_0}\}$$

and let

$$V_{\text{fib}}(\theta_0) = \inf_{\beta : Z \rightarrow [l, u]} \sup_{\theta \in \mathcal{F}(\theta_0)} R_\theta(\beta).$$

The fiber is nonempty because it contains θ_0 . By the previous bound, the worst-case risk of the constant decision b_0 over this fiber is at most $(u - l)^2$, hence $V_{\text{fib}}(\theta_0) \leq (u - l)^2$. Thus the set of all fiber values is bounded above.

For an exact-side rule $\psi : Z \times \text{Prob}(\mathbf{C}) \rightarrow [l, u]$, write

$$R_\theta^{\text{ex}}(\psi) = \sum_{z \in Z} \Lambda_\theta(z) (\psi(z, \Xi_\theta) - v_\theta)^2, \quad \mathfrak{R}_\star = \inf_{\psi} \sup_{\theta \in \Theta} R_\theta^{\text{ex}}(\psi).$$

For any such ψ , define the fiber decision

$$\beta_{\psi, \theta_0}(z) = \psi(z, \Xi_{\theta_0}).$$

If $\theta \in \mathcal{F}(\theta_0)$, then $\Xi_\theta = \Xi_{\theta_0}$, so

$$R_\theta(\beta_{\psi, \theta_0}) = R_\theta^{\text{ex}}(\psi).$$

Consequently

$$V_{\text{fib}}(\theta_0) \leq \sup_{\theta \in \mathcal{F}(\theta_0)} R_\theta(\beta_{\psi, \theta_0}) \leq \sup_{\theta \in \Theta} R_\theta^{\text{ex}}(\psi).$$

Taking the infimum over ψ gives

$$V_{\text{fib}}(\theta_0) \leq \mathfrak{R}_\star, \quad \sup_{\vartheta \in \Theta} V_{\text{fib}}(\vartheta) \leq \mathfrak{R}_\star.$$

For the reverse inequality, set

$$\mathfrak{s} = \sup_{\vartheta \in \Theta} V_{\text{fib}}(\vartheta).$$

Let $\mathfrak{a}$ be any real number with $\mathfrak{s} < \mathfrak{a}$. Then every fiber value is strictly below $\mathfrak{a}$. Since the fiber risks are nonnegative, the strict infimum property gives, for each θ_0 , a decision $\beta_{\theta_0} : \mathbf{Z} \rightarrow [l, u]$ such that

$$\sup_{\theta \in \mathcal{F}(\theta_0)} R_\theta(\beta_{\theta_0}) < \mathfrak{a}.$$

Define a representative map on the whole simplex by

$$\text{rep}(\nu) = \begin{cases} \text{one } \theta \in \Theta \text{ with } \Xi_\theta = \nu, & \text{if such a } \theta \text{ exists,} \\ \text{an arbitrary point of } \Theta, & \text{otherwise.} \end{cases}$$

This is defined for every $\nu \in \text{Prob}(\mathbf{C})$ because Θ is nonempty. Now assemble the exact-side rule

$$\psi_{\mathfrak{a}}(z, \nu) = \beta_{\text{rep}(\nu)}(z).$$

For each $\theta \in \Theta$, the representative $\text{rep}(\Xi_\theta)$ has side law Ξ_θ , so $\theta \in \mathcal{F}(\text{rep}(\Xi_\theta))$. Hence

$$R_\theta^{\text{ex}}(\psi_{\mathfrak{a}}) = R_\theta(\beta_{\text{rep}(\Xi_\theta)}) < \mathfrak{a}.$$

Therefore

$$\mathfrak{R}_\star \leq \sup_{\theta \in \Theta} R_\theta^{\text{ex}}(\psi_{\mathfrak{a}}) \leq \mathfrak{a}.$$

Since this holds for every $\mathfrak{a} > \mathfrak{s}$,

$$\mathfrak{R}_\star \leq \mathfrak{s}.$$

Combining the two inequalities yields

$$\mathfrak{R}_\star = \sup_{\vartheta \in \Theta} V_{\text{fib}}(\vartheta),$$

and in particular

$$V_{\text{fib}}(\theta_0) \leq \mathfrak{R}_\star.$$

3. For $\alpha > 0$, define the closed side-law neighborhood value

$$V_\alpha(\theta_0) = \inf_{\beta : \mathbf{Z} \rightarrow [l, u]} \sup_{\theta : D_\infty(\Xi_\theta, \Xi_{\theta_0}) \leq \alpha} R_\theta(\beta).$$

The required shrinking-neighborhood fact is the reciprocal-radius convergence

$$V_{1/(N+1)}(\theta_0) \longrightarrow V_{\text{fib}}(\theta_0) \quad \text{as } N \rightarrow \infty.$$

Indeed, for every $\alpha \geq 0$, the exact fiber $\mathcal{F}(\theta_0)$ is contained in the closed α -neighborhood, and therefore

$$V_{\text{fib}}(\theta_0) \leq V_\alpha(\theta_0).$$

This gives the lower side of the limit. For the upper side, fix a real number $\mathfrak{b} > V_{\text{fib}}(\theta_0)$ and put

$$\mathfrak{c} = \frac{V_{\text{fib}}(\theta_0) + \mathfrak{b}}{2}, \quad V_{\text{fib}}(\theta_0) < \mathfrak{c} < \mathfrak{b}.$$

Since the fiber risks are nonnegative, the strict infimum property gives a decision $\beta^\sharp : Z \rightarrow [l, u]$ with

$$\sup_{\theta \in \mathcal{F}(\theta_0)} R_\theta(\beta^\sharp) < \mathfrak{c},$$

so

$$\theta \in \mathcal{F}(\theta_0) \implies R_\theta(\beta^\sharp) < \mathfrak{c}.$$

For this fixed decision the map $\theta \mapsto R_\theta(\beta^\sharp)$ is continuous, because Z is finite and the coordinates $\Lambda_\theta(z)$ and v_θ are continuous. Hence the set

$$\mathcal{B} = \{\theta \in \Theta : \mathfrak{c} \leq R_\theta(\beta^\sharp)\}$$

is closed, and compactness of Θ makes $\mathcal{B}$ compact. The side-law map $\theta \mapsto \Xi_\theta$ is continuous, so its image $\Xi(\mathcal{B}) = \{\Xi_\theta : \theta \in \mathcal{B}\}$ is compact and therefore closed in the finite simplex. The preceding fiber inequality implies

$$\Xi_{\theta_0} \notin \Xi(\mathcal{B}),$$

for otherwise some $\theta \in \mathcal{B}$ would satisfy $\Xi_\theta = \Xi_{\theta_0}$ and hence both $\mathfrak{c} \leq R_\theta(\beta^\sharp)$ and $R_\theta(\beta^\sharp) < \mathfrak{c}$. Since a closed set misses Ξ_{θ_0} , there is $\alpha_0 > 0$ such that

$$D_\infty(\Xi_\theta, \Xi_{\theta_0}) < \alpha_0 \implies R_\theta(\beta^\sharp) < \mathfrak{c}.$$

For all sufficiently large N , $1/(N+1) < \alpha_0$. For such N , every θ with

$$D_\infty(\Xi_\theta, \Xi_{\theta_0}) \leq \frac{1}{N+1}$$

satisfies $R_\theta(\beta^\sharp) < \mathfrak{c}$, and therefore

$$V_{1/(N+1)}(\theta_0) \leq \sup_{\theta: D_\infty(\Xi_\theta, \Xi_{\theta_0}) \leq 1/(N+1)} R_\theta(\beta^\sharp) < \mathfrak{c} < \mathfrak{b}.$$

Together with $V_{\text{fib}}(\theta_0) \leq V_{1/(N+1)}(\theta_0)$, this proves the displayed convergence.

Apply the convergence with $\mathfrak{b} = V_{\text{fib}}(\theta_0) + \delta/2$. Choose an integer $N_{\theta_0} \geq 0$ with

$$V_{1/(N_{\theta_0}+1)}(\theta_0) < V_{\text{fib}}(\theta_0) + \delta/2.$$

Set

$$r(\theta_0) = \frac{1}{N_{\theta_0} + 1} > 0.$$

Together with $V_{\text{fib}}(\theta_0) \leq \mathfrak{R}_*$, this gives

$$V_{r(\theta_0)}(\theta_0) < \mathfrak{R}_* + \delta.$$

Since all risks are nonnegative, the range of worst-case risks over admissible decisions is bounded below by 0. Hence the strict inequality for the infimum yields a decision

$$\beta_{\theta_0} : Z \rightarrow [l, u]$$

such that

$$\sup_{\theta: D_\infty(\Xi_\theta, \Xi_{\theta_0}) \leq r(\theta_0)} R_\theta(\beta_{\theta_0}) < \mathfrak{R}_\star + \delta.$$

Consequently, whenever

$$D_\infty(\Xi_\theta, \Xi_{\theta_0}) < r(\theta_0),$$

the same θ belongs to the closed neighborhood, and therefore

$$R_\theta(\beta_{\theta_0}) \leq \mathfrak{R}_\star + \delta.$$

4. The map $\theta \mapsto \Xi_\theta$ is continuous because each coordinate $\theta \mapsto \Xi_\theta(c)$ is continuous and $\mathbf{C}$ is finite. For each θ_0 , define

$$U_{\theta_0} = \left\{ \theta \in \Theta : D_\infty(\Xi_\theta, \Xi_{\theta_0}) < \frac{r(\theta_0)}{2} \right\}.$$

Each U_{θ_0} is open, and $\theta_0 \in U_{\theta_0}$. Thus $\{U_{\theta_0} : \theta_0 \in \Theta\}$ is an open cover of Θ . Compactness gives finitely many centers

$$\theta_1^\circ, \dots, \theta_k^\circ$$

such that these sets still cover Θ . Since Θ is nonempty, this finite subcover has $k \geq 1$.

5. For each selected center set

$$\text{rad}_j = r(\theta_j^\circ), \quad \beta_j = \beta_{\theta_j^\circ}.$$

Define

$$\eta = \min_{1 \leq j \leq k} \frac{\text{rad}_j}{2}.$$

Every rad_j is positive, so $\eta > 0$, and

$$\eta \leq \frac{\text{rad}_j}{2} < \text{rad}_j \quad \text{for every } j.$$

If $\theta \in \Theta$, the finite subcover gives an index j with

$$D_\infty(\Xi_\theta, \Xi_{\theta_j^\circ}) < \frac{\text{rad}_j}{2}.$$

Since $\eta \leq \text{rad}_j / 2$, this implies

$$D_\infty(\Xi_\theta, \Xi_{\theta_j^\circ}) < \text{rad}_j - \eta.$$

Finally, if for some j and θ ,

$$D_\infty(\Xi_\theta, \Xi_{\theta_j^\circ}) < \text{rad}_j,$$

then the local guarantee from the construction at θ_j° gives

$$\sum_{z \in \mathbf{Z}} \Lambda_\theta(z) (\beta_j(z) - v_\theta)^2 = R_\theta(\beta_j) \leq \mathfrak{R}_\star + \delta.$$

These are exactly the required centers, decisions, radii, and margin. □

[Lemma 26](#) supplies the finite selection step used in the side-information limit. Compactness and continuity allow the exact-side decision problem to be represented, up to any tolerance δ , by finitely many neighborhoods of side laws. The positive margin η gives room for an empirical side table to select one of these neighborhoods uniformly once the auxiliary sample is large.

Lemma 27. [\[lem:finite-side-information-convergence\]](#) (Finite side-information convergence) *Let the overlap parameter ϵ satisfy $0 < \epsilon < 1/2$, let $n \geq 0$, and let $d \geq 2$. Let*

$$\Theta_{\epsilon,d} = \left\{ (\pi, \varrho, \omega) : \pi \in \text{Prob}([d]), \varrho_x \in [\epsilon, 1 - \epsilon], \omega_{\alpha x} \in [0, 1] \text{ for } x \in [d], \alpha \in \{0, 1\} \right\}.$$

For $\theta = (\pi, \varrho, \omega) \in \Theta_{\epsilon,d}$, set

$$P_\theta(x, \alpha, y) = \pi_x \varrho_x^\alpha (1 - \varrho_x)^{1-\alpha} \omega_{\alpha x}^y (1 - \omega_{\alpha x})^{1-y},$$

let $\lambda_\theta = P_\theta^{\otimes n}$ be the labeled-sample law on $([d] \times \{0, 1\}^2)^n$, define

$$\xi_\theta(x, \alpha) = \sum_{y \in \{0,1\}} P_\theta(x, \alpha, y), \quad v_\theta = \sum_{x \in [d]} \pi_x (\omega_{1x} - \omega_{0x}).$$

For each $m \geq 0$, define

$$\mathfrak{R}_m = \inf_{\varphi: ([d] \times \{0,1\}^2)^n \times ([d] \times \{0,1\})^m \rightarrow [-1,1]} \sup_{\theta \in \Theta_{\epsilon,d}} \int (\varphi(Z, U) - v_\theta)^2 d(\lambda_\theta \otimes \xi_\theta^{\otimes m})(Z, U),$$

and define

$$\mathfrak{R}_\star = \inf_{\psi: ([d] \times \{0,1\}^2)^n \times \text{Prob}([d] \times \{0,1\}) \rightarrow [-1,1]} \sup_{\theta \in \Theta_{\epsilon,d}} \int (\psi(Z, \xi_\theta) - v_\theta)^2 d\lambda_\theta(Z).$$

Then

$$\lim_{m \rightarrow \infty} \mathfrak{R}_m = \mathfrak{R}_\star.$$

Proof of [Lemma 27](#). Let

$$\mathbf{Z}_n = ([d] \times \{0, 1\}^2)^n, \quad \mathbf{C}_d = [d] \times \{0, 1\}.$$

For $\theta \in \Theta_{\epsilon,d}$, write

$$\Lambda_\theta(z) = \lambda_\theta(\{z\}), \quad \Xi_\theta(c) = \xi_\theta(c), \quad c \in \mathbf{C}_d.$$

The assumptions $d \geq 2$ and $0 < \epsilon < 1/2$ make $[d]$ nonempty and make $[\epsilon, 1 - \epsilon]$ nonempty. The coordinate map

$$(\pi, \varrho, \omega) \mapsto \left(\pi, (\varrho_x)_{x \in [d]}, (\omega_{\alpha x})_{(\alpha,x) \in \{0,1\} \times [d]} \right)$$

identifies $\Theta_{\epsilon,d}$, with its coordinate topology, with the finite product

$$\text{Prob}([d]) \times [\epsilon, 1 - \epsilon]^{[d]} \times [0, 1]^{\{0,1\} \times [d]}.$$

Here $\text{Prob}([d])$ is the closed simplex

$$\left\{ \pi \in [0, 1]^{[d]} : \sum_{x \in [d]} \pi_x = 1 \right\},$$

which is compact as a closed subset of the finite cube $[0, 1]^{[d]}$. The two interval powers are finite products of compact intervals. Hence $\Theta_{\epsilon, d}$ is nonempty and compact.

For each θ , Λ_θ is a probability law on Z_n , because it is the n -fold product of the one-record law P_θ . Also Ξ_θ is a probability law on C_d . Indeed, every summand is nonnegative, and the normalization expands as

$$\begin{aligned} \sum_{x \in [d]} \sum_{\alpha \in \{0, 1\}} \Xi_\theta(x, \alpha) &= \sum_{x \in [d]} \sum_{\alpha \in \{0, 1\}} \sum_{y \in \{0, 1\}} \pi_x \varrho_x^\alpha (1 - \varrho_x)^{1-\alpha} \omega_{\alpha x}^y (1 - \omega_{\alpha x})^{1-y} \\ &= \sum_{x \in [d]} \pi_x \left(\sum_{\alpha \in \{0, 1\}} \varrho_x^\alpha (1 - \varrho_x)^{1-\alpha} \left[\sum_{y \in \{0, 1\}} \omega_{\alpha x}^y (1 - \omega_{\alpha x})^{1-y} \right] \right) \\ &= \sum_{x \in [d]} \pi_x ((1 - \varrho_x) + \varrho_x) = 1. \end{aligned}$$

The maps $\theta \mapsto \Lambda_\theta(z)$, $\theta \mapsto \Xi_\theta(c)$, and $\theta \mapsto v_\theta$ are continuous, since each is obtained from finitely many coordinate projections by finite sums and products. Finally,

$$-1 \leq \omega_{1x} - \omega_{0x} \leq 1 \quad \text{for every } x,$$

and the weights π_x are nonnegative with total mass one, so

$$-1 = \sum_x \pi_x(-1) \leq \sum_x \pi_x(\omega_{1x} - \omega_{0x}) = v_\theta \leq \sum_x \pi_x(1) = 1.$$

With this notation, the empirical-side minimax value at auxiliary size m is

$$\mathfrak{R}_m = \inf_{\varphi: Z_n \times C_d^m \rightarrow [-1, 1]} \sup_{\theta \in \Theta_{\epsilon, d}} \sum_{z \in Z_n} \Lambda_\theta(z) \sum_{u \in C_d^m} \Xi_\theta^{\otimes m}(u) (\varphi(z, u) - v_\theta)^2,$$

and the exact-side minimax value is

$$\mathfrak{R}_\star = \inf_{\psi: Z_n \times \text{Prob}(C_d) \rightarrow [-1, 1]} \sup_{\theta \in \Theta_{\epsilon, d}} \sum_{z \in Z_n} \Lambda_\theta(z) (\psi(z, \Xi_\theta) - v_\theta)^2.$$

We now use the finite side-information convergence argument for these data. First, for any empirical estimator $\varphi: Z_n \times C_d^m \rightarrow [-1, 1]$, define the corresponding exact-side estimator

$$\bar{\varphi}(z, \nu) = \sum_{u \in C_d^m} \nu^{\otimes m}(u) \varphi(z, u), \quad z \in Z_n, \quad \nu \in \text{Prob}(C_d).$$

Convexity of $v \mapsto (v - v_\theta)^2$ gives, for every θ ,

$$\int (\bar{\varphi}(Z, \Xi_\theta) - v_\theta)^2 d\lambda_\theta(Z) \leq \int (\varphi(Z, U) - v_\theta)^2 d(\lambda_\theta \otimes \Xi_\theta^{\otimes m})(Z, U).$$

Taking suprema over θ and then infima over φ yields

$$\mathfrak{R}_\star \leq \mathfrak{R}_m \quad \text{for every } m \geq 0.$$

It remains to prove the reverse asymptotic inequality. Put

$$D_\infty(\nu, \nu') = \max_{c \in C_d} |\nu(c) - \nu'(c)|$$

for probability laws on $\mathbb{C}_d$. Fix $\delta > 0$. With $\Theta = \Theta_{\epsilon,d}$, $\mathbf{Z} = \mathbf{Z}_n$, $\mathbf{C} = \mathbf{C}_d$, $\Lambda_\theta = \lambda_\theta$, $\Xi_\theta = \xi_\theta$, $l = -1$, and $u = 1$, the compactness, simplex, continuity, and target-range checks above give the analytic hypotheses of Lemma 26. Its remaining structural hypotheses are immediate but worth recording: $\mathbf{Z}_n$ is a finite power of a finite set, $\mathbf{C}_d$ is finite and nonempty because $d \geq 2$, and $-1 \leq 1$. Thus every hypothesis of Lemma 26 holds. Hence, for the present δ , there are an integer $k \geq 1$, centers $\theta_1^\circ, \dots, \theta_k^\circ \in \Theta_{\epsilon,d}$, bounded decisions $\beta_1, \dots, \beta_k : \mathbf{Z}_n \rightarrow [-1, 1]$, radii $\text{rad}_1, \dots, \text{rad}_k$, and a margin $\eta > 0$ such that

$$\eta < \text{rad}_j \quad (1 \leq j \leq k),$$

$$\forall \theta \in \Theta_{\epsilon,d} \quad \exists j \in \{1, \dots, k\}, \quad D_\infty(\Xi_\theta, \Xi_{\theta_j^\circ}) < \text{rad}_j - \eta,$$

and, for every $j \in \{1, \dots, k\}$ and every $\theta \in \Theta_{\epsilon,d}$,

$$D_\infty(\Xi_\theta, \Xi_{\theta_j^\circ}) < \text{rad}_j \implies \int (\beta_j(Z) - v_\theta)^2 d\lambda_\theta(Z) \leq \mathfrak{R}_\star + \delta.$$

For $m \geq 1$ and $u \in \mathbb{C}_d^m$, let $\widehat{\Xi}_u \in \text{Prob}(\mathbb{C}_d)$ be the empirical law of u , and put $\mathcal{J} := \{1, \dots, k\}$. Since $k \geq 1$, this set is nonempty. Choose a selector $\sigma : \text{Prob}(\mathbb{C}_d) \rightarrow \mathcal{J}$ as follows: if at least one $j \in \mathcal{J}$ satisfies

$$D_\infty(\nu, \Xi_{\theta_j^\circ}) < \text{rad}_j - \eta/2,$$

let $\sigma(\nu)$ be the least such index; if there is no such index, set $\sigma(\nu) = 1$. Define

$$\Phi_{\delta,m}(z, u) = \beta_{\sigma(\widehat{\Xi}_u)}(z).$$

On the event

$$G_{\theta,m} = \left\{ u \in \mathbb{C}_d^m : D_\infty(\widehat{\Xi}_u, \Xi_\theta) < \eta/2 \right\},$$

choose a cover index $j \in \mathcal{J}$ with

$$D_\infty(\Xi_\theta, \Xi_{\theta_j^\circ}) < \text{rad}_j - \eta.$$

The good-event bound and the triangle inequality give

$$D_\infty(\widehat{\Xi}_u, \Xi_{\theta_j^\circ}) < \eta/2 + \text{rad}_j - \eta = \text{rad}_j - \eta/2,$$

so j is eligible for the selector. Hence

$$D_\infty(\widehat{\Xi}_u, \Xi_{\theta_{\sigma(\widehat{\Xi}_u)}^\circ}) < \text{rad}_{\sigma(\widehat{\Xi}_u)} - \eta/2.$$

Applying the triangle inequality once more, now through $\widehat{\Xi}_u$, yields

$$D_\infty(\Xi_\theta, \Xi_{\theta_{\sigma(\widehat{\Xi}_u)}^\circ}) < \eta/2 + \text{rad}_{\sigma(\widehat{\Xi}_u)} - \eta/2 = \text{rad}_{\sigma(\widehat{\Xi}_u)}.$$

Therefore

$$\int (\Phi_{\delta,m}(Z, U) - v_\theta)^2 d(\lambda_\theta \otimes \Xi_\theta^{\otimes m})(Z, U) \leq \mathfrak{R}_\star + \delta + 4 \Xi_\theta^{\otimes m}(G_{\theta,m}^c),$$

because all estimates and all targets lie in $[-1, 1]$. For the finite-alphabet concentration step, fix θ . For $c \in \mathbb{C}_d$ and $u = (u_1, \dots, u_m) \in \mathbb{C}_d^m$, write

$$N_c(u) = \sum_{i=1}^m \mathbf{1}\{u_i = c\}, \quad \widehat{\Xi}_u(c) = \frac{N_c(u)}{m}.$$

Under $\Xi_\theta^{\otimes m}$, the variables $\mathbf{1}\{U_i = c\}$ are independent Bernoulli variables with mean $\Xi_\theta(c)$. Thus

$$\mathbb{E}_\theta \widehat{\Xi}_U(c) = \Xi_\theta(c)$$

and, since a Bernoulli variance is at most 1/4,

$$\text{Var}_\theta \left(\widehat{\Xi}_U(c) \right) = \frac{1}{m^2} \sum_{i=1}^m \text{Var}_\theta(\mathbf{1}\{U_i = c\}) \leq \frac{1}{4m}.$$

Chebyshev's inequality gives, for every $t > 0$,

$$\Xi_\theta^{\otimes m} \left(\left| \widehat{\Xi}_U(c) - \Xi_\theta(c) \right| \geq t \right) \leq \frac{1}{4mt^2}.$$

If $D_\infty(\widehat{\Xi}_U, \Xi_\theta) \geq \eta/2$, then the defining identity

$$D_\infty(\widehat{\Xi}_U, \Xi_\theta) = \max_{c \in \mathcal{C}_d} \left| \widehat{\Xi}_U(c) - \Xi_\theta(c) \right|$$

implies that some $c \in \mathcal{C}_d$ satisfies

$$\left| \widehat{\Xi}_U(c) - \Xi_\theta(c) \right| \geq \eta/2.$$

Hence the bad event satisfies

$$G_{\theta,m}^c = \left\{ D_\infty(\widehat{\Xi}_U, \Xi_\theta) \geq \eta/2 \right\} \subseteq \bigcup_{c \in \mathcal{C}_d} \left\{ \left| \widehat{\Xi}_U(c) - \Xi_\theta(c) \right| \geq \eta/2 \right\}.$$

The union bound over the finite alphabet gives

$$\Xi_\theta^{\otimes m}(G_{\theta,m}^c) \leq \sum_{c \in \mathcal{C}_d} \Xi_\theta^{\otimes m} \left(\left| \widehat{\Xi}_U(c) - \Xi_\theta(c) \right| \geq \eta/2 \right) \leq \frac{|\mathcal{C}_d|}{m\eta^2}.$$

The right-hand side is independent of θ and tends to zero, hence

$$\sup_{\theta \in \Theta_{\epsilon,d}} \Xi_\theta^{\otimes m}(G_{\theta,m}^c) \leq \frac{|\mathcal{C}_d|}{m\eta^2} \longrightarrow 0.$$

Taking the supremum over θ and then using the displayed estimator $\Phi_{\delta,m}$ in the minimax infimum yields

$$\limsup_{m \rightarrow \infty} \mathfrak{R}_m \leq \mathfrak{R}_\star + \delta.$$

Since $\delta > 0$ was arbitrary,

$$\limsup_{m \rightarrow \infty} \mathfrak{R}_m \leq \mathfrak{R}_\star.$$

Combining $\mathfrak{R}_\star \leq \mathfrak{R}_m$ for every m with the last display gives

$$\liminf_{m \rightarrow \infty} \mathfrak{R}_m \geq \mathfrak{R}_\star \quad \text{and} \quad \limsup_{m \rightarrow \infty} \mathfrak{R}_m \leq \mathfrak{R}_\star.$$

Thus $\lim_{m \rightarrow \infty} \mathfrak{R}_m = \mathfrak{R}_\star$. □

Lemma 27 states the population-side-information approximation directly: as the number of auxiliary treatment–covariate records grows, the finite-sample minimax risk converges to the risk obtained when the treatment–covariate marginal is observed exactly. This supplies the finite-dimensional continuity step behind the known-marginal limit in [Theorem 5](#) and keeps the benchmark comparison on the same labeled-sample scale as the annotation experiment.

Finite-cover selection input. The finite-cover mechanism in [Lemma 26](#) applies to the discrete causal parametrization because the parameter set is compact and its labeled probabilities, side-law probabilities, and target are continuous functions of the cell probabilities. Concentration of the empirical side table then selects a valid neighborhood with probability tending uniformly to one, while [Lemma 28](#) gives the complementary comparison from an empirical side sample to its exact probability vector.

Lemma 28. [lem:finite-side-measurable-comparison] (Empirical side comparison) *Let Θ be a nonempty parameter set, let $\mathcal{Z}$ and $\mathcal{C}$ be nonempty finite sets, and let $\beta_- \leq \beta_+$ be real numbers. Suppose that, for each $\theta \in \Theta$, the following objects are given:*

- *(Observed side law.)* A probability vector λ_θ on $\mathcal{Z}$.
- *(Empirical side law.)* A probability vector ξ_θ on $\mathcal{C}$.
- *(Target.)* A number $v_\theta \in [\beta_-, \beta_+]$.

For each integer $m \geq 0$, define

$$\mathfrak{R}_m = \inf_{\varphi: \mathcal{Z} \times \mathcal{C}^m \rightarrow [\beta_-, \beta_+]} \sup_{\theta \in \Theta} \sum_{z \in \mathcal{Z}} \lambda_\theta(z) \sum_{v \in \mathcal{C}^m} \left(\prod_{j=1}^m \xi_\theta(v_j) \right) (\varphi(z, v) - v_\theta)^2.$$

Define also

$$\tilde{\mathfrak{R}}_\star = \inf_{\psi} \sup_{\theta \in \Theta} \sum_{z \in \mathcal{Z}} \lambda_\theta(z) (\psi(z, \xi_\theta) - v_\theta)^2,$$

where the infimum ranges over all maps

$$\psi: \mathcal{Z} \times \mathbb{R}^{\mathcal{C}} \rightarrow [\beta_-, \beta_+]$$

such that, for each $z \in \mathcal{Z}$, the section $w \mapsto \psi(z, w)$ is Borel measurable. Then, for every integer $m \geq 0$,

$$\tilde{\mathfrak{R}}_\star \leq \mathfrak{R}_m.$$

[Lemma 28](#) gives the corresponding monotone comparison for finite spaces: replacing an empirical side sample by its exact probability vector weakly lowers the minimax risk. In the known-marginal application, this comparison expresses the benchmark as the risk associated with observing the treatment–covariate law itself, rather than a random finite sample from that law.

A complementary asymptotic comparison spells out when auxiliary annotation improves the rate in [Definition 12](#) relative to the labeled-only case. It expresses the improvement entirely through the alphabet growth and the effective treatment–covariate sample size.

Lemma 29. [lem:annotation-frontier-strict-improvement] (Strict frontier improvement) *Let $(m_n)_{n \geq 1}$ and $(d_n)_{n \geq 1}$ be sequences of nonnegative integers such that $d_n \geq 2$ for every $n \geq 1$. For each $n \geq 1$, set $N_n = n + m_n$ and $\ell_n = \log(en)$, and let $r_\epsilon(n, m, d)$ be the annotation-frontier rate in [Definition 12](#). Then, as $n \rightarrow \infty$,*

$$r_\epsilon(n, m_n, d_n) = o(r_\epsilon(n, 0, d_n))$$

if and only if

$$\frac{d_n}{\sqrt{n} \ell_n} \rightarrow \infty$$

and

$$\frac{d_n/(N_n \ell_n)}{\min\{1, d_n/(n \ell_n)\}} \rightarrow 0.$$

Proof of Lemma 29. 1. All limits are taken as $n \rightarrow \infty$, and changing finitely many initial values is immaterial. Put

$$\iota_n := \frac{1}{n}, \quad \alpha_n := \frac{d_n}{n \ell_n}, \quad \beta_n := \frac{d_n}{N_n \ell_n}.$$

For every sufficiently large n , $\ell_n = \log(en) > 0$, $0 < \iota_n \leq 1$, $\alpha_n > 0$, $\beta_n \geq 0$, and $\beta_n \leq \alpha_n$. The last inequality uses $N_n = n + m_n \geq n$, while positivity of α_n uses $d_n \geq 2$. By Definition 12,

$$r_\epsilon(n, m_n, d_n) = \min\{1, \iota_n + \beta_n^2\}, \quad r_\epsilon(n, 0, d_n) = \min\{1, \iota_n + \alpha_n^2\}.$$

2. We use the following numerical reduction. Suppose sequences $\iota_n, \alpha_n, \beta_n$ satisfy, eventually,

$$0 < \iota_n \leq 1, \quad 0 < \alpha_n, \quad 0 \leq \beta_n \leq \alpha_n.$$

Define

$$\mathcal{R}_n := \min\{1, \iota_n + \beta_n^2\}, \quad \mathcal{R}_n^0 := \min\{1, \iota_n + \alpha_n^2\}, \quad \mathcal{H}_n := \min\{1, \alpha_n\}.$$

Then

$$\mathcal{R}_n = o(\mathcal{R}_n^0) \iff \frac{\iota_n}{\mathcal{R}_n^0} \rightarrow 0 \text{ and } \frac{\beta_n}{\mathcal{H}_n} \rightarrow 0.$$

Indeed, if $\mathcal{R}_n = o(\mathcal{R}_n^0)$, then $\mathcal{R}_n/\mathcal{R}_n^0 \rightarrow 0$. Since $\iota_n \leq \mathcal{R}_n$ and $\mathcal{R}_n^0 > 0$ eventually,

$$0 \leq \frac{\iota_n}{\mathcal{R}_n^0} \leq \frac{\mathcal{R}_n}{\mathcal{R}_n^0},$$

so $\iota_n/\mathcal{R}_n^0 \rightarrow 0$. Also $\mathcal{R}_n/\mathcal{R}_n^0 < 1$ eventually; hence $\mathcal{R}_n < 1$ eventually, and therefore

$$\mathcal{R}_n = \iota_n + \beta_n^2$$

eventually. Thus

$$0 \leq \frac{\beta_n^2}{\mathcal{R}_n^0} \leq \frac{\mathcal{R}_n}{\mathcal{R}_n^0},$$

so $\beta_n^2/\mathcal{R}_n^0 \rightarrow 0$. To replace $\mathcal{R}_n^0$ by $\mathcal{H}_n^2$, observe that

$$\mathcal{H}_n^2 = \min\{1, \alpha_n^2\},$$

and hence

$$\mathcal{H}_n^2 \leq \mathcal{R}_n^0 \leq \iota_n + \mathcal{H}_n^2.$$

Writing $\mathcal{Q}_n := \iota_n/\mathcal{R}_n^0$, we have $\mathcal{Q}_n \rightarrow 0$, and whenever $\mathcal{Q}_n < 1$,

$$0 \leq \frac{\iota_n}{\mathcal{H}_n^2} \leq \frac{\mathcal{Q}_n}{1 - \mathcal{Q}_n}.$$

Consequently $\iota_n/\mathcal{H}_n^2 \rightarrow 0$, and the displayed two-sided bound gives

$$\frac{\mathcal{R}_n^0}{\mathcal{H}_n^2} \rightarrow 1.$$

Therefore

$$\frac{\beta_n^2}{\mathcal{H}_n^2} = \frac{\beta_n^2}{\mathcal{R}_n^0} \frac{\mathcal{R}_n^0}{\mathcal{H}_n^2} \rightarrow 0,$$

and taking square roots, using $\beta_n \geq 0$ and $\mathcal{H}_n > 0$, yields $\beta_n/\mathcal{H}_n \rightarrow 0$.

Conversely, assume

$$\frac{\iota_n}{\mathcal{R}_n^0} \rightarrow 0, \quad \frac{\beta_n}{\mathcal{H}_n} \rightarrow 0.$$

Then $\beta_n^2/\mathcal{H}_n^2 \rightarrow 0$, and since $\mathcal{H}_n^2 \leq \mathcal{R}_n^0$,

$$0 \leq \frac{\beta_n^2}{\mathcal{R}_n^0} \leq \frac{\beta_n^2}{\mathcal{H}_n^2} \rightarrow 0.$$

Finally,

$$0 \leq \frac{\mathcal{R}_n}{\mathcal{R}_n^0} \leq \frac{\iota_n + \beta_n^2}{\mathcal{R}_n^0} = \frac{\iota_n}{\mathcal{R}_n^0} + \frac{\beta_n^2}{\mathcal{R}_n^0} \rightarrow 0,$$

which is exactly $\mathcal{R}_n = o(\mathcal{R}_n^0)$.

3. We also need the supervised floor equivalence

$$\frac{(1/n)}{r_\epsilon(n, 0, d_n)} \rightarrow 0 \quad \Longleftrightarrow \quad \frac{d_n}{\sqrt{n} \ell_n} \rightarrow \infty.$$

Set

$$\sigma_n := \frac{d_n}{\sqrt{n} \ell_n}.$$

For $n \geq 1$,

$$r_\epsilon(n, 0, d_n) = \min\{1, \iota_n + \alpha_n^2\},$$

and direct algebra gives

$$\frac{\iota_n}{\iota_n + \alpha_n^2} = \frac{1}{1 + \sigma_n^2}.$$

Assume first that $\iota_n/r_\epsilon(n, 0, d_n) \rightarrow 0$. Given a real threshold Y , put

$$Y_+ := \max\{Y, 0\} + 1.$$

Eventually

$$\frac{\iota_n}{r_\epsilon(n, 0, d_n)} < \frac{1}{1 + Y_+^2}.$$

Since $r_\epsilon(n, 0, d_n) \leq \iota_n + \alpha_n^2$,

$$\frac{1}{1 + \sigma_n^2} = \frac{\iota_n}{\iota_n + \alpha_n^2} \leq \frac{\iota_n}{r_\epsilon(n, 0, d_n)} < \frac{1}{1 + Y_+^2}.$$

Because $\sigma_n \geq 0$, this forces $\sigma_n \geq Y_+$, and hence $\sigma_n \geq Y$ eventually. Thus $\sigma_n \rightarrow \infty$.

Conversely, if $\sigma_n \rightarrow \infty$, then $\iota_n \rightarrow 0$ and $1/\sigma_n^2 \rightarrow 0$. With

$$\mathcal{H}_n := \min\{1, \alpha_n\},$$

the same identity $\mathcal{H}_n^2 = \min\{1, \alpha_n^2\}$ gives

$$\mathcal{H}_n^2 \leq r_\epsilon(n, 0, d_n).$$

Therefore

$$\frac{\iota_n}{r_\epsilon(n, 0, d_n)} \leq \frac{\iota_n}{\mathcal{H}_n^2}.$$

If $\alpha_n \leq 1$, then

$$\frac{\iota_n}{\mathcal{H}_n^2} = \frac{\iota_n}{\alpha_n^2} = \frac{1}{\sigma_n^2};$$

if $\alpha_n > 1$, then $\mathcal{H}_n = 1$ and

$$\frac{\iota_n}{\mathcal{H}_n^2} = \iota_n \leq \iota_n + \frac{1}{\sigma_n^2}.$$

Thus, in all cases,

$$0 \leq \frac{\iota_n}{r_\epsilon(n, 0, d_n)} \leq \iota_n + \frac{1}{\sigma_n^2} \rightarrow 0,$$

which proves the equivalence.

4. Applying the reduction from the second step to the sequences in the first step yields

$$r_\epsilon(n, m_n, d_n) = o(r_\epsilon(n, 0, d_n))$$

if and only if

$$\frac{\iota_n}{r_\epsilon(n, 0, d_n)} \rightarrow 0 \quad \text{and} \quad \frac{\beta_n}{\min\{1, \alpha_n\}} \rightarrow 0.$$

Substituting the definitions of α_n and β_n , the second condition is exactly

$$\frac{d_n/(N_n \ell_n)}{\min\{1, d_n/(n \ell_n)\}} \rightarrow 0.$$

By the supervised floor equivalence from the third step, the first condition is exactly

$$\frac{d_n}{\sqrt{n} \ell_n} \rightarrow \infty.$$

Combining the two equivalences proves the claimed characterization. □

[Lemma 29](#) identifies the regimes in which the annotation rate is asymptotically smaller than the labeled-only rate. The first condition places the alphabet beyond the labeled-only parametric scale, so treatment–covariate information can affect the leading rate. The second condition requires the combined treatment–covariate sample size to reduce the alphabet-dependent term relative to its labeled-only analogue. Together these conditions give an exact rate-level criterion for when auxiliary annotation changes the asymptotic order delivered by [Theorem 4](#).

Verification note. The displayed theorem, proposition, lemma, definition, and assumption statements, together with their mathematical proofs, are machine-checked in Lean 4. Displayed objects marked *presentation-synthesized* organize or restate the checked layer for exposition. Causal interpretation, bibliographic comparison, and surrounding econometric prose are reader-facing synthesis built around that mathematical layer.

The stable statement-to-declaration manifest is `presentation_crosswalk.json` in this presentation directory; it records the source file, fully qualified declaration, mapping status, and sorry-free status for each mapped object. The corresponding extracted statement layer is recorded in `formal_layer.json`, and proof-audit receipts are recorded in `proof_audit_cache.json`. Verification used repository commit `db544e5fa3f974dbfb320a2ecc9407c400aae9d2`, Lean 4.33.0 (release commit `d8b18978322de05a8f3dba51ef03cf5461676c17`), and Lake 5.0.0.

D Proofs of the main results

Proof of Theorem 1. Fix $0 < \epsilon < 1/2$. We prove the result with

$$c_\epsilon = \frac{1}{100}, \quad C_\epsilon = 2\epsilon^{-1}.$$

The first constant is positive, and $c_\epsilon \leq C_\epsilon$ because $0 < \epsilon < 1/2$ gives

$$\frac{1}{100} \leq \frac{2}{\epsilon} \iff \frac{\epsilon}{100} \leq 2.$$

It remains to verify the two risk inequalities uniformly over $n \geq 1$ and $d \geq 2$.

1. For the lower bound, put

$$\delta := \frac{2}{5\sqrt{n}}.$$

Then $\delta \geq 0$, $\delta \leq 2/5$, and

$$\delta^2 = \frac{4}{25n}.$$

Consider two laws on $[d] \times \{0, 1\}^2$. Under $P_\circ$, the covariate is uniform on $[d]$, the propensity is $1/2$, and both outcome regressions equal $1/2$. Under P_δ , the covariate is again uniform, the propensity is $1/2$, the control regression is $1/2$, and the treated regression is $1/2 + \delta$. Since $\delta \leq 2/5$, the perturbed treated regression lies in $[0, 1]$. Both laws belong to $\mathcal{M}_{d,\epsilon}$ from Definition 6, because their occupied-cell propensity is $1/2$, which satisfies the overlap condition in Assumption 1.

2. The two laws have the same supplied treatment-covariate table:

$$(P_\circ)_{XA}(x, a) = \frac{1}{2d} = (P_\delta)_{XA}(x, a), \quad x \in [d], \ a \in \{0, 1\}.$$

Their ATE values, computed from Definition 8, are

$$\tau(P_\circ) = 0, \quad \tau(P_\delta) = \sum_{x=1}^d \frac{1}{d} \left(\frac{1}{2} + \delta - \frac{1}{2} \right) = \delta.$$

Thus, with the testing half-separation

$$\eta_{\text{test}} := \frac{\delta}{2},$$

we have

$$2\eta_{\text{test}} = \delta = |\tau(P_{\circ}) - \tau(P_{\delta})|.$$

3. The product laws are sufficiently close in total variation. For one labeled observation, with the perturbed law compared to the null law, the chi-squared increment is

$$\chi_1^2 := \frac{(1/2)\delta^2}{(1/2)(1-1/2)} = 2\delta^2 = \frac{8}{25n}.$$

Thus

$$n\chi_1^2 = \frac{8}{25} \leq \log 2.$$

The product chi-squared divergence tensorizes as

$$1 + \chi^2(P_{\delta}^{\otimes n} \| P_{\circ}^{\otimes n}) = (1 + \chi_1^2)^n.$$

Since $1 + u \leq e^u$ for $u \geq 0$, the preceding calibration gives

$$1 + \chi^2(P_{\delta}^{\otimes n} \| P_{\circ}^{\otimes n}) \leq \exp(n\chi_1^2) \leq \exp(\log 2) = 2,$$

and hence

$$\chi^2(P_{\delta}^{\otimes n} \| P_{\circ}^{\otimes n}) \leq 1.$$

The perturbed product law is absolutely continuous with respect to the null product law because every one-observation atom under the null has positive mass. Therefore the standard Cauchy–Schwarz comparison between total variation and chi-squared divergence yields

$$\text{TV}(P_{\circ}^{\otimes n}, P_{\delta}^{\otimes n}) = \text{TV}(P_{\delta}^{\otimes n}, P_{\circ}^{\otimes n}) \leq \frac{1}{2} \sqrt{\chi^2(P_{\delta}^{\otimes n} \| P_{\circ}^{\otimes n})} \leq \frac{1}{2}.$$

4. Let $T \in \mathcal{T}_{n,d}^{\text{known}}$ be any known-marginal estimator, and define

$$f(\mathcal{L}_n) := T(\mathcal{L}_n, (P_{\circ})_{XA}) = T(\mathcal{L}_n, (P_{\delta})_{XA}),$$

where the equality uses the common supplied table. Since the two targets are separated by $2\eta_{\text{test}}$, for every sample point at least one of

$$|f - \tau(P_{\circ})| \geq \eta_{\text{test}}, \quad |f - \tau(P_{\delta})| \geq \eta_{\text{test}}$$

holds. The total-variation bound therefore gives

$$\max\{P_{\circ}^{\otimes n}(|f - \tau(P_{\circ})| \geq \eta_{\text{test}}), P_{\delta}^{\otimes n}(|f - \tau(P_{\delta})| \geq \eta_{\text{test}})\} \geq \frac{1 - \text{TV}(P_{\circ}^{\otimes n}, P_{\delta}^{\otimes n})}{2} \geq \frac{1}{4}.$$

For either law $P_{\star} \in \{P_{\circ}, P_{\delta}\}$,

$$\eta_{\text{test}}^2 P_{\star}^{\otimes n}(|f - \tau(P_{\star})| \geq \eta_{\text{test}}) \leq \mathbb{E}_{P_{\star}^{\otimes n}} \left[(T(\mathcal{L}_n, (P_{\star})_{XA}) - \tau(P_{\star}))^2 \right].$$

Taking the larger of the two risks, then the supremum over $\mathcal{M}_{d,\epsilon}$ and the infimum in [Definition 15](#), yields

$$R_{\epsilon}^{\text{known}}(n, d) \geq \frac{\eta_{\text{test}}^2}{4} = \frac{\delta^2}{16} = \frac{1}{100n}.$$

5. For the upper bound, take any $P \in \mathcal{M}_{d,\epsilon}$. For a supplied table $\Lambda = (\Lambda_{x,a})_{x,a}$, define the known-marginal score on a record (x, a, y) by

$$\psi_\Lambda(x, a, y) := \begin{cases} (\Lambda_{x,0} + \Lambda_{x,1})/\Lambda_{x,1}, & y = 1, a = 1, \\ -(\Lambda_{x,0} + \Lambda_{x,1})/\Lambda_{x,0}, & y = 1, a = 0, \\ 0, & y = 0, \end{cases}$$

with any ratio having denominator 0 assigned value 0. The raw estimator is

$$\bar{\psi}_\Lambda(\mathcal{L}_n) := \frac{1}{n} \sum_{i=1}^n \psi_\Lambda(X_i, A_i, Y_i),$$

and the estimator used in the risk is its projection to $[-1, 1]$. This projection is measurable on the finite sample space and weakly decreases squared loss because $\tau(P) \in [-1, 1]$.

6. Now specialize the table to the true table P_{XA} . Write

$$p_x = P(X = x), \quad s_{ax} = P(X = x, A = a), \quad q_{ax} = P(X = x, A = a, Y = 1).$$

If $p_x > 0$, overlap gives

$$\epsilon p_x \leq s_{1x}, \quad \epsilon p_x \leq s_{0x};$$

if $p_x = 0$, then $s_{0x} = s_{1x} = q_{0x} = q_{1x} = 0$. Hence

$$\mathbb{E}_P[\psi_{P_{XA}}(X, A, Y)] = \sum_x \left(q_{1x} \frac{p_x}{s_{1x}} - q_{0x} \frac{p_x}{s_{0x}} \right) = \sum_x p_x (\mu_{1x} - \mu_{0x}) = \tau(P),$$

with zero-mass cells contributing 0. Similarly,

$$\begin{aligned} \mathbb{E}_P[\psi_{P_{XA}}(X, A, Y)^2] &= \sum_x \left\{ q_{1x} \left(\frac{p_x}{s_{1x}} \right)^2 + q_{0x} \left(\frac{p_x}{s_{0x}} \right)^2 \right\} \\ &\leq \sum_x \left\{ \frac{p_x^2}{s_{1x}} + \frac{p_x^2}{s_{0x}} \right\} \leq \sum_x 2\epsilon^{-1} p_x = 2\epsilon^{-1}. \end{aligned}$$

Thus

$$\text{Var}_P(\psi_{P_{XA}}(X, A, Y)) \leq 2\epsilon^{-1}.$$

7. By the labeled product sampling in [Assumption 2](#),

$$\mathbb{E}_{P^{\otimes n}}[\bar{\psi}_{P_{XA}}(\mathcal{L}_n)] = \tau(P), \quad \text{Var}_{P^{\otimes n}}(\bar{\psi}_{P_{XA}}(\mathcal{L}_n)) \leq \frac{2\epsilon^{-1}}{n}.$$

After clipping, the squared risk is bounded by the same quantity. Taking the supremum over $P \in \mathcal{M}_{d,\epsilon}$ and then the infimum over estimators in [Definition 15](#) gives

$$R_\epsilon^{\text{known}}(n, d) \leq \frac{2\epsilon^{-1}}{n} = \frac{C_\epsilon}{n}.$$

Together with the lower bound, this proves the claimed inequalities for all $n \geq 1$ and $d \geq 2$. □

Proof of Theorem 2. Fix $0 < \epsilon < 1/2$.

1. By Lemma 14, choose a universal constant $C_{\text{tr}} > 0$ such that, for every $n \geq 1$, $m \geq 0$, $d \geq 2$, and $P \in \mathcal{M}_{d,\epsilon}$,

$$\mathbb{E}_{P^{\otimes n} \otimes P_{XA}^{\otimes m}} \left[\left\{ \hat{\tau}_{n,m,d}^{\text{mix}} - \tau(P) \right\}^2 \right] \leq 2 \mathcal{R}_{\text{hyb}}^{\text{Pois}}(P; n, m, \epsilon) + \frac{C_{\text{tr}}}{n}.$$

By Lemma 13, choose $C_{\text{Pois},\epsilon} > 0$, depending only on ϵ , such that, for the same range of parameters and laws,

$$\mathcal{R}_{\text{hyb}}^{\text{Pois}}(P; n, m, \epsilon) \leq C_{\text{Pois},\epsilon} r_{\epsilon}(n, m, d).$$

Set

$$C_{\epsilon} := 2C_{\text{Pois},\epsilon} + C_{\text{tr}}.$$

Then $C_{\epsilon} > 0$.

2. It remains to verify the displayed bound uniformly over P . Fix $n \geq 1$, $m \geq 0$, and $d \geq 2$. For any $P \in \mathcal{M}_{d,\epsilon}$, the two estimates above give

$$\begin{aligned} \mathbb{E}_{P^{\otimes n} \otimes P_{XA}^{\otimes m}} \left[\left\{ \hat{\tau}_{n,m,d}^{\text{mix}} - \tau(P) \right\}^2 \right] &\leq 2 \mathcal{R}_{\text{hyb}}^{\text{Pois}}(P; n, m, \epsilon) + \frac{C_{\text{tr}}}{n} \\ &\leq 2C_{\text{Pois},\epsilon} r_{\epsilon}(n, m, d) + C_{\text{tr}} r_{\epsilon}(n, m, d). \end{aligned}$$

The last inequality uses the elementary rate comparison

$$\frac{1}{n} \leq r_{\epsilon}(n, m, d),$$

which follows directly from Definition 12, since

$$\frac{1}{n} \leq \min \left\{ 1, \frac{1}{n} + \frac{d^2}{(n+m)^2 \ell_n^2} \right\} = r_{\epsilon}(n, m, d)$$

for $n \geq 1$. Therefore

$$\mathbb{E}_{P^{\otimes n} \otimes P_{XA}^{\otimes m}} \left[\left\{ \hat{\tau}_{n,m,d}^{\text{mix}} - \tau(P) \right\}^2 \right] \leq (2C_{\text{Pois},\epsilon} + C_{\text{tr}}) r_{\epsilon}(n, m, d) = C_{\epsilon} r_{\epsilon}(n, m, d).$$

3. Since the preceding bound holds for every $P \in \mathcal{M}_{d,\epsilon}$, taking the supremum over the model class yields

$$\sup_{P \in \mathcal{M}_{d,\epsilon}} \mathbb{E}_{P^{\otimes n} \otimes P_{XA}^{\otimes m}} \left[\left\{ \hat{\tau}_{n,m,d}^{\text{mix}} - \tau(P) \right\}^2 \right] \leq C_{\epsilon} r_{\epsilon}(n, m, d).$$

The rate is positive for $n \geq 1$ by Definition 12, so the right-hand side is a well-defined finite real number. The asserted uniform bound follows for all $n \geq 1$, $m \geq 0$, and $d \geq 2$. □

Proof of Theorem 3. 1. Fix $0 < \epsilon < 1/2$. By Lemma 25, there are positive numerical constants

$$C_{\text{fix}} > 0, \quad c_{\text{risk}} > 0,$$

and, at this ϵ , calibration constants as in [Definition 19](#), together with

$$\rho \in (0, 1), \quad c_{\text{floor}} > 0, \quad c_{\text{floor}} \leq c_{\text{risk}}.$$

Write $C_d > 0$ for the dimension constant in this calibration, and write $c_{\text{gap}} > 0$ for the calibration's dual-gap constant. For every n, m, d with $n \geq 1$ and $d \geq 2$, [Lemma 25](#) supplies a handle H satisfying

$$\mathcal{L}_{\Pi_0(H)}(P_{XA}) = \mathcal{L}_{\Pi_1(H)}(P_{XA})$$

and the floor certificate

$$c_{\text{floor}} \frac{1}{n} \leq R_\epsilon(n, m, d).$$

For this handle, put

$$D(n, m, d) := \min \left\{ 1, \frac{d^2}{(n+m)^2 \log(en)^2} \right\}$$

and

$$\Delta_H := c_{\text{gap}} k(H) a(H),$$

where $k(H)$ and $a(H)$ are the rare-cell count and shift in [Definition 19](#). In the finite-atom branch, the same cited certificate gives the nonparametric dimension comparison

$$C_d D(n, m, d) \leq \Delta_H^2$$

and the fixed-sample lower bound attached to the canonical handle,

$$c_{\text{floor}} \Delta_H^2 \leq R_\epsilon(n, m, d).$$

The displayed lower bound follows once the handle's common-marginal witness, total-variation bound, center separation, and variance absorption satisfy the hypotheses of [Lemma 21](#). In the complementary branch, [Lemma 25](#) gives directly

$$C_d D(n, m, d) \leq \frac{c_{\text{floor}}}{n}.$$

2. Define

$$c_{\text{lb}} := \min \left\{ \frac{c_{\text{floor}}}{2}, \frac{C_d}{2}, \frac{c_{\text{floor}} C_d}{2} \right\}.$$

Since $c_{\text{floor}} > 0$ and $C_d > 0$, this constant is positive.

3. Fix n, m, d with $n \geq 1$ and $d \geq 2$, and take the handle H from the certificate package. Put

$$Q(n, m, d) := \frac{d^2}{(n+m)^2 \log(en)^2}, \quad D(n, m, d) = \min\{1, Q(n, m, d)\}.$$

Both $Q(n, m, d)$ and $D(n, m, d)$ are nonnegative. The frontier rate in [Definition 12](#) satisfies

$$r_\epsilon(n, m, d) = \min \left\{ 1, \frac{1}{n} + Q(n, m, d) \right\} \leq \frac{1}{n} + D(n, m, d).$$

Indeed, if $Q(n, m, d) \leq 1$, the right side is $1/n + Q(n, m, d)$, which dominates the minimum. If $Q(n, m, d) > 1$, then $D(n, m, d) = 1$, and the left side is at most $1 \leq 1/n + 1$.

4. Suppose first that H is in the finite-atom branch. Since $c_{\text{floor}} > 0$, multiplying the finite-branch dimension comparison by c_{floor} and then using the transferred fixed-sample lower bound gives

$$c_{\text{floor}} C_d D(n, m, d) \leq c_{\text{floor}} \Delta_H^2 \leq R_\epsilon(n, m, d).$$

Together with the floor inequality,

$$c_{\text{floor}} \frac{1}{n} \leq R_\epsilon(n, m, d),$$

this supplies the two lower bounds used in the averaging step. Using the definition of c_{lb} and the bound from the previous step,

$$\begin{aligned} c_{\text{lb}} r_\epsilon(n, m, d) &\leq c_{\text{lb}} \left(\frac{1}{n} + D(n, m, d) \right) \\ &= c_{\text{lb}} \frac{1}{n} + c_{\text{lb}} D(n, m, d) \\ &\leq \frac{c_{\text{floor}}}{2} \frac{1}{n} + \frac{c_{\text{floor}} C_d}{2} D(n, m, d) \\ &= \frac{1}{2} \left(c_{\text{floor}} \frac{1}{n} + c_{\text{floor}} C_d D(n, m, d) \right) \\ &\leq R_\epsilon(n, m, d). \end{aligned}$$

Thus the asserted lower bound holds in the finite-atom branch.

5. It remains to treat the complementary branch. In this branch, the certificate clause in [Lemma 25](#) gives directly

$$C_d D(n, m, d) \leq \frac{c_{\text{floor}}}{n} = c_{\text{floor}} \frac{1}{n}.$$

Together with the floor inequality

$$c_{\text{floor}} \frac{1}{n} \leq R_\epsilon(n, m, d),$$

we obtain

$$\begin{aligned} c_{\text{lb}} r_\epsilon(n, m, d) &\leq c_{\text{lb}} \left(\frac{1}{n} + D(n, m, d) \right) \\ &= c_{\text{lb}} \frac{1}{n} + c_{\text{lb}} D(n, m, d) \\ &\leq \frac{c_{\text{floor}}}{2} \frac{1}{n} + \frac{C_d}{2} D(n, m, d) \\ &\leq \frac{1}{2} \left(c_{\text{floor}} \frac{1}{n} + c_{\text{floor}} \frac{1}{n} \right) \\ &= c_{\text{floor}} \frac{1}{n} \\ &\leq R_\epsilon(n, m, d). \end{aligned}$$

This proves the risk lower bound in the complementary branch as well.

6. Combining the two branches, for every n, m, d with $n \geq 1$, $m \geq 0$, and $d \geq 2$, the handle H supplies priors $\Pi_0(H)$ and $\Pi_1(H)$ with common random treatment-covariate marginal law and satisfies

$$c_{\text{lb}} r_\epsilon(n, m, d) \leq R_\epsilon(n, m, d).$$

Taking the theorem's lower-bound constant c_ϵ to be c_{lb} , and using $c_{\text{lb}} > 0$ with the fixed calibration constants chosen above, gives the asserted conclusion. $\square$

Proof of Theorem 4. 1. By Theorem 3, choose a constant $c_\epsilon > 0$ such that, for every $n \geq 1$, $m \geq 0$, and $d \geq 2$,

$$c_\epsilon r_\epsilon(n, m, d) \leq R_\epsilon(n, m, d).$$

By Theorem 2, choose $C_\epsilon^{\text{up}} > 0$ such that

$$\sup_{P \in \mathcal{M}_{d,\epsilon}} E_{P^{\otimes n} \otimes P_{XA}^{\otimes m}} \left[\left\{ \hat{\tau}_{n,m,d}^{\text{mix}} - \tau(P) \right\}^2 \right] \leq C_\epsilon^{\text{up}} r_\epsilon(n, m, d)$$

for the same range of n, m, d . Set

$$C_\epsilon := \max\{c_\epsilon, C_\epsilon^{\text{up}}\}.$$

Then $0 < c_\epsilon \leq C_\epsilon$. The estimator $\hat{\tau}_{n,m,d}^{\text{mix}}$ is measurable, so the definition of $R_\epsilon(n, m, d)$ in Definition 11 gives

$$R_\epsilon(n, m, d) \leq \sup_{P \in \mathcal{M}_{d,\epsilon}} E_{P^{\otimes n} \otimes P_{XA}^{\otimes m}} \left[\left\{ \hat{\tau}_{n,m,d}^{\text{mix}} - \tau(P) \right\}^2 \right].$$

Combining the last two displays, and using $r_\epsilon(n, m, d) > 0$, yields

$$R_\epsilon(n, m, d) \leq C_\epsilon^{\text{up}} r_\epsilon(n, m, d) \leq C_\epsilon r_\epsilon(n, m, d).$$

Together with the lower bound from Theorem 3, this proves the finite-sample bracket.

2. The unannotated endpoint is the specialization $m = 0$ of Definition 12. Since $N = n$ in that case,

$$r_\epsilon(n, 0, d) = \min \left\{ 1, \frac{1}{n} + \frac{d^2}{n^2 \log^2(en)} \right\}.$$

3. The known-marginal limit is exactly the conclusion of Theorem 5: for fixed $n \geq 1$ and $d \geq 2$,

$$\lim_{m \rightarrow \infty} R_\epsilon(n, m, d) = R_\epsilon^{\text{known}}(n, d).$$

4. It remains to record the sequence consequences. Fix sequences (m_n) and (d_n) with $d_n \geq 2$, and set

$$\begin{aligned} \rho_n &:= R_\epsilon(n, m_n, d_n), & q_n &:= r_\epsilon(n, m_n, d_n), \\ \rho_n^0 &:= R_\epsilon(n, 0, d_n), & q_n^0 &:= r_\epsilon(n, 0, d_n). \end{aligned}$$

The finite-sample bracket gives, for all $n \geq 1$,

$$c_\epsilon q_n \leq \rho_n \leq C_\epsilon q_n, \quad c_\epsilon q_n^0 \leq \rho_n^0 \leq C_\epsilon q_n^0.$$

Since all four sequences are nonnegative, these two-sided bounds imply

$$(\rho_n) = \Theta(q_n), \quad (\rho_n^0) = \Theta(q_n^0).$$

Indeed, the upper inequalities give the O -bounds in one direction, and the lower inequalities give $q_n \leq c_\epsilon^{-1} \rho_n$ and $q_n^0 \leq c_\epsilon^{-1} \rho_n^0$.

With

$$N_n = n + m_n, \quad b_n := \frac{d_n}{N_n \ell_n},$$

[Definition 12](#) gives

$$q_n = \min \left\{ 1, \frac{1}{n} + b_n^2 \right\}.$$

Because $n^{-1} \rightarrow 0$, this display satisfies

$$q_n \rightarrow 0 \iff b_n \rightarrow 0.$$

The implication from right to left follows by continuity of $x \mapsto \min\{1, x\}$; the implication from left to right follows because eventually $q_n < 1/2$, hence

$$q_n = \frac{1}{n} + b_n^2$$

eventually, so $b_n^2 \rightarrow 0$. Using $(\rho_n) = \Theta(q_n)$, we obtain

$$R_\epsilon(n, m_n, d_n) \rightarrow 0 \iff \frac{d_n}{N_n \ell_n} \rightarrow 0.$$

Next,

$$q_n = O(n^{-1}) \iff b_n^2 = O(n^{-1}) \iff d_n = O\left(\frac{N_n \ell_n}{\sqrt{n}}\right).$$

For the first equivalence, the term n^{-1} is already present in q_n , and the same eventual equality $q_n = n^{-1} + b_n^2$ applies after the $O(n^{-1})$ bound forces $q_n \rightarrow 0$. The second equivalence is obtained by taking square roots of nonnegative sequences and multiplying by $N_n \ell_n$. Transferring through $(\rho_n) = \Theta(q_n)$ gives

$$R_\epsilon(n, m_n, d_n) = O(n^{-1}) \iff d_n = O\left(\frac{N_n \ell_n}{\sqrt{n}}\right).$$

Finally put

$$a_n := \frac{d_n}{n \ell_n}.$$

Then

$$q_n = \min \left\{ 1, \frac{1}{n} + b_n^2 \right\}, \quad q_n^0 = \min \left\{ 1, \frac{1}{n} + a_n^2 \right\}, \quad 0 \leq b_n \leq a_n.$$

By [Lemma 29](#), the numerical strict-improvement criterion for these two rates is

$$q_n = o(q_n^0)$$

if and only if both

$$\frac{d_n}{\sqrt{n} \ell_n} \rightarrow \infty$$

and

$$\frac{b_n}{\min\{1, a_n\}} \rightarrow 0.$$

To see this, observe first that $q_n \geq n^{-1}$, so $q_n = o(q_n^0)$ forces $n^{-1} = o(q_n^0)$, which is equivalent to $a_n \sqrt{n} = d_n / (\sqrt{n} \ell_n) \rightarrow \infty$. Under this condition the floor term n^{-1} is negligible in q_n^0 , and

$$q_n^0 \asymp \min\{1, a_n^2\}.$$

Since $0 \leq b_n \leq a_n$, the remaining requirement $\min\{1, n^{-1} + b_n^2\} = o(q_n^0)$ is equivalent to

$$b_n^2 = o(\min\{1, a_n^2\}),$$

which, for nonnegative sequences, is the same as

$$\frac{b_n}{\min\{1, a_n\}} \rightarrow 0.$$

Substituting the definitions of a_n and b_n , this last condition is exactly

$$\frac{d_n / (N_n \ell_n)}{\min\{1, d_n / (n \ell_n)\}} \rightarrow 0.$$

The two-sided bounds $(\rho_n) = \Theta(q_n)$ and $(\rho_n^0) = \Theta(q_n^0)$ transfer $q_n = o(q_n^0)$ precisely to

$$R_\epsilon(n, m_n, d_n) = o(R_\epsilon(n, 0, d_n)),$$

which proves the stated strict-improvement equivalence. □

Proof of Theorem 5. Let

$$\mathcal{Z}_n := ([d] \times \{0, 1\}^2)^n, \quad \mathcal{C}_d := [d] \times \{0, 1\}.$$

We use the compact coordinate space

$$\Theta_{\epsilon, d} = \left\{ (\pi, \varrho, \omega) : \pi \in \mathbf{Prob}([d]), \varrho_x \in [\epsilon, 1 - \epsilon], \omega_{\alpha x} \in [0, 1] \right\}.$$

The assumptions $d \geq 2$ and $0 < \epsilon < 1/2$ make this space nonempty. For $\theta = (\pi, \varrho, \omega) \in \Theta_{\epsilon, d}$, set

$$P_\theta(x, \alpha, y) = \pi_x \varrho_x^\alpha (1 - \varrho_x)^{1-\alpha} \omega_{\alpha x}^y (1 - \omega_{\alpha x})^{1-y},$$

and define

$$\lambda_\theta(z) = \prod_{i=1}^n P_\theta(z_i), \quad \xi_\theta(x, \alpha) = \sum_{y \in \{0, 1\}} P_\theta(x, \alpha, y), \quad v_\theta = \sum_{x \in [d]} \pi_x (\omega_{1x} - \omega_{0x}).$$

Also,

$$-1 = \sum_x \pi_x (-1) \leq \sum_x \pi_x (\omega_{1x} - \omega_{0x}) \leq \sum_x \pi_x (1) = 1,$$

so $v_\theta \in [-1, 1]$ for every θ .

1. Define the finite-side values

$$\mathfrak{R}_m = \inf_{\varphi: \mathcal{Z}_n \times \mathcal{C}_d^m \rightarrow [-1,1]} \sup_{\theta \in \Theta_{\epsilon,d}} \sum_{z \in \mathcal{Z}_n} \lambda_\theta(z) \sum_{u \in \mathcal{C}_d^m} \left(\prod_{j=1}^m \xi_\theta(u_j) \right) (\varphi(z, u) - v_\theta)^2$$

and

$$\mathfrak{R}_\star = \inf_{\psi: \mathcal{Z}_n \times \text{Prob}(\mathcal{C}_d) \rightarrow [-1,1]} \sup_{\theta \in \Theta_{\epsilon,d}} \sum_{z \in \mathcal{Z}_n} \lambda_\theta(z) (\psi(z, \xi_\theta) - v_\theta)^2.$$

With the preceding coordinates, [Lemma 27](#) gives

$$\lim_{m \rightarrow \infty} \mathfrak{R}_m = \mathfrak{R}_\star.$$

2. Let $\tilde{\mathfrak{R}}_\star$ be the measurable ambient exact-table value

$$\tilde{\mathfrak{R}}_\star = \inf_{\psi} \sup_{\theta \in \Theta_{\epsilon,d}} \sum_{z \in \mathcal{Z}_n} \lambda_\theta(z) (\psi(z, \xi_\theta) - v_\theta)^2,$$

where the infimum ranges over all rules $\psi: \mathcal{Z}_n \times \mathbb{R}^{\mathcal{C}_d} \rightarrow [-1, 1]$ whose section $w \mapsto \psi(z, w)$ is Borel measurable for each z . The target range established above gives $v_\theta \in [-1, 1]$, and [Lemma 27](#) supplies the convergence

$$\lim_{m \rightarrow \infty} \mathfrak{R}_m = \mathfrak{R}_\star.$$

The measurable ambient exact-table value is tied to this limiting sequence by two comparisons. The exact-side value infimizes over all rules on $\text{Prob}(\mathcal{C}_d)$, while the ambient value infimizes over rules on $\mathbb{R}^{\mathcal{C}_d}$ with Borel-measurable sections. Restricting any ambient rule to $\text{Prob}(\mathcal{C}_d)$ gives an admissible exact-side rule, so the exact-side class is at least as large after restriction, and therefore

$$\mathfrak{R}_\star \leq \tilde{\mathfrak{R}}_\star.$$

Conversely, [Lemma 28](#) gives $\tilde{\mathfrak{R}}_\star \leq \mathfrak{R}_m$ for every $m \geq 0$. For completeness, its measurable conditional-average construction in the present coordinates is as follows. Fix an integer $m \geq 0$ and an empirical-side rule $\varphi: \mathcal{Z}_n \times \mathcal{C}_d^m \rightarrow [-1, 1]$. Define first the polynomial conditional average on the ambient table space by

$$\psi_\varphi^\circ(z, w) = \sum_{u \in \mathcal{C}_d^m} \left(\prod_{j=1}^m w(u_j) \right) \varphi(z, u), \quad w \in \mathbb{R}^{\mathcal{C}_d}.$$

This function is Borel measurable in w for each fixed z . Use its clipped version as the ambient exact-table rule,

$$\psi_\varphi(z, w) = \min\{1, \max\{-1, \psi_\varphi^\circ(z, w)\}\}, \quad w \in \mathbb{R}^{\mathcal{C}_d}.$$

Then $\psi_\varphi: \mathcal{Z}_n \times \mathbb{R}^{\mathcal{C}_d} \rightarrow [-1, 1]$ is admissible for $\tilde{\mathfrak{R}}_\star$. When $w = r \in \text{Prob}(\mathcal{C}_d)$, the coefficients $\prod_{j=1}^m r(u_j)$ form a probability vector on $\mathcal{C}_d^m$, and since φ takes values in $[-1, 1]$, the polynomial average already lies in $[-1, 1]$. Hence, on probability tables,

$$\psi_\varphi(z, r) = \sum_{u \in \mathcal{C}_d^m} \left(\prod_{j=1}^m r(u_j) \right) \varphi(z, u).$$

For every θ and z , Jensen's inequality for the product law $\xi_\theta^{\otimes m}$ gives

$$(\psi_\varphi(z, \xi_\theta) - v_\theta)^2 = \left(\sum_{u \in \mathcal{C}_d^m} \left(\prod_{j=1}^m \xi_\theta(u_j) \right) (\varphi(z, u) - v_\theta) \right)^2 \leq \sum_{u \in \mathcal{C}_d^m} \left(\prod_{j=1}^m \xi_\theta(u_j) \right) (\varphi(z, u) - v_\theta)^2.$$

Multiplying by $\lambda_\theta(z)$, summing over z , and then taking the supremum over θ gives

$$\sup_\theta \sum_{z \in \mathcal{Z}_n} \lambda_\theta(z) (\psi_\varphi(z, \xi_\theta) - v_\theta)^2 \leq \sup_\theta \sum_{z \in \mathcal{Z}_n} \lambda_\theta(z) \sum_{u \in \mathcal{C}_d^m} \left(\prod_{j=1}^m \xi_\theta(u_j) \right) (\varphi(z, u) - v_\theta)^2.$$

Taking the infimum over φ yields

$$\tilde{\mathfrak{R}}_\star \leq \mathfrak{R}_m \quad \text{for every } m \geq 0.$$

Taking the limit along the displayed convergence yields

$$\tilde{\mathfrak{R}}_\star \leq \lim_{m \rightarrow \infty} \mathfrak{R}_m = \mathfrak{R}_\star.$$

The two inequalities identify the ambient measurable exact-table value with the exact-side value:

$$\tilde{\mathfrak{R}}_\star = \mathfrak{R}_\star.$$

3. We use here the standing theorem hypotheses $d \geq 2$ and $0 < \epsilon < 1/2$, which are also the parameter-range clauses in the definition of $\mathcal{M}_{d,\epsilon}$. Under these hypotheses, the compact coordinates parametrize the laws in $\mathcal{M}_{d,\epsilon}$ from [Definition 6](#). For every $\theta \in \Theta_{\epsilon,d}$, the law P_θ satisfies occupied-cell overlap because, whenever $\pi_x > 0$, its treatment propensity is

$$\frac{\sum_y P_\theta(x, 1, y)}{\sum_{\alpha, y} P_\theta(x, \alpha, y)} = \varrho_x \in [\epsilon, 1 - \epsilon].$$

Conversely, given any $P \in \mathcal{M}_{d,\epsilon}$, choose coordinates by

$$\pi_x = p_x, \quad \varrho_x = \begin{cases} e_x, & p_x > 0, \\ \epsilon, & p_x = 0, \end{cases} \quad \omega_{\alpha x} = \mu_{\alpha x}.$$

The value of ϱ_x in an empty cell is immaterial because every atom in that cell is multiplied by $\pi_x = 0$. For occupied cells, the overlap condition places ϱ_x in $[\epsilon, 1 - \epsilon]$, and the binary outcome regressions place $\omega_{\alpha x}$ in $[0, 1]$. These coordinates recover the atom probabilities,

$$P(x, \alpha, y) = \pi_x \varrho_x^\alpha (1 - \varrho_x)^{1-\alpha} \omega_{\alpha x}^y (1 - \omega_{\alpha x})^{1-y},$$

with the empty-cell case giving zero on both sides. Thus the map $\theta \mapsto P_\theta$ is surjective onto the model class.

Under this parametrization, the three quantities entering the known-marginal risk are transported by

$$P_\theta^{\otimes n}(z) = \lambda_\theta(z), \quad (P_\theta)_{XA}(x, \alpha) = \xi_\theta(x, \alpha), \quad \tau(P_\theta) = v_\theta.$$

The last identity follows from [Definition 8](#): on occupied cells the outcome regressions are ω_{1x} and ω_{0x} , while empty cells have coefficient $\pi_x = 0$. Hence, for every bounded measurable known-table rule ψ ,

$$\int (\psi(Z, (P_\theta)_{XA}) - \tau(P_\theta))^2 dP_\theta^{\otimes n}(Z) = \sum_{z \in \mathcal{Z}_n} \lambda_\theta(z) (\psi(z, \xi_\theta) - v_\theta)^2.$$

The known-table sample space in [Definition 15](#) is $\mathcal{Z}_n \times \mathbb{R}^{\mathcal{C}_d}$. Its bounded measurable procedures are in bijection with the ambient procedures used in $\tilde{\mathfrak{R}}_\star$. Explicitly, a paper procedure f is sent to

$$\psi_f(z, w) := f(z, w),$$

and an ambient procedure ψ is sent back to $f_\psi(z, w) := \psi(z, w)$. Because $\mathcal{Z}_n$ is finite and discrete, measurability of every section $w \mapsto \psi(z, w)$ is equivalent to measurability of the uncurried map; the two assignments are pointwise inverse and preserve the range $[-1, 1]$.

For corresponding f and ψ_f , evaluation at the known table $w = (P_\theta)_{XA} = \xi_\theta$, followed by expansion of the finite labeled product law, gives

$$\begin{aligned} \int (f(Z, (P_\theta)_{XA}) - \tau(P_\theta))^2 dP_\theta^{\otimes n}(Z) \\ = \sum_{z \in \mathcal{Z}_n} \lambda_\theta(z) (\psi_f(z, \xi_\theta) - v_\theta)^2. \end{aligned}$$

Thus the estimator bijection preserves every coordinate risk exactly. Reindexing the supremum over $P \in \mathcal{M}_{d,\epsilon}$ by the surjective coordinate map, and then taking the same infimum over the bijective procedure classes, gives

$$R_\epsilon^{\text{known}}(n, d) = \tilde{\mathfrak{R}}_\star.$$

Together with the equality $\tilde{\mathfrak{R}}_\star = \mathfrak{R}_\star$ above, this gives

$$R_\epsilon^{\text{known}}(n, d) = \mathfrak{R}_\star.$$

4. For each m , the minimax risk in [Definition 11](#) equals the empirical finite-side value:

$$R_\epsilon(n, m, d) = \mathfrak{R}_m.$$

First reindex the supremum over $P \in \mathcal{M}_{d,\epsilon}$ by the surjective map $\theta \mapsto P_\theta$. The annotation sample space is the finite product $\mathcal{Z}_n \times \mathcal{C}_d^m$, so every real-valued map on it is measurable. For any measurable estimator T in the annotation experiment, write its two-sample mean-squared risk at P_θ as

$$\mathcal{V}_T(\theta) = \int (T(Z, U) - \tau(P_\theta))^2 d(P_\theta^{\otimes n} \otimes (P_\theta)_{XA}^{\otimes m})(Z, U).$$

Expanding both finite product measures, and using $(P_\theta)_{XA} = \xi_\theta$, $P_\theta^{\otimes n} = \lambda_\theta$, and $\tau(P_\theta) = v_\theta$, gives the exact table identity

$$\mathcal{V}_{T_\varphi}(\theta) = \sum_{z \in \mathcal{Z}_n} \lambda_\theta(z) \sum_{u \in \mathcal{C}_d^m} \left(\prod_{j=1}^m \xi_\theta(u_j) \right) (\varphi(z, u) - v_\theta)^2$$

for $T_\varphi(z, u) := \varphi(z, u)$.

Thus every empirical-side procedure $\varphi : \mathcal{Z}_n \times \mathcal{C}_d^m \rightarrow [-1, 1]$ induces the paper estimator

$$T_\varphi(z, u) = \varphi(z, u),$$

viewed as an estimator in the annotation experiment. For this induced estimator, the preceding expansion gives equality of the coordinate risks for every θ , hence equality of the corresponding worst-case risks after the reindexing by θ . Since the paper infimum ranges over all measurable estimators, including these induced estimators, taking the infimum over φ gives

$$R_\epsilon(n, m, d) \leq \mathfrak{R}_m.$$

For the reverse inequality, let T be any measurable paper estimator and define the clipped empirical-side rule

$$\varphi_T(z, u) = \min\{1, \max\{-1, T(z, u)\}\}.$$

The transported identity $\tau(P_\theta) = v_\theta$, together with the target-range bound, gives the pointwise clipping inequality

$$(\varphi_T(z, u) - \tau(P_\theta))^2 \leq (T(z, u) - \tau(P_\theta))^2 \quad \text{for all } (z, u), \theta.$$

Integrating this inequality against $P_\theta^{\otimes n} \otimes (P_\theta)_{XA}^{\otimes m}$ yields

$$\int (\varphi_T(Z, U) - \tau(P_\theta))^2 d(P_\theta^{\otimes n} \otimes (P_\theta)_{XA}^{\otimes m})(Z, U) \leq \mathcal{V}_T(\theta).$$

By the same finite product expansion as above, the left side is exactly the empirical-side table risk of φ_T at θ . The comparison of worst-case risks also requires a finite bound for the unclipped estimator. Set

$$B_T := \sum_{(z, u) \in \mathcal{Z}_n \times \mathcal{C}_d^m} (|T(z, u)| + 1)^2.$$

Since $|v_\theta| \leq 1$,

$$(T(z, u) - v_\theta)^2 \leq (|T(z, u)| + 1)^2 \leq B_T$$

for every (z, u) and θ . Hence the worst-case risk of T over the reindexed model is bounded, and the integrated clipping inequality implies

$$\sup_{\theta \in \Theta_{\epsilon, d}} \int (\varphi_T(Z, U) - \tau(P_\theta))^2 d(P_\theta^{\otimes n} \otimes (P_\theta)_{XA}^{\otimes m})(Z, U) \leq \sup_{\theta \in \Theta_{\epsilon, d}} \mathcal{V}_T(\theta).$$

Taking infima over all measurable paper estimators T gives

$$\mathfrak{R}_m \leq R_\epsilon(n, m, d).$$

The two inequalities prove $R_\epsilon(n, m, d) = \mathfrak{R}_m$.

5. The preceding displayed identities give

$$\lim_{m \rightarrow \infty} R_\epsilon(n, m, d) = \lim_{m \rightarrow \infty} \mathfrak{R}_m = \mathfrak{R}_\star = R_\epsilon^{\text{known}}(n, d).$$

□

Proof of Theorem 6. 1. By Lemma 2, choose $C_0 > 0$, depending only on ϵ , such that for every d , every $u, t > 0$, and every $P \in \mathcal{M}_{d,\epsilon}$,

$$\text{Var}_P(Z^H) \leq C_0(u^{-1} + t^{-1}), \quad |E_P Z^H - \tau(P)| \leq C_0 \min\{1, d/t\}.$$

Hence, in the independent Poisson count experiment,

$$E_P\{Z^H - \tau(P)\}^2 = \text{Var}_P(Z^H) + \{E_P Z^H - \tau(P)\}^2 \leq C_0(u^{-1} + t^{-1}) + C_0^2 \min\{1, d/t\}^2.$$

Set

$$C_\epsilon = 100000(1 + C_0 + C_0^2) > 0.$$

2. Fix $n \geq 1$, $m \geq 0$, and $d \geq 2$. Let

$$M_0 = \lfloor n/3 \rfloor, \quad n_p = \left\lfloor \frac{n - M_0}{2} \right\rfloor, \quad n_f = n - M_0 - n_p, \quad m_f = m - \lfloor m/2 \rfloor,$$

and put

$$u = \frac{M_0}{8}, \quad t = \frac{n_f + m_f}{8}, \quad \mathfrak{q} = \frac{1}{n} + \frac{d^2}{(n + m)^2}.$$

The estimator is a finite sum of measurable prefix statistics with deterministic weights, so it is measurable. The class $\mathcal{M}_{d,\epsilon}$ is nonempty: the law with uniform covariate distribution, propensity 1/2, and equal outcome regressions in the two arms satisfies Assumption 1 because $0 < \epsilon < 1/2$.

3. First suppose $24 \leq n$. The block arithmetic gives

$$n \leq 4M_0, \quad n + m \leq 4(n_f + m_f),$$

and therefore $M_0 > 0$, $n_f + m_f > 0$, $u > 0$, and $t > 0$. Consequently

$$\frac{n}{32} \leq u, \quad \frac{n + m}{32} \leq t,$$

so

$$u^{-1} \leq \frac{32}{n}, \quad t^{-1} \leq \frac{32}{n + m} \leq \frac{32}{n}.$$

Also,

$$\frac{d}{t} \leq \frac{32d}{n + m},$$

and hence

$$\min\{1, d/t\}^2 \leq (d/t)^2 \leq 1024 \frac{d^2}{(n + m)^2}.$$

4. For the Poisson prefix overflow terms, the elementary bound

$$\Pr\{\text{Poi}(\Lambda_{\text{tail}}/8) > \Lambda_{\text{tail}}\} \leq \frac{8}{\Lambda_{\text{tail}}}, \quad \Lambda_{\text{tail}} \geq 1,$$

is obtained as follows. Bernstein's upper-tail inequality for a Poisson variable with mean $\Lambda_{\text{tail}}/8$, applied with deviation parameter $\Lambda_{\text{tail}}/4$, yields

$$\Pr\{\text{Poi}(\Lambda_{\text{tail}}/8) > \Lambda_{\text{tail}}\} \leq \exp(-\Lambda_{\text{tail}}/4).$$

Since $\exp(x) \geq 1 + x$,

$$\exp(-\Lambda_{\text{tail}}/4) = \frac{1}{\exp(\Lambda_{\text{tail}}/4)} \leq \frac{1}{\Lambda_{\text{tail}}/4} = \frac{4}{\Lambda_{\text{tail}}} \leq \frac{8}{\Lambda_{\text{tail}}}.$$

Applying this with $\Lambda_{\text{tail}} = M_0$ and $\Lambda_{\text{tail}} = n_f + m_f$ gives

$$\Pr\{\text{Poi}(u) > M_0\} \leq \frac{8}{M_0}, \quad \Pr\{\text{Poi}(t) > n_f + m_f\} \leq \frac{8}{n_f + m_f}.$$

Together with $n \leq 4M_0$ and $n + m \leq 4(n_f + m_f)$, this implies

$$\frac{8}{M_0} \leq \frac{32}{n}, \quad \frac{8}{n_f + m_f} \leq \frac{32}{n + m} \leq \frac{32}{n}.$$

5. Fix $P \in \mathcal{M}_{d,\epsilon}$. Under the probability law $\mathbb{P}_{P,u,t}^{\text{pref}}$, draw two independent finite prefixes: an outcome-marked prefix from P with Poisson length of mean u , and a treatment-covariate prefix from P_{XA} with Poisson length of mean t . If S_{ax} is the number of outcome-marked prefix records with $(X, A, Y) = (x, a, 1)$ and K_{ax} is the number of treatment-covariate prefix records with $(X, A) = (x, a)$, set

$$\hat{\tau}_{u,t}^{\text{IC,pref}} = \text{clip}_{[-1,1]} \left[\sum_{x=1}^d \left\{ \frac{S_{1x}}{u} \frac{K_{0x} + K_{1x} + 1}{K_{1x} + 1} - \frac{S_{0x}}{u} \frac{K_{0x} + K_{1x} + 1}{K_{0x} + 1} \right\} \right],$$

and define

$$\mathcal{R}_{\text{pref}}(P; u, t) := E_{\mathbb{P}_{P,u,t}^{\text{pref}}} \left[\left\{ \hat{\tau}_{u,t}^{\text{IC,pref}} - \tau(P) \right\}^2 \right].$$

The prefix transfer in [Lemma 4](#), applied with the present block sizes and intensities, gives

$$E_P \left[\left\{ \hat{\tau}_{n,m,d}^{\text{IC}} - \tau(P) \right\}^2 \right] \leq \mathcal{R}_{\text{pref}}(P; u, t) + 4 [\Pr\{\text{Poi}(u) > M_0\} + \Pr\{\text{Poi}(t) > n_f + m_f\}].$$

Here the fixed-sample estimator is the finite Rao–Blackwell average over admissible prefix lengths in [Definition 20](#); the displayed inequality accounts for the two overflow events, and the coefficient 4 is the squared diameter of $[-1, 1]$ for the clipped statistic and the target $\tau(P)$.

6. The Poisson-prefix counts have the Poisson count law by [Lemma 3](#): for every $x \in [d]$ and $a \in \{0, 1\}$,

$$S_{ax} \sim \text{Poi}(uq_{ax}), \quad K_{ax} \sim \text{Poi}(ts_{ax}),$$

and these coordinates are mutually independent. Thus the pushforward of $\mathbb{P}_{P,u,t}^{\text{pref}}$ under the count map is exactly the product law used in [Lemma 2](#). We first record the range of the target. The cell masses satisfy $p_x \geq 0$ and $\sum_{x=1}^d p_x = 1$, and the binary-outcome regressions satisfy $0 \leq \mu_{ax} \leq 1$. Hence

$$|\tau(P)| = \left| \sum_{x=1}^d p_x (\mu_{1x} - \mu_{0x}) \right| \leq \sum_{x=1}^d p_x |\mu_{1x} - \mu_{0x}| \leq \sum_{x=1}^d p_x = 1.$$

Thus $\tau(P) \in [-1, 1]$. Since projection onto $[-1, 1]$ cannot increase distance from a point of $[-1, 1]$, clipping gives

$$\mathcal{R}_{\text{pref}}(P; u, t) \leq E_P\{Z^H - \tau(P)\}^2.$$

Combining this with Step 1 gives

$$\mathcal{R}_{\text{pref}}(P; u, t) \leq C_0(u^{-1} + t^{-1}) + C_0^2 \min\{1, d/t\}^2.$$

7. The preceding displays imply

$$E_P\left[\{\hat{\tau}_{n,m,d}^{\text{IC}} - \tau(P)\}^2\right] \leq (64C_0 + 1024C_0^2 + 256) \left\{ \frac{1}{n} + \frac{d^2}{(n+m)^2} \right\}.$$

Indeed, the variance contribution is bounded by

$$C_0(u^{-1} + t^{-1}) \leq 64C_0 \frac{1}{n},$$

the squared-bias contribution by

$$C_0^2 \min\{1, d/t\}^2 \leq 1024C_0^2 \frac{d^2}{(n+m)^2},$$

and the overflow contribution by

$$4 \left(\frac{8}{M_0} + \frac{8}{n_f + m_f} \right) \leq 256 \frac{1}{n}.$$

Since

$$64C_0 + 1024C_0^2 + 256 \leq C_\epsilon,$$

the preceding display gives the large-sample estimate

$$E_P\left[\{\hat{\tau}_{n,m,d}^{\text{IC}} - \tau(P)\}^2\right] \leq C_\epsilon \mathfrak{q}.$$

If $\mathfrak{q} \leq 1$, then $\min\{1, \mathfrak{q}\} = \mathfrak{q}$, and hence

$$E_P\left[\{\hat{\tau}_{n,m,d}^{\text{IC}} - \tau(P)\}^2\right] \leq C_\epsilon \min\{1, \mathfrak{q}\}.$$

When $\mathfrak{q} > 1$, both the estimator and $\tau(P)$ lie in $[-1, 1]$, so

$$E_P\left[\{\hat{\tau}_{n,m,d}^{\text{IC}} - \tau(P)\}^2\right] \leq 4 \leq C_\epsilon \min\{1, \mathfrak{q}\}.$$

8. It remains to cover $n < 24$. Then $n \leq 23$, and

$$\frac{1}{23} \leq \frac{1}{n} \leq \mathfrak{q}, \quad \frac{1}{23} \leq \min\{1, \mathfrak{q}\}.$$

Again using the boundedness of the estimator and of $\tau(P)$,

$$E_P\left[\{\hat{\tau}_{n,m,d}^{\text{IC}} - \tau(P)\}^2\right] \leq 4 \leq C_\epsilon \min\{1, \mathfrak{q}\},$$

because $C_\epsilon \geq 92$.

9. Taking the supremum over $P \in \mathcal{M}_{d,\epsilon}$ in the preceding bound gives

$$\sup_{P \in \mathcal{M}_{d,\epsilon}} E_{P^{\otimes n} \otimes P_{XA}^{\otimes m}} \left[\left\{ \hat{\tau}_{n,m,d}^{\text{IC}} - \tau(P) \right\}^2 \right] \leq C_\epsilon \min \left\{ 1, \frac{1}{n} + \frac{d^2}{(n+m)^2} \right\}.$$

Finally, $\hat{\tau}_{n,m,d}^{\text{IC}}$ is an admissible measurable estimator in the class defining $R_\epsilon(n, m, d)$ in [Definition 11](#). Therefore

$$R_\epsilon(n, m, d) \leq C_\epsilon \min \left\{ 1, \frac{1}{n} + \frac{d^2}{(n+m)^2} \right\}.$$

□

Proof of [Theorem 7](#). Fix $0 < \epsilon < 1/2$.

1. By [Lemma 22](#), choose $c_{\text{lab}} > 0$ such that, for every $n \geq 1$, $m \geq 0$, and $d \geq 2$,

$$\frac{c_{\text{lab}}}{n} \leq R_\epsilon(n, m, d).$$

By [Theorem 6](#), choose $C_{\text{IC}} > 0$ such that, for every $n \geq 1$, $m \geq 0$, and $d \geq 2$,

$$R_\epsilon(n, m, d) \leq C_{\text{IC}} \min \left\{ 1, \frac{1}{n} + \frac{d^2}{(n+m)^2} \right\}.$$

Set

$$c_\epsilon := c_{\text{lab}}, \quad C_\epsilon := \max\{c_{\text{lab}}, 5C_{\text{IC}}\}.$$

Then $0 < c_\epsilon \leq C_\epsilon$.

2. For $d = 2$, the lower bound supplied by [Lemma 22](#) gives, for every $n \geq 1$ and $m \geq 0$,

$$\frac{c_\epsilon}{n} = \frac{c_{\text{lab}}}{n} \leq R_\epsilon(n, m, 2).$$

3. It remains to bound the right-hand side. Fix $n \geq 1$ and $m \geq 0$, and put

$$N := n + m.$$

Then $N \geq n \geq 1$, hence $N > 0$, and

$$n \leq N \leq N^2,$$

because $N \geq 1$. Therefore

$$\frac{4}{N^2} \leq \frac{4}{n}.$$

Using [Theorem 6](#) with $d = 2$,

$$\begin{aligned} R_\epsilon(n, m, 2) &\leq C_{\text{IC}} \min \left\{ 1, \frac{1}{n} + \frac{4}{N^2} \right\} \\ &\leq C_{\text{IC}} \left(\frac{1}{n} + \frac{4}{N^2} \right) \\ &\leq C_{\text{IC}} \frac{5}{n} = \frac{5C_{\text{IC}}}{n} \leq \frac{C_\epsilon}{n}. \end{aligned}$$

This proves the asserted two-sided bound for all $n \geq 1$ and $m \geq 0$.

□

References

- Anastasios N. Angelopoulos, Stephen Bates, Clara Fannjiang, Michael I. Jordan, and Tijana Zrnic. Prediction-powered inference. *Science*, 382(6671):669–674, 2023. doi: 10.1126/science.adi6000.
- Yilizhati Anniwaer and Yuqian Zhang. Semi-supervised inference for treatment heterogeneity. *arXiv preprint arXiv:2509.05048*, 2025.
- Susan Athey, Guido W. Imbens, and Stefan Wager. Approximate residual balancing: Debiased inference of average treatment effects in high dimensions. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 80(4):597–623, 2018. doi: 10.1111/rssb.12268.
- Heejung Bang and James M. Robins. Doubly robust estimation in missing data and causal inference models. *Biometrics*, 61(4):962–973, 2005. doi: 10.1111/j.1541-0420.2005.00377.x.
- Abhishek Chakraborty and Tianxi Cai. Efficient and adaptive linear regression in semi-supervised settings. *The Annals of Statistics*, 46(4):1541–1572, 2018. doi: 10.1214/17-AOS1594.
- Abhishek Chakraborty and Guorong Dai. A general framework for treatment effect estimation in semi-supervised and high dimensional settings. *arXiv preprint arXiv:2201.00468*, 2024.
- Abhishek Chakraborty, Jiarui Lu, T. Tony Cai, and Hongzhe Li. High dimensional m-estimation with missing outcomes: A semi-parametric framework. *arXiv preprint arXiv:1911.11345*, 2019.
- Abhishek Chakraborty, Guorong Dai, and Raymond J. Carroll. Semi-supervised quantile estimation: Robust and efficient inference in high dimensional settings. *arXiv preprint arXiv:2201.10208*, 2022.
- David Cheng, Ashwin N. Ananthakrishnan, and Tianxi Cai. Robust and efficient semi-supervised estimation of average treatment effects with application to electronic health records data. *Biometrics*, 77(2):413–423, 2021. doi: 10.1111/biom.13298.
- Victor Chernozhukov, Denis Chetverikov, Mert Demirer, Esther Duflo, Christian Hansen, Whitney Newey, and James Robins. Double/debiased machine learning for treatment and structural parameters. *The Econometrics Journal*, 21(1):C1–C68, 2018. doi: 10.1111/ectj.12097.
- Siyi Deng, Yang Ning, Jiwei Zhao, and Heping Zhang. Optimal and safe estimation for high-dimensional semi-supervised learning. *Journal of the American Statistical Association*, 119(548):2748–2759, 2024. doi: 10.1080/01621459.2023.2277409.
- Max H. Farrell. Robust inference on average treatment effects with possibly more covariates than observations. *Journal of Econometrics*, 189(1):1–23, 2015.
- Jinyong Hahn. On the role of the propensity score in efficient semiparametric estimation of average treatment effects. *Econometrica*, 66(2):315–331, 1998. doi: 10.2307/2998560.
- Keisuke Hirano, Guido W. Imbens, and Geert Ridder. Efficient estimation of average treatment effects using the estimated propensity score. *Econometrica*, 71(4):1161–1189, 2003. doi: 10.1111/1468-0262.00442.

- Jue Hou, Rajarshi Mukherjee, and Tianxi Cai. Efficient and robust semi-supervised estimation of average treatment effect with partially annotated treatment and response. *Journal of Machine Learning Research*, 26(40):1–77, 2025. URL <https://www.jmlr.org/papers/v26/23-1587.html>.
- Guido W. Imbens and Donald B. Rubin. *Causal Inference for Statistics, Social, and Biomedical Sciences: An Introduction*. Cambridge University Press, Cambridge, 2015. doi: 10.1017/CBO9781139025751.
- Jiantao Jiao, Kartik Venkat, Yanjun Han, and Tsachy Weissman. Minimax estimation of functionals of discrete distributions. *IEEE Transactions on Information Theory*, 61(5):2835–2885, 2015. doi: 10.1109/TIT.2015.2412945.
- Nathan Kallus and Xiaojie Mao. On the role of surrogates in the efficient estimation of treatment effects with limited outcome data. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 87(2):480–509, 2025.
- Masahiro Kato. Semi-supervised treatment effect estimation with unlabeled covariates via generalized riesz regression. *arXiv preprint arXiv:2511.08303*, 2025.
- Edward H. Kennedy. Semiparametric theory and empirical processes in causal inference. In *Statistical Causal Inferences and Their Applications in Public Health Research*, pages 141–167. Springer, 2016. doi: 10.1007/978-3-319-41259-7_8.
- Alon Orlitsky, Ananda Theertha Suresh, and Yihong Wu. Optimal prediction of the number of unseen species. *Proceedings of the National Academy of Sciences*, 113(47):13283–13288, 2016. doi: 10.1073/pnas.1607774113.
- James M. Robins, Andrea Rotnitzky, and Lue Ping Zhao. Estimation of regression coefficients when some regressors are not always observed. *Journal of the American Statistical Association*, 89(427):846–866, 1994. doi: 10.1080/01621459.1994.10476818.
- Paul R. Rosenbaum and Donald B. Rubin. The central role of the propensity score in observational studies for causal effects. *Biometrika*, 70(1):41–55, 1983. doi: 10.1093/biomet/70.1.41.
- Donald B. Rubin. Estimating causal effects of treatments in randomized and nonrandomized studies. *Journal of Educational Psychology*, 66(5):688–701, 1974. doi: 10.1037/h0037350.
- Gregory Valiant and Paul Valiant. Estimating the unseen: Improved estimators for entropy and other properties. *Journal of the ACM*, 64(6):37:1–37:41, 2017. doi: 10.1145/3125643.
- Yihong Wu and Pengkun Yang. Minimax rates of entropy estimation on large alphabets via best polynomial approximation. *IEEE Transactions on Information Theory*, 62(6):3702–3720, 2016. doi: 10.1109/TIT.2016.2548468.
- Zichun Xu, Daniela Witten, and Ali Shojaie. A unified framework for semiparametrically efficient semi-supervised learning. *arXiv preprint arXiv:2502.17741*, 2025.
- Zhenghao Zeng, Sivaraman Balakrishnan, Yanjun Han, and Edward H. Kennedy. Causal inference with high-dimensional discrete covariates. *arXiv preprint arXiv:2405.00118*, 2026.

Anru Zhang, Lawrence D. Brown, and T. Tony Cai. Semi-supervised inference: General theory and estimation of means. *The Annals of Statistics*, 47(5):2538–2566, 2019. doi: 10.1214/18-AOS1756.